

General Curriculum in Computer Science

General Curriculum in Computer Science - BS in Computer Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CSC/CPE 1000	Computing Majors Orientation	1
CSC/CPE 1024	Introduction to Computing	2
MATH 1261	Calculus I (2) ¹	4
or MATH 1264	or Calculus for Data Science I	
Life Science Support Elective (5B)	¹	4

General Education Requirement (1A)	3
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Units	14
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Term 2

CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory	
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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or PHYS 1141		
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or General Physics I		
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MATH 1262	Calculus II	4
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or MATH 1265		
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or Calculus for Data Science II		
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General Education Requirement (1B) 3

Units 15

Second Year

Term 1

CSC 2001 Data Structures 4

& 2001L and Data Structures Laboratory

MATH 1151 Linear Algebra 3

General Education Requirement (1C) 3

General Education Requirement (3A) 3

Units

13

Term 2

CPE 2300	Introduction to Computer Systems	3
CSC/CPE 2050	System Software Mechanics	3
MATH 2031	Transition to Advanced Mathematics	3
General Education Requirement (3B)		3
General Education Requirement (4A)		3

Units

15

Third Year

Term 1

CSC 3001	Modern Application Development	4
CSC/CPE 3201	Introduction to Computer Security	3
CSC 3449	Algorithms and Complexity	4
General Education Requirement (4B)		3
General Education Requirement (6)		3
Units		17

Term 2

CSC 3100	Software Engineering	4
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CSC 3300	Programming Languages	3
STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
Technical Elective	^{2, 3, 4}	3-4
General Education Requirement (Upper-Division 4)		3
Units		17

Fourth Year

Term 1

CSC/CPE 4553	Introduction to Operating Systems	3
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PHIL 3323	Ethics, Science, and Technology (Upper-Division 3)	1	3
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WGQS 3350	Gender, Race, Culture, Science, and Technology		4
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or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies		
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Technical Elective	2, 3, 4		3-4
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Units			14
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Term 2

Technical Elective	2, 3, 4		3-4
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Technical Elective	2, 3, 4		0-4
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Approved External Elective	2, 3, 4	3
Approved External Elective	2, 3, 4	2-3
Senior Project	2	2-8
Units		15
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of 23 units is required to complete the concentration. Unit selection for requirements will vary based on students' selection of senior project, technical electives, and approved external electives.

3

A combined maximum of 6 units may be taken from CPE 4491, CPE 4492, CPE 4493, CSC 4091, CSC 4092, CSC 4093, CSC 4191, CSC 4192, CSC 4193, CSC 4291, CSC 4292, CSC 4293, CSC 4400, CSC 4495, CSC 4691, CSC 4692, CSC 4693, CSC 4791, CSC 4792, CSC 4793, CSC 4891, CSC 4892, CSC 4893, CSC 4991, CSC 4992, CSC 4993, DATA 4720, ENGR 2995, and ENGR 4995; of which up to a combined 4 units may be taken from CPE 4491, CSC 4091, CSC 4191, CSC 4291, CSC 4400, CSC 4495, CSC 4691, CSC 4791, CSC 4891, CSC 4991, and DATA 4720

4

Courses used to satisfy any other Major or Support requirement are not allowed to count toward this electives requirement. Credit/No Credit grading is not allowed.

Artificial Intelligence and Machine Learning

Artificial Intelligence and Machine Learning Concentration - BS in Computer Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CSC/CPE 1000	Computing Majors Orientation	1
CSC/CPE 1024	Introduction to Computing	2
MATH 1261	Calculus I (2) ¹	4
or MATH 1264	or Calculus for Data Science I	
Life Science Support Elective (5B) ¹		4
General Education Requirement (1A)		3
Units		14

Term 2

CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory	
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
or PHYS 1141	or General Physics I	
MATH 1262	Calculus II	4
or MATH 1265	or Calculus for Data Science II	
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 1151	Linear Algebra	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		13

Term 2

CPE 2300	Introduction to Computer Systems	3
CSC/CPE 2050	System Software Mechanics	3

MATH 2031	Transition to Advanced Mathematics	3
General Education Requirement (3B)		3
General Education Requirement (4A)		3
Units		15

Third Year

Term 1

CSC 3001	Modern Application Development	4
CSC 3449	Algorithms and Complexity	4
MATH 2621	Introduction to Mathematical Optimization ²	3

STAT 3210	Engineering Statistics (Upper-Division 2/5)	¹	3
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General Education Requirement (4B)			3
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Units			17
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Term 2

CSC 3100	Software Engineering		4
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CSC/CPE 3201	Introduction to Computer Security		3
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DATA 3301	Introduction to Data Science		4
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PHIL 3323	Ethics, Science, and Technology (Upper-Division 3)	¹	3
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General Education Requirement (6)			3
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Fourth Year**Term 1**

CSC/CPE 4553	Introduction to Operating Systems	3
CSC 3300	Programming Languages	3
DATA/CSC 4610	Fundamentals of Machine Learning	4
or CSC 4667	or Deep Learning	
or CSC 4820	or Natural Language Processing	
or CSC 4880	or Artificial Intelligence	
WGQS 3350	Gender, Race, Culture, Science, and Technology	4

or WGQS
3351

or Gender, Race, Class, Nation: Critical Computing and
Engineering Studies

Units

14

Term 2

CSC 4460

Senior Project

2

or CSC 4461

or Senior Project - Research

Concentration Elective^{3,4}

4

Concentration Elective^{3, 4}

3-4

External Elective^{3, 4}

2-3

General Education Requirement (Upper-Division 4)

3

Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If [MATH 1262](#) was taken in the core requirements, then [MATH 2263](#) needs to be taken from the External Electives to satisfy the prerequisites for [MATH 2621](#).

3

A minimum of 23 units is required to complete the concentration. Unit selection for requirements will vary based on students' selection of concentration electives and external electives.

4

A combined maximum of 6 units may be taken from [CSC 4495](#), [CSC 4891](#), [CSC 4892](#), and [CSC 4893](#); of which up to a combined 4 units may be taken from [CSC 4495](#) and [CSC 4891](#).

Data Engineering

Data Engineering Concentration - BS in Computer Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CSC/CPE 1000	Computing Majors Orientation	1
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CSC/CPE 1024	Introduction to Computing	2
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MATH 1261	Calculus I (2)	4
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or MATH 1264	or Calculus for Data Science I
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Life Science Support Elective (5B)	¹	4
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General Education Requirement (1A)		3
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Units		14
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Term 2

CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory	
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C)	4
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or PHYS
1141

or General Physics I

MATH 1262	Calculus II	4
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or MATH 1265 or Calculus for Data Science II

General Education Requirement (1B) 3

Units 15

Second Year

Term 1

CSC 2001 Data Structures 4

& 2001L and Data Structures Laboratory

MATH 1151 Linear Algebra 3

General Education Requirement (1C) 3

General Education Requirement (3A)	3
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Units	13
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Term 2

CPE 2300	Introduction to Computer Systems	3
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CSC/CPE 2050	System Software Mechanics	3
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MATH 2031	Transition to Advanced Mathematics	3
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General Education Requirement (3B)	3
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General Education Requirement (4A)	3
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Units	15
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Third Year

Term 1

CSC 3001	Modern Application Development	4
CSC/CPE 3201	Introduction to Computer Security	3
CSC 3449	Algorithms and Complexity	4
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ¹	3
General Education Requirement (4B)		3
Units		17

Term 2

CSC 3100	Software Engineering	4
CSC 3300	Programming Languages	3
CSC 3665	Introduction to Database Management Systems	4
STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
General Education Requirement (6)		3
Units		17

Fourth Year

Term 1

CSC/CPE 4553	Introduction to Operating Systems	3
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DATA 3301	Introduction to Data Science	4
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WGQS 3350	Gender, Race, Culture, Science, and Technology	4
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or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies	
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General Education Requirement (Upper-Division 4)	3
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Units	14
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Term 2

CSC 4460	Senior Project	2
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or CSC 4461	or Senior Project - Research	
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CSC 4665	Database Management Systems Organization	4
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Concentration Elective	2, 3	3-4
External Elective	2, 3	3
External Elective	2, 3	2-3
Units		15
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of 23 units is required to complete the concentration. Unit selection for requirements will vary based on students' selection in the notated requirement and technical electives.

A combined maximum of 6 units may be taken from [CSC 4495](#), [CSC 4691](#), [CSC 4692](#), and [CSC 4693](#); of which up to a combined 4 units may be taken from [CSC 4495](#) and [CSC 4691](#).

Game Development

Game Development Concentration - BS in Computer Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CSC/CPE 1000	Computing Majors Orientation	1
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CSC/CPE 1024	Introduction to Computing	2
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MATH 1261	Calculus I (2)	¹	4
or MATH 1264	or Calculus for Data Science I		
Life Science Support Elective (5B)		¹	4
General Education Requirement (1A)			3
Units			14

Term 2

CSC 1001	Fundamentals of Computer Science		4
& 1001L	and Fundamentals of Computer Science Laboratory		
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C)	¹	4

or PHYS 1141	or General Physics I	
MATH 1262	Calculus II	4
or MATH 1265	or Calculus for Data Science II	
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	

MATH 1151	Linear Algebra	3
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General Education Requirement (1C)	3
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General Education Requirement (3A)	3
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Units	13
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Term 2

CPE 2300	Introduction to Computer Systems	3
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CSC/CPE 2050	System Software Mechanics	3
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MATH 2031	Transition to Advanced Mathematics	3
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General Education Requirement (3B)	3
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General Education Requirement (4A)	3
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Units	15
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Third Year

Term 1

CSC 3001	Modern Application Development	4
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CSC/CPE 3201	Introduction to Computer Security	3
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CSC 3449	Algorithms and Complexity	4
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General Education Requirement (4B)	3
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General Education Requirement (6)	3
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Units

17

Term 2

CSC 3100	Software Engineering	4
CSC 3300	Programming Languages	3
CSC 3710	Game Design and Development	4
STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
General Education Requirement (Upper-Division 4)		3

Units

17

Fourth Year

Term 1

CSC 3780	Game Engineering and Critical Analysis	4
CSC/CPE 4553	Introduction to Operating Systems	3
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ¹	3
WGQS 3350	Gender, Race, Culture, Science, and Technology	4
or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies	
Units		14

Term 2

CSC 4460	Senior Project	2
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or CSC 4461 or Senior Project - Research

COMS 4404	Video Games and Society	3
Computer Science Elective	2, 3	4
Computer Science Elective	2, 3	3-4
External Elective	2, 3	2-3
Units		15
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of 23 units is required to complete the concentration. Unit selection for requirements will vary based on students' selection of computer science electives and external electives.

3

A combined maximum of 6 units may be taken from [CSC 4495](#), [CSC 4791](#), [CSC 4792](#), [CSC 4793](#), [CSC 4991](#), [CSC 4992](#), and [CSC 4993](#); of which up to a combined 4 units may be taken from [CSC 4495](#), [CSC 4791](#), and [CSC 4991](#).

Graphics

Graphics Concentration - BS in Computer Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CSC/CPE 1000	Computing Majors Orientation	1
CSC/CPE 1024	Introduction to Computing	2
MATH 1261	Calculus I (2) ¹	4
or MATH 1264	or Calculus for Data Science I	
Life Science Support Elective (5B) ¹		4
General Education Requirement (1A)		3
Units		14

Term 2

CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory	
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
or PHYS 1141	or General Physics I	
MATH 1262	Calculus II	4
or MATH 1265	or Calculus for Data Science II	
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 1151	Linear Algebra	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		13

Term 2

CPE 2300	Introduction to Computer Systems	3
CSC/CPE 2050	System Software Mechanics	3

MATH 2031	Transition to Advanced Mathematics	3
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General Education Requirement (3B)		3
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General Education Requirement (4A)		3
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Units		15
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Third Year

Term 1

CSC 3001	Modern Application Development	4
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CSC/CPE 3201	Introduction to Computer Security	3
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CSC 3449	Algorithms and Complexity	4
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General Education Requirement (4B)	3
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General Education Requirement (6)	3
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Units	17
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Term 2

CSC 3100	Software Engineering	4
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CSC 3300	Programming Languages	3
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CSC 4710	Introduction to Computer Graphics	3
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STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
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External Elective ^{2, 3}	2-3
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Fourth Year

Term 1

CSC/CPE 4553	Introduction to Operating Systems	3
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ¹	3
WGQS 3350	Gender, Race, Culture, Science, and Technology	4
or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies	
Graphics Elective		3
General Education Requirement (Upper-Division 4)		3

Units

16

Term 2

CSC 4460

Senior Project

2

or CSC 4461

or Senior Project - Research

Graphics Elective

3

Concentration Elective^{2, 3}

3

Concentration Elective^{2,3}

3-4

External Elective^{2, 3}

3

Units

15

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of 23 units is required to complete the concentration. Unit selection for requirements will vary based on students' selection of concentration electives and external electives.

3

A combined maximum of 6 units may be taken from [CSC 4495](#), [CSC 4791](#), [CSC 4792](#), [CSC 4793](#), [CSC 4991](#), [CSC 4992](#), [CSC 4993](#), [ENGR 2995](#), and [ENGR 4995](#); of which up to a combined 4 units may be taken from [CSC 4495](#), [CSC 4791](#), and [CSC 4991](#).

Privacy and Security

Privacy and Security Concentration - BS in Computer Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CSC/CPE 1000 Computing Majors Orientation 1

CSC/CPE 1024 Introduction to Computing 2

MATH 1261 Calculus I (2) ¹ 4

or MATH
1264 or Calculus for Data Science I

Life Science Support Elective (5B) ¹ 4

General Education Requirement (1A) 3

Units 14

Term 2

CSC 1001 Fundamentals of Computer Science 4

& 1001L and Fundamentals of Computer Science Laboratory

CHEM 1120 Fundamentals of Chemical Structure and Properties (5A & 5C) ¹ 4

or PHYS
1141

or General Physics I

MATH 1262 Calculus II 4

or MATH
1265

or Calculus for Data Science II

General Education Requirement (1B) 3

Units 15

Second Year

Term 1

CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 1151	Linear Algebra	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		13

Term 2

CPE 2300	Introduction to Computer Systems	3
CSC/CPE 2050	System Software Mechanics	3
MATH 2031	Transition to Advanced Mathematics	3
General Education Requirement (3B)		3
General Education Requirement (4A)		3
Units		15

Third Year

Term 1

CSC 3001	Modern Application Development	4
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CSC/CPE 3201	Introduction to Computer Security	3
CSC 3449	Algorithms and Complexity	4
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ¹	3
General Education Requirement (4B)		3
Units		17

Term 2

CSC 3100	Software Engineering	4
CSC 3250	Introduction to Privacy: Policy and Technology	3
CSC 3300	Programming Languages	3

STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
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General Education Requirement (6)	¹	3
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Units	16
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Fourth Year

Term 1

CSC 4260	Interdisciplinary Privacy and Security Capstone I	3
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CSC/CPE 4553	Introduction to Operating Systems	3
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WGQS 3350	Gender, Race, Culture, Science, and Technology	4
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or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies	
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Concentration Elective

2, 3, 4

3-4

General Education Requirement (Upper-Division 4)

3

Units

16

Term 2

CSC 4261

Senior Project - Privacy and Security Capstone II

3

Concentration Elective

2, 3, 4

3-4

Concentration Elective

2, 3, 4

0-3

Mathematics Elective

3

External Elective	2, 3	2-3
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of 23 units is required to complete the concentration. Unit selection for requirements will vary based on students' selection of concentration electives and external electives.

3

A combined maximum of 6 units may be taken from [CSC 4495](#), [CSC 4291](#), [CSC 4292](#), [CSC 4293](#), [ENGR 2995](#), and [ENGR 4995](#); of which up to a combined 4 units may be taken from [CSC 4495](#) and [CSC 4291](#).

4

Topic courses require department approval. Please contact department for a list of approved topics.

Aerospace Engineering

Concentration Not Yet Declared - BS in Aerospace Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
AERO 1121	Aerospace Fundamentals	2
IME 1143	Introduction to Design and Manufacturing	3
& IME 1140	and Technical Graphics Communication for Design and Manufacturing	
MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I	4

General Education Requirement (1A)	3
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Units	16
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Term 2

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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MATH 1262	Calculus II	4
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PHYS 1143	General Physics II	4
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General Education Requirement (1B)	3
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General Education Requirement (3A)	3
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Units	18
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Second Year

Term 1

ENGR 2211	Introduction to Mechanics	4
MATE 1220	Principles of Materials Engineering for Non-Majors	2
MATH 2263	Calculus III	3
General Education Requirement (1C)		3
General Education Requirement (6)		3
Units		15

Term 2

AERO 2220	Aerospace Engineering Dynamics	3
MATH 2341	Linear Analysis	4
General Education Requirement (3B)		3
General Education Requirement (4A)		3
General Education Requirement (4B)		3
Units		16

Third Year

Term 1

AERO 3300	Engineering Numerical Analysis (Upper-Division 2/5)	1	4
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AERO 3301	Thermo Fluid Dynamics	4
AERO 3302	Thermo Fluids Laboratory	1
AERO 3331	Aerospace Structural Analysis I	4
AERO 3460	Aerospace Engineering Professional Preparation	1
General Education Requirement (5B)		3
Units		17

Term 2

AERO 3303	Heat and Mass Transfer	2
AERO 3304	Aerospace Propulsion Systems	3

AERO 3320	System Dynamics	3
AERO 4431	Aerospace Structural Analysis II	3
AERO 4433	Experimental Stress Analysis	1
Concentration Course		3
Units		15

Fourth Year

Term 1

AERO 4403	Propulsion Laboratory	1
Concentration Course		3

Concentration Course	3
Concentration Course	3
Concentration Course	3
Concentration Course	3
Units	16

Term 2

Concentration Course	3
Concentration Technical Elective	3
Concentration Technical Elective	3

General Education Requirement (Upper-Division 3)	3
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General Education Requirement (Upper-Division 4)	3
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Units	15
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Total Units	128
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Aeronautics

Aeronautics Concentration - BS in Aerospace Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
AERO 1121	Aerospace Fundamentals	2
IME 1143	Introduction to Design and Manufacturing	3
& IME 1140	and Technical Graphics Communication for Design and Manufacturing	
MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I	4
General Education Requirement (1A)		3
Units		16
Term 2		

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
MATH 1262	Calculus II	4
PHYS 1143	General Physics II	4
General Education Requirement (1B)		3
General Education Requirement (3A)		3
Units		18

Second Year

Term 1

ENGR 2211	Introduction to Mechanics	4
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MATE 1220	Principles of Materials Engineering for Non-Majors	2
MATH 2263	Calculus III	3
General Education Requirement (1C)		3
General Education Requirement (6)		3
Units		15

Term 2		
AERO 2220	Aerospace Engineering Dynamics	3
MATH 2341	Linear Analysis	4
General Education Requirement (3B)		3

General Education Requirement (4A) 3

General Education Requirement (4B) 3

Units 16

Third Year

Term 1

AERO 3300 Engineering Numerical Analysis (Upper-Division 2/5) ¹ 4

AERO 3301 Thermo Fluid Dynamics 4

AERO 3302 Thermo Fluids Laboratory 1

AERO 3331 Aerospace Structural Analysis I 4

AERO 3460	Aerospace Engineering Professional Preparation	1
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General Education Requirement (5B)		3
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Units		17
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Term 2

AERO 3303	Heat and Mass Transfer	2
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AERO 3304	Aerospace Propulsion Systems	3
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AERO 3305	Introduction to Aerodynamics	3
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AERO 3320	System Dynamics	3
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AERO 4431	Aerospace Structural Analysis II	3
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AERO 4433	Experimental Stress Analysis	1
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Units	15
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Fourth Year

Term 1

AERO 4401	Ground to Space Propulsion	2
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AERO 4403	Propulsion Laboratory	1
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AERO 4405	Advanced Aerodynamics	3
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AERO 4406	Experimental Aerodynamics	1
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AERO 4420	Aircraft Dynamics and Control	3
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AERO 4461	Aircraft Design Senior Project I	3
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Concentration Technical Elective	3
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Units	16
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Term 2

AERO 4462	Aircraft Design Senior Project II	3
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Concentration Technical Elective	3
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Concentration Technical Elective	3
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General Education Requirement (Upper-Division 3)	3
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General Education Requirement (Upper-Division 4)	3
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Units	15
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Total Units	128
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Astronautics

Astronautics Concentration - BS in Aerospace Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1	Units
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AERO 1121	Aerospace Fundamentals
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2

IME 1143	Introduction to Design and Manufacturing	3
& IME 1140	and Technical Graphics Communication for Design and Manufacturing	
MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I	4
General Education Requirement (1A)		3
Units		16

Term 2

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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MATH 1262	Calculus II	4
PHYS 1143	General Physics II	4
General Education Requirement (1B)		3
General Education Requirement (3A)		3
Units		18

Second Year		
Term 1		

ENGR 2211	Introduction to Mechanics	4
MATE 1220	Principles of Materials Engineering for Non-Majors	2

MATH 2263	Calculus III	3
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General Education Requirement (1C)	3
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General Education Requirement (6)	3
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Units	15
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Term 2

AERO 2220	Aerospace Engineering Dynamics	3
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MATH 2341	Linear Analysis	4
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General Education Requirement (3B)	3
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General Education Requirement (4A)	3
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General Education Requirement (4B)	3
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Units	16
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Third Year

Term 1

AERO 3300	Engineering Numerical Analysis (Upper-Division 2/5) ¹	4
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AERO 3301	Thermo Fluid Dynamics	4
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AERO 3302	Thermo Fluids Laboratory	1
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AERO 3331	Aerospace Structural Analysis I	4
-----------	---------------------------------	---

AERO 3460	Aerospace Engineering Professional Preparation	1
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General Education Requirement (5B)	3
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Units	17
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Term 2

AERO 3303	Heat and Mass Transfer	2
-----------	------------------------	---

AERO 3304	Aerospace Propulsion Systems	3
-----------	------------------------------	---

AERO 3320	System Dynamics	3
-----------	-----------------	---

AERO 3351	Introduction to Orbital Mechanics	3
-----------	-----------------------------------	---

AERO 4431	Aerospace Structural Analysis II	3
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AERO 4433	Experimental Stress Analysis	1
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Fourth Year

Term 1

AERO 4402	Spaceflight Propulsion	2
AERO 4403	Propulsion Laboratory	1
AERO 4421	Spacecraft Attitude Dynamics and Control	3
AERO 4446	Spacecraft Electrical and Electric Systems	3
AERO 4455	Space Environments	4
& AERO 4456	and Space Environments Laboratory	

AERO 4463	Spacecraft Design Senior Project I	3
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Units	16
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Term 2

AERO 4464	Spacecraft Design Senior Project II	3
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Concentration Technical Elective	3
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Concentration Technical Elective	3
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General Education Requirement (Upper-Division 3)	3
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General Education Requirement (Upper-Division 4)	3
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Units	15
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Total Units

128

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Agricultural Business

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

AGB 1101	Introduction to Agribusiness	3
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AGB 2202	Introduction to Sales	3
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BUS 2207	Legal Responsibilities of Business	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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Units	15
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Term 2

AGB 2212	Agricultural Economics	3
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AGB 2214	Agribusiness Financial Accounting	3
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MATH 1267	Business Calculus (2) ¹	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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Units 15

Second Year

Term 1

AGB 2260 Agribusiness Data Literacy 3

ECON 2040 Macroeconomics (4B) ¹ 3

STAT 1210 Business Statistics I 3

Agricultural Elective 3-4

General Education Requirement 3

Units 15

Term 2

AGB 3301	Food Marketing	3
AGB 3308	Introduction to Agribusiness Finance	3
AGB 3327	Agribusiness Data Analysis	3
General Education Requirement		3
General Education Requirement		3
Units		15

Third Year

Term 1

AGB 3322	Principles of Agribusiness Management	3
AGB 3328	Decision Tools for Agribusiness	3
Agribusiness General Elective		3
CAFES Prefix Elective		3
General Education Requirement		3
General Education Requirement		1
Units		16

Term 2

AGB 3312	Agricultural Policy	3
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AGB 3369	Agricultural Personnel Management	3
Agribusiness General Elective		3
General Education Requirement		3
General Education Requirement		3
Units		15

Fourth Year

Term 1

Agribusiness General Elective	3
Agribusiness General Elective	3

CAFES Prefix Elective	3
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General Education Requirement	3
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Free Elective ²	3-4
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Units	16
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Term 2

AGB 4462	Senior Project - Applied Agribusiness Problems	3
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or AGB 4463	or Senior Project - Agribusiness Consulting
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Agribusiness General Elective	3
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General Education Requirement	3
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Free Elective ² 3-4

Units 13

Total Units 120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Agricultural Communication

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
AGC 1102	Orientation to Agricultural Communication & Agricultural Science	1
AGC 2207	Software Applications for Agricultural Publications	2
BIO 1111	General Biology	3
STAT/DATA 1000	Statistical and Data Literacy	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		15
Term 2		

ASCI 1101	Principles of Animal Physiology	3
ASCI 1102	Animal Management Systems	3
ASCI 1103	Animal Science Laboratory	1
CHEM 1110	World of Chemistry (5A & 5C) ¹	4
or CHEM 1120	or Fundamentals of Chemical Structure and Properties	
MATH 1006	College Algebra (2) ¹	3
General Education Requirement (1B)		3
Units		17

Second Year

Term 1

AGC 2205 Agricultural Communications 3

FSN 1111 Elements of Food Processing 3

or FSN 2245 or Elements of Food Safety

JOUR 2203 News Reporting and Writing 3

PLSC 1120 Principles of Plant Sciences 3

& 1120L and Principles of Plant Sciences Lab

General Education Requirement 3

Units 15

Term 2

AGC 2225	Digital Communication in Agriculture and Science	3
AGB 2212	Agricultural Economics	3
AGB 2260	Agribusiness Data Literacy	3
ECON 2040	Macroeconomics (4B) ¹	3
SS 1120	Introductory Soil Science	4
	Units	16

Third Year		
Term 1		

AGC 3301	New Media Communication Strategies in Agriculture	3
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AGB 3301	Food Marketing	3
or WVIT 3343	or Branded Wine Marketing	
AGB 3312	Agricultural Policy	3
COMS 3316	Intercultural Communication (USCP)	3
General Education Requirement		3
Units		15

Term 2

AGC 3339	Internship in Agricultural Communications	3
BRAE 3340	Irrigation Water Management (Upper-Division 2/5) ¹	3

NR 3308	Fire and Society	3
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General Education Requirement		3
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Units		12
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Fourth Year

Term 1

AGC 4404	Applications of Agricultural Leadership	3
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AGC 4407	Agricultural Publications	3
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AGC 4425	Multimedia Storytelling in Agriculture and Science	3
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ENGL 3310	Corporate Communication (Upper-Division 4, GWR) ¹	3
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General Education Requirement	3
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Units	15
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Term 2

AGC 4426	Presentation Methods in Agricultural Communication	3
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AGC 4452	Current Trends and Issues in Agricultural Communication	3
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AGC 4463	Senior Project	3
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AGC 4475	Crisis Communication in Food and Agriculture	3
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General Education Requirement	3
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Units	15
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Agricultural Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year**Term 1****Units**

AGC 1102	Orientation to Agricultural Communication & Agricultural Science	1
ASCI 1101	Principles of Animal Physiology	3

BIO 1111	General Biology (5B)	1	3-4
or BIO 1151	or Life: Molecules and Cells		
or BOT 1121	or General Botany		
or MCRO 2221	or Introduction to Microbiology		
BRAE 1141	Agricultural Machinery Safety		2
General Education Requirement (1A)			3
General Education Requirement (1C)			3
Units			15

Term 2

ASCI 1102	Animal Management Systems		4
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& ASCI 1103 and Animal Science Laboratory

CHEM 1110 World of Chemistry (5A & 5C) ¹ 4

PLSC 1120 Principles of Plant Sciences 3

& 1120L and Principles of Plant Sciences Lab

General Education Requirement (1B) 3

Units 14

Second Year

Term 1

AGB 2202 Introduction to Sales 3

MATH 1006	College Algebra (2)	¹	3-4
or MATH 1007	or Precalculus		
or MATH 1261	or Calculus I		
or MATH 1264	or Calculus for Data Science I		
or MATH 1267	or Business Calculus		
SS 1120	Introductory Soil Science		3-4
or SS 1130	or Soils in Environmental and Agricultural Systems		
General Education Requirement			3
Free Elective		²	1-2
Units			14

Term 2

AGB 2212	Agricultural Economics	3
DSCI 2229	General Dairy Manufacturing	3-4
or FSN 2245	or Elements of Food Safety	
General Education Requirement		3
General Education Requirement		3
Free Elective ²		3
Units		15

Third Year

Term 1

AGC 3314 California Fairgrounds and Expositions 1-3

or AGED or Computer Applications in Agricultural Education
4410

AGB 3301 Food Marketing 3

or WVIT or Branded Wine Marketing
3343

AGED 4421 Agricultural Mechanics 3

Approved Elective 3

Free Elective² 1-4

Units **15**

Term 2

BRAE 3340	Irrigation Water Management (Upper-Division 2/5)	¹	3
NR 3308	Fire and Society (Upper-Division 4)	¹	3
or NR 3323	or Human Dimensions in Natural Resources Management		
PLSC 3301	Horticultural Production Techniques		3
Approved Elective			3
General Education Requirement			3
Units			15

Fourth Year

Term 1

AGC 4404 Applications of Agricultural Leadership 3

AGC 4452 Current Trends and Issues in Agricultural Communication 3-4

or AG or Leadership Seminar on Issues Affecting California
4452 Agriculture, Food Systems, and Natural Resources

Approved Elective 3

Approved Elective 3

General Education Requirement 3

Units 15

Term 2

AGC 4426	Presentation Methods in Agricultural Communication	3
AGC 4463	Senior Project	3
Approved Elective		3
Approved Elective		3
General Education Requirement		3
Free Elective ²		0-4
Units		17
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Agricultural Systems Management

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BRAE 1128	Careers in BioResource and Agricultural Engineering	2
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BRAE 1150	Design Graphics and CAD for Agricultural Engineering	2
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MATH 1007	Precalculus (2) ¹	3
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or STAT 1110

or Applied Statistical Concepts and Methods

PHYS 1121	College Physics I	4
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General Education Requirement (1A)	3
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Units	14
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Term 2

BRAE 1239	Engineering Surveying	3
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CHEM 1110	World of Chemistry	4
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MATH 1267	Business Calculus	3
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General Education Requirement (1B)	3
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General Education Requirement (1C)	3
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Units

16

Second Year

Term 1

BRAE 2203	Systems Management I	4
AGB 2260	Agribusiness Data Literacy	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3

Units

16

Term 2

BRAE 2142	Agricultural Power and Machinery Management	3
AGB 2214	Agribusiness Financial Accounting	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
Units		15

Third Year

Term 1

BRAE 3301	Hydraulic and Mechanical Power Systems	3
BRAE 3340	Irrigation Water Management (Upper-Divison 2/5) ¹	3
AGB 2212	Agricultural Economics	3
AGB 3308	Introduction to Agribusiness Finance	3
Approved Elective		3
Units		15

Term 2

BRAE 3317	Systems Management II	4
BRAE 3343	Mechanical Systems Analysis	4

BRAE 3348	Energy for a Sustainable Society	3
AGB 3369	Agricultural Personnel Management	3
Units		14

Fourth Year

Term 1

BRAE 4419	Systems Management III	3
BRAE 4425	Agricultural Mechatronics	5
BRAE 4432	Agricultural Buildings	4
BRAE 4440	Agricultural Irrigation Systems	4

BRAE 4460	Senior Project I	1
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Units	17
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Term 2

BRAE 4461	Senior Project II	2
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Approved Elective	3
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Approved Elective	3
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General Education Requirement	3
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General Education Requirement	3
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Units	14
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Animal Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.
Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ASCI 1100	Introduction to the Animal Sciences	1
ASCI 1101	Principles of Animal Physiology	3
MATH 1006	College Algebra (2) ²	3

General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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General Education Requirement	3
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Units	16
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Term 2

ASCI 1102	Animal Management Systems	3
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ASCI 1103	Animal Science Laboratory	1
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BIO 1151	Life: Molecules and Cells (5B & 5C) ²	4
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 2	4
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General Education Requirement (1B) 3

Units 15

Second Year

Term 1

ASCI 2210 Meat Science 3

& ASCI 2211 and Meat Science Laboratory

STAT 1110 Applied Statistical Concepts and Methods 3

Animal Management Elective 3

General Education Requirement 3

General Education Requirement 3

Units

15

Term 2

ASCI 2220 Animal Nutrition and Feeding 3

ASCI 2229 Anatomy and Physiology of Farm Animals 3

Animal Management Elective 3

Enterprise Elective¹ 3

Free Elective^{5,6} 3

Units

15

Third Year

Term 1

ASCI 3302 Animal Genetics 3

CHEM 2240 Organic Chemistry: Fundamentals and Applications 4

Approved Elective^{3,4} 3

Free Elective^{5, 6} 3-4

General Education Requirement 3

Units 17

Term 2

ASCI 3304 Animal Genomics 2

ASCI 3340	Animal Welfare and Ethics	3
ASCI 3351	Mechanisms of Hormone Action and Reproductive Physiology	3
ASCI 3363	Professional Development in the Animal Sciences	1
General Education Requirement		3
Free Elective	5,6	4
Units		16

Fourth Year

Term 1

Nutrition Elective	3
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Physiology Elective	3-4
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Approved Elective ^{3,4}	3
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Approved Elective ^{3,4}	3
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General Education Requirement	3
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Units	15
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Term 2

ASCI 4477	Senior Project - Research Experience in Animal Science	2
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or ASCI 4478	or Senior Project - Advanced Internship Experience in Animal Science
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or ASCI 4479	or Senior Project - Current Issues in Animal Science
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Biochemistry Elective ²	3-4
Approved Elective ^{3,4}	3
General Education Requirement	3
Units	11
Total Units	120

1

Repeatable courses can only be used once for Major degree credit.

2

Required in Major or Support; also satisfies General Education (GE) requirement.

3

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Approved Elective.

4

A combined maximum of 3 units may count from [ASCI 3339](#), [ASCI 4001](#), [ASCI 4002](#), [ASCI 4003](#), [ASCI 4004](#), [ASCI 4005](#), [ASCI 4006](#), [ASCI 4007](#), [ASCI 4008](#), [ASCI 4009](#), [ASCI 4010](#), [ASCI 4011](#), [ASCI 4012](#), [ASCI 4013](#), [ASCI 4014](#), and [ASCI 4015](#) towards the Approved Electives requirement.

5

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

6

Students should consult faculty advisor before electing Credit/No Credit grading for any course. Courses elected for Credit/No Credit grading may not be recognized by post-graduate academic programs. If a 5000-level course is used to satisfy free elective requirements it cannot be used towards a post-graduate academic program.

Anthropology and Geography

Concentration Not Yet Declared - BS in Anthropology and Geography

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ANT 2201	Cultural Anthropology (4B) ¹	3
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ANT 2250	Biological Anthropology (5B) ¹	3
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GEOG/ERSC 2250	Physical Geography	3
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		15
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Term 2

GEOG 1150	Human Geography	3
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STAT 1110	Applied Statistical Concepts and Methods (2)	¹	3
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General Education Requirement (1B)			3
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General Education Requirement			3
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General Education Requirement			3
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Units			15
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Second Year

Term 1

ANT 2202	World History Before Writing		3
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ANT/GEOG 3384	Professional Preparation for Anthropologists/Geographers		2
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GEOG 2218	Applications in GIS	3
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Free Elective	3	3
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Free Elective	3	3
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Units	14
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Term 2

Methodological Elective	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	1
Free Elective ³	3
Units	16

Third Year

Term 1

ANT/GEOG 4465	Internship ²	2
GEOG 3308	Global Geography	3
ANT Elective		3
General Education Requirement		3

Free Elective	3	3
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Units	14
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Term 2

ANT 3360	Human Cultural Adaptations (Upper-Division 4)	1	3
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GEOG 3340	Geography of California		3
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or GEOG 3370	or Geography of Latin America
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or GEOG 3380	or Geography of the Caribbean
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GEOG 3350	The Global Environment (Upper-Division 2/5)	1	3
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Concentration Course or Elective	3
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Concentration Course or Elective 3

Units 15

Fourth Year

Term 1

ANT/GEOG 4455 Anthropology-Geography Research Design and Methods 4

ANT 4461 Senior Project I 1

or GEOG 4461 or Senior Project I

GEOG Elective 3

Concentration Course or Elective 3

Concentration Course or Elective 4

Units 15

Term 2

ANT 4462 Senior Project II 2

or GEOG 4462 or Senior Project II

ANT Elective 3

ANT, GEOG, or SOC Elective 3

Concentration Course or Elective 3

Free Elective³ 2

Free Elective³ 3

Units 16

Total Units 120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Approved study abroad course may substitute.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Environmental Studies and Sustainability

Environmental Studies and Sustainability Concentration - BS in Anthropology and Geography

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ANT 2201	Cultural Anthropology (4B) ¹	3
ANT 2250	Biological Anthropology (5B) ¹	3
GEOG/ERSC 2250	Physical Geography	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3

Units15

Term 2

GEOG 1150Human Geography3

STAT 1110Applied Statistical Concepts and Methods (2)¹3

General Education Requirement (1B)3

General Education Requirement3

General Education Requirement3

Units15

Second Year

Term 1

ANT 2202	World History Before Writing	3
ANT/GEOG 3384	Professional Preparation for Anthropologists/Geographers	2
GEOG 2218	Applications in GIS	3
Free Elective ³		3
Free Elective ³		3
Units		14

Term 2

Methodological Elective	3
General Education Requirement	3
General Education Requirement	3
General Education Requirement	3
General Education Requirement	1
Free Elective ³	3
Units	16

Third Year

Term 1

ANT/GEOG 4465	Internship ²	2
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GEOG 3308	Global Geography	3
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ANT Elective		3
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General Education Requirement		3
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Free Elective ³		3
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Units		14
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Term 2

ANT 3360	Human Cultural Adaptations (Upper-Division 4) ¹	3
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GEOG/ERSC 3325	Climate and Humanity	3
or GEOG/ERSC 4414	or Global and Regional Climatology	
or GEOG/ERSC 4415	or Applied Meteorology and Climatology	
GEOG 3340	Geography of California	3
or GEOG 3370	or Geography of Latin America	
or GEOG 3380	or Geography of the Caribbean	
GEOG 3350	The Global Environment (Upper-Division 2/5) ¹	3
GEOG 4435	Biodiversity and Biogeography Methods	3
	Units	15

Fourth Year

Term 1

ANT/GEOG 4455	Anthropology-Geography Research Design and Methods	4
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ANT 4461	Senior Project I	1
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or GEOG 4461	or Senior Project I	
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GEOG 3328	Applications in Remote Sensing and GIS	3
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or GEOG 4441	or Advanced Applications in Geospatial Technologies	
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GEOG Elective		3
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Problems and Issues Course		4
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Units		15
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Term 2

ANT 4462 Senior Project II 2

or GEOG 4462 or Senior Project II

ANT Elective 3

ANT, GEOG, or SOC Elective 3

Problems and Issues Course 3

Free Elective³ 2

Free Elective³ 3

Units 16

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Approved study abroad course may substitute.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Global Studies and International Development

Global Studies and International Development Concentration - BS in Anthropology and Geography

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ANT 2201	Cultural Anthropology (4B) ¹	3
ANT 2250	Biological Anthropology (5B) ¹	3
GEOG/ERSC 2250	Physical Geography	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		15

Term 2

GEOG 1150	Human Geography	3
STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1B)		3
General Education Requirement		3
General Education Requirement		3
Units		15

Second Year

Term 1

ANT 2202	World History Before Writing	3
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ANT/GEOG 3384	Professional Preparation for Anthropologists/Geographers	2
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GEOG 2218	Applications in GIS	3
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Free Elective ³		3
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Free Elective ³		3
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Units	14
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Term 2

Methodological Elective	3
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General Education Requirement	3
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General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Free Elective³ 3

Units 16

Third Year

Term 1

ANT/GEOG 4465 Internship² 2

GEOG 3308 Global Geography 3

ANT Elective	3
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General Education Requirement	3
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Free Elective ³	3
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Units	14
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Term 2

ANT 3360	Human Cultural Adaptations (Upper-Division 4) ¹	3
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GEOG 3340	Geography of California	3
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or GEOG 3370	or Geography of Latin America
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or GEOG 3380	or Geography of the Caribbean
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GEOG 3350	The Global Environment (Upper-Division 2/5) ¹	3
GEOG 4408	Geography of International Development	3
Problems and Issues Course	^{4, 5}	3-4
Units		15

Fourth Year

Term 1

ANT 4401	Culture and Health	4
ANT/GEOG 4455	Anthropology-Geography Research Design and Methods	4

ANT 4461	Senior Project I	1
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or GEOG 4461	or Senior Project I	
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GEOG Elective		3
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Problems and Issues Course		3
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Units		15
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Term 2

ANT 4462	Senior Project II	2
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or GEOG 4462	or Senior Project II	
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ANT Elective		3
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ANT, GEOG, or SOC Elective		3
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Problems and Issues Course	3
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Free Elective ³	2
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Free Elective ³	3
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Units	16
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Approved study abroad course may substitute.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Units in excess of total will be applied towards major's free electives.

5

Credit for a comparable level of proficiency may be substituted up to 3 units max.

Human Ecology

Human Ecology Concentration - BS in Anthropology and Geography

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ANT 2201	Cultural Anthropology (4B) ¹	3
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ANT 2250	Biological Anthropology (5B) ¹	3
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GEOG/ERSC 2250	Physical Geography	3
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		15
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Term 2

GEOG 1150	Human Geography	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1B)		3
General Education Requirement		3
General Education Requirement		3
Units		15

Second Year

Term 1

ANT 2202	World History Before Writing	3
ANT/GEOG 3384	Professional Preparation for Anthropologists/Geographers	2

GEOG 2218	Applications in GIS	3
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Free Elective	3	3
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Free Elective	3	3
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Units	14
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Term 2

Methodological Elective	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	1
Free Elective ³	3
Units	16

Third Year

Term 1

ANT/GEOG 4465	Internship ²	2
GEOG 3308	Global Geography	3
ANT Elective		3
General Education Requirement		3

Free Elective

3

3

Units

14

Term 2

ANT 3309

Elements of Archaeology

3

or ANT 3320

or California's Indigenous Past

ANT 3345

Human Behavioral Ecology

3

ANT 3360

Human Cultural Adaptations (Upper-Division 4)

13

GEOG 3340

Geography of California

3

or GEOG 3370

or Geography of Latin America

or GEOG 3380

or Geography of the Caribbean

GEOG 3350	The Global Environment (Upper-Division 2/5)	¹	3
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Units	15
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Fourth Year

Term 1

ANT 4401	Culture and Health	4
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ANT/GEOG 4455	Anthropology-Geography Research Design and Methods	4
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ANT 4461	Senior Project I	1
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or GEOG 4461	or Senior Project I
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GEOG Elective	3
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Concentration Elective	4	3-4
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Units	15
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Term 2

ANT 4462	Senior Project II	2
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or GEOG 4462	or Senior Project II
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GEOG/ERSC 3325	Climate and Humanity	3
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or GEOG 4408	or Geography of International Development
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ANT Elective	3
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ANT, GEOG, or SOC Elective	3
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Free Elective	3	2
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Free Elective	3	3
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Units	16
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Approved study abroad course may substitute.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Units in excess of total will be applied towards major's free electives.

Individualized Course of Study

Individualized Course of Study - BS in Anthropology and Geography

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year			
Term 1			Units
ANT 2201	Cultural Anthropology (4B)	¹	3
ANT 2250	Biological Anthropology (5B)	¹	3
GEOG/ERSC 2250	Physical Geography		3

General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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Units	15
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Term 2

GEOG 1150	Human Geography	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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General Education Requirement	3
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Units

15

Second Year

Term 1

ANT 2202	World History Before Writing	3
ANT/GEOG 3384	Professional Preparation for Anthropologists/Geographers	2
GEOG 2218	Applications in GIS	3
Free Elective ³		3
Free Elective ³		3

Units

14

Term 2

Methodological Elective

3

General Education Requirement

3

General Education Requirement

3

General Education Requirement

3

General Education Requirement

1

Free Elective³

3

Units

16

Third Year

Term 1

ANT/GEOG 4465	Internship ²	2
GEOG 3308	Global Geography	3
ANT Elective		3
General Education Requirement		3
Free Elective ³		3
Units		14

Term 2

ANT 3360	Human Cultural Adaptations (Upper-Division 4)	¹	3
GEOG 3340	Geography of California		3
or GEOG 3370	or Geography of Latin America		
or GEOG 3380	or Geography of the Caribbean		
GEOG 3350	The Global Environment (Upper-Division 2/5)	¹	3
Individual Course of Study Elective			3
Individual Course of Study Elective			3
Units			15

Fourth Year

Term 1

ANT/GEOG 4455	Anthropology-Geography Research Design and Methods	4
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ANT 4461	Senior Project I	1
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or GEOG 4461	or Senior Project I	
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GEOG Elective		3
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Individual Course of Study Elective		3
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Individual Course of Study Elective		4
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Units		15
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Term 2

ANT 4462	Senior Project II	2
or GEOG 4462	or Senior Project II	
ANT Elective		3
ANT, GEOG, or SOC Elective		3
Individual Course of Study Elective		3
Free Elective ³		2
Free Elective ³		3
	Units	16
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Approved study abroad course may substitute.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Architectural Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ARCE 1110	Introduction to Architectural Engineering	2
ARCH 1101	Architectural Design I	6

& ARCH 1131 and Architectural Representation I

MATH 1261 Calculus I (2)¹ 4

PHYS 1141 General Physics I 4

Units **16**

Term 2

ARCE 1121 Structural Principles I 3

MATH 1262 Calculus II 4

PHYS 1143 General Physics II 4

General Education Requirement (1A) 3

General Education Requirement (1C)	3
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Units	17
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Second Year

Term 1

ARCE 2211	Structural Principles II	3
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ARCE 2212	Structural Principles II Laboratory	1
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CM 1115	Fundamentals of Construction Management	4
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CSC 1031	Programming for Engineers	2
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MATH 2341	Linear Analysis	4
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General Education Requirement (1B)	3
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Units	17
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Term 2

ARCE 2222	Structural Systems Laboratory	2
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ARCE 2223	Structural Drawings	2
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ARCH 2221	History of World Architecture I: Prehistory to 17th Century (3A) ¹	3
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or ARCH 2222

or ARCE 2280 or History of World Architecture II: 17th Century to the Present

or History of Structures

FE Technical Elective	2-3
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General Education Requirement	3
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General Education Requirement	3
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Units	15-16
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Third Year

Term 1

ARCE 3311	Structural Analysis	3
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ARCE 3312	Structural Analysis Laboratory	1
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ARCE 3331	Timber Design	2
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ARCE 3353	Soil Mechanics and Foundation Design	4
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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General Education Requirement		3
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Units		17
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Term 2

ARCE 3332	Timber Design and Constructability Laboratory	2
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ARCE 3341	Steel Design	2
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ARCE 4411	Structural Dynamics	3
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ARCE 4412	Structural Dynamics Computing Laboratory	1
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General Education Requirement		3
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General Education Requirement 3

General Education Requirement 3

Units 17

Fourth Year

Term 1

ARCE 4413 Seismic Analysis and Design 3

ARCE 4442 Steel Structures Design Laboratory 2

ARCE 4461 Reinforced Concrete and Masonry Design 4

FE/PE Surveying Elective 2-3

FE Technical Elective	3
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Units	14-15
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Term 2

ARCE 4421	Architectural Engineering Building Systems	2
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ARCE 4462	Senior Project - Reinforced Concrete and Masonry Laboratory	2
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STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
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ARCE Elective	3
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CAED Interdisciplinary Elective	2-3
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General Education Requirement	3
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Units 15-16

Total Units 128-131

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Architecture

Suggested Five-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ARCH 1101 Architectural Design I 4

ARCH 1131 Architectural Representation I 2

EDES 1123	Place, People, and the Built Environment	3
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MATH 1007	Precalculus (2) ¹	3-4
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or MATH 1261	or Calculus I	
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General Education Requirement (1A)		3
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Units		15
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Term 2

ARCH 1102	Architectural Design II	4
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ARCH 1121	Equity, Social Justice, and Architecture (3A) ¹	3
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ARCH 1132	Architectural Representation II	2
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PHYS 1121	College Physics I (5A & 5C) ¹	4
or PHYS 1141	or General Physics I	
General Education Requirement (1C)		3
Units		16

Second Year

Term 1

ARCH 2201	Architectural Design III	5
ARCH 2221	History of World Architecture I: Prehistory to 17th Century ¹	3
ARCH 2231	Architectural Representation III	1

ARCH 2241	Architectural Technology Fundamentals I	4
ARCE 1121	Structural Principles I	3
Units		16

Term 2

ARCH 2202	Architectural Design IV	5
ARCH 2222	History of World Architecture II: 17th Century to the Present (4B)	3
ARCH 2232	Architectural Representation IV	1
ARCH 2242	Architectural Technology Fundamentals II	4
ARCE 3301	Introduction to Structural Systems	4

Units 17

Third Year

Term 1

ARCH 3331 Building Information Modeling 2

ARCH 3341 Architectural Systems Integration I 4

ARCH 4401 Advanced Architectural Design 5

General Education Requirement (1B) 3

General Education Requirement 3

Units 17

Term 2

ARCH 3301	Integrated Architectural Design	5
ARCH 3342	Architectural Systems Integration II	5
ARCH 4425	Seminar in Architectural History, Theory and Criticism	3
General Education Requirement		3
Units		16

Fourth Year

Term 1

ARCH 4401	Advanced Architectural Design	5
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Professional Elective	2	3
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Professional Elective	2	3
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General Education Requirement		3
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Units		14
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Term 2

ARCH 4401	Advanced Architectural Design	5
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Professional Elective	2	3
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Professional Elective	2	2
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General Education Requirement 3

Units 13

Fifth Year

Term 1

ARCH 4460 Senior Architectural Thesis Theory and Research Seminar 3

ARCH 4461 Senior Project: Architectural Thesis I 5

General Education Requirement 3

General Education Requirement 3

Units 14

Term 2

ARCH 4441	Issues in Contemporary Professional Practice	4
ARCH 4462	Senior Project: Architectural Thesis II	5
General Education Requirement		3
Units		12
Total Units		150

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of 9 units must be taken at the 3000-4000 level.

Art and Design

Concentration Not Yet Declared - BFA in Art and Design

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ART 1101	Fundamentals of Drawing (3A) ¹	3
ART 1102	2D Design	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
General Education Requirement (2)		3
Units		15

Term 2

ART 1103	3D Design	3
ART 1141	Design Thinking and Methods	3
ART 2260	Camera and Light	3
General Education Requirement (1B)		3
General Education Requirement		3
Units		15

Second Year

Term 1

ART 1104	4D Design	3
ART 1184	Beginning Sculpture ²	3
or ART 2282	or Beginning Painting	
ART 2212	Renaissance to Modern Art	3
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

ART 2201	Visual Culture and Society: Analysis and Practice (4B) 1	3
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ART 2215	Global Contemporary Art	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
Units		15

Third Year

Term 1

Concentration Course	3
Concentration Course	3

Concentration Course	3
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General Education Requirement	3
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General Education Requirement	1
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Units	13
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Term 2

ART 3359	Portfolio: Graphic Design	3
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or ART 3379	or Portfolio: Photo Video
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or ART 3399	or Portfolio: Studio Art
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Concentration Course	3
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Concentration Course	3
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Concentration Course 3

Concentration Course 3

Units 15

Fourth Year

Term 1

Upper-Division Art History Elective (Upper-Division 3)¹ 3

3000-4000 level Art Course³ 4

Concentration Course 3

Concentration Course 3

Concentration Course	3
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Units	16
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Term 2

3000-4000 level Art Course ³	4
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Concentration Course	3
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Concentration Course	3
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Concentration Course	3
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Free Elective ⁴	3
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Units	16
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

[ART 1184](#) and [ART 2282](#) are required in the major and the Studio Art concentration. The course taken to fulfill the major course requirement cannot be double-counted in the concentration.

3

A maximum of 3 units from Art History courses.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Graphic Design

Graphic Design Concentration - BFA in Art and Design

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ART 1101	Fundamentals of Drawing (3A) ¹	3
ART 1102	2D Design	3
ART 2212	Renaissance to Modern Art	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3

Units

15

Term 2

ART 1103

3D Design

3

ART 1141

Design Thinking and Methods

3

ART 2215

Global Contemporary Art

3

General Education Requirement (1B)

3

General Education Requirement (2)

3

Units

15

Second Year

Term 1

ART 1104	4D Design	3
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ART 2241	Graphic Design I	3
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ART 2242	Typography I	3
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General Education Requirement		3
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General Education Requirement		3
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Units		15
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Term 2

ART 2201	Visual Culture and Society: Analysis and Practice (4B) 1	3
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ART 2243	Interaction Design I	3
ART 3359	Portfolio: Graphic Design	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		1
Units		16

Third Year

Term 1

ART 1184	Beginning Sculpture ²	3
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or ART 2282	or Beginning Painting	
ART 3342	Typography II	3
ART 3343	Interaction Design II	3
Design Elective		3
General Education Requirement		3
Units		15

Term 2

ART 2260	Camera and Light	3
ART 3313	Design History	3

ART 3341	Graphic Design II	3
Design Elective		3
General Education Requirement		3
Units		15

Fourth Year

Term 1

ART 4441	Design Collaborative Studio	3
or ART 4444	or Motion Design	
3000-4000 level Art Course	3	4

Design Elective	3
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General Education Requirement	3
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Free Elective ⁴	3
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Units	16
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Term 2

ART 4441	Design Collaborative Studio	3
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or ART 4444	or Motion Design
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ART 4459	Senior Project - Design Portfolio	3
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Upper-Division Art History Elective (Upper-Division 3) ¹	3
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3000-4000 level Art Course	3	4
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Units	13
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

[ART 1184](#) and [ART 2282](#) are required in the major and the Studio Art concentration. The course taken to fulfill the major course requirement cannot be double-counted in the concentration.

3

A maximum of 3 units from Art History courses.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Photography and Video

Photography and Video Concentration - BFA in Art and Design

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ART 1102	2D Design	3
ART 2212	Renaissance to Modern Art	3
ART 2260	Camera and Light	3
General Education Requirement (1A)		3

General Education Requirement (1C)	3
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Units	15
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Term 2

ART 1101	Fundamentals of Drawing (3A) ¹	3
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ART 1103	3D Design	3
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ART 2215	Global Contemporary Art	3
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General Education Requirement (1B)	3
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General Education Requirement (2)	3
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Units	15
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Second Year

Term 1

ART 1104	4D Design	3
ART 2201	Visual Culture and Society: Analysis and Practice (4B) 1	3
ART 2263	Black and White Photography	3
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

ART 1141	Design Thinking and Methods	3
ART 2261	Creative Lighting	3
ART 3379	Portfolio: Photo Video	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		1
Units		16

Third Year

Term 1

ART 1184	Beginning Sculpture ²	3
or ART 2282	or Beginning Painting	
ART 3314	History and Contemporary Practices of Photography	3
ART 3370	Digital Video	4
ENGL 3371	Film Styles and Genres	3
or ENGL 3372	or Film Directors	
or ENGL 3387	or Fiction Writing	
General Education Requirement		3
Units		16

Term 2

ART 3364	Editorial Storytelling	4
or ART 3365	or Advertising Photography	
3000-4000 level Art Course		4
General Education Requirement		3
General Education Requirement		3
Units		14

Fourth Year

Term 1

ART 3362	Photographic Expression	4
or ART 3363	or Large Format Photography	

Upper-Division Art History Elective (Upper-Division 3)	¹	3
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3000-4000 level Art Course	³	3
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Concentration Elective		4
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Units		14
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Term 2

ART 4472	Video Art and Expanded Cinema	4
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or ART 4473	or Documentary Video	
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or ART 4474	or Narrative Video	
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ART 4479	Senior Project - Portfolio Photo Video	4
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3000-4000 level Art Course	³	4
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Free Elective	⁴	3
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Units		15
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Total Units		120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

[ART 1184](#) and [ART 2282](#) are required in the major and the Studio Art concentration. The course taken to fulfill the major course requirement cannot be double-counted in the concentration.

3

A maximum of 3 units from Art History courses.

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Studio Art

Studio Art Concentration - BFA in Art and Design

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ART 1102	2D Design	3
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ART 1184	Beginning Sculpture ²	3
----------	----------------------------------	---

ART 2212	Renaissance to Modern Art	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

Units	15
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Term 2

ART 1101	Fundamentals of Drawing (3A) ¹	3
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ART 1103	3D Design	3
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ART 2215	Global Contemporary Art	3
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General Education Requirement (1B)	3
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General Education Requirement (2) 3

Units 15

Second Year

Term 1

ART 1104 4D Design 3

ART 2201 Visual Culture and Society: Analysis and Practice (4B) 3
1

ART 2282 Beginning Painting 2 3

General Education Requirement 3

General Education Requirement 3

Units 15

Term 2

ART 2281 Intermediate Drawing 3

ART 3382 Intermediate Painting 3

ART 3399 Portfolio: Studio Art 3

General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Units

16

Third Year

Term 1

ART 2260 Camera and Light 3

ART 3384 Intermediate Sculpture 4

ART 3386 Junior Studio Art Practice 4

General Education Requirement 3

Units

14

Term 2

ART 1141	Design Thinking and Methods	3
ART 4481	Advanced Figure in Studio Art	4
or ART 4482	or Advanced 2-D	
Concentration Art Elective		3
General Education Requirement		3
General Education Requirement		3
Units		16

Fourth Year
Term 1

ART 3385	Time-Based Art	4
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or ART 4484	or Advanced Sculpture	
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ART 4486	Senior Studio Art Practice	4
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Upper-Division Art History Elective (Upper-Division 3)	¹	3
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3000-4000 level Art Course	³	4
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Units	15
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Term 2

ART 4499	Senior Project - Studio Portfolio	4
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3000-4000 level Art Course	³	4
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Concentration Art History Upper-Division Elective	3
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Free Elective ⁴	3
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

[ART 1184](#) and [ART 2282](#) are required in the major and the Studio Art concentration. The course taken to fulfill the major course requirement cannot be double-counted in the concentration.

3

A maximum of 3 units from Art History courses.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Biochemistry

Concentration Not Yet Declared - BS in Biochemistry

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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BIO 1151	Life: Molecules and Cells (5B) ¹	4
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MATH 1261	Calculus I (2) ¹	4
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General Education Requirement (1A)		3
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Free Elective	¹	1
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Units	16
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Term 2

CHEM 1103	Research Methods I	1
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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MATH 1262	Calculus II	4
-----------	-------------	---

PHYS 1141	General Physics I	4
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General Education Requirement (1C)	3
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Units	16
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Second Year

Term 1

CHEM 2242	Organic Chemistry I	5
MCRO 2224	General Microbiology I	4
PHYS 1143	General Physics II	4
General Education Requirement (1B)		3
Units		16

Term 2

CHEM 2201	Undergraduate Research	1
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or CHEM 2203 or Research Methods II

CHEM 3330	Foundations of Chemical Analysis	4
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CHEM 3352	Biochemistry (Upper-Division 2/5) ¹	4
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Third Year

Term 1

CHEM 3302	Undergraduate Seminar II	1
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CHEM 3356	Genetic Information Processing	4
CHEM 3390	Physical Chemistry for Life Sciences	3
CHEM 3391	Physical Chemistry for Life Sciences Laboratory	1
Concentration or Advanced Elective Course	2,3,4,5,6	4
General Education Requirement		3
Units		16

Term 2

CHEM 3354	Metabolism	3
CHEM 4453	Molecular Biology Techniques	2

or CHEM 4454 or Protein Techniques

Concentration or Advanced Elective Course^{2,3,4,5,6} 4

General Education Requirement 3

General Education Requirement 3

Units 15

Fourth Year

Term 1

CHEM 4461 Senior Project I 1

Biochemistry Advanced Elective 2-3

Concentration or Advanced Elective Course	2,3,4,5,6	5
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General Education Requirement		3
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Units		11
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Term 2

CHEM 4462	Senior Project II	1
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BIO/MCRO Advanced Elective		3-4
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Concentration or Advanced Elective Course	2,3,4,5,6	4
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General Education Requirement		3
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Free Elective	7	2-4
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major requirement cannot be double-counted in the Advanced Electives.

3

Consultation with advisor is recommended prior to selecting advanced electives; bear in mind your selections may impact pursuit of post-baccalaureate studies and/or goals.

4

A minimum of 5 units must be taken at the 3000-4000 level.

5

Maximum of 4 units may be applied toward Advanced Electives from CHEM 4401.

6

Maximum of 2 units may be applied toward Advanced Electives from the following: CHEM 4404, CHEM 4485, CHEM 4495, or SCM/ENGR 3302.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Polymers and Coatings

Polymers and Coatings Concentration - BS in Biochemistry

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CHEM 1120 Fundamentals of Chemical Structure and Properties (5A & 5C) ¹ 4

BIO 1151 Life: Molecules and Cells (5B) ¹ 4

MATH 1261 Calculus I (2) ¹ 4

General Education Requirement (1A) 3

Free Elective ¹ 1

Units 16

Term 2

CHEM 1103	Research Methods I	1
CHEM 1122	Fundamentals of Chemical Reactivity	4
MATH 1262	Calculus II	4
PHYS 1141	General Physics I	4
General Education Requirement (1C)		3
Units		16

Second Year

Term 1

CHEM 2242	Organic Chemistry I	5
CHEM 3390	Physical Chemistry for Life Sciences	3
CHEM 3391	Physical Chemistry for Life Sciences Laboratory	1
PHYS 1143	General Physics II	4
General Education Requirement (1B)		3
Units		16

Term 2

CHEM 2201	Undergraduate Research	1
or CHEM 2203	or Research Methods II	

CHEM 3352	Biochemistry (Upper-Division 2/5) ¹	4
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CHEM 3380	Foundations of Macromolecular Chemistry	4
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General Education Requirement		3
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General Education Requirement		3
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Units		15
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Third Year

Term 1

CHEM 3302	Undergraduate Seminar II	1
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CHEM 3356	Genetic Information Processing	4
-----------	--------------------------------	---

CHEM 4480	Polymer Synthesis and Characterization	3
CHEM 4481	Polymer Synthesis and Characterization Laboratory	2
MCRO 2224	General Microbiology I	4
General Education Requirement		3
Units		17

Term 2

CHEM 3330	Foundations of Chemical Analysis	4
CHEM 3354	Metabolism	3
CHEM 4482	Coatings and Formulations	3

CHEM 4483	Coatings and Formulations Laboratory	2
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General Education Requirement	3
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Units	15
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Fourth Year

Term 1

CHEM 4461	Senior Project I	1
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CHEM 4453	Molecular Biology Techniques	2
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or CHEM 4454	or Protein Techniques
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Biochemistry Advanced Elective	2-3
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General Education Requirement	3
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General Education Requirement	3
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Units	11
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Term 2

CHEM 4462	Senior Project II	1
-----------	-------------------	---

CHEM 4486	Surface Chemistry of Materials	3
-----------	--------------------------------	---

BIO/MCRO Advanced Elective	3-4
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General Education Requirement	3
-------------------------------	---

Free Elective ⁷	2-3
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major requirement cannot be double-counted in the Advanced Electives.

3

Consultation with advisor is recommended prior to selecting advanced electives; bear in mind your selections may impact pursuit of post-baccalaureate studies and/or goals.

4

A minimum of 5 units must be taken at the 3000-4000 level.

5

Maximum of 4 units may be applied toward Advanced Electives from [CHEM 4401](#).

6

Maximum of 2 units may be applied toward Advanced Electives from the following: [CHEM 4404](#), [CHEM 4485](#), [CHEM 4495](#), or SCM/[ENGR 3302](#).

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

General Curriculum in Biology

General Curriculum in Biology - BS in Biological Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
BIO 1150	Life: History and Diversity (5B & 5C) ¹	4
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
STAT 1110	Applied Statistical Concepts and Methods	3

General Education Requirement (1A)	3
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Units	14
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Term 2

BIO 1151	Life: Molecules and Cells	4
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CHEM 1122	Fundamentals of Chemical Reactivity	4
-----------	-------------------------------------	---

MATH 1264	Calculus for Data Science I (2) ¹	4
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General Education Requirement (1B)	3
------------------------------------	---

Units	15
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Second Year

Term 1

BIO 2253 Principles of Ecology and Evolution 4

CHEM 2240 Organic Chemistry: Fundamentals and Applications 3 4-5

or CHEM 2242
or Organic Chemistry I

PHYS 1121 College Physics I 4

or PHYS 1141 or General Physics I

General Education Requirement (1C) 3

Units 15

Term 2

BIO 3351 Principles of Genetics 3

BIO 3352	Principles of Animal Physiology	4
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Technical Elective	4	3-4
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General Education Requirement		3
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Units		13
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Third Year

Term 1

Bioscience Elective	4,6	4
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Approved Elective	2,4,7,8,9,10	4
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Free Elective ⁵	1-2
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Units	16
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Term 2

4000-level Elective ^{4,6}	3
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Bioscience Elective ^{4,6}	4
------------------------------------	---

Approved Elective ^{2,4,7,8,9,10}	3
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General Education Requirement 3

General Education Requirement 3

Units 16

Fourth Year

Term 1

4000-level Elective^{4,6} 4

4000-level Elective^{4,6} 3

Approved Elective^{2,4,7,8,9,10} 3

General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

BIO 4461	Senior Project - Research Proposal	2
----------	------------------------------------	---

or BIO 4462	or Senior Project - Research Experience
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or BIO 4463	or Senior Project - Meta-analysis in Biology
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Approved Elective	2,4,7,8,9,10	3
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Approved Elective	2,4,7,8,9,10	3
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General Education Requirement	3
-------------------------------	---

Free Elective ⁵	4
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major or Support requirement cannot be double-counted in a concentration or the General curriculum.

3

Students in the Molecular and Cellular Biology concentration should take [CHEM 2242](#) to satisfy this requirement.

4

Consultation with an advisor is recommended prior to selecting courses; the best choice for an individual will depend on area of interest, and may impact post-baccalaureate studies and career plans.

5

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

6

Excess units will be applied to Bioscience Electives or Approved Electives.

7

Taking a General Education (GE) course that double-counts as an elective may cause an upper-division unit shortage. Use care to ensure that you have taken enough 3000-4000 level courses to meet the required 40 units of upper-division coursework.

8

If [BIO 4461](#), [BIO 4462](#), or [BIO 4463](#) is used to meet the senior project requirement, it cannot also be counted as an elective.

9

Maximum of 6 units may be applied toward Approved Electives from the following: [BIO 2200](#), [BIO 3300](#), [BIO 4400](#), [BIO 4450](#), [BIO 4466](#), [BIO 4485](#), [BIO 4495](#), or [MSCI 4401](#).

10

Maximum of 2 units may be applied toward Approved Electives from ENGR/[SCM 3302](#).

Anatomy and Physiology

Anatomy and Physiology Concentration - BS in Biological Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
BIO 1150	Life: History and Diversity (5B & 5C) ¹	4

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1A)		3
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Units		14
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Term 2

BIO 1151	Life: Molecules and Cells	4
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CHEM 1122	Fundamentals of Chemical Reactivity	4
-----------	-------------------------------------	---

MATH 1264	Calculus for Data Science I (2) ¹	4
-----------	--	---

General Education Requirement (1B)	3
------------------------------------	---

Units	15
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Second Year

Term 1

BIO 2253	Principles of Ecology and Evolution	4
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CHEM 2240	Organic Chemistry: Fundamentals and Applications	3	4-5
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or CHEM 2242	or Organic Chemistry I
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PHYS 1121	College Physics I	4
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or PHYS 1141	or General Physics I
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General Education Requirement (1C)	3
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Units

15

Term 2

BIO 3351

Principles of Genetics

3

BIO 3352

Principles of Animal Physiology

4

Technical Elective⁴

3-4

General Education Requirement

3

Units

13

Third Year

Term 1

BIO 4431	Advanced Anatomy and Physiology I	4
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Approved Elective	2,7,8,9	4
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General Education Requirement		3
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General Education Requirement		3
-------------------------------	--	---

Free Elective	5	1-2
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Units		16
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Term 2

BIO 4432	Advanced Anatomy and Physiology II	4
----------	------------------------------------	---

CHEM 3352	Biochemistry	4
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Approved Elective ^{2,7,8,9}	3
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General Education Requirement	3
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Free Elective ⁵	2
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Units	16
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Fourth Year

Term 1

Anatomy and Physiology Elective ⁶	4
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Anatomy and Physiology Elective ⁶	4
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Free Elective ⁵	0-2
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Units	16
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Term 2

BIO 4461	Senior Project - Research Proposal	2
----------	------------------------------------	---

or BIO 4462	or Senior Project - Research Experience
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or BIO 4463	or Senior Project - Meta-analysis in Biology
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Anatomy and Physiology Elective ⁶	3
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Approved Elective	2,7,8,9	4
General Education Requirement		3
General Education Requirement		3
	Units	15
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major or Support requirement cannot be double-counted in a concentration or the General curriculum.

3

Students in the Molecular and Cellular Biology concentration should take [CHEM 2242](#) to satisfy this requirement.

4

Consultation with an advisor is recommended prior to selecting electives; the best choice for an individual will depend on area of interest and career plans.

5

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

6

Excess Anatomy and Physiology Elective units will be applied to Approved Electives.

7

Maximum of 6 units may be applied toward Approved Electives from: [BIO 3300](#), [BIO 4400](#), [BIO 4450](#), [BIO 4466](#), [BIO 4485](#), [BIO 4495](#).

8

Maximum of 2 units may be applied toward Approved Electives from ENGR/[SCM 3302](#) or [MSCI 4401](#).

9

If [BIO 4461](#), [BIO 4462](#), or [BIO 4463](#) is used to meet the senior project requirement, it cannot be double-counted as an approved elective.

Ecology, Evolution, Biodiversity, and Conservation

Ecology, Evolution, Biodiversity, and Conservation Concentration - BS in Biological Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BIO 1150

Life: History and Diversity (5B & 5C) ¹

4

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
-----------	--	---

STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1A)		3
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Units		14
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Term 2

BIO 1151	Life: Molecules and Cells	4
----------	---------------------------	---

CHEM 1122	Fundamentals of Chemical Reactivity	4
-----------	-------------------------------------	---

MATH 1264	Calculus for Data Science I (2) ¹	4
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General Education Requirement (1B)	3
------------------------------------	---

Units	15
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Second Year

Term 1

BIO 2253	Principles of Ecology and Evolution	4
----------	-------------------------------------	---

CHEM 2240	Organic Chemistry: Fundamentals and Applications	3	4-5
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or CHEM 2242	or Organic Chemistry I
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PHYS 1121	College Physics I	4
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or PHYS 1141	or General Physics I
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General Education Requirement (1C)	3
------------------------------------	---

Units

15

Term 2

BIO 3351

Principles of Genetics

3

Biodiversity Course^{6,7}

4

Technical Elective⁴

3-4

General Education Requirement

3

General Education Requirement

3

Units

16

Third Year

Term 1

BIO 3352 Principles of Animal Physiology 4

Ecology Course⁶ 3

General Education Requirement 3

General Education Requirement 3

Free Elective⁵ 1-2

Units 15

Term 2

BIO 4413 Evolutionary Medicine 4

or BIO 4414	or Evolution	
Ecology Course	6	4
Approved Elective	2,4,8,9,10	4
General Education Requirement		3
	Units	15

Fourth Year

Term 1

BIO 3343	Principles of Conservation Biology	6	3
or MSCI 4439	or Marine Fisheries and Conservation		

Approved Elective	2,4,8,9,10	4
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General Education Requirement		3
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General Education Requirement		3
-------------------------------	--	---

Free Elective	5	2
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Units		15
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Term 2

BIO 4461	Senior Project - Research Proposal	2
----------	------------------------------------	---

or BIO 4462	or Senior Project - Research Experience	
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or BIO 4463	or Senior Project - Meta-analysis in Biology	
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Biodiversity Course	^{6,7}	4
Approved Elective	^{2,4,8,9,10}	4
General Education Requirement		3
Free Elective	⁵	2
	Units	15
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major or Support requirement cannot be double-counted in a concentration or the General curriculum.

3

Students in the Molecular and Cellular Biology concentration should take [CHEM 2242](#) to satisfy this requirement.

4

Consultation with an advisor is recommended prior to selecting courses; the best choice for an individual will depend on area of interest, and may impact post-baccalaureate studies and career plans.

5

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

6

Excess units will be applied to Approved Electives.

7

Students seeking certification (e.g., as an Associate Wildlife Biologist from the Wildlife Society) should see their faculty advisor for guidance.

8

Maximum of 6 units may be applied toward Approved Electives from the following: [BIO 3300](#), [BIO 4400](#), [BIO 4450](#), [BIO 4466](#), [BIO 4485](#), [BIO 4495](#).

9

If [BIO 4461](#), [BIO 4462](#), or [BIO 4463](#) is used to meet the senior project requirement, it cannot be double-counted as an approved elective.

10

Maximum of 2 units may be applied toward the Approved Electives from ENGR/[SCM 3302](#) or [MSCI 4401](#).

Molecular and Cellular Biology

Molecular and Cellular Biology Concentration - BS in Biological Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BIO 1150	Life: History and Diversity (5B & 5C) ¹	4
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
-----------	--	---

STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1A)		3
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Units	14
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Term 2

BIO 1151	Life: Molecules and Cells	4
CHEM 1122	Fundamentals of Chemical Reactivity	4
MATH 1264	Calculus for Data Science I (2) ¹	4
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

BIO 2255	Molecular and Cellular Biology Lab Skills	1
BIO 3351	Principles of Genetics	3

CHEM 2240	Organic Chemistry: Fundamentals and Applications	3	4-5
or CHEM 2242	or Organic Chemistry I		

General Education Requirement (1C)			3
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General Education Requirement			3
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Units			14
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Term 2

BIO 2253	Principles of Ecology and Evolution		4
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BIO 3352	Principles of Animal Physiology		4
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PHYS 1121	College Physics I		4
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or PHYS 1141	or General Physics I		
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General Education Requirement	3
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Units	15
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Third Year

Term 1

BIO 4457	Molecular Biology Laboratory	3
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Approved Elective	2,7,8,9	4
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Technical Elective	4	3-4
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General Education Requirement	3
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General Education Requirement	3
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Units

16

Term 2

BIO 4452

Cell Biology

3

Advanced Elective 2,4,6

3

Approved Elective 2,7,8,9

4

General Education Requirement

3

General Education Requirement

3

Units

16

Fourth Year

Term 1

Advanced Elective	2,4,6	4
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Approved Elective	2,7,8,9	4
-------------------	---------	---

Approved Elective	2,7,8,9	2
-------------------	---------	---

General Education Requirement		3
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Free Elective	5	3
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Units		16
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Term 2

BIO 4461	Senior Project - Research Proposal	2
or BIO 4462	or Senior Project - Research Experience	
or BIO 4463	or Senior Project - Meta-analysis in Biology	
Advanced Elective	2,4,6	3
Approved Elective	2,7,8,9	3
General Education Requirement		3
Free Elective	5	2-3
	Units	14
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major or Support requirement cannot be double-counted in a concentration or the General curriculum.

3

Students in the Molecular and Cellular Biology concentration should take [CHEM 2242](#) to satisfy this requirement.

4

Consultation with an advisor is recommended prior to selecting courses; the best choice for an individual will depend on area of interest, and may impact post-baccalaureate studies and career plans.

5

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

6

Excess units from Advanced Electives applied to Approved Electives.

7

Consult with your faculty advisor for approval to use other relevant upper-division coursework from other departments.

8

Maximum of 6 units may be applied toward Approved Electives from the following: [BIO 3300](#), [BIO 4400](#), [BIO 4450](#), [BIO 4466](#), [BIO 4485](#), [BIO 4495](#).

9

Maximum of 2 units may be applied toward Approved Electives from ENGR/[SCM 3302](#) or [MSCI 4401](#).

Biomedical Engineering

Concentration Not Yet Declared - BS in Biomedical Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BMED 1101	Introduction to Biomedical Engineering	1
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4

MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I	4
General Education Requirement (1A)		3
Units		16

Term 2

BMED 2310	Introduction to Electrical Design in Biomedical Engineering	2
BMED 2311	Introduction to Electrical Design in Biomedical Engineering Lab	1
CHEM 1122	Fundamentals of Chemical Reactivity	4

MATH 1262	Calculus II	4
PHYS 1143	General Physics II	4
General Education Requirement (1B)		3
Units		18

Second Year

Term 1

BMED 2212	Introduction to Mechanical Design in Biomedical Engineering	3
BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
ENGR 2211	Introduction to Mechanics	4

MATH 2263	Calculus III	3
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General Education Requirement (1C)		3
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Units		17
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Term 2

BMED 2420	Principles and Applications of Biomaterials	4
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BIO 2231	Human Anatomy and Physiology I	4
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or BIO 2232	or Human Anatomy and Physiology II	
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ENGR 2212	Introduction to Engineering Dynamics	2
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MATH 2341	Linear Analysis	4
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General Education Requirement (3A)	3
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Units	17
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Third Year

Term 1

BMED 3102	Biomedical Engineering Professional Development	1
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BMED 3430	Biomedical Modeling and Simulation	2
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STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
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Concentration Course or Elective	3
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General Education Requirement (3B)	3
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General Education Requirement (4B)	3
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Units	15
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Term 2

BMED 3410	Biomechanics	3
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BMED 4440	Bioelectronics and Instrumentation	3
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ME 3341	Fluid Mechanics	3
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Concentration Course or Elective	3-4
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General Education Requirement (4A)	3
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Units	15
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Fourth Year

Term 1

BMED 3425	Biomedical Engineering Transport	4
BMED 4460	Medical Physiology for Engineers	3
BMED 4465	Senior Project: Design I	2
Concentration Course or Elective		3
General Education Requirement (Upper-Division 3)		3
General Education Requirement (6)		3
Units		18

Term 2

BMED 4466 Senior Project: Design II 2

Concentration Course or Elective 3

Concentration Course or Elective 3

Concentration Course or Elective 3-4

General Education Requirement (Upper-Division 4) 3

Units 14-15

Total Units 130-131

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Bioinstrumentation

Bioinstrumentation Concentration - BS in Biomedical Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BMED 1101	Introduction to Biomedical Engineering	1
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I	4

General Education Requirement (1A)	3
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Units	16
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Term 2

BMED 2310	Introduction to Electrical Design in Biomedical Engineering	2
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BMED 2311	Introduction to Electrical Design in Biomedical Engineering Lab	1
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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MATH 1262	Calculus II	4
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PHYS 1143	General Physics II	4
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General Education Requirement (1B)	3
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Units

18

Second Year

Term 1

BMED 2212	Introduction to Mechanical Design in Biomedical Engineering	3
BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
General Education Requirement (1C)		3

Units

17

Term 2

BMED 2420	Principles and Applications of Biomaterials	4
BIO 2231	Human Anatomy and Physiology I	4
or BIO 2232	or Human Anatomy and Physiology II	
ENGR 2212	Introduction to Engineering Dynamics	2
MATH 2341	Linear Analysis	4
General Education Requirement (3A)		3
Units		17

Third Year

Term 1

BMED 3102	Biomedical Engineering Professional Development	1
BMED 3430	Biomedical Modeling and Simulation	2
BMED 3355	Electrical Engineering Concepts for Biomedical Engineering	3
MATH 3351	Differential Equations and Boundary Value Problems	3
STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
General Education Requirement (3B)		3
General Education Requirement (4B)		3
Units		18

Term 2

BMED 3410	Biomechanics	3
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BMED 4440	Bioelectronics and Instrumentation	3
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ME 3341	Fluid Mechanics	3
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Concentration Technical Elective		3-4
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General Education Requirement (4A)		3
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Units		15-16
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Fourth Year

Term 1

BMED 3425	Biomedical Engineering Transport	4
BMED 4460	Medical Physiology for Engineers	3
BMED 4465	Senior Project: Design I	2
Concentration Technical Elective		3
General Education Requirement (6)		3
Units		15

Term 2

BMED 4445	Biopotential Instrumentation	3
BMED 4466	Senior Project: Design II	2

Concentration Technical Elective	3
General Education Requirement (Upper-Division 3)	3
General Education Requirement (Upper-Division 4)	3
Units	14
Total Units	130-131

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Cell Therapy

Cell Therapy Concentration - BS in Biomedical Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BMED 1101 Introduction to Biomedical Engineering 1

CHEM 1120 Fundamentals of Chemical Structure and Properties (5A)
1 4

MATH 1261 Calculus I (2)¹ 4

PHYS 1141 General Physics I 4

General Education Requirement (1A) 3

Units 16

Term 2

BMED 2310	Introduction to Electrical Design in Biomedical Engineering	2
BMED 2311	Introduction to Electrical Design in Biomedical Engineering Lab	1
CHEM 1122	Fundamentals of Chemical Reactivity	4
MATH 1262	Calculus II	4
PHYS 1143	General Physics II	4
General Education Requirement (1B)		3
Units		18

Second Year

Term 1

BMED 2212	Introduction to Mechanical Design in Biomedical Engineering	3
BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
General Education Requirement (1C)		3
Units		17

Term 2

BMED 2420	Principles and Applications of Biomaterials	4
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BIO 3351	Principles of Genetics	3
MATH 2341	Linear Analysis	4
ENGR 2212	Introduction to Engineering Dynamics	2
General Education Requirement (3A)		3
Units		16

Third Year

Term 1

BMED 3102	Biomedical Engineering Professional Development	1
BMED 3360	Cellular Immunotherapy	3

STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
ME 3341	Fluid Mechanics	3
BIO 2231	Human Anatomy and Physiology I	4
or BIO 2232	or Human Anatomy and Physiology II	
General Education Requirement (4A)		3
Units		17

Term 2

BMED 3362	Regenerative Medicine Therapies	3
BMED 3410	Biomechanics	3

BMED 4460	Medical Physiology for Engineers	3
BIO 4457	Molecular Biology Laboratory	3
General Education Requirement (3B)		3
General Education Requirement (4B)		3
Units		18

Fourth Year

Term 1

BMED 3430	Biomedical Modeling and Simulation	2
BMED 4465	Senior Project: Design I	2

BIO 4456	Immunology	4
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General Education Requirement (Upper-Division 3)	3
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General Education Requirement (6)	3
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Units	14
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Term 2

BMED 3425	Biomedical Engineering Transport	4
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BMED 4440	Bioelectronics and Instrumentation	3
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BMED 4466	Senior Project: Design II	2
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Concentration Technical Elective	3
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General Education Requirement (Upper-Division 4)	3
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Units	15
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Total Units	131
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Mechanical Design

Mechanical Design Concentration - BS in Biomedical Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BMED 1101	Introduction to Biomedical Engineering	1
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I	4
General Education Requirement (1A)		3
Units		16

Term 2

BMED 2310	Introduction to Electrical Design in Biomedical Engineering	2
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BMED 2311	Introduction to Electrical Design in Biomedical Engineering Lab	1
CHEM 1122	Fundamentals of Chemical Reactivity	4
MATH 1262	Calculus II	4
PHYS 1143	General Physics II	4
General Education Requirement (1B)		3
Units		18

Second Year

Term 1

BMED 2212	Introduction to Mechanical Design in Biomedical Engineering	3
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BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
General Education Requirement (1C)		3
Units		17

Term 2

BMED 2420	Principles and Applications of Biomaterials	4
BIO 2231	Human Anatomy and Physiology I	4
or BIO 2232	or Human Anatomy and Physiology II	

ENGR 2212	Introduction to Engineering Dynamics	2
MATH 2341	Linear Analysis	4
General Education Requirement (3A)		3
Units		17

Third Year

Term 1

BMED 3102	Biomedical Engineering Professional Development	1
BMED 3410	Biomechanics	3
IME 1140	Technical Graphics Communication for Design and Manufacturing	3

& IME 1143 and Introduction to Design and Manufacturing

ME 3328 Design for Strength and Stiffness 4

ME 3341 Fluid Mechanics 3

General Education Requirement (4A) 3

Units **17**

Term 2

BMED 3330 Intermediate Biomedical Design 3

or ME 3329 or Mechanical Systems Design

BMED 3425 Biomedical Engineering Transport 4

BMED 4460 Medical Physiology for Engineers 3

STAT 3210	Engineering Statistics	3
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General Education Requirement (3B)		3
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Units		16
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Fourth Year

Term 1

BMED 3430	Biomedical Modeling and Simulation	2
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BMED 4465	Senior Project: Design I	2
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Concentration Support Elective		3-4
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Concentration Technical Elective		2-3
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General Education Requirement (4B)	3
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General Education Requirement (6)	3
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Units	15-16
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Term 2

BMED 4440	Bioelectronics and Instrumentation	3
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BMED 4466	Senior Project: Design II	2
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Concentration Technical Elective	3
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General Education Requirement (Upper-Division 3)	3
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General Education Requirement (Upper-Division 4)	3
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Units	14
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Total Units	130-131
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Individualized Course of Study

Individualized Course of Study - BS in Biomedical Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BMED 1101	Introduction to Biomedical Engineering	1
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I	4
General Education Requirement (1A)		3
Units		16

Term 2

BMED 2310	Introduction to Electrical Design in Biomedical Engineering	2
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BMED 2311	Introduction to Electrical Design in Biomedical Engineering Lab	1
CHEM 1122	Fundamentals of Chemical Reactivity	4
MATH 1262	Calculus II	4
PHYS 1143	General Physics II	4
General Education Requirement (1B)		3
Units		18

Second Year		
Term 1		

BMED 2212	Introduction to Mechanical Design in Biomedical Engineering	3
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BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
General Education Requirement (1C)		3
Units		17

Term 2

BMED 2420	Principles and Applications of Biomaterials	4
BIO 2231	Human Anatomy and Physiology I	4
or BIO 2232	or Human Anatomy and Physiology II	

ENGR 2212	Introduction to Engineering Dynamics	2
MATH 2341	Linear Analysis	4
General Education Requirement (3A)		3
Units		17

Third Year

Term 1

BMED 3102	Biomedical Engineering Professional Development	1
BMED 3410	Biomechanics (Support Elective)	3
ME 3341	Fluid Mechanics	3

STAT 3210	Engineering Statistics (Upper-Division 2/5)	¹	3
Support Elective			3
General Education Requirement (4A)			3
Units			16

Term 2

BMED 3425	Biomedical Engineering Transport		4
BMED 4460	Medical Physiology for Engineers		3
Support Elective			3-4
General Education Requirement (3B)			3

General Education Requirement (4B)

3

Units

16-17

Fourth Year

Term 1

BMED 3430

Biomedical Modeling and Simulation

2

BMED 4440

Bioelectronics and Instrumentation

3

BMED 4465

Senior Project: Design I

2

Technical Elective

3

Support Elective

3

General Education Requirement (6)	3
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Units	16
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Term 2

BMED 4466	Senior Project: Design II	2
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Technical Elective	3
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Technical Elective	3-4
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General Education Requirement (Upper-Division 3)	3
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General Education Requirement (Upper-Division 4)	3
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Units	14
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

BioResource and Agricultural Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.
Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
BRAE 1128	Careers in BioResource and Agricultural Engineering	2
MATH 1261	Calculus I (2)	4
PHYS 1141	General Physics I	4

General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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Units	16
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Term 2

BRAE 1150	Design Graphics and CAD for Agricultural Engineering	2
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MATH 1262	Calculus II	4
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PHYS 1143	General Physics II	4
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General Education Requirement (1B)	3
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General Education Requirement	3
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Units

16

Second Year

Term 1

BRAE 2236 Principles of Irrigation 4

CHEM 1122 Fundamentals of Chemical Reactivity (5A)¹ 4

MATH 2263 Calculus III 3

General Education Requirement 3

General Education Requirement 3

Units

17

Term 2

BRAE 2216	Fundamentals of Electricity	4
BRAE 2220	Introduction to Biological Systems (5B & 5C) ¹	4
BRAE 2221	Engineering Mechanics with Agricultural Applications I	4
CSC 1032	Programming for Scientists and Engineers	3
General Education Requirement		3
Units		18

Third Year

Term 1

BRAE 1239	Engineering Surveying	3
BRAE 2222	Engineering Mechanics with Agricultural Applications II	4
BRAE 3312	Hydraulics	3
BRAE 3332	Environmental Controls for Agricultural Structures	3
ECON 2001	Survey of Economics ¹	3
or ECON 2040	or Macroeconomics	

Units 16

Term 2

BRAE 3234	Equipment Engineering I	4
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BRAE 3320	Bioresource Engineering	3
BRAE 4414	Irrigation Engineering	3
BRAE 4428	Agricultural Robotics and Automation	4
Approved Elective ²		3
Units		17

Fourth Year

Term 1

BRAE 4403	Agricultural Engineering Ethics, Economics, and Optimization	3
BRAE 4422	Equipment Engineering II	3

BRAE 4433	Agricultural Structures Design	4
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BRAE 4460	Senior Project I	1
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BRAE Focus Area Elective		3
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Units		14
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Term 2

BRAE 4461	Senior Project II	2
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STAT 3210	Engineering Statistics ¹	3
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BRAE Focus Area Elective		3
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General Education Requirement		3
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General Education Requirement	3
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Units	14
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Total Units	128
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major requirement, it cannot be double-counted as an Approved Elective.

Business Administration

Concentration Not Yet Declared - BS in Business Administration

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BUS 1100	Career Readiness I	1
BUS 2214	Financial Accounting	3
ECON 2001	Survey of Economics (4B) ⁴	3
STAT 1210	Business Statistics I	3
General Education Requirement (1A)		3
Units		13

Term 2

BUS 2207	Legal Responsibilities of Business	3
BUS 2215	Managerial Accounting	3

MATH 1264	Calculus for Data Science I (2)	4, 5	3-4
or MATH 1267	or Business Calculus		
STAT 1220	Business Statistics II		3
General Education Requirement (1C)			3
	Units		16

Second Year

Term 1

BUS 1342	Financial Institutions	1	3
or BUS 3343	or Quantitative Methods in Finance		

BUS 2206	Career Readiness II	1
BUS 3391	Information Systems	3
General Education Requirement (1B)		3
General Education Requirement (3A)		3
General Education Requirement (4A)		3
Units		16

Term 2

BUS 3346	Principles of Marketing	3
BUS 3387	Organizational Behavior	3

Technology Management Course	3
General Education Requirement (3B)	3
General Education Requirement (4B)	3
Units	15

Third Year

Term 1

BUS 3306	Career Readiness III	1
International Business Course		3-4
Concentration Course		3

Concentration Course	3
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General Education Requirement (6)	3
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Free Elective ^{7, 8}	3
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Units	17
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Term 2

BUS 4404	Law, Governmental, and Social Influences on Business	3
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Concentration Course	3
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Concentration Course	3
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General Education Requirement (5A) ⁶	3
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General Education Requirement (5C)	6	1
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General Education Requirement (Upper-Division 4)		3
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Units		16
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Fourth Year

Term 1

Senior Project Course	2, 3	3-4
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Concentration Course		3-4
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Concentration Course		3-4
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General Education Requirement (Upper-Division 2/5)		3
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General Education Requirement (5B) 3

Units 16

Term 2

BUS 4401 Strategic Management 3

& BUS 4411 and BSBA Assurance of Learning Assessment

Concentration Course 3-4

General Education Requirement (Upper-Division 3) 3

Free Elective^{7, 8} 2-3

Free Elective^{7, 8} 0-7

Units	11
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Total Units	120
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1

[BUS 3343](#) is a substitute for [BUS 1342](#). Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take [BUS 3343](#). Students interested in the Entrepreneurship concentration are recommended to take [BUS 1342](#).

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

MATH 1264 is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking BUS 1204 for free electives.

Accounting

Accounting Concentration - BS in Business Administration

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BUS 1100	Career Readiness I	1
BUS 2214	Financial Accounting	3
ECON 2001	Survey of Economics (4B) ⁴	3
STAT 1210	Business Statistics I	3
General Education Requirement (1A)		3
Units		13

Term 2

BUS 2207	Legal Responsibilities of Business	3
BUS 2215	Managerial Accounting	3
MATH 1264	Calculus for Data Science I (2)	4, 5
or MATH 1267	or Business Calculus	3-4
STAT 1220	Business Statistics II	3
General Education Requirement (1C)		3
Units		16

Second Year

Term 1

BUS 1342	Financial Institutions	1	3
or BUS 3343	or Quantitative Methods in Finance		
BUS 2206	Career Readiness II		1
BUS 3391	Information Systems		3
General Education Requirement (1B)			3
General Education Requirement (3A)			3
General Education Requirement (4A)			3
Units			16

Term 2

BUS 3346	Principles of Marketing	3
BUS 3387	Organizational Behavior	3
Technology Management Course		3
General Education Requirement (3B)		3
General Education Requirement (5A) ⁶		3
General Education Requirement (5C) ⁶		1
Units		16

Third Year

Term 1

BUS 3306	Career Readiness III	1
BUS 3319	Data Analytics and Accounting Information Systems	3
BUS 3320	Federal Income Taxation for Individuals	3
International Business Course		3-4
General Education Requirement (6)		3
Free Elective	^{7, 8}	2-3
Units		16

Term 2

BUS 3321	Intermediate Accounting I	3
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BUS 4404	Law, Governmental, and Social Influences on Business	3
Accounting Elective		3
General Education Requirement (5B)		3
General Education Requirement (Upper-Division 4)		3
Units		15

Fourth Year

Term 1

BUS 3322	Intermediate Accounting II	3
Senior Project Course	2, 3	3-4

Accounting Elective 3

General Education Requirement (Upper-Division 2/5) 3

Free Elective^{7, 8} 3

Units 16

Term 2

BUS 3323 Financial Statement Auditing 3

BUS 4401 Strategic Management 3

& BUS 4411 and BSBA Assurance of Learning Assessment

General Education Requirement (Upper-Division 3) 3

Free Elective ^{7, 8} 2-4

Units 12

Total Units 120

1

BUS 3343 is a substitute for BUS 1342. Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take BUS 3343. Students interested in the Entrepreneurship concentration are recommended to take BUS 1342.

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

[MATH 1264](#) is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

[CHEM 1120](#) is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking [BUS 1204](#) for free electives.

Consumer Packaging

Consumer Packaging Concentration - BS in Business Administration

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
BUS 1100	Career Readiness I	1
BUS 2214	Financial Accounting	3
ECON 2001	Survey of Economics (4B) ⁴	3
STAT 1210	Business Statistics I	3
General Education Requirement (1A)		3

Units

13

Term 2

BUS 2207

Legal Responsibilities of Business

3

BUS 2215

Managerial Accounting

3

MATH 1264

Calculus for Data Science I (2) ^{4, 5}

3-4

or MATH 1267

or Business Calculus

STAT 1220

Business Statistics II

3

General Education Requirement (1C)

3

Units

16

Second Year

Term 1

BUS 1342	Financial Institutions	1	3
or BUS 3343	or Quantitative Methods in Finance		
BUS 2206	Career Readiness II		1
BUS 3391	Information Systems		3
ITP 2234	Packaging Design Fundamentals		3
General Education Requirement (1B)			3
General Education Requirement (3A)			3

General Education Requirement (4A)	3
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Units	19
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Term 2

BUS 3346	Principles of Marketing	3
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BUS 3387	Organizational Behavior	3
----------	-------------------------	---

ITP 3334	Structural Packaging Design	3
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Technology Management Course	3
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General Education Requirement (3B)	3
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Units	15
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Third Year

Term 1

BUS 3306	Career Readiness III	1
BUS 3396	Consumer Insights	3
ITP 3330	Packaging Fundamentals	3
International Business Course		3-4
General Education Requirement (6)		3
Units		14

Term 2

BUS 4404	Law, Governmental, and Social Influences on Business	3
ITP 4475	Distribution Packaging Dynamics	3
General Education Requirement (5A)		3
General Education Requirement (5B)		3
General Education Requirement (5C)		1
General Education Requirement (Upper-Division 4)		3
Units		16

Fourth Year

Term 1

BUS 4418	Marketing Research	3
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Senior Project Course	2, 3	3-4
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General Education Requirement (Upper-Division 2/5)		3
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Free Elective	7, 8	3
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Units		13
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Term 2

BUS 4401	Strategic Management	3
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& BUS 4411	and BSBA Assurance of Learning Assessment	
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Approved Elective	9	3
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General Education Requirement (Upper-Division 3)	3
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Free Elective ^{7, 8}	2-3
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Free Elective ^{7, 8}	2-7
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Units	14
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Total Units	120
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1

BUS 3343 is a substitute for BUS 1342. Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take BUS 3343. Students interested in the Entrepreneurship concentration are recommended to take BUS 1342.

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

MATH 1264 is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking [BUS 1204](#) for free electives.

9

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

Entrepreneurship

Entrepreneurship Concentration - BS in Business Administration

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BUS 1100

Career Readiness I

1

BUS 2214	Financial Accounting	3
ECON 2001	Survey of Economics (4B) ⁴	3
STAT 1210	Business Statistics I	3
General Education Requirement (1A)		3
Units		13

Term 2

BUS 2207	Legal Responsibilities of Business	3
BUS 2215	Managerial Accounting	3
MATH 1264	Calculus for Data Science I (2) ^{4, 5}	3-4

or MATH 1267 or Business Calculus

STAT 1220	Business Statistics II	3
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General Education Requirement (1C)	3
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Units	16
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Second Year

Term 1

BUS 1342	Financial Institutions ¹	3
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or BUS 3343
or Quantitative Methods in Finance

BUS 2206	Career Readiness II	1
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BUS 3391	Information Systems	3
General Education Requirement (1B)		3
General Education Requirement (3A)		3
General Education Requirement (4A)		3
Units		16

Term 2		
BUS 3346	Principles of Marketing	3
BUS 3387	Organizational Behavior	3
Technology Management Course		3

General Education Requirement (3B)	3
------------------------------------	---

General Education Requirement (5A) ⁶	3
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General Education Requirement (5C) ⁶	1
---	---

Units	16
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Third Year

Term 1

BUS 3306	Career Readiness III	1
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BUS 3310	Introduction to Entrepreneurship	4
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International Business Course	3-4
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General Education Requirement (Upper-Division 4)	3
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General Education Requirement (6)	3
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Units	15
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Term 2

BUS 3330	Funding and Managing Startup Companies	4
----------	--	---

BUS 4404	Law, Governmental, and Social Influences on Business	3
----------	--	---

ITP 4406	Business-to-Business Complex Solutions Selling	3
----------	--	---

Approved Elective	9, 10	3
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General Education Requirement (5B)	3
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Units

16

Fourth Year

Term 1

BUS 4401 Strategic Management 3

& BUS 4411 and BSBA Assurance of Learning Assessment

BUS 4447 Defining and Building Technology Startups 4

or BUS 4488 or Building a Startup Skill Set

Approved Elective 9,10 3-4

General Education Requirement (Upper-Division 2/5) 3

Units

13

Term 2

BUS 4465 Senior Project: Building and Launching the Technology 4
Startup²
or BUS 4467

or Senior Project: Growing the Early Stage Startup

General Education Requirement (Upper-Division 3) 3

Free Elective^{7, 8} 3

Free Elective^{7, 8} 2-3

Free Elective^{7, 8} 2-7

Units 15

1

BUS 3343 is a substitute for BUS 1342. Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take BUS 3343. Students interested in the Entrepreneurship concentration are recommended to take BUS 1342.

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

MATH 1264 is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking BUS 1204 for free electives.

9

Units in excess of total will be applied towards major's free electives.

10

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

Financial Management

Financial Management Concentration - BS in Business Administration

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BUS 1100	Career Readiness I	1
BUS 2214	Financial Accounting	3
ECON 2001	Survey of Economics (4B) ⁴	3
STAT 1210	Business Statistics I	3

General Education Requirement (1A)	3
------------------------------------	---

Units	13
-------	----

Term 2

BUS 2207	Legal Responsibilities of Business	3
----------	------------------------------------	---

BUS 2215	Managerial Accounting	3
----------	-----------------------	---

MATH 1264	Calculus for Data Science I (2)	3-4
-----------	---------------------------------	-----

or MATH 1267

or Business Calculus

STAT 1220	Business Statistics II	3
-----------	------------------------	---

General Education Requirement (1C)	3
------------------------------------	---

Units

15

Second Year

Term 1

BUS 1342	Financial Institutions	1	3
or BUS 3343	or Quantitative Methods in Finance		
BUS 2206	Career Readiness II		1
BUS 3391	Information Systems		3
General Education Requirement (1B)			3
General Education Requirement (3A)			3

General Education Requirement (4A)	3
------------------------------------	---

Units	16
-------	----

Term 2

BUS 3346	Principles of Marketing	3
----------	-------------------------	---

BUS 3387	Organizational Behavior	3
----------	-------------------------	---

Technology Management Course	3
------------------------------	---

General Education Requirement (3B)	3
------------------------------------	---

General Education Requirement (5A) ⁶	3
---	---

General Education Requirement (5C) ⁶	1
---	---

Units

16

Third Year

Term 1

BUS 3306 Career Readiness III 1

BUS 3431 Security Analysis and Portfolio Management⁹ 4

International Business Course 3-4

Concentration Track Course^{9, 10} 3-4

General Education Requirement (6) 3

Units

15

Term 2

BUS 3438	Corporate Finance ⁹	4
BUS 4404	Law, Governmental, and Social Influences on Business	3
Concentration Track Course	^{9, 10}	3
General Education Requirement (5B)		3
General Education Requirement (Upper-Division 4)		3
	Units	16

Fourth Year

Term 1

BUS 3441	Financial Modeling and Analytics in Python or R	9	4
or BUS 3440	or Financial Modeling and Visualization in Excel		

Senior Project Course	2, 3		3-4
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Concentration Track Course	9, 10		3-4
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General Education Requirement (Upper-Division 2/5)			3
--	--	--	---

Units			14
-------	--	--	----

Term 2

BUS 3411	Finance Practicum	9	1
----------	-------------------	---	---

BUS 4401	Strategic Management	3
& BUS 4411	and BSBA Assurance of Learning Assessment	
General Education Requirement (Upper-Division 3)		3
Free Elective	7, 8	3
Free Elective	7, 8	2-3
Free Elective	7, 8	2-7
Units		15
Total Units		120

[BUS 3343](#) is a substitute for [BUS 1342](#). Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take [BUS 3343](#). Students interested in the Entrepreneurship concentration are recommended to take [BUS 1342](#).

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

[MATH 1264](#) is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking BUS 1204 for free electives.

9

All students need an approval from the finance chair to enroll in the Financial Management concentration.

10

Courses taken to meet a major course requirement cannot be double-counted in the concentration. Units in excess of total will be applied towards major's free electives.

Information Systems and Analytics

Information Systems and Analytics Concentration - BS in Business Administration
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
BUS 1100	Career Readiness I	1
BUS 2214	Financial Accounting	3
ECON 2001	Survey of Economics (4B) ⁴	3
STAT 1210	Business Statistics I	3
General Education Requirement (1A)		3

Units

13

Term 2

BUS 2207

Legal Responsibilities of Business

3

BUS 2215

Managerial Accounting

3

MATH 1264

Calculus for Data Science I (2) ^{4, 5}

3-4

or MATH 1267

or Business Calculus

STAT 1220

Business Statistics II

3

General Education Requirement (1C)

3

Units

16

Second Year

Term 1

BUS 1342	Financial Institutions	1	3
or BUS 3343	or Quantitative Methods in Finance		
BUS 2206	Career Readiness II		1
BUS 3391	Information Systems		3
General Education Requirement (1B)			3
General Education Requirement (3A)			3
General Education Requirement (4A)			3

Units

16

Term 2

BUS 3346

Principles of Marketing

3

BUS 3387

Organizational Behavior

3

BUS 3392

Business Application Development

3

Concentration Project Course

1

Technology Management Course

3

General Education Requirement (3B)

3

Units

16

Third Year

Term 1

BUS 3306	Career Readiness III	1
BUS 3393	Database Systems in Business	3
International Business Course		3-4
Concentration Project Course		0-1
General Education Requirement (6)		3
Free Elective	^{7,8}	3
Units		14

Term 2

BUS 3394	Systems Analysis and Design	3
BUS 4404	Law, Governmental, and Social Influences on Business	3
BUS 4497	Business Analytics	3
Concentration Project Course		1
General Education Requirement (5A)		3
General Education Requirement (5C)		1
General Education Requirement (Upper-Division 4)		3
Units		17

Fourth Year

Term 1

BUS 3399	IT Infrastructure and Security Management	3
Senior Project Course	2, 3	3-4
Concentration Project Course		0-1
General Education Requirement (Upper-Division 2/5)		3
General Education Requirement (5B)		3
Units		14

Term 2

BUS 4401	Strategic Management	3
& BUS 4411	and BSBA Assurance of Learning Assessment	
Approved Elective		3
General Education Requirement (Upper-Division 3)		3
Free Elective ^{7, 8}		2-3
Free Elective ^{7, 8}		2-7
	Units	14
	Total Units	120

[BUS 3343](#) is a substitute for [BUS 1342](#). Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take [BUS 3343](#). Students interested in the Entrepreneurship concentration are recommended to take [BUS 1342](#).

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

[MATH 1264](#) is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking BUS 1204 for free electives.

Management and Human Resources

Management and Human Resources Concentration - BS in Business Administration
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BUS 1100	Career Readiness I	1
BUS 2214	Financial Accounting	3
ECON 2001	Survey of Economics (4B) ⁴	3
STAT 1210	Business Statistics I	3
General Education Requirement (1A)		3
Units		13

Term 2

BUS 2207	Legal Responsibilities of Business	3
BUS 2215	Managerial Accounting	3

MATH 1264	Calculus for Data Science I (2)	4, 5	3-4
or MATH 1267	or Business Calculus		
STAT 1220	Business Statistics II		3
General Education Requirement (1C)			3
Units			16

Second Year

Term 1

BUS 1342	Financial Institutions	1	3
or BUS 3343	or Quantitative Methods in Finance		

BUS 2206	Career Readiness II	1
BUS 3391	Information Systems	3
General Education Requirement (1B)		3
General Education Requirement (3A)		3
General Education Requirement (4A)		3
Units		16

Term 2

BUS 3346	Principles of Marketing	3
BUS 3387	Organizational Behavior	3

Technology Management Course	3
General Education Requirement (3B)	3
General Education Requirement (5A) ⁶	3
General Education Requirement (5C) ⁶	1
Units	16

Third Year

Term 1

BUS 3306	Career Readiness III	1
BUS 3384	Human Resources Management	3

International Business Course	3-4
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Concentration Project Course	1
------------------------------	---

Approved Elective ^{9, 10}	3
------------------------------------	---

General Education Requirement (6)	3
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Units	15
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Term 2

BUS 3388	Training, Developing, and Managing Employees	3
----------	--	---

BUS 4404	Law, Governmental, and Social Influences on Business	3
----------	--	---

BUS 4477	Management Consulting, Change, and Development	3
----------	--	---

Concentration Project Course	1
------------------------------	---

General Education Requirement (Upper-Division 4)	3
--	---

Units	13
-------	----

Fourth Year

Term 1

BUS 4489	Negotiation	3
----------	-------------	---

Senior Project Course	3-4
-----------------------	-----

Concentration Project Course	1
------------------------------	---

General Education Requirement (Upper-Division 2/5)	3
--	---

General Education Requirement (5B)	3
------------------------------------	---

Units	14
-------	----

Term 2

BUS 4401	Strategic Management	3
----------	----------------------	---

& BUS 4411	and BSBA Assurance of Learning Assessment
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Approved Elective	9, 10	3-4
-------------------	-------	-----

General Education Requirement (Upper-Division 3)	3
--	---

Free Elective	7, 8	3
---------------	------	---

Free Elective	7, 8	2-3
---------------	------	-----

Free Elective ^{7, 8} 2-7

Units 17

Total Units 120

1

[BUS 3343](#) is a substitute for [BUS 1342](#). Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take [BUS 3343](#). Students interested in the Entrepreneurship concentration are recommended to take [BUS 1342](#).

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

MATH 1264 is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking BUS 1204 for free electives.

9

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

10

Units in excess of total will be applied towards major's free electives.

Marketing Management

Marketing Management Concentration - BS in Business Administration

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BUS 1100	Career Readiness I	1
BUS 2214	Financial Accounting	3

ECON 2001	Survey of Economics (4B) ⁴	3
-----------	---------------------------------------	---

STAT 1210	Business Statistics I	3
-----------	-----------------------	---

General Education Requirement (1A)		3
------------------------------------	--	---

Units		13
-------	--	----

Term 2

BUS 2207	Legal Responsibilities of Business	3
----------	------------------------------------	---

BUS 2215	Managerial Accounting	3
----------	-----------------------	---

MATH 1264	Calculus for Data Science I (2) ^{4, 5}	3-4
-----------	---	-----

or MATH 1267	or Business Calculus	
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STAT 1220	Business Statistics II	3
-----------	------------------------	---

General Education Requirement (1C)		3
------------------------------------	--	---

Units		16
-------	--	----

Second Year

Term 1

BUS 1342	Financial Institutions ¹	3
----------	-------------------------------------	---

or BUS 3343	or Quantitative Methods in Finance	
-------------	------------------------------------	--

BUS 2206	Career Readiness II	1
----------	---------------------	---

BUS 3391	Information Systems	3
----------	---------------------	---

General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement (3A)	3
------------------------------------	---

General Education Requirement (4A)	3
------------------------------------	---

Units	16
--------------	-----------

Term 2

BUS 3346	Principles of Marketing	3
----------	-------------------------	---

BUS 3387	Organizational Behavior	3
----------	-------------------------	---

Technology Management Course	3
------------------------------	---

General Education Requirement (3B)	3
------------------------------------	---

General Education Requirement (5A)	6	3
------------------------------------	---	---

General Education Requirement (5C)	6	1
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Units		16
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Third Year

Term 1

BUS 3306	Career Readiness III	1
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BUS 3396	Consumer Insights	4
----------	-------------------	---

& 3396A	and Consumer Insights Project	
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International Business Course		3-4
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General Education Requirement (Upper-Division 4) 3

General Education Requirement (6) 3

Units 15

Term 2

BUS 4404 Law, Governmental, and Social Influences on Business 3

BUS 4418 Marketing Research 4

& 4418A and Marketing Research Project

Approved Elective 9, 10 4

General Education Requirement (5B) 6 3

Units

14

Fourth Year

Term 1

Senior Project Course^{2, 3}

3-4

Approved Elective^{9, 10}

4

General Education Requirement (Upper-Division 2/5)

3

Free Elective^{7, 8}

3

Units

14

Term 2

BUS 4401 Strategic Management 3

& BUS 4411 and BSBA Assurance of Learning Assessment

Approved Elective^{9, 10} 4

General Education Requirement (Upper-Division 3) 3

Free Elective^{7, 8} 3-4

Free Elective^{7, 8} 2-7

Units 16

Total Units 120

1

[BUS 3343](#) is a substitute for [BUS 1342](#). Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take [BUS 3343](#). Students interested in the Entrepreneurship concentration are recommended to take [BUS 1342](#).

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

[MATH 1264](#) is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking BUS 1204 for free electives.

9

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

10

This course does not have an activity. Marketing requires this course to be four units.

Real Estate Finance

Real Estate Finance Concentration - BS in Business Administration

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BUS 1100	Career Readiness I	1
BUS 2214	Financial Accounting	3
ECON 2001	Survey of Economics (4B) ⁴	3
STAT 1210	Business Statistics I	3
General Education Requirement (1A)		3

Units

13

Term 2

BUS 2207

Legal Responsibilities of Business

3

BUS 2215

Managerial Accounting

3

MATH 1264

Calculus for Data Science I (2) ^{4, 5}

3-4

or MATH 1267

or Business Calculus

STAT 1220

Business Statistics II

3

General Education Requirement (1C)

3

Units

16

Second Year

Term 1

BUS 1342	Financial Institutions	1	3
or BUS 3343	or Quantitative Methods in Finance		
BUS 2206	Career Readiness II		1
BUS 3391	Information Systems		3
General Education Requirement (1B)			3
General Education Requirement (3A)			3
General Education Requirement (4A)			3

Units

16

Term 2

BUS 3346

Principles of Marketing

3

BUS 3387

Organizational Behavior

3

Technology Management Course

3

General Education Requirement (3B)

3

General Education Requirement (5A) ⁶

3

General Education Requirement (5C) ⁶

1

Units

16

Third Year

Term 1

BUS 3306	Career Readiness III	1
BUS 3431	Security Analysis and Portfolio Management ^{9, 10}	4
International Business Course		3-4
General Education Requirement (6)		3
General Education Requirement (5B)		3
Units		15

Term 2

BUS 3440	Financial Modeling and Visualization in Excel	9, 10	4
----------	---	-------	---

BUS 4404	Law, Governmental, and Social Influences on Business		3
----------	--	--	---

Approved Elective	11, 12		3
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General Education Requirement (Upper-Division 4)			3
--	--	--	---

Units			13
-------	--	--	----

Fourth Year

Term 1

Senior Project Course	2, 3		3-4
-----------------------	------	--	-----

Approved Elective	11, 12	3
-------------------	--------	---

General Education Requirement (Upper-Division 2/5)		3
--	--	---

Free Elective	7, 8	3
---------------	------	---

Units		13
-------	--	----

Term 2

BUS 4401	Strategic Management	3
----------	----------------------	---

& BUS 4411	and BSBA Assurance of Learning Assessment	
------------	---	--

BUS 4434	Real Estate Finance	9, 10	4
----------	---------------------	-------	---

Approved Elective	11, 12	3
General Education Requirement (Upper-Division 3)		3
Free Elective	7, 8	2-3
Free Elective	7, 8	2-7
	Units	18
	Total Units	120

1

BUS 3343 is a substitute for BUS 1342. Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take BUS 3343. Students interested in the Entrepreneurship concentration are recommended to take BUS 1342.

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

MATH 1264 is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking [BUS 1204](#) for free electives.

9

All students need an approval from the finance department chair to enroll in the concentration.

10

All students enrolled in the concentration are recommended to pursue a minor with the CAED department. The most popular minor is the Real Estate Property Development minor.

11

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

12

Highly recommended for students concentrating in real estate finance. This course also counts toward the real estate development minor.

Supply Chain Management

Supply Chain Management Concentration - BS in Business Administration
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.
Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year			
Term 1			Units
BUS 1100	Career Readiness I		1
BUS 2214	Financial Accounting		3
ECON 2001	Survey of Economics (4B)	4	3
STAT 1210	Business Statistics I		3

General Education Requirement (1A)	3
------------------------------------	---

Units	13
-------	----

Term 2

BUS 2207	Legal Responsibilities of Business	3
----------	------------------------------------	---

BUS 2215	Managerial Accounting	3
----------	-----------------------	---

MATH 1264	Calculus for Data Science I (2)	3-4
-----------	---------------------------------	-----

or MATH 1267	or Business Calculus
--------------	----------------------

STAT 1220	Business Statistics II	3
-----------	------------------------	---

General Education Requirement (1C)	3
------------------------------------	---

Units

16

Second Year

Term 1

BUS 1342	Financial Institutions	1	3
or BUS 3343	or Quantitative Methods in Finance		
BUS 2206	Career Readiness II		1
BUS 3391	Information Systems		3
General Education Requirement (1B)			3
General Education Requirement (3A)			3

General Education Requirement (4A)	3
------------------------------------	---

Units	16
-------	----

Term 2

BUS 3346	Principles of Marketing	3
----------	-------------------------	---

BUS 3387	Organizational Behavior	3
----------	-------------------------	---

Technology Management Course	9	3
------------------------------	---	---

General Education Requirement (3B)	3
------------------------------------	---

General Education Requirement (5A)	6	3
------------------------------------	---	---

General Education Requirement (5C)⁶

1

Units

16

Third Year

Term 1

BUS 2208

Shipping and Port Management

3

BUS 3306

Career Readiness III

1

ITP 3371

Supply Chain Management in Manufacturing and
Services⁹

3

International Business Course

3-4

General Education Requirement (6)	3
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Units	14
-------	----

Term 2

BUS 3308	Logistics and Intermodal Transportation	3
----------	---	---

BUS 4404	Law, Governmental, and Social Influences on Business	3
----------	--	---

BUS 4416	Strategic Sourcing and Procurement Management	3
----------	---	---

ITP 3303	Lean Six Sigma Green Belt	3
----------	---------------------------	---

9

General Education Requirement (5B)	3
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Units	15
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Fourth Year

Term 1

BUS 4417 Supply Chain Analytics 3

Senior Project Course^{2, 3} 3-4

General Education Requirement (Upper-Division 2/5) 3

General Education Requirement (Upper-Division 4) 3

Free Elective^{7, 8} 3

Units 16

Term 2

BUS 4401	Strategic Management	3
& BUS 4411	and BSBA Assurance of Learning Assessment	
ITP 4410	Operations Planning and Control	3
Approved Elective ⁹		3
General Education Requirement (Upper-Division 3)		3
Free Elective ^{7, 8}		2-3
	Units	14
	Total Units	120

[BUS 3343](#) is a substitute for [BUS 1342](#). Students interested in the Financial Management and/or Real Estate Finance concentrations are strongly recommended to take [BUS 3343](#). Students interested in the Entrepreneurship concentration are recommended to take [BUS 1342](#).

2

Recommended for Entrepreneurship concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

3

Recommended for Accounting concentration students. For other concentration students, additional prerequisites may be needed to complete this requirement.

4

Required in Major or Support; also satisfies General Education (GE) requirement.

5

[MATH 1264](#) is recommended for those pursuing the Cross-Disciplinary Studies Data Science Minor or Actuarial Preparation Minor.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

8

Students should consider taking BUS 1204 for free electives.

9

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

Chemistry

Concentration Not Yet Declared - BS in Chemistry

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CHEM 1120 Fundamentals of Chemical Structure and Properties (5A & 5C) ¹ 4

BIO 1151 Life: Molecules and Cells (5B) ¹ 4

MATH 1261 Calculus I (2) ¹ 4

General Education Requirement (1A) 3

Free Elective ^{6,7} 1

Units 16

Term 2

CHEM 1103	Research Methods I	1
CHEM 1122	Fundamentals of Chemical Reactivity	4
MATH 1262	Calculus II	4
PHYS 1141	General Physics I	4
General Education Requirement (1C)		3
Units		16

Second Year

Term 1

CHEM 2220	Inorganic Chemistry I: D-Block Chemistry	3
CHEM 2221	Inorganic Chemistry I Laboratory	1
CHEM 2242	Organic Chemistry I	5
MATH 2263	Calculus III	3
General Education Requirement (1B)		3
Units		15

Term 2

CHEM 2201	Undergraduate Research	1
or CHEM 2203	or Research Methods II	

CHEM 3330	Foundations of Chemical Analysis	4
PHYS 1143	General Physics II	4
General Education Requirement		3
General Education Requirement		3
Units		15

Third Year

Term 1

CHEM 3302	Undergraduate Seminar II	1
CHEM 3352	Biochemistry (Upper-Division 2/5) ¹	4

CHEM 3392	Physical Chemistry I	3
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CHEM 3393	Physical Chemistry Laboratory I	1
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Advanced Lecture/Laboratory Subdiscipline Extension Elective	4-5
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General Education Requirement	3
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Units	17
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Term 2

CHEM 3380	Foundations of Macromolecular Chemistry	4
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Advanced Lecture/Laboratory Subdiscipline Extension Elective	4-5
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Fourth Year

Term 1

CHEM 4461	Senior Project I	1
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Concentration or Advanced Elective Course	2,3,4,5	4
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Concentration or Advanced Elective Course	2,3,4,5	4
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General Education Requirement	3
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Free Elective	6,7	2
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Units	14
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Term 2

CHEM 4462	Senior Project II	1
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Concentration or Advanced Elective Course	2,3,4,5	4
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Concentration or Advanced Elective Course	2,3,4,5	1-4
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General Education Requirement		3
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Free Elective	6,7	0-5
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Units	12
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major requirement cannot be double-counted in the concentration or in the Advanced Electives.

3

Consultation with advisor is recommended prior to selecting advanced electives; bear in mind your selections may impact pursuit of post-baccalaureate studies and/or goals.

4

Maximum of 4 units may be applied toward Advanced Electives from [CHEM 4401](#).

5

Maximum of 2 units may be applied toward Advanced Electives from the following:
CHEM 4404, CHEM 4485, CHEM 4495, or SCM/ENGR 3302.

6

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

7

Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Polymers and Coatings

Polymers and Coatings Concentration - BS in Chemistry

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
BIO 1151	Life: Molecules and Cells (5B) ¹	4
MATH 1261	Calculus I (2) ¹	4
General Education Requirement (1A)		3
Free Elective ^{6,7}		1
Units		16
Term 2		

CHEM 1103	Research Methods I	1
CHEM 1122	Fundamentals of Chemical Reactivity	4
MATH 1262	Calculus II	4
PHYS 1141	General Physics I	4
General Education Requirement (1C)		3
Units		16

Second Year

Term 1

CHEM 2220	Inorganic Chemistry I: D-Block Chemistry	3
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CHEM 2221	Inorganic Chemistry I Laboratory	1
CHEM 2242	Organic Chemistry I	5
MATH 2263	Calculus III	3
General Education Requirement (1B)		3
Units		15

Term 2

CHEM 2201	Undergraduate Research	1
or CHEM 2203	or Research Methods II	
CHEM 3330	Foundations of Chemical Analysis	4

CHEM 3380	Foundations of Macromolecular Chemistry	4
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PHYS 1143	General Physics II	4
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General Education Requirement		3
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Units		16
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Third Year

Term 1

CHEM 3302	Undergraduate Seminar II	1
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CHEM 3352	Biochemistry (Upper-Division 2/5) ¹	4
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CHEM 3392	Physical Chemistry I	3
-----------	----------------------	---

CHEM 3393	Physical Chemistry Laboratory I	1
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Advanced Lecture/Laboratory Subdiscipline Extension Elective	4-5
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General Education Requirement	3
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Units	17
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Term 2

CHEM 4486	Surface Chemistry of Materials	3
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Advanced Lecture/Laboratory Subdiscipline Extension Elective	4-5
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General Education Requirement	3
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General Education Requirement	3
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Units

14

Fourth Year

Term 1

CHEM 4461	Senior Project I	1
CHEM 4480	Polymer Synthesis and Characterization	3
CHEM 4481	Polymer Synthesis and Characterization Laboratory	2
General Education Requirement		3
General Education Requirement		3
Free Elective	6,7	2

Units **14**

Term 2

CHEM 4462 Senior Project II 1

CHEM 4482 Coatings and Formulations 3

CHEM 4483 Coatings and Formulations Laboratory 2

General Education Requirement 3

Free Elective^{6,7} 3

Units **12**

Total Units **120**

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major requirement cannot be double-counted in the concentration or in the Advanced Electives.

3

Consultation with advisor is recommended prior to selecting advanced electives; bear in mind your selections may impact pursuit of post-baccalaureate studies and/or goals.

4

Maximum of 4 units may be applied toward Advanced Electives from [CHEM 4401](#).

5

Maximum of 2 units may be applied toward Advanced Electives from the following: [CHEM 4404](#), [CHEM 4485](#), [CHEM 4495](#), or SCM/[ENGR 3302](#).

6

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

7

Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Child Development

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
CD 1102	Orientation to the Child Development Major	2
PSY 2201	Introductory Psychology (4B) ¹	3
STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3

Units 14

Term 2

CD 1131 Observing and Interacting with Children 3

or CD 2202 or Developmental Science Technology Activity

or CD 2254 or Child, Family, and Community

CD/PSY 2229 Research Methods in Psychology 3

CD/PSY 2256 Developmental Psychology 3

General Education Requirement (1B) 3

General Education Requirement 3

Units 15

Second Year

Term 1

CD 2230	Preschool Laboratory	3
CD 3304	Infant and Toddler Development	4
or CD 3305	or Early and Middle Childhood Development	
or CD/PSY 3306	or Adolescence	
General Education Requirement (5A)		3
General Education Requirement (5C)		1
General Education Requirement		3
Units		14

Term 2

CD 3304	Infant and Toddler Development	4
or CD 3305	or Early and Middle Childhood Development	
or CD/PSY 3306	or Adolescence	
CD 3329	Advanced Research Methods in Developmental Science	4
PSY 2240	Biopsychology	3
General Education Requirement		3
Units		14

Third Year

Term 1

Child Development Elective	4
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Professional Skills Course	3-4
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Diversity, Equity and Inclusion Course (Upper-Division 4) ¹	3-4
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General Education Requirement	3
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Free Elective ²	3
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Units	16
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Term 2

CD 4424	Children's Development in Diverse Cultures	4
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Child Development Elective	4
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Upper-Division Science Course (Upper-Division 2/5)	¹	3-4
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General Education Requirement (Upper-Division 3)		3
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Units		14
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Fourth Year

Term 1

CD 4448	Research Internship I	3
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or CD 4453	or Supervised Fieldwork Internship I	
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CD 4461	Senior Project Seminar	2
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Child Development Elective		4
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Free Elective	2	3
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Free Elective	2	3
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Free Elective	2	3
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Units	18
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Term 2

CD 4462	Senior Project	2
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Child Development Elective	4
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Free Elective	2	3
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Free Elective	2	3
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Free Elective	2	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

City and Regional Planning

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
CRP 1211	Urban Planning History	4
CRP 1212	Introduction to City Planning	4
DATA 1000	Statistical and Data Literacy (2) ¹	3
or STAT 1110	or Applied Statistical Concepts and Methods	
General Education Requirement (4A)		3
	Units	14
Term 2		

CRP 1213	Methods of Population and Housing Analysis	4
CRP/ES 1215	Planning Approaches to a Just City (USCP)	4
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		14

Second Year

Term 1

General Education Requirement (1B)	3
General Education Requirement	3

General Education Requirement	3
Free Elective ^{2,3}	4
Free Elective ^{2,3}	4
Units	17

Term 2

CRP 2214	Methods in Land Use and Transportation	4
CRP 2216	Representing the Built Environment	3
CRP 2457	GIS Applications in Planning	2
General Education Requirement		3

General Education Requirement	1
Free Electives ^{2,3}	4
Units	17

Third Year

Term 1

CRP 3202	Urban Design Studio	4
CRP 3336	Introduction to Environmental Planning	4
General Education Requirement		3
General Education Requirement		3

Units

14

Term 2

CRP 3315

Public and Private Real Estate Development

4

CRP 3341

Urban Development Studio

4

General Education Requirement

3

General Education Requirement

3

Units

14

Fourth Year

Term 1

CRP 4410	Urban Planning Studio	4
CRP 4420	Land Use Law	4
CAED Elective ^{2,3}		3
General Education Requirement		3
Units		14

Term 2

CRP 4461	Senior Project	2
or CRP 4463	or Senior Project - Studio	
CAED Elective ^{2,3}		3

General Education Requirement	3
Free Elective ^{2,3}	4
Free Elective ^{2,3}	4
Units	16
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Civil Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
CE 1111	Introduction to Civil Engineering	1
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I	4
General Education Requirement (1A)		3

Units

16

Term 2

CE/ENVE 1112 Spatial Visualization and Drawing 3

MATH 1262 Calculus II 4

PHYS 1143 General Physics II 4

General Education Requirement (1B) 3

General Education Requirement (1C) 3

Units

17

Second Year

Term 1

CE 2251	Computational Applications in Civil and Environmental Engineering	2
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
General Education Requirement (3A)		3
General Education Requirement (3B)		3
Units		15

Term 2

CE 2259	Civil Engineering Materials	3
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ENGR 2212	Introduction to Engineering Dynamics	2
GEOL 2240	Physical Geology	3
MATH 2341	Linear Analysis	4
General Education Requirement (4A)		3
Units		15

Third Year

Term 1

CE 3321	Fundamentals of Transportation Engineering	4
CE/ENVE 3336	Environmental Fluid Mechanics and Hydraulic Systems	4

CE 3352	Structural Analysis	4
STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
General Education Requirement (4B)		3
Units		18

Term 2

CE/ENVE 3337	Water Resources & Environmental Engineering	4
CE 3355	Reinforced Concrete Design	3
CE 3381	Geotechnical Engineering	4
CE/ENVE 3465	Infrastructure Systems	2

General Education Requirement (5B)	3
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Units	16
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Fourth Year

Term 1

CE 3375	Fundamentals of Construction Engineering and Management	4
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CE 4466	Senior Design Project I	1
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Technical Elective	3
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Technical Elective	3
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Technical Elective	3
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General Education Requirement (6)	3
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Units	17
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Term 2

CE 4467	Senior Design Project II	3
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Technical Elective	3
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Technical Elective	3
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Technical Elective	3
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General Education Requirement (Upper-Division 3)	3
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General Education Requirement (Upper-Division 4)	3
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Units 18

Total Units 132

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Communication Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

COMS 1100 Introduction to the Communication Studies Major 2

COMS 1101 Public Speaking (1C) ¹ 3

or COMS 1102 or Public Speaking in Digital Spaces

General Education Requirement (1A)	3
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General Education Requirement	3
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General Education Requirement	3
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Units	14
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Term 2

COMS 2205	Rhetorical Studies	3
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COMS 2206	Communication Theory	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1B) 3

Free Elective ^{6, 7} 3

Units 15

Second Year

Term 1

COMS 2211 Interpersonal Communication ⁷ 3

or COMS 2213

or Organizational Communication

or COMS 2215

or Media Studies

or COMS 2217

or Small Group Collaboration and Creativity

COMS 2250	Building Advocacy Skills ²	3
or COMS 2208	or Performance, Literature, and Culture	
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		1
Units		16

Term 2

COMS 2211	Interpersonal Communication ⁷	3
or COMS 2213		

or COMS 2215 or Organizational Communication

or COMS 2217 or Media Studies

or Small Group Collaboration and Creativity

Focus Area Course	2, 3	3
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General Education Requirement		3
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General Education Requirement		3
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Free Elective	6, 7	3
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Units		15
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Third Year

Term 1

COMS 3312	Quantitative Research Methods in Communication Studies	3
or COMS 3313	or Qualitative Research Methods in Communication Studies	
COMS 3316	Intercultural Communication (Upper-Division 4, USCP) 1	3
Focus Area Course	2, 3	3
General Education Requirement		3
Free Elective	6, 7	3
Units		15

Term 2

COMS 3332	Rhetorical Criticism (GWR)	3
or COMS 3385	or Media Criticism	
Focus Area Course	2, 3	3
Focus Area Course	2, 3	3
COMS Elective	4, 5	3
Free Elective	6, 7	3
Units		15

Fourth Year

Term 1

COMS 4460	Undergraduate Seminar	1
Focus Area Course	2, 3	3
Focus Area Course	2, 3	3
General Education Requirement		3
Free Elective	6, 7	3
Free Elective	6, 7	3
Units		16

Term 2

COMS 4461	Senior Project	2
COMS Elective	4, 5	3
Free Elective	6, 7	3
Free Elective	6, 7	3
Free Elective	6, 7	3
Units		14
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Repeatable course; Major credit limited to 3 units.

3

Repeatable course; Major credit limited to 6 units.

4

Courses taken to meet the focus area requirement cannot be double-counted in the COMS Electives.

5

A maximum of 3 units may be applied toward COMS electives: [COMS 4400](#), [COMS 4480](#), [COMS 4485](#).

6

Free electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division.

7

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Comparative Ethnic Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ES 1112 Race, Culture, and Politics in the United States (4A) ¹ 3

ES 1134 Writing and Rhetoric (1A) ¹ 3

General Education Requirement (1C) 3

General Education Requirement (2) 3

Units **12**

Term 2

ES 1114 Introduction to Ethnic Studies 4

ES 1145	Writing, Reasoning, and Argumentation (1B)	¹	3
ES Course (6)		¹	3
General Education Requirement			3
Free Elective		²	3
Units			16

Second Year

Term 1

ES Course (3A)		¹	3
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ES Course (6)¹

3

General Education Requirement

3

General Education Requirement

3

General Education Requirement

1

Free Elective²

3

Units

16

Term 2

ES Course (3A)¹

3

ES Course (3A)	¹	3
ES Course (6)	¹	3
General Education Requirement (4B)		3
Free Elective	²	3
Units		15

Third Year

Term 1

ES/WGQS 3345	Queer Ethnic Studies (Upper-Division 4, USCP)	¹	4
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ES/WGQS 3350	Gender, Race, Culture, Science, and Technology (Upper-Division 2/5) ¹	4
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ES 3380	Critical Race Theory	4
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Free Elective ²		3
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Units	15
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Term 2

ES 3382	Racial Capitalism	4
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ES 4401	Ethnic Studies: Theoretical Foundations	4
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or ES 4402	or Decolonial Theory
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or ES 4403	or Feminist and Queer/Trans of Color Theories
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ES Course (Upper-Division 3) ¹	4
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3000-4000 level ES Course	4
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Units	16
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Fourth Year

Term 1

ES 3390	Research Methodology in Comparative Ethnic Studies	4
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ES 4401	Ethnic Studies: Theoretical Foundations	4
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or ES 4402	or Decolonial Theory
------------	----------------------

or ES 4403	or Feminist and Queer/Trans of Color Theories
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3000-4000 level ES Course	4
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Free Elective

2

3

Units

15

Term 2

ES 4401

Ethnic Studies: Theoretical Foundations

4

or ES 4402

or Decolonial Theory

or ES 4403

or Feminist and Queer/Trans of Color Theories

ES 4461

Senior Project

4

3000-4000 level ES Course

4

Free Elective

2

3

Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

General Curriculum in Computer Engineering

General Curriculum in Computer Engineering - BS in Computer Engineering
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1	Units
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CPE/CSC 1000	Computing Majors Orientation	1
CPE/CSC 1024	Introduction to Computing	2
BIO 1111	General Biology (5B) ²	3-4
or BIO 2213	or Life Science for Engineers	
or BOT 1121	or General Botany	
or MCRO 2221	or Introduction to Microbiology	
MATH 1261	Calculus I (2) ²	4
General Education Requirement (1A)		3
General Education Requirement (3A)		3
Units		16-17

Term 2

CSC 1001	Fundamentals of Computer Science	4
& 1001L	and Fundamentals of Computer Science Laboratory	
MATH 1262	Calculus II	4
PHYS 1141	General Physics I (5A & 5C) ²	4
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

CPE 2300	Introduction to Computer Systems	3
CPE 2301	Introduction to HDL and Digital Design Laboratory	1
CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 2031	Transition to Advanced Mathematics	3
PHYS 1143	General Physics II	4
General Education Requirement (1C)		3
Units		18

Term 2

CPE/CSC 2050	System Software Mechanics	3
EE 2211	Electric Circuit Analysis I	4
& EE 2241	and Electric Circuit Analysis Laboratory I	
MATH 1151	Linear Algebra	3
or MATH 2341	or Linear Analysis	
General Education Requirement (3B)		3
General Education Requirement (4A)		3
Units		16

Third Year	
Term 1	

CPE/CSC 3201	Introduction to Computer Security	3
CPE 3300	Computer Architecture	4
CPE/CSC 4553	Introduction to Operating Systems	3
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ²	3
General Education Requirement (4B)		3
Units		16

Term 2

CPE 3160	Microcontrollers and Embedded Applications	3
CPE 4464	Introduction to Computer Networks	3

EE 3306	Electronics I	4
& 3306L	or Signals and Systems	
or EE 2328		
STAT 3210	Engineering Statistics (Upper-Division 2/5)	2
or STAT 3310	or Probability and Random Processes for Engineers	
General Education Requirement (6)		3
Units		16

Fourth Year

Term 1

WGQS 3350 Gender, Race, Culture, Science, and Technology 4

or WGQS
3351

or Gender, Race, Class, Nation: Critical Computing and
Engineering Studies

Math/Science Support Elective³ 3

Senior Project⁵ 2-3

Technical Elective⁴ 3

Technical Elective⁴ 3

Units 15-16

Term 2

Senior Project⁵ 2-3

Technical Elective	4	4
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Technical Elective	4	4
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General Education Requirement (Upper-Division 4)		3
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Units		13-14
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Total Units		125-128
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1

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

2

Required in Major or Support; also satisfies General Education (GE) requirement.

3

Excess units at the 2000-5000 level may be applied toward technical electives.

4

A combined maximum of 8 units may be taken from [CPE 4491](#), [CPE 4492](#), [CPE 4493](#), [CPE 4495](#), [CSC 4091](#), [CSC 4092](#), [CSC 4093](#), [CSC 4191](#), [CSC 4192](#), [CSC 4193](#), [CSC 4291](#), [CSC 4292](#), [CSC 4293](#), [CSC 4400](#), [CSC 4495](#), [CSC 4691](#), [CSC 4692](#), [CSC 4693](#), [CSC 4791](#), [CSC 4792](#), [CSC 4793](#), [CSC 4891](#), [CSC 4892](#), [CSC 4893](#), [CSC 4991](#), [CSC 4992](#), [CSC 4993](#), [DATA 4720](#), and [ENGR 4995](#); of which up to a combined 4 units may be taken from [CPE 4400](#) and [CSC 4400](#).

5

Excess units completed for senior project may be applied toward technical electives.

Computer Architecture

Computer Architecture Concentration - BS in Computer Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
CPE/CSC 1000	Computing Majors Orientation	1
CPE/CSC 1024	Introduction to Computing	2
BIO 1111	General Biology (5B) ¹	3-4
or BIO 2213	or Life Science for Engineers	
or BOT 1121	or General Botany	
or MCRO 2221	or Introduction to Microbiology	
MATH 1261	Calculus I (2) ¹	4
General Education Requirement (1A)		3
General Education Requirement (3A)		3

Units

16-17

Term 2

CSC 1001 Fundamentals of Computer Science 4

& 1001L and Fundamentals of Computer Science Laboratory

MATH 1262 Calculus II 4

PHYS 1141 General Physics I (5A & 5C)¹ 4

General Education Requirement (1B) 3

Units

15

Second Year

Term 1

CPE 2300	Introduction to Computer Systems	3
CPE 2301	Introduction to HDL and Digital Design Laboratory	1
CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 2031	Transition to Advanced Mathematics	3
PHYS 1143	General Physics II	4
General Education Requirement (1C)		3
Units		18

Term 2

CPE/CSC 2050	System Software Mechanics	3
EE 2211	Electric Circuit Analysis I	4
& EE 2241	and Electric Circuit Analysis Laboratory I	
MATH 1151	Linear Algebra	3
or MATH 2341	or Linear Analysis	
General Education Requirement (3B)		3
General Education Requirement (4A)		3
Units		16

Third Year

Term 1

CPE/CSC 3201	Introduction to Computer Security	3
CPE 3300	Computer Architecture	4
CPE/CSC 4553	Introduction to Operating Systems	3
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ¹	3
General Education Requirement (4B)		3
Units		16

Term 2

CPE 3160	Microcontrollers and Embedded Applications	3
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CPE 4300	Advanced Computer Architecture	4
EE 3306	Electronics I	4
& 3306L	and Electronics Laboratory I	
STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
or STAT 3310	or Probability and Random Processes for Engineers	
General Education Requirement (6)		3
Units		17

Fourth Year

Term 1

CPE 4464	Introduction to Computer Networks	3
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MATH 2263	Calculus III	3
PHYS 2211	General Physics III: Modern Physics	4
Senior Project		3
Units		13

Term 2

WGQS 3350	Gender, Race, Culture, Science, and Technology	4
Senior Project		3
Concentration Elective		3-4
General Education Requirement (Upper-Division 4)		3

Units 13-14

Total Units 124-126

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Computer Hardware Engineering

Computer Hardware Engineering Concentration - BS in Computer Engineering
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CPE/CSC 1000 Computing Majors Orientation

1

CPE/CSC 1024	Introduction to Computing	2
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BIO 1111	General Biology (5B)	1	3-4
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or BIO 2213	or Life Science for Engineers
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or BOT 1121	or General Botany
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or MCRO 2221	or Introduction to Microbiology
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MATH 1261	Calculus I (2)	1	4
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General Education Requirement (1A)	3
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General Education Requirement (3A)	3
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Units	16-17
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Term 2

CSC 1001	Fundamentals of Computer Science	4
& 1001L	and Fundamentals of Computer Science Laboratory	
MATH 1262	Calculus II	4
PHYS 1141	General Physics I (5A & 5C) ¹	4
General Education Requirement (1B)		3
Units		15

Second Year		
Term 1		

CPE 2300	Introduction to Computer Systems	3
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CPE 2301	Introduction to HDL and Digital Design Laboratory	1
CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 2031	Transition to Advanced Mathematics	3
PHYS 1143	General Physics II	4
General Education Requirement (1C)		3
Units		18

Term 2

CPE/CSC 2050	System Software Mechanics	3
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EE 2211	Electric Circuit Analysis I	4
& EE 2241	and Electric Circuit Analysis Laboratory I	
Math and Science Concentration Course ²		3
General Education Requirement (3B)		3
General Education Requirement (4A)		3
Units		16

Third Year

Term 1

CPE/CSC 3201	Introduction to Computer Security	3
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CPE 3300	Computer Architecture	4
CPE/CSC 4553	Introduction to Operating Systems	3
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ¹	3
General Education Requirement (4B)		3
Units		16

Term 2

CPE 3160	Microcontrollers and Embedded Applications	3
CPE 4464	Introduction to Computer Networks	3
EE 3306	Electronics I	4

& 3306L and Electronics Laboratory I

STAT 3210 Engineering Statistics (Upper-Division 2/5) ¹ 3

or STAT
3310 or Probability and Random Processes for Engineers

Math and Science Concentration Course ² 3

Units 16

Fourth Year

Term 1

WGQS 3350 Gender, Race, Culture, Science, and Technology 4

or WGQS
3351 or Gender, Race, Class, Nation: Critical Computing and
Engineering Studies

Senior Project	5	2-3
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Technical Elective	3, 4	3
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Technical Elective	3, 4	3
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General Education Requirement (6)		3
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Units		15
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Term 2		
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EE 4431	Computer-Aided Design of VLSI Devices	3
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Senior Project	5	2-3
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Technical Elective	3, 4	3
Technical Elective	3, 4	3
General Education Requirement (Upper-Division 4)		3
Units		14-15
Total Units		126-128

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Excess units may be applied toward technical electives.

3

A minimum of 3 units must be taken at the 3000-5000 level.

4

A combined maximum of 8 units may be taken from [CPE 4491](#), [CPE 4492](#), [CPE 4493](#), [CPE 4495](#), [ENGR 2995](#), [ENGR 4995](#), [ENGR 5995](#); of which up to a combined 4 units may be taken from [CPE 4495](#).

5

Excess units completed for senior project may be applied toward technical electives.

Computer Systems

Computer Systems Concentration - BS in Computer Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CPE/CSC 1000	Computing Majors Orientation	1
CPE/CSC 1024	Introduction to Computing	2
BIO 1111	General Biology (5B) ²	3-4
or BIO 2213	or Life Science for Engineers	
or BOT 1121	or General Botany	
or MCRO 2221	or Introduction to Microbiology	
MATH 1261	Calculus I (2) ²	4
General Education Requirement (1A)		3
General Education Requirement (3A)		3
Units		16-17

Term 2

CSC 1001	Fundamentals of Computer Science	4
& 1001L	and Fundamentals of Computer Science Laboratory	
MATH 1262	Calculus II	4
PHYS 1141	General Physics I (5A & 5C) ²	4
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

CPE 2300	Introduction to Computer Systems	3
CPE 2301	Introduction to HDL and Digital Design Laboratory	1
CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 2031	Transition to Advanced Mathematics	3
PHYS 1143	General Physics II	4
General Education Requirement (1C)		3
Units		18

Term 2

CPE/CSC 2050	System Software Mechanics	3
EE 2211	Electric Circuit Analysis I	4
& EE 2241	and Electric Circuit Analysis Laboratory I	
MATH 1151	Linear Algebra	3
or MATH 2341	or Linear Analysis	
General Education Requirement (3B)		3
General Education Requirement (4A)		3
Units		16

Third Year

Term 1

CPE/CSC 3201	Introduction to Computer Security	3
CPE 3300	Computer Architecture	4
CPE/CSC 4553	Introduction to Operating Systems	3
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ²	3
General Education Requirement (4B)		3
Units		16

Term 2

CPE 3160	Microcontrollers and Embedded Applications	3
CPE 4464	Introduction to Computer Networks	3

EE 3306	Electronics I	¹	4
& 3306L	or Signals and Systems		
or EE 2328			
STAT 3210	Engineering Statistics (Upper-Division 2/5)	²	3
or STAT 3310	or Probability and Random Processes for Engineers		
General Education Requirement (6)			3
Units			16

Fourth Year

Term 1

CSC 4214	Binary Exploitation: Tools and Techniques	3
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or CPE 4220 or Network Security

or CPE 4280 or Introduction to Hardware Security

CSC 4310 Compiler Construction 3-4

or CSC 4554 or Implementation of Operating Systems

or CPE 4465 or Advanced Computer Networks

WGQS 3350 Gender, Race, Culture, Science, and Technology 4

or WGQS 3351 or Gender, Race, Class, Nation: Critical Computing and Engineering Studies

Math/Science Elective¹ 3

Senior Project 3

Units 16-17

Term 2

CPE 4650	Scalable Server Implementation and Testing	3
CPE 4669	Distributed Systems	4
Senior Project		3
General Education Requirement (Upper-Division 4)		3
	Units	13
	Total Units	126-128

1

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

Required in Major or Support; also satisfies General Education (GE) requirement.

Embedded Systems

Embedded Systems Concentration - BS in Computer Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
CPE/CSC 1000	Computing Majors Orientation	1
CPE/CSC 1024	Introduction to Computing	2
BIO 1111	General Biology (5B) ²	3-4

or BIO 2213 or Life Science for Engineers

or BOT 1121 or General Botany

or MCRO 2221 or Introduction to Microbiology

MATH 1261 Calculus I (2)² 4

General Education Requirement (1A) 3

General Education Requirement (3A) 3

Units 16-17

Term 2

CSC 1001 Fundamentals of Computer Science 4

& 1001L and Fundamentals of Computer Science Laboratory

MATH 1262	Calculus II	4
PHYS 1141	General Physics I (5A & 5C) ²	4
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

CPE 2300	Introduction to Computer Systems	3
CPE 2301	Introduction to HDL and Digital Design Laboratory	1
CSC 2001	Data Structures	4

& 2001L and Data Structures Laboratory

MATH 2031 Transition to Advanced Mathematics 3

PHYS 1143 General Physics II 4

General Education Requirement (1C) 3

Units **18**

Term 2

CPE/CSC 2050 System Software Mechanics 3

EE 2211 Electric Circuit Analysis I 4

& EE 2241 and Electric Circuit Analysis Laboratory I

Linear Algebra + Differential Equations Course	¹	3
General Education Requirement (3B)		3
General Education Requirement (4A)		3
Units		16

Third Year

Term 1

CPE/CSC 3201	Introduction to Computer Security	3
CPE 3300	Computer Architecture	4
CPE/CSC 4553	Introduction to Operating Systems	3

PHIL 3323	Ethics, Science, and Technology (Upper-Division 3)	2	3
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General Education Requirement (4B)			3
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Units			16
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Term 2

CPE 3160	Microcontrollers and Embedded Applications		3
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CPE 4464	Introduction to Computer Networks		3
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EE 2328	Signals and Systems		4
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STAT 3210	Engineering Statistics (Upper-Division 2/5)	2	3
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or STAT 3310	or Probability and Random Processes for Engineers		
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General Education Requirement (6)	3
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Units	16
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Fourth Year

Term 1

CPE 4180	Advanced Microcontrollers and Embedded Applications	3
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CPE 4390	Introduction to Real-Time Operating Systems	3
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WGQS 3350	Gender, Race, Culture, Science, and Technology	4
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or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies
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Linear Algebra + Differential Equations Course ¹	3
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Senior Project

2-3

Units

15-16

Term 2

CPE 4420

High-Performance Embedded Systems

3

CPE 4455

Design of Fault-Tolerant Systems

4

Senior Project

2-3

General Education Requirement (Upper-Division 4)

3

Units

12-13

Total Units

124-127

1

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

2

Required in Major or Support; also satisfies General Education (GE) requirement.

Robotics and Autonomous Systems

*Robotics and Autonomous Systems Concentration - BS in Computer Engineering
Suggested Four-Year Flowcharts are planning guides and do not represent a required
course sequence.*

*Courses may be completed in a different order, provided requisite requirements are
satisfied.*

First Year

Term 1

Units

CPE/CSC 1000 Computing Majors Orientation

1

CPE/CSC 1024	Introduction to Computing	2
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BIO 1111	General Biology (5B)	2	3-4
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or BIO 2213

or Life Science for Engineers

or BOT 1121

or General Botany

or MCRO
2221

or Introduction to Microbiology

MATH 1261	Calculus I (2)	2	4
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General Education Requirement (1A)	3
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General Education Requirement (3A)	3
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Units	16-17
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Term 2

CSC 1001	Fundamentals of Computer Science	4
& 1001L	and Fundamentals of Computer Science Laboratory	
MATH 1262	Calculus II	4
PHYS 1141	General Physics I (5A & 5C) ²	4
General Education Requirement (1B)		3
Units		15

Second Year		
Term 1		

CPE 2300	Introduction to Computer Systems	3
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CPE 2301	Introduction to HDL and Digital Design Laboratory	1
CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 2031	Transition to Advanced Mathematics	3
PHYS 1143	General Physics II	4
General Education Requirement (1C)		3
Units		18

Term 2

CPE/CSC 2050	System Software Mechanics	3
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EE 2211	Electric Circuit Analysis I	4
& EE 2241	and Electric Circuit Analysis Laboratory I	
MATH 2341	Linear Analysis	4
General Education Requirement (3B)		3
General Education Requirement (4A)		3
Units		17

Third Year

Term 1

CPE/CSC 3201	Introduction to Computer Security	3
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CPE 3300	Computer Architecture	4
CPE/CSC 4553	Introduction to Operating Systems	3
ENGR 2211	Introduction to Mechanics	4
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ²	3
Units		17

Term 2

CPE 3160	Microcontrollers and Embedded Applications	3
CPE 4464	Introduction to Computer Networks	3
EE 2328	Signals and Systems	4

STAT 3210	Engineering Statistics (Upper-Division 2/5)	2	3
or STAT 3310	or Probability and Random Processes for Engineers		
General Education Requirement (6)			3
Units			16

Fourth Year

Term 1

ME 2212	Engineering Dynamics		3
WGQS 3350	Gender, Race, Culture, Science, and Technology		4
or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies		

Robotic Elective^{3, 4} 3

Senior Project⁴ 2-3

General Education Requirement (4B) 3

Units 16

Term 2

Robotic Elective^{3, 4} 3-4

Robotic Elective^{3, 4} 3-4

Senior Project⁴ 2-3

General Education Requirement (Upper-Division 4) 3

Units 12

Total Units 127-128

1

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

2

Required in Major or Support; also satisfies General Education (GE) requirement.

3

A maximum of 4 units from [CPE 4491](#), [CPE 4492](#), [CPE 4493](#), [CPE 4495](#), [CSC 4091](#), [CSC 4092](#), [CSC 4093](#), [CSC 4192](#), [CSC 4193](#), [CSC 4291](#), [CSC 4292](#), [CSC 4293](#), [CSC 4400](#), [CSC 4495](#), [CSC 4691](#), [CSC 4692](#), [CSC 4693](#), [CSC 4791](#), [CSC 4792](#), [CSC 4793](#), [CSC 4891](#), [CSC 4892](#), [CSC 4893](#), [CSC 4991](#), [CSC 4992](#), [CSC 4993](#), [DATA 4720](#), and [ENGR 4995](#).

Excess units completed for senior project may be applied toward technical electives.

Security

Security Concentration - BS in Computer Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CPE/CSC 1000	Computing Majors Orientation	1
CPE/CSC 1024	Introduction to Computing	2
BIO 1111	General Biology (5B) ²	3-4

or BIO 2213 or Life Science for Engineers

or BOT 1121 or General Botany

or MCRO 2221 or Introduction to Microbiology

MATH 1261 Calculus I (2)² 4

General Education Requirement (1A) 3

General Education Requirement (3A) 3

Units 16-17

Term 2

CSC 1001 Fundamentals of Computer Science 4

& 1001L and Fundamentals of Computer Science Laboratory

MATH 1262	Calculus II	4
PHYS 1141	General Physics I (5A & 5C) ²	4
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

CPE 2300	Introduction to Computer Systems	3
CPE 2301	Introduction to HDL and Digital Design Laboratory	1
CSC 2001	Data Structures	4

CSCI 2001L	and Data Structures Laboratory	
MATH 2031	Transition to Advanced Mathematics	3
PHYS 1143	General Physics II	4
General Education Requirement (1C)		3
Units		18

General Education Requirement (3B)	3
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General Education Requirement (4A)	3
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Units	16
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Third Year

Term 1

CPE/CSC 3201	Introduction to Computer Security	3
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CPE 3300	Computer Architecture	4
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CPE/CSC 4553	Introduction to Operating Systems	3
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PHIL 3323	Ethics, Science, and Technology (Upper-Division 3) ²	3
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General Education Requirement (4B)	3
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Units	16
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Term 2

CPE 3160	Microcontrollers and Embedded Applications	3
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EE 3306	Electronics I	4
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& 3306L	and Electronics Laboratory I
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STAT 3210	Engineering Statistics (Upper-Division 2/5) ²	3
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or STAT 3310	or Probability and Random Processes for Engineers
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Concentration Elective	4
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General Education Requirement (6)	3
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Fourth Year**Term 1**

CSC 4260	Interdisciplinary Privacy and Security Capstone I	3
WGQS 3350	Gender, Race, Culture, Science, and Technology	4
or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies	
Math Elective		3
CPE Security Elective		3
Concentration Elective		3

Units	16
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Term 2

CSC 4261	Senior Project - Privacy and Security Capstone II	3
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CPE 4464	Introduction to Computer Networks	3
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Concentration Elective	4
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General Education Requirement (Upper-Division 4)	3
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Units	13
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Total Units	127-128
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1

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

2

Required in Major or Support; also satisfies General Education (GE) requirement.

Construction Management

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
CM 1113	Construction Materials and Assemblies	2
EDES 1123	Place, People, and the Built Environment	3
MATH 1261	Calculus I (2) ¹	4
General Education Requirement (1A)		3

General Education Requirement (1C)	3
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Units	15
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Term 2

CM 1114	Mechanical, Electrical, and Plumbing Construction Materials and Assemblies	2
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CM 1115	Fundamentals of Construction Management	4
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CM 1413	Jobsite Construction Management	3
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PHYS 1141	General Physics I (5A & 5C) ¹	4
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General Education Requirement (1B)	3
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Units	16
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Second Year

Term 1

CM 2113	Soil Analysis and Construction of Foundations & Temporary Structures	2
CM 2114	Principles of Mechanical, Electrical, and Plumbing Building Systems	2
ARCE 1121	Structural Principles I	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
Units		16

Term 2

CM 1280	Fundamentals of Virtual Design and Construction Management	3
ARCE 3301	Introduction to Structural Systems	4
ECON 2001	Survey of Economics (4B) ¹	3
General Education Requirement		3
General Education Requirement		3
Units		16

Third Year

Term 1

CM 2239	Construction Surveying	3
CM 3214	Residential Construction Management	4
CM 3334	Construction Law	2
BUS 2207	Legal Responsibilities of Business	3
BUS 2212	Financial Accounting for Nonbusiness Majors	3
or BUS 2214	or Financial Accounting	
Units		15

Term 2

CM 3313	Commercial Construction Management	4
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CM 3314	Heavy Civil Construction Management	4
CM 3317	Sustainability and the Built Environment (Upper-Division 2/5) ¹	3
CM 3335	Construction Economics, Finance, and Accounting	3
CM 4420	Professional Preparation	2
Units		16

Fourth Year

Term 1

CM 3318	Housing and Communities (Upper-Division 4) ¹	3
CM 4411	Specialty Contracting Construction Management	4

CM 4443	Management of the Construction Firm	2
CM 4461	Senior Project I	3
General Education Requirement		3
Units		15

Term 2

CM 4420	Professional Preparation	2
CM 4450	Integrated Facility Development, Design, Construction, and Operations	3
CM 4462	Senior Project II	3
BUS 2215	Managerial Accounting	3

or any 3000-4000 level BUS course

Major ARCH, CM, CRP, or LA Elective	3
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Units	14
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Total Units	123
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Dairy Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1	Units
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ASCI 1100	Introduction to the Animal Sciences	1
ASCI 1101	Principles of Animal Physiology	3
DSCI 2230	General Dairy Husbandry	3
BIO 1151	Life: Molecules and Cells	4
General Education Requirement (1A)		3
Units		14

Term 2

DSCI 2229	General Dairy Manufacturing (5B & 5C) ¹	4
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4

MATH 1006	College Algebra (2)	1	3
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General Education Requirement (1C)			3
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Units			14
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Second Year

Term 1

DSCI 2202	Dairy Promotion and Marketing		3
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STAT 1110	Applied Statistical Concepts and Methods		3
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Approved Elective		2	3
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General Education Requirement (1B)			3
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General Education Requirement	3
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Units	15
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Term 2

ASCI 2220	Animal Nutrition and Feeding	3
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DSCI 3321	Lactation Physiology	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Third Year

Term 1

ASCI 3340	Animal Welfare and Ethics	3
ASCI 3355	Ruminant Nutrition	3
Approved Elective ²		3
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

ASCI 3302	Animal Genetics	3
ASCI 3363	Professional Development in the Animal Sciences	1
DSCI 3330	Dairy Cattle Reproductive Management and Artificial Insemination	3
DSCI 3344	Dairy Microbiology	3
Approved Elective ²		3
General Education Requirement		3
Units		16

Fourth Year	
Term 1	

DSCI 4432	Advanced Dairy Herd Management	4
Upper-Division DSCI Elective		3
Approved Elective ²		3
General Education Requirement		3
Free Elective ³		4
Units		17

Term 2

DSCI 4422	Breeding and Genetics of Dairy Cattle	3
ASCI 4477	Senior Project - Research Experience in Animal Science	2

or ASCI 4478 or Senior Project - Advanced Internship Experience in
Animal Science

or ASCI 4479 or Senior Project - Current Issues in Animal Science

Upper Division DSCI Elective	3
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Approved Elective ²	3
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Approved Elective ²	3
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Approved Elective.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Economics

Concentration Not Yet Declared - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001

Survey of Economics (4B) ¹

3

or ECON 2030

or Microeconomics

BUS 2207

Legal Responsibilities of Business ³

3

or BUS 2214

or Financial Accounting

MATH 1264	Calculus for Data Science I (2)	¹	4
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General Education Requirement (1A)			3
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General Education Requirement (4A)			3
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Units			16
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Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems		3
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or ECON 2040	or Macroeconomics		
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MATH 1265	Calculus for Data Science II		4
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STAT 1510	Statistics I	3
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General Education Requirement (1C)	3
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General Education Requirement (3A)	3
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Units	16
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Second Year

Term 1

ECON 3021	Econometrics	4
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ECON 3030	Intermediate Microeconomics	4
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General Education Requirement (1B)	3
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General Education Requirement (3B)	3
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Units	14
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Term 2

ECON 3040	Intermediate Macroeconomics	4
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ECON 3050	The Economics of Equity and Social Welfare	4
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Major Support Course Elective	4, 5, 6, 7, 8, 9, 10	3
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General Education Requirement (5A)	3
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General Education Requirement (5C)	1
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Units	15
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Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
& 3015A	and Programming for Economics and Analytics Project	
Concentration Course		3
Concentration Course		3
General Education Requirement (6)		3
General Education Requirement (Upper-Division 4)		3
Units		16

Term 2

Economics and Project Elective ²	4
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Concentration Course	3
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Concentration Course	3
----------------------	---

General Education Requirement (5B)	3
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Units	13
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Fourth Year

Term 1

Economics and Project Elective ²	4
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Economics and Project Elective ²	4
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Concentration Course	3
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General Education Requirement (Upper-Division 2/5)	3
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Units	14
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Term 2

ECON 4460	Applied Senior Project	4
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Concentration Course	3
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Concentration Course	3
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General Education Requirement (Upper-Division 3)	3
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Free Elective	11	2-4
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Units	16
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

Required course for students pursuing the Entrepreneurship concentration.

5

Required course for students pursuing the Accounting concentration.

6

Required course for students pursuing Financial Management and/or Real Estate Finance concentrations.

7

Required course for students pursuing Marketing Management and/or Consumer Packaging concentrations.

8

Required course for students pursuing Management and Human Resources concentration.

9

Required course for students pursuing Information Systems and Analytics concentration.

10

Option for students pursuing the General Curriculum in Economics or Economics in Data Science concentration.

11

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

General Curriculum in Economics

General Curriculum in Economics - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001	Survey of Economics (4B)	¹	3
or ECON 2030	or Microeconomics		
BUS 2207	Legal Responsibilities of Business	³	3
or BUS 2214	or Financial Accounting		
MATH 1264	Calculus for Data Science I (2)	¹	4
General Education Requirement (1A)			3
General Education Requirement (4A)			3
Units			16

Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
or ECON 2040	or Macroeconomics	
MATH 1265	Calculus for Data Science II	4
STAT 1510	Statistics I	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		16

Second Year

Term 1

ECON 3021	Econometrics	4
ECON 3030	Intermediate Microeconomics	4
General Education Requirement (1B)		3
General Education Requirement (3B)		3
Units		14

Term 2

ECON 3040	Intermediate Macroeconomics	4
ECON 3050	The Economics of Equity and Social Welfare	4
Major Support Course Elective		3

General Education Requirement (5A)	3
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General Education Requirement (5C)	1
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Units	15
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Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
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& 3015A	and Programming for Economics and Analytics Project	
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General Curriculum Course	3
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General Curriculum Course	3
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General Education Requirement (6)	3
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General Education Requirement (Upper-Division 4)	3
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Units	16
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Term 2

Economics and Project Elective ²	4
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General Curriculum Course	3
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General Curriculum Course	3
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General Education Requirement (5B)	3
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Units	13
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Fourth Year

Term 1

Economics and Project Elective² 4

Economics and Project Elective² 4

General Curriculum Course 3

General Education Requirement (Upper-Division 2/5) 3

Units 14

Term 2

ECON 4460 Applied Senior Project 4

General Curriculum Course	3
General Curriculum Course	3
General Education Requirement (Upper-Division 3)	3
Free Elective ⁴	3
Units	16
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Accounting

Accounting Concentration - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001	Survey of Economics (4B)	¹	3
or ECON 2030	or Microeconomics		
BUS 2207	Legal Responsibilities of Business	³	3
or BUS 2214	or Financial Accounting		
MATH 1264	Calculus for Data Science I (2)	¹	4
General Education Requirement (1A)			3
General Education Requirement (4A)			3
Units			16

Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
or ECON 2040	or Macroeconomics	
MATH 1265	Calculus for Data Science II	4
STAT 1510	Statistics I	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		16

Second Year

Term 1

ECON 3021	Econometrics	4
ECON 3030	Intermediate Microeconomics	4
General Education Requirement (1B)		3
General Education Requirement (3B)		3
Units		14

Term 2		
ECON 3040	Intermediate Macroeconomics	4
ECON 3050	The Economics of Equity and Social Welfare	4
BUS 2215	Managerial Accounting	3

General Education Requirement (5A)	3
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General Education Requirement (5C)	1
------------------------------------	---

Units	15
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Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
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& 3015A	and Programming for Economics and Analytics Project	
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BUS 3319	Data Analytics and Accounting Information Systems	3
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BUS 3320	Federal Income Taxation for Individuals	3
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General Education Requirement (6)	3
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General Education Requirement (Upper-Division 4)	3
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Units	16
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Term 2

BUS 3321	Intermediate Accounting I	3
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Economics and Project Elective ²	4
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Accounting Elective	3
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General Education Requirement (5B)	3
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Units	13
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Fourth Year

Term 1

BUS 3322	Intermediate Accounting II	3
Economics and Project Elective ²		4
Economics and Project Elective ²		4
General Education Requirement (Upper-Division 2/5)		3
Units		14

Term 2

ECON 4460	Applied Senior Project	4
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BUS 3323	Financial Statement Auditing	3
Accounting Elective		3
General Education Requirement (Upper-Division 3)		3
Free Elective ⁴		3
	Units	16
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Consumer Packaging

Consumer Packing Concentration - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001	Survey of Economics (4B)	¹	3
or ECON 2030	or Microeconomics		
BUS 2207	Legal Responsibilities of Business	³	3
or BUS 2214	or Financial Accounting		
MATH 1264	Calculus for Data Science I (2)	¹	4
General Education Requirement (1A)			3
General Education Requirement (4A)			3
Units			16

Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
or ECON 2040	or Macroeconomics	
MATH 1265	Calculus for Data Science II	4
STAT 1510	Statistics I	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		16

Second Year

Term 1

ECON 3021	Econometrics	4
ECON 3030	Intermediate Microeconomics	4
General Education Requirement (1B)		3
General Education Requirement (3B)		3
Units		14

Term 2		
ECON 3040	Intermediate Macroeconomics	4
ECON 3050	The Economics of Equity and Social Welfare	4
BUS 3346	Principles of Marketing	3

General Education Requirement (5A)	6	3
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General Education Requirement (5C)	6	1
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Units		15
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Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
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& 3015A	and Programming for Economics and Analytics Project	
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BUS 3396	Consumer Insights	3
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ITP 2234	Packaging Design Fundamentals	3
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General Education Requirement (6)	3
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General Education Requirement (Upper-Division 4)	3
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Units	16
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Term 2

ITP 3330	Packaging Fundamentals ⁵	3
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Approved Elective ⁵	3
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Economics and Project Elective ²	4
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General Education Requirement (5B)	3
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Units

13

Fourth Year

Term 1

ITP 3334 Structural Packaging Design 3

Economics and Project Elective² 4

Economics and Project Elective² 4

General Education Requirement (Upper-Division 2/5) 3

Units

14

Term 2

ECON 4460	Applied Senior Project	4
BUS 4418	Marketing Research	3
ITP 4475	Distribution Packaging Dynamics	3
General Education Requirement (Upper-Division 3)		3
Free Elective ⁴		3
Units		16
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

6

CHEM 1120 is recommended for those pursuing the Consumer Packaging concentration to fulfill GE Area 5A & 5C.

Economics for Data Science

Economics for Data Science Concentration - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001	Survey of Economics (4B) ¹	3
or ECON 2030	or Microeconomics	
BUS 2207	Legal Responsibilities of Business ³	3
or BUS 2214	or Financial Accounting	
MATH 1264	Calculus for Data Science I (2) ¹	4

General Education Requirement (1A)	3
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General Education Requirement (4A)	3
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Units	16
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Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
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or ECON 2040	or Macroeconomics
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MATH 1265	Calculus for Data Science II	4
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STAT 1510	Statistics I	3
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General Education Requirement (1C)	3
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General Education Requirement (3A)	3
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Units	16
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Second Year

Term 1

ECON 3021	Econometrics	4
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ECON 3030	Intermediate Microeconomics	4
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General Education Requirement (1B)	3
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General Education Requirement (3B)	3
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Units	14
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Term 2

ECON 3040	Intermediate Macroeconomics	4
ECON 3050	The Economics of Equity and Social Welfare	4
Major Support Course Elective		3
General Education Requirement (5A)		3
General Education Requirement (5C)		1
Units		15

Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
& 3015A	and Programming for Economics and Analytics Project	
ECON 4010	Mathematical Economics ⁵	4
& 4010A	or Optimization in Julia for Economics and Data Science <i>and</i> Julia for Data Analysis Project	
or ECON 4011 <i>and</i> ECON 4011A		
Approved Elective ⁵		3
General Education Requirement (6)		3
General Education Requirement (Upper-Division 4)		3
	Units	17

Term 2

ECON 4021	Advanced Econometrics	5	4
& 4021A	or Applied Forecasting <i>and</i> Applied Forecasting Project		
or ECON 4024 <i>and</i> ECON 4024A			
Approved Elective		5	3
Economics and Project Elective		2	4
General Education Requirement (5B)			3
	Units		14

Fourth Year

Term 1

Economics and Project Elective² 4

Economics and Project Elective² 4

Approved Elective⁵ 3

General Education Requirement (Upper-Division 2/5) 3

Units 14

Term 2

ECON 4025 Causal Machine Learning for Economics 4

& 4025A

and Causal Machine Learning for Economics
Project

ECON 4460	Applied Senior Project	4
General Education Requirement (Upper-Division 3)		3
Free Elective ⁴		2-4
Units		14
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

Entrepreneurship

Entrepreneurship Concentration - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
ECON 2001	Survey of Economics (4B) ¹	3
or ECON 2030	or Microeconomics	
BUS 2207	Legal Responsibilities of Business ³	3
or BUS 2214	or Financial Accounting	
MATH 1264	Calculus for Data Science I (2) ¹	4
General Education Requirement (1A)		3
General Education Requirement (4A)		3
Units		16

Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
or ECON 2040	or Macroeconomics	
MATH 1265	Calculus for Data Science II	4
STAT 1510	Statistics I	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		16

Second Year

Term 1

ECON 3021	Econometrics	4
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ECON 3030	Intermediate Microeconomics	4
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General Education Requirement (1B)	3
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General Education Requirement (3B)	3
------------------------------------	---

Units	14
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Term 2

ECON 3040	Intermediate Macroeconomics	4
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ECON 3050	The Economics of Equity and Social Welfare	4
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BUS 1342	Financial Institutions	3
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General Education Requirement (5A)	3
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General Education Requirement (5C)	1
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Units	15
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Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
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& 3015A	and Programming for Economics and Analytics Project	
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BUS 3310	Introduction to Entrepreneurship	4
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Approved Elective	3-4
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General Education Requirement (6)	3
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General Education Requirement (Upper-Division 4)	3
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Units	17
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Term 2

BUS 3330	Funding and Managing Startup Companies	4
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ITP 4406	Business-to-Business Complex Solutions Selling	3
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Economics and Project Elective ²	4
---	---

General Education Requirement (5B)	3
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Units

14

Fourth Year

Term 1

BUS 4447 Defining and Building Technology Startups 4

or BUS 4488 or Building a Startup Skill Set

Economics and Project Elective² 4

Economics and Project Elective² 4

General Education Requirement (Upper-Division 2/5) 3

Units

15

Term 2

ECON 4460	Applied Senior Project	4
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Approved Elective	3-4
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General Education Requirement (Upper-Division 3)	3
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Free Elective ⁴	3
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Units	13
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Financial Management

Financial Management Concentration - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001	Survey of Economics (4B)	¹	3
or ECON 2030	or Microeconomics		
BUS 2207	Legal Responsibilities of Business	³	3
or BUS 2214	or Financial Accounting		
MATH 1264	Calculus for Data Science I (2)	¹	4
General Education Requirement (1A)			3
General Education Requirement (4A)			3
	Units		16

Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
or ECON 2040	or Macroeconomics	
MATH 1265	Calculus for Data Science II	4
STAT 1510	Statistics I	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		16

Second Year

Term 1

ECON 3021	Econometrics	4
ECON 3030	Intermediate Microeconomics	4
General Education Requirement (1B)		3
General Education Requirement (3B)		3
Units		14

Term 2

ECON 3040	Intermediate Macroeconomics	4
ECON 3050	The Economics of Equity and Social Welfare	4
BUS 3343	Quantitative Methods in Finance	3

General Education Requirement (5A) 3

General Education Requirement (5C) 1

Units 15

Third Year

Term 1

ECON 3015 Programming for Economics and Analytics 4

& 3015A and Programming for Economics and Analytics Project

BUS 3431 Security Analysis and Portfolio Management⁵ 4

Concentration Track Course^{6, 7, 8, 8} 3

General Education Requirement (6)	3
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Units	14
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Term 2

BUS 3441	Financial Modeling and Analytics in Python or R	5	4
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or BUS 3440

or Financial Modeling and Visualization in Excel

Economics and Project Elective	2	4
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Concentration Track Course	6, 7, 8, 9	3-4
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General Education Requirement (5B)	3
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Units	14
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Fourth Year

Term 1

BUS 3411	Finance Practicum	5	1
BUS 3438	Corporate Finance	5	4
Economics and Project Elective		2	4
Economics and Project Elective		2	4
General Education Requirement (Upper-Division 2/5)			3
	Units		16

Term 2

ECON 4460 Applied Senior Project 4

Concentration Track Course^{6, 7, 8, 9} 3-4

General Education Requirement (Upper-Division 3) 3

General Education Requirement (Upper-Division 4) 3

Free Elective⁴ 2

Units 15

Total Units 120

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

All students need an approval from the finance chair to enroll in the Financial Management concentration.

6

Courses taken to meet a major course requirement cannot be double-counted in the concentration. Units in excess of total will be applied towards major's free electives.

7

Highly recommended for students in the Financial Analytics track.

8

Highly recommended for students in the Financial Management track.

9

Highly recommended for students in the Financial Planning and Wealth Management track.

Information Systems and Analytics

Information Systems and Analytics Concentration - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
ECON 2001	Survey of Economics (4B) ¹	3
or ECON 2030	or Microeconomics	
BUS 2207	Legal Responsibilities of Business ³	3
or BUS 2214	or Financial Accounting	
MATH 1264	Calculus for Data Science I (2) ¹	4
General Education Requirement (1A)		3
General Education Requirement (4A)		3
Units		16

Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
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or ECON 2040	or Macroeconomics	
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MATH 1265	Calculus for Data Science II	4
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STAT 1510	Statistics I	3
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General Education Requirement (1C)		3
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General Education Requirement (3A)		3
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Units		16
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Second Year

Term 1

ECON 3021	Econometrics	4
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ECON 3030	Intermediate Microeconomics	4
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General Education Requirement (1B)	3
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General Education Requirement (3B)	3
------------------------------------	---

Units	14
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Term 2

ECON 3040	Intermediate Macroeconomics	4
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ECON 3050	The Economics of Equity and Social Welfare	4
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BUS 3391	Information Systems	3
General Education Requirement (5A)		3
General Education Requirement (5C)		1
Units		15

Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
& 3015A	and Programming for Economics and Analytics Project	
BUS 3392	Business Application Development	3

Concentration Project Course 1

General Education Requirement (6) 3

General Education Requirement (Upper-Division 4) 3

Units 14

Term 2

BUS 3393 Database Systems in Business 3

BUS 3394 Systems Analysis and Design 3

Concentration Project Course 1

Economics and Project Elective² 4

General Education Requirement (5B) 3

Units 14

Fourth Year

Term 1

BUS 3399 IT Infrastructure and Security Management 3

Approved Elective 3

Concentration Project Course 1

Economics and Project Elective² 4

General Education Requirement (Upper-Division 2/5) 3

	Units	14
Term 2		
ECON 4460	Applied Senior Project	4
BUS 4497	Business Analytics	3
Economics and Project Elective ²		4
General Education Requirement (Upper-Division 3)		3
Free Elective ⁴		3
	Units	17
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Management and Human Resources

Management and Human Resources Concentration - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001	Survey of Economics (4B) ¹	3
or ECON 2030	or Microeconomics	
BUS 2207	Legal Responsibilities of Business ³	3
or BUS 2214	or Financial Accounting	
MATH 1264	Calculus for Data Science I (2) ¹	4
General Education Requirement (1A)		3
General Education Requirement (4A)		3

Units

16

Term 2

ECON 2021 Using Big Data to Solve Economic and Social Problems 3

or ECON 2040 or Macroeconomics

MATH 1265 Calculus for Data Science II 4

STAT 1510 Statistics I 3

General Education Requirement (1C) 3

General Education Requirement (3A) 3

Units

16

Second Year

Term 1

ECON 3021	Econometrics	4
ECON 3030	Intermediate Microeconomics	4
General Education Requirement (1B)		3
General Education Requirement (3B)		3
Units		14

Term 2

ECON 3040	Intermediate Macroeconomics	4
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ECON 3050	The Economics of Equity and Social Welfare	4
BUS 3387	Organizational Behavior	3
General Education Requirement (5A)		3
General Education Requirement (5C)		1
Units		15

Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
& 3015A	and Programming for Economics and Analytics Project	

BUS 3384	Human Resources Management	3
Concentration Project Course		1
General Education Requirement (6)		3
General Education Requirement (Upper-Division 4)		3
Units		14

Term 2

BUS 3388	Training, Developing, and Managing Employees	3
BUS 4477	Management Consulting, Change, and Development	3
Economics and Project Elective ²		4

Concentration Project Course	1
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General Education Requirement (5B)	3
------------------------------------	---

Units	14
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Fourth Year

Term 1

BUS 4489	Negotiation	3
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Concentration Project Course	1
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Economics and Project Elective ²	4
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Economics and Project Elective ²	4
---	---

General Education Requirement (Upper-Division 2/5)	3
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Units	15
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Term 2

ECON 4460	Applied Senior Project	4
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Approved Elective	5, 6	3
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Approved Elective	5, 6	3
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General Education Requirement (Upper-Division 3)	3
--	---

Free Elective	4	3
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Units	16
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

6

Units in excess of total will be applied towards major's free electives.

Marketing Management

Marketing Management Concentration in - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001

Survey of Economics (4B) ¹

3

or ECON 2030

	or Microeconomics	
BUS 2207	Legal Responsibilities of Business	3
or BUS 2214	or Financial Accounting	
MATH 1264	Calculus for Data Science I (2)	4
General Education Requirement (1A)		3
General Education Requirement (4A)		3
	Units	16

Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
-----------	--	---

or ECON 2040 or Macroeconomics

MATH 1265	Calculus for Data Science II	4
STAT 1510	Statistics I	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		16

Second Year

Term 1

ECON 3021	Econometrics	4
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ECON 3030	Intermediate Microeconomics	4
-----------	-----------------------------	---

General Education Requirement (1B)		3
------------------------------------	--	---

General Education Requirement (3B)		3
------------------------------------	--	---

Units		14
-------	--	----

Term 2

ECON 3040	Intermediate Macroeconomics	4
-----------	-----------------------------	---

ECON 3050	The Economics of Equity and Social Welfare	4
-----------	--	---

BUS 3346	Principles of Marketing	3
----------	-------------------------	---

General Education Requirement (5A)		3
------------------------------------	--	---

General Education Requirement (5C)	1
------------------------------------	---

Units	15
-------	----

Third Year

Term 1

ECON 3015	Programming for Economics and Analytics	4
-----------	---	---

& 3015A	and Programming for Economics and Analytics Project	
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BUS 3396	Consumer Insights	4
----------	-------------------	---

& 3396A	and Consumer Insights Project	
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General Education Requirement (6)	3
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General Education Requirement (Upper-Division 4)	3
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Units

14

Term 2

BUS 4418

Marketing Research

4

& 4418A

and Marketing Research Project

Economics and Project Elective²

4

Approved Elective⁵

4

General Education Requirement (5B)

3

Units

15

Fourth Year

Term 1

Economics and Project Elective² 4

Economics and Project Elective² 4

Approved Elective⁵ 4

General Education Requirement (Upper-Division 2/5) 3

Units 15

Term 2

ECON 4460 Applied Senior Project 4

Approved Elective ⁵	4
General Education Requirement (Upper-Division 3)	3
Free Elective ⁴	4
Units	15
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

Real Estate Finance

Real Estate Finance Concentration - BS in Economics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ECON 2001	Survey of Economics (4B)	¹	3
or ECON 2030	or Microeconomics		
BUS 2207	Legal Responsibilities of Business	³	3
or BUS 2214	or Financial Accounting		
MATH 1264	Calculus for Data Science I (2)	¹	4
General Education Requirement (1A)			3
General Education Requirement (4A)			3
Units			16

Term 2

ECON 2021	Using Big Data to Solve Economic and Social Problems	3
or ECON 2040	or Macroeconomics	
MATH 1265	Calculus for Data Science II	4
STAT 1510	Statistics I	3
General Education Requirement (1C)		3
General Education Requirement (3A)		3
Units		16

Second Year

Term 1

ECON 3021	Econometrics	4
ECON 3030	Intermediate Microeconomics	4
General Education Requirement (1B)		3
General Education Requirement (3B)		3
Units		14

Term 2

ECON 3040	Intermediate Macroeconomics	4
ECON 3050	The Economics of Equity and Social Welfare	4
BUS 1342	Financial Institutions	3

or BUS 3343 or Quantitative Methods in Finance

General Education Requirement (5A) 3

General Education Requirement (5C) 1

Units 15

Third Year

Term 1

ECON 3015 Programming for Economics and Analytics 4

& 3015A and Programming for Economics and Analytics Project

BUS 3431 Security Analysis and Portfolio Management 5, 6 4

Approved Elective⁷

3

General Education Requirement (6)

3

General Education Requirement (Upper-Division 4)

3

Units

17

Term 2

BUS 3440

Financial Modeling and Visualization in Excel^{5, 6}

4

Economics and Project Elective²

4

Approved Elective⁷

3

General Education Requirement (5B)	3
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Units	14
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Fourth Year

Term 1

Economics and Project Elective ²	4
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Economics and Project Elective ²	4
---	---

Approved Elective ⁷	3
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General Education Requirement (Upper-Division 2/5)	3
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	Units	14
Term 2		
ECON 4460	Applied Senior Project	4
BUS 4434	Real Estate Finance ^{5, 6}	4
General Education Requirement (Upper-Division 3)		3
Free Elective ⁴		2-4
	Units	14
Total Units		120

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Lecture and activity courses should be taken in the same term.

3

BUS 2214 is required for Accounting, Management and Human Resources, and/or Information Systems Management concentrations or additional coursework might be needed to complete the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

All students need an approval from the finance department chair to enroll in the concentration.

6

All students enrolled in the concentration are recommended to pursue a minor with the CAED department. The most popular minor is the Real Estate Property Development minor.

7

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

General Curriculum in Electrical Engineering

General Curriculum in Electrical Engineering - BS in Electrical Engineering
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
EE 1111	Introduction to Electrical Engineering	2
& 1111L	and Introduction to Electrical Engineering Laboratory	

MATH 1261	Calculus I (2) ¹	4
PHYS 1141	General Physics I (5A & 5C) ¹	4
General Education Requirement (1A)		3
General Education Requirement (6)		3
Units		16

Term 2

CSC 1001	Fundamentals of Computer Science	4
& 1001L	and Fundamentals of Computer Science Laboratory	
MATH 1262	Calculus II	4

PHYS 1143	General Physics II	4
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General Education Requirement (1B)	3
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Units	15
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Second Year

Term 1

EE 2211	Electric Circuit Analysis I	3
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EE 2241	Electric Circuit Analysis Laboratory I	1
---------	--	---

BIO 2213	Life Science for Engineers (5B) ¹	3
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CPE 2300	Introduction to Computer Systems	3
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CPE 2301	Introduction to HDL and Digital Design Laboratory	1
MATH 2341	Linear Analysis	4
General Education Requirement (1C)		3
Units		18

Term 2

EE 2212	Electric Circuit Analysis II	3
EE 2328	Signals and Systems	4
MATH 2263	Calculus III	3
General Education Requirement (3B)		3

General Education Requirement (4A)	3
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Units	16
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Third Year

Term 1

EE 3302	Classical Control Systems	4
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& 3302L	and Classical Control Systems Laboratory	
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EE 3306	Electronics I	4
---------	---------------	---

& 3306L	and Electronics Laboratory I	
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EE 3335	Electromagnetic Fields and Transmission	4
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& 3335L	and Electromagnetic Fields and Transmission Laboratory	
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General Education Requirement (3A)	3
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General Education Requirement (4B)	3
------------------------------------	---

Units	18
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Term 2

EE 3255	Electric Machines and Power Systems	4
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& 3255L	and Electric Machines and Power Systems Laboratory	
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EE 3308	Electronics II	4
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& 3308L	and Electronics Laboratory II	
---------	-------------------------------	--

EE 3329	Cyber-Physical Systems	4
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General Education Requirement (Upper-Division 4)	3
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Fourth Year

Term 1

EE 4314	Communication Systems	4
& 4314L	and Communication Systems Laboratory	
EE 4459	Electrical Engineering Fundamentals of Engineering Exam	1
EE 4461	Senior Project I	1
EE 4463	Senior Project Design Laboratory I	1
or EE 4465	or Senior Design: Individual Project I	

STAT 3310	Probability and Random Processes for Engineers (Upper-Division 2/5) ¹	3
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Technical Elective ²		4
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Units		14
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Term 2

EE 4462	Senior Project II	1
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EE 4464	Senior Project Design Laboratory II	1
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or EE 4466	or Senior Design: Individual Project II	
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Technical Elective ²		4
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Lower-Division Elective or Technical Elective ³	4
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Lower-Division Elective or Technical Elective ³	3
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General Education Requirement (Upper-Division 3)	3
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Units	16
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Total Units	128
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of 8 units from the EE Technical Electives list is required.

3

A maximum of 4 units may come from either the Non-EE Technical Electives or the Lower-Division Electives list, with a combined maximum of 7 units allowed across both lists.

Electronics, Controls, and Communications

Electronics, Controls, and Communications Concentration - BS in Electrical Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

EE 1111	Introduction to Electrical Engineering	2
& 1111L	and Introduction to Electrical Engineering Laboratory	
MATH 1261	Calculus I (2) ¹	4

PHYS 1141	General Physics I (5A & 5C) ¹	4
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General Education Requirement (1A)	3
------------------------------------	---

General Education Requirement (6)	3
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Units	16
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Term 2

CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory	
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MATH 1262	Calculus II	4
-----------	-------------	---

PHYS 1143	General Physics II	4
-----------	--------------------	---

General Education Requirement (1B)	3
------------------------------------	---

Units	15
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Second Year

Term 1

EE 2211	Electric Circuit Analysis I	3
---------	-----------------------------	---

EE 2241	Electric Circuit Analysis Laboratory I	1
---------	--	---

BIO 2213	Life Science for Engineers (5B) ¹	3
----------	--	---

CPE 2300	Introduction to Computer Systems	3
----------	----------------------------------	---

CPE 2301	Introduction to HDL and Digital Design Laboratory	1
----------	---	---

MATH 2341	Linear Analysis	4
-----------	-----------------	---

General Education Requirement (1C)	3
------------------------------------	---

Units	18
-------	----

Term 2

EE 2212	Electric Circuit Analysis II	3
---------	------------------------------	---

EE 2328	Signals and Systems	4
---------	---------------------	---

MATH 2263	Calculus III	3
-----------	--------------	---

General Education Requirement (3B)	3
------------------------------------	---

General Education Requirement (4A)	3
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Third Year**Term 1**

EE 3302	Classical Control Systems	4
& 3302L	and Classical Control Systems Laboratory	
EE 3306	Electronics I	4
& 3306L	and Electronics Laboratory I	
EE 3335	Electromagnetic Fields and Transmission	4
& 3335L	and Electromagnetic Fields and Transmission Laboratory	
General Education Requirement (3A)		3

General Education Requirement (4B)	3
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Units	18
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Term 2

EE 3255	Electric Machines and Power Systems	4
---------	-------------------------------------	---

& 3255L	and Electric Machines and Power Systems Laboratory	
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EE 3308	Electronics II	4
---------	----------------	---

& 3308L	and Electronics Laboratory II	
---------	-------------------------------	--

EE 3329	Cyber-Physical Systems	4
---------	------------------------	---

General Education Requirement (Upper-Division 4)	3
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Units	15
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Fourth Year

Term 1

EE 4314	Communication Systems	4
& 4314L	and Communication Systems Laboratory	
EE 4459	Electrical Engineering Fundamentals of Engineering Exam	1
EE 4461	Senior Project I	1
EE 4463	Senior Project Design Laboratory I	1
or EE 4465	or Senior Design: Individual Project I	
STAT 3310	Probability and Random Processes for Engineers (Upper-Division 2/5) ¹	3

Technical Elective²

4

Units

14

Term 2

EE 4462

Senior Project II

1

EE 4464

Senior Project Design Laboratory II

1

or EE 4466

or Senior Design: Individual Project II

Technical Elective²

4

Technical Elective²

3-4

Lower-Division Elective or Technical Elective ³	3-4
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General Education Requirement (Upper-Division 3)	3
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Units	16
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Total Units	128
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of 11 units from the EE Technical Electives list is required.

3

A maximum of 4 units may come from the Lower-Division Electives list.

Power

Power Concentration - BS in Electrical Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

EE 1111 Introduction to Electrical Engineering 2

& 1111L and Introduction to Electrical Engineering Laboratory

MATH 1261 Calculus I (2) ¹ 4

PHYS 1141 General Physics I (5A & 5C) ¹ 4

General Education Requirement (1A) 3

General Education Requirement (6)	3
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Units	16
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Term 2

CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory	
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MATH 1262	Calculus II	4
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PHYS 1143	General Physics II	4
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General Education Requirement (1B)	3
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Units	15
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Second Year

Term 1

EE 2211	Electric Circuit Analysis I	3
EE 2241	Electric Circuit Analysis Laboratory I	1
BIO 2213	Life Science for Engineers (5B) ¹	3
CPE 2300	Introduction to Computer Systems	3
CPE 2301	Introduction to HDL and Digital Design Laboratory	1
MATH 2341	Linear Analysis	4
General Education Requirement (1C)		3

Units

18

Term 2

EE 2212 Electric Circuit Analysis II 3

EE 2328 Signals and Systems 4

MATH 2263 Calculus III 3

General Education Requirement (3B) 3

General Education Requirement (4A) 3

Units

16

Third Year

Term 1

EE 3302 Classical Control Systems 4

& 3302L and Classical Control Systems Laboratory

EE 3306 Electronics I 4

& 3306L and Electronics Laboratory I

EE 3335 Electromagnetic Fields and Transmission 4

& 3335L and Electromagnetic Fields and Transmission Laboratory

General Education Requirement (3A) 3

General Education Requirement (4B) 3

Units 18

Term 2

EE 3255 Electric Machines and Power Systems 4

& 3255L and Electric Machines and Power Systems Laboratory

EE 3308 Electronics II 4

& 3308L and Electronics Laboratory II

EE 3329 Cyber-Physical Systems 4

General Education Requirement (Upper-Division 4) 3

Units **15**

Fourth Year

Term 1

EE 4314	Communication Systems	4
& 4314L	and Communication Systems Laboratory	
EE 4406	Power System Analysis I	3
or EE 4410	or Fundamentals of Power Electronics	
EE 4459	Electrical Engineering Fundamentals of Engineering Exam	1
EE 4461	Senior Project I	1
EE 4463	Senior Project Design Laboratory I	1
or EE 4465	or Senior Design: Individual Project I	
STAT 3310	Probability and Random Processes for Engineers (Upper-Division 2/5) ¹	3
Units		13

Term 2

EE 4462 Senior Project II 1

EE 4464 Senior Project Design Laboratory II 1

or EE 4466 or Senior Design: Individual Project II

Technical Elective² 4

Technical Elective² 4

Lower-Division Elective or Technical Elective³ 4

General Education Requirement (Upper-Division 3) 3

Units 17

Total Units

128

¹ Required in Major or Support; also satisfies General Education (GE) requirement. ²

A minimum of 8 units from the EE Technical Electives list is required.

³ A maximum of 4 units may come from the Lower-Division Electives list.

English

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ENGL 1101 Introduction to English Studies

4

World Languages Support Elective ²

4

General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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Units	14
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Term 2

2000-level ENGL Elective	4
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Literature Elective (3B) ¹	3
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General Education Requirement (1B)	3
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General Education Requirement (2)	3
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Free Elective ³	3
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Units

16

Second Year

Term 1

2000-level ENGL Elective

4

General Education Requirement

3

General Education Requirement

3

General Education Requirement

3

General Education Requirement

3

Units

16

Term 2

ENGL 3611	Literary Themes (Upper-Division 3, GWR)	1	4
or ENGL 3618	or Research Topics in Diversity in Twentieth- and Twenty-First Century U.S. Literature		
or ENGL 3625	or Research Topics in Queer and Trans Literature and Media		
or ENGL 3626	or Intermediate Topics in Film		
2000-level ENGL Elective			4
General Education Requirement			3
General Education Requirement			3
General Education Requirement			1
Units			15

Third Year

Term 1

3000-level ENGL Elective 4

3000-level ENGL Elective 4

Free Elective³ 3

Free Elective³ 3

Units 14

Term 2

3000-level ENGL Elective 4

4000-level ENGL Elective	4
ENGL 4000-level Diversity Elective	3-4
General Education Requirement	3
Units	15

Fourth Year

Term 1

4000-level ENGL Elective	4
4000-level ENGL Elective	4
General Education Requirement	3

Free Elective

3

4

Units

15

Term 2

ENGL 4461

Senior Project

4

4000-level ENGL Elective

4

Free Elective

3

3

Free Elective

3

4

Units

15

1

Required in Major or Support; also satisfied General Education (GE) requirement.

2

Students who demonstrate a comparable level of proficiency in a foreign language can substitute a 4-unit ENGL course for this requirement.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Environmental Earth and Soil Sciences

Concentration Not Yet Declared - BS in Environmental Earth and Soil Sciences
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ERSC 1144	Introduction to Earth Science	3
GEOL 2240	Physical Geology	3
BOT 1121	General Botany (5B) ¹	4
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		16

Term 2

SS 1120	Introductory Soil Science (5A & 5C) ¹	4
GEOL 2241	Physical Geology Laboratory	1

MATH 1261	Calculus I (2)	1, 3	4
or MATH 1264	or Calculus for Data Science I		

General Education Requirement (1B)			3
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General Education Requirement			3
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Units			15
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Second Year

Term 1

SS 3321	Soil Morphology		4
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CHEM 1120	Fundamentals of Chemical Structure and Properties		4
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PHYS 1121	College Physics I	3	4
or PHYS 1141	or General Physics I		

General Education Requirement			3
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Units			15
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Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)		3
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STAT 1110	Applied Statistical Concepts and Methods		3
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Concentration Course or Approved Elective	2, 3, 4		3
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General Education Requirement			3
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General Education Requirement	3
-------------------------------	---

Units	15
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Third Year

Term 1

GEOL 3330	Principles of Stratigraphy	2, 3	3-4
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or SS 2221

or Soil Health and Plant Nutrition

or SS 3444

or Climate Smart Agriculture

CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
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Concentration Course or Approved Elective	2, 3, 4	3
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Concentration Course or Approved Elective	2, 3, 4	3
---	---------	---

General Education Requirement		3
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Units		16
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Term 2

NR 3310	Global Climate Change (Upper-Division 2/5)	¹ 3
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NR 3363	Career Preparation and Practices in Natural Resources Fields	2
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SS 4422	Soil Ecology	^{2, 3} 3-4
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or ERSC 4450	or Geomorphology	
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Concentration Course or Approved Elective

2, 3, 4

3

General Education Requirement

3

Units

15

Fourth Year

Term 1

SS 4423

Environmental Soil and Water Chemistry

4

BRAE 3345

Photogrammetry and Remote Sensing with GIS
Applications²

2-4

or GEOG 4441

or NR 4418

or Advanced Applications in Geospatial
Technologies

or SS/NR 4431

or Applied Geographic Information System

or Spatial Data Analysis and Environmental Mapping

Concentration Course or Approved Elective^{2, 3, 4} 3

General Education Requirement 3

Units 13

Term 2

SS 4424 Environmental Soil Physics³ 4

or ERSC 4442

or Applied Groundwater Hydrology

or ERSC 4443

or Applied Environmental Contaminant Transport

ERSC/NR 4478 Senior Project - Current Topics in Environmental Science 2

or ERSC/NR 4479

or Senior Project - Independent Study

Major ERSC/NR/SS Elective ²	3-4
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Concentration Course or Approved Elective ^{2, 3, 4}	2-4
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Free Elective ⁵	0-3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major or Support requirement cannot be double-counted in the concentration or the Approved Electives.

3

ERSC 4450, GEOL 3330, MATH 1261, and PHYS 1141 are prerequisite courses for those pursuing the Geology concentration.

4

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

5

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Geology

Geology Concentration - BS in Environmental Earth and Soil Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ERSC 1144	Introduction to Earth Science	3
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GEOL 2240	Physical Geology	3
-----------	------------------	---

BOT 1121	General Botany (5B) ¹	4
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		16
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Term 2

SS 1120	Introductory Soil Science (5A & 5C) ¹	4
GEOL 2241	Physical Geology Laboratory	1
MATH 1261	Calculus I (2) ^{1, 3}	4
General Education Requirement (1B)		3
General Education Requirement		3
Units		15

Second Year		
Term 1		
SS 3321	Soil Morphology	4

GEOL 3310	Igneous and Metamorphic Petrology	4
CHEM 1120	Fundamentals of Chemical Structure and Properties	4
PHYS 1121	College Physics I	4
or PHYS 1141	or General Physics I	
Units		16

Term 2

GEOL 3330	Principles of Stratigraphy ^{2, 3}	3
NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
STAT 1110	Applied Statistical Concepts and Methods	3

General Education Requirement	3
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General Education Requirement	3
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Units	15
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Third Year

Term 1

ERSC 4450	Geomorphology ^{2, 3}	3
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GEOL 4420	Field Geophysics	3
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NR 3318	Introduction to Environmental Data Science	3
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CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
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General Education Requirement	3
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Units	16
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Term 2

NR 3310	Global Climate Change (Upper-Division 2/5) ¹	3
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NR 3363	Career Preparation and Practices in Natural Resources Fields	2
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	14
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Fourth Year

Term 1

BRAE 3345	Photogrammetry and Remote Sensing with GIS Applications ²	2-4
or GEOG 4441		
or NR 4418	or Advanced Applications in Geospatial Technologies	
or SS/NR 4431	or Applied Geographic Information System	
	or Spatial Data Analysis and Environmental Mapping	
ERSC 4442	Applied Groundwater Hydrology ²	4
or ERSC 4443		
	or Applied Environmental Contaminant Transport	
or SS 4424		
	or Environmental Soil Physics	
GEOL 4415	Structural Geology	3

SS 4423	Environmental Soil and Water Chemistry	4
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	Units	14
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Term 2

ERSC/NR 4478	Senior Project - Current Topics in Environmental Science	2
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or ERSC/NR 4479

or Senior Project - Independent Study

GEOL 4417	Field Geology Methods and Mapping	5
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SS 4424	Environmental Soil Physics ²	4
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or ERSC 4442

or Applied Groundwater Hydrology

or ERSC 4443

or Applied Environmental Contaminant Transport

Major ERSC/NR/SS Elective ²	3-4
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Free Elective ⁵	0-1
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses taken to meet a Major or Support requirement cannot be double-counted in the concentration or the Approved Electives.

3

ERSC 4450, GEOL 3330, MATH 1261, and PHYS 1141 are prerequisite courses for those pursuing the Geology concentration.

4

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

5

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Environmental Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ENVE 1111	Introduction to Environmental Engineering	1
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
MATH 1261	Calculus I (2) ¹	4

PHYS 1141	General Physics I	4
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General Education Requirement (1A)	3
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Units	16
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Term 2

ENVE/CE 1112	Spatial Visualization and Drawing	3
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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MATH 1262	Calculus II	4
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PHYS 1143	General Physics II	4
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General Education Requirement (1B)	3
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Units

18

Second Year

Term 1

ENVE 2325 Air Quality Engineering and Measurements 4

ENVE 3309 Noise Control and Occupational Safety and Health 3

MCRO 2221 Introduction to Microbiology (5B) ¹ 4

General Education Requirement (1C) 3

General Education Requirement (3A) 3

Units

17

Term 2

ENVE/CE 2251	Computational Applications in Civil and Environmental Engineering	2
ENVE 2331	Fundamentals of Environmental Engineering	2
ENGR 2211	Introduction to Mechanics	4
MATH 2341	Linear Analysis	4
General Education Requirement (3B)		3
Units		15

Third Year

Term 1

ENVE/CE 3336	Environmental Fluid Mechanics and Hydraulic Systems	4
ENVE 3434	Chemistry of Environmental Systems	4
ENVE 3450	Sustainable Systems Engineering	3
General Education Requirement (4A)		3
General Education Requirement (4B)		3
Units		17

Term 2

ENVE/CE 3337	Water Resources & Environmental Engineering	4
ENVE 3421	Transfer Phenomena and Process Thermodynamics	4

ENVE 3438	Water and Wastewater Treatment Design	4
STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
General Education Requirement (Upper-Division 4)		3
Units		18

Fourth Year

Term 1

ENVE 3465	Infrastructure Systems	2
ENVE 4437	Fate, Transport & Control of Environmental Pollutants	3
ENVE 4466	Senior Project Design I	2

CE Technical Elective	3
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General Education Requirement (6)	3
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General Education Requirement (Upper-Division 3)	3
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Units	16
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Term 2

ENVE 4439	Solid and Hazardous Waste Management	3
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ENVE 4467	Senior Project Design II	2
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ENVE Technical Elective ²	3
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ENVE Technical Elective ²	3-4
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CE Technical Elective	3-4
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Units	15
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Total Units	132
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A combined maximum of 4 units from [ENVE 4400](#), [ENVE 4405](#), and [ENVE 4407](#) may count towards technical electives.

Environmental Management and Protection

Concentration Not Yet Declared - BS in Environmental Management and Protection

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
NR 1142	Environmental Management	4
BIO 1114	Plant Diversity and Ecology	4
or BOT 1121	or General Botany	
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Free Elective	3, 4	3
Units		17
Term 2		

MATH 1264	Calculus for Data Science I (2)	¹	3-4
or MATH 1267	or Business Calculus		
SS 1120	Introductory Soil Science		4
General Education Requirement (1B)			3
Free Elective	^{3, 4}		3
Free Elective	^{3, 4}		3
Units			16

Second Year			
Term 1			

NR 2208	Dendrology		3
or BIO 2215	or Biodiversity of California		
or BIO 2217	or Wildlife Conservation Biology		
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	¹	4
or CHEM 1122	or Fundamentals of Chemical Reactivity		
STAT 1110	Applied Statistical Concepts and Methods		3
General Education Requirement			3
General Education Requirement			3
Units			16

Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
BIO 1111	General Biology (5B & 5C) ¹	4
& BIO 1112	or Life: History and Diversity	
or BIO 1150	or Life: Molecules and Cells	
or BIO 1151		
General Education Requirement		3
Free Elective ^{3, 4}		3-4
Units		14

Third Year

Term 1

NR 3308	Fire and Society (Upper-Division 4)	¹	3-4
or NR 3323	or Human Dimensions in Natural Resources Management		
or NR 3324	or Social Dimensions of Sustainable Food Systems		
or NR 3328	or Environmental Leadership and Community Engagement		
or NR 3335	or Conflict Management in Natural Resources		
NR 3310	Global Climate Change (Upper-Division 2/5)	¹	3-4
or BOT 3311	or Plants, People and Civilization		
or BRAE 3348	or Energy for a Sustainable Society		
or ENVE 3324	or Introduction to Air Pollution		
NR 3363	Career Preparation and Practices in Natural Resources Fields		2

Concentration Course

2

3

General Education Requirement

3

Units

14

Term 2

NR 3304

Agroecology

4

or NR 3305

or Forest and Fire Ecology

or NR 3306

or Natural Resource Ecology and Habitat Management

NR 3319

Watershed Processes and Management

3-4

& NR 3320

or Forest Health and Disturbance Ecology

or NR 4402

or Wetlands

or NR/SS
4421

or Soil Morphology

or SS 3321

or Soil Erosion and Water Conservation

or ERSC
3303

Concentration Course	2	3
General Education Requirement		3
Units		13

Fourth Year		
Term 1		

NR/AGB 3326	Natural Resources Economics and Valuation	2-4
or NR 4418	or Applied Geographic Information System	

or NR/SS 4431	or Spatial Data Analysis and Environmental Mapping	
or NR 4442	or Environmental Life-Cycle Analysis	
or NR 4445	or Systems Thinking in Environmental Management	
NR/CRP 4404	Environmental Law	4
or NR/CRP 4408	or Water Resource Law and Policy	
or NR 4416	or Environmental Impact Analysis and Management	
or AG 4452	or Leadership Seminar on Issues Affecting California Agriculture, Food Systems, and Natural Resources	
Concentration Course	²	3
General Education Requirement		3
Free Elective	^{3, 4}	3-4

Units

16

Term 2

NR 4460	Senior Project - Watershed Assessment and Protection	3-4
or NR 4462	or Senior Project - Applied Resource Analysis and Assessment	
or NR 4463	or Senior Project - Ecological Restoration	
or NR 4464	or Senior Project - Environmental Policy Analysis	
or NR 4465	or Senior Project - Ecosystem Management	
Concentration Course	2	3
Concentration Course	2	3

Free Elective	3, 4	3-5
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in a concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Conservation Science and Management

Conservation Science and Management Concentration - BS in Environmental Management and Protection

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NR 1142	Environmental Management	4
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BIO 1114	Plant Diversity and Ecology	4
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or BOT 1121	or General Botany
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General Education Requirement (1A)	3
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General Education Requirement (1C) 3

Free Elective^{3, 4} 3

Units 17

Term 2

MATH 1264 Calculus for Data Science I (2)¹ 3-4

or MATH
1267 or Business Calculus

SS 1120 Introductory Soil Science 4

General Education Requirement (1B) 3

Free Elective^{3, 4} 3

Free Elective	3, 4	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology	3
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or BIO 2215 or Biodiversity of California

or BIO 2217 or Wildlife Conservation Biology

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	¹ 4
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or CHEM
1122 or Fundamentals of Chemical Reactivity

STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
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BIO 1111	General Biology (5B & 5C) ¹	4
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& BIO 1112	or Life: History and Diversity
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or BIO 1150	or Life: Molecules and Cells
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or BIO 1151

General Education Requirement	3
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Free Elective

3, 4

3-4

Units

14

Third Year

Term 1

NR 3308

Fire and Society (Upper-Division 4)

1

3-4

or NR 3323

or Human Dimensions in Natural Resources Management

or NR 3324

or Social Dimensions of Sustainable Food Systems

or NR 3328

or Environmental Leadership and Community Engagement

or NR 3335

or Conflict Management in Natural Resources

NR 3310	Global Climate Change (Upper-Division 2/5)	¹	3-4
or BOT 3311	or Plants, People and Civilization		
or BRAE 3348	or Energy for a Sustainable Society		
or ENVE 3324	or Introduction to Air Pollution		
NR 3363	Career Preparation and Practices in Natural Resources Fields		2
Approved Elective	^{2, 4}		3
General Education Requirement			3
	Units		14

Term 2

NR 3304	Agroecology	4
or NR 3305	or Forest and Fire Ecology	
or NR 3306	or Natural Resource Ecology and Habitat Management	
NR 3319	Watershed Processes and Management	3-4
& NR 3320	or Forest Health and Disturbance Ecology	
or NR 4402	or Wetlands	
or NR/SS 4421	or Soil Morphology	
or SS 3321	or Soil Erosion and Water Conservation	
or ERSC 3303		
Approved Elective	2, 4	3
General Education Requirement		3

Fourth Year**Term 1**

NR/AGB 3326 Natural Resources Economics and Valuation 2-4

or NR 4418 or Applied Geographic Information System

or NR/SS
4431 or Spatial Data Analysis and Environmental Mapping

or NR 4442 or Environmental Life-Cycle Analysis

or NR 4445 or Systems Thinking in Environmental Management

NR/CRP 4404 Environmental Law 4

or NR/CRP
4408 or Water Resource Law and Policy

or NR 4416 or Environmental Impact Analysis and Management

or AG 4452 or Leadership Seminar on Issues Affecting California
Agriculture, Food Systems, and Natural Resources

Approved Elective 2, 4 3

General Education Requirement 3

Free Elective 3, 4 3-4

Units 16

Term 2

NR 4460 Senior Project - Watershed Assessment and Protection 3-4

or NR 4462 or Senior Project - Applied Resource Analysis and
Assessment

or NR 4463 or Senior Project - Ecological Restoration

or NR 4464

or NR 4465 or Senior Project - Environmental Policy Analysis

or Senior Project - Ecosystem Management

Approved Elective	2, 4	3
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Approved Elective	2, 4	3
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Free Elective	3, 4	3-5
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in a concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses/Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Corporate Environmental Management

Corporate Environmental Management Concentration - BS in Environmental Management and Protection

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NR 1142	Environmental Management	4
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BIO 1114	Plant Diversity and Ecology	4
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or BOT 1121	or General Botany	
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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Free Elective	3, 4	3
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Units	17
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Term 2

MATH 1264	Calculus for Data Science I (2)	1	3-4
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or MATH 1267 or Business Calculus

SS 1120	Introductory Soil Science	4
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General Education Requirement (1B)	3
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Free Elective	3, 4	3
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Free Elective	3, 4	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology		3
or BIO 2215	or Biodiversity of California		
or BIO 2217	or Wildlife Conservation Biology		
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	¹	4
or CHEM 1122	or Fundamentals of Chemical Reactivity		
STAT 1110	Applied Statistical Concepts and Methods		3
General Education Requirement			3
General Education Requirement			3
Units			16

Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
BIO 1111	General Biology (5B & 5C) ¹	4
& BIO 1112	or Life: History and Diversity	
or BIO 1150	or Life: Molecules and Cells	
or BIO 1151		
General Education Requirement		3
Free Elective ^{3, 4}		3-4
Units		14

Third Year

Term 1

NR 3308	Fire and Society (Upper-Division 4)	¹	3-4
or NR 3323	or Human Dimensions in Natural Resources Management		
or NR 3324	or Social Dimensions of Sustainable Food Systems		
or NR 3328	or Environmental Leadership and Community Engagement		
or NR 3335	or Conflict Management in Natural Resources		
NR 3310	Global Climate Change (Upper-Division 2/5)	¹	3-4
or BOT 3311	or Plants, People and Civilization		
or BRAE 3348	or Energy for a Sustainable Society		
or ENVE 3324	or Introduction to Air Pollution		
NR 3363	Career Preparation and Practices in Natural Resources Fields		2

Approved Elective	2, 4	3
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General Education Requirement		3
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Units		14
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Term 2

NR 3304	Agroecology	4
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or NR 3305 or Forest and Fire Ecology

or NR 3306 or Natural Resource Ecology and Habitat Management

NR 3319	Watershed Processes and Management	3-4
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& NR 3320 or Forest Health and Disturbance Ecology

or NR 4402 or Wetlands

or NR/SS
4421

or Soil Morphology

or SS 3321

or Soil Erosion and Water Conservation

or ERSC
3303

Approved Elective 2, 4 3

General Education Requirement 3

Units 13

Fourth Year

Term 1

NR/AGB 3326 Natural Resources Economics and Valuation 2-4

or NR 4418 or Applied Geographic Information System

or NR/SS 4431	or Spatial Data Analysis and Environmental Mapping	
or NR 4442	or Environmental Life-Cycle Analysis	
or NR 4445	or Systems Thinking in Environmental Management	
NR/CRP 4404	Environmental Law	4
or NR/CRP 4408	or Water Resource Law and Policy	
or NR 4416	or Environmental Impact Analysis and Management	
or AG 4452	or Leadership Seminar on Issues Affecting California Agriculture, Food Systems, and Natural Resources	
Approved Elective	2, 4	3
General Education Requirement		3
Free Elective	3, 4	3-4

Units

16

Term 2

NR 4460	Senior Project - Watershed Assessment and Protection	3-4
or NR 4462	or Senior Project - Applied Resource Analysis and Assessment	
or NR 4463	or Senior Project - Ecological Restoration	
or NR 4464	or Senior Project - Environmental Policy Analysis	
or NR 4465	or Senior Project - Ecosystem Management	
Approved Elective	2, 4	3
Approved Elective	2, 4	3

Free Elective	3, 4	3-5
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in a concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses/Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Environmental Data Science

Environmental Data Science Concentration - BS in Environmental Management and Protection

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NR 1142	Environmental Management	4
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BIO 1114	Plant Diversity and Ecology	4
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or BOT 1121	or General Botany
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General Education Requirement (1A)	3
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General Education Requirement (1C) 3

Free Elective^{3, 4} 3

Units 17

Term 2

MATH 1264 Calculus for Data Science I (2)¹ 3-4

or MATH 1267 or Business Calculus

SS 1120 Introductory Soil Science 4

General Education Requirement (1B) 3

Free Elective^{3, 4} 3

Free Elective	3, 4	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology	3
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or BIO 2215 or Biodiversity of California

or BIO 2217 or Wildlife Conservation Biology

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	¹ 4
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or CHEM 1122 or Fundamentals of Chemical Reactivity

STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
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BIO 1111	General Biology (5B & 5C) ¹	4
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& BIO 1112	or Life: History and Diversity
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or BIO 1150	or Life: Molecules and Cells
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or BIO 1151

General Education Requirement	3
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Free Elective

3, 4

3-4

Units

14

Third Year

Term 1

NR 3308

Fire and Society (Upper-Division 4)

1

3-4

or NR 3323

or Human Dimensions in Natural Resources Management

or NR 3324

or Social Dimensions of Sustainable Food Systems

or NR 3328

or Environmental Leadership and Community Engagement

or NR 3335

or Conflict Management in Natural Resources

NR 3310	Global Climate Change (Upper-Division 2/5)	¹	3-4
or BOT 3311	or Plants, People and Civilization		
or BRAE 3348	or Energy for a Sustainable Society		
or ENVE 3324	or Introduction to Air Pollution		
NR 3363	Career Preparation and Practices in Natural Resources Fields		2
Approved Elective	^{2, 4}		3
General Education Requirement			3
	Units		14

Term 2

NR 3304	Agroecology	4
or NR 3305	or Forest and Fire Ecology	
or NR 3306	or Natural Resource Ecology and Habitat Management	
NR 3319	Watershed Processes and Management	3-4
& NR 3320	or Forest Health and Disturbance Ecology	
or NR 4402	or Wetlands	
or NR/SS 4421	or Soil Morphology	
or SS 3321	or Soil Erosion and Water Conservation	
or ERSC 3303		
Approved Elective	2, 4	3
General Education Requirement		3

Fourth Year**Term 1**

NR/AGB 3326 Natural Resources Economics and Valuation 2-4

or NR 4418 or Applied Geographic Information System

or NR/SS
4431 or Spatial Data Analysis and Environmental Mapping

or NR 4442 or Environmental Life-Cycle Analysis

or NR 4445 or Systems Thinking in Environmental Management

NR/CRP 4404 Environmental Law 4

or NR/CRP
4408 or Water Resource Law and Policy

or NR 4416 or Environmental Impact Analysis and Management

or AG 4452 or Leadership Seminar on Issues Affecting California
Agriculture, Food Systems, and Natural Resources

Approved Elective 2, 4 3

General Education Requirement 3

Free Elective 3, 4 3-4

Units 16

Term 2

NR 4460 Senior Project - Watershed Assessment and Protection 3-4

or NR 4462 or Senior Project - Applied Resource Analysis and
Assessment

or NR 4463 or Senior Project - Ecological Restoration

or NR 4464

or NR 4465 or Senior Project - Environmental Policy Analysis

or Senior Project - Ecosystem Management

Approved Elective	2, 4	3
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Approved Elective	2, 4	3
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Free Elective	3, 4	3-5
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in a concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses/Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Environmental Law, Justice and Policy

Environmental Law, Justice and Policy Concentration - BS in Environmental Management and Protection

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NR 1142	Environmental Management	4
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BIO 1114	Plant Diversity and Ecology	4
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or BOT 1121	or General Botany	
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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Free Elective	3, 4	3
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Units	17
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Term 2

MATH 1264	Calculus for Data Science I (2)	1	3-4
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or MATH 1267 or Business Calculus

SS 1120	Introductory Soil Science	4
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General Education Requirement (1B)	3
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Free Elective	3, 4	3
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Free Elective	3, 4	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology		3
or BIO 2215	or Biodiversity of California		
or BIO 2217	or Wildlife Conservation Biology		
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	1	4
or CHEM 1122	or Fundamentals of Chemical Reactivity		
STAT 1110	Applied Statistical Concepts and Methods		3
General Education Requirement			3
General Education Requirement			3
Units			16

Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
BIO 1111	General Biology (5B & 5C) ¹	4
& BIO 1112	or Life: History and Diversity	
or BIO 1150	or Life: Molecules and Cells	
or BIO 1151		
General Education Requirement		3
Free Elective ^{3, 4}		3-4
Units		14

Third Year

Term 1

NR 3308	Fire and Society (Upper-Division 4)	¹	3-4
or NR 3323	or Human Dimensions in Natural Resources Management		
or NR 3324	or Social Dimensions of Sustainable Food Systems		
or NR 3328	or Environmental Leadership and Community Engagement		
or NR 3335	or Conflict Management in Natural Resources		
NR 3310	Global Climate Change (Upper-Division 2/5)	¹	3-4
or BOT 3311	or Plants, People and Civilization		
or BRAE 3348	or Energy for a Sustainable Society		
or ENVE 3324	or Introduction to Air Pollution		
NR 3363	Career Preparation and Practices in Natural Resources Fields		2

Approved Elective	2, 4	3
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General Education Requirement		3
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Units		14
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Term 2

NR 3304	Agroecology	4
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or NR 3305 or Forest and Fire Ecology

or NR 3306 or Natural Resource Ecology and Habitat Management

NR 3319	Watershed Processes and Management	3-4
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& NR 3320 or Forest Health and Disturbance Ecology

or NR 4402 or Wetlands

or NR/SS
4421

or Soil Morphology

or SS 3321

or Soil Erosion and Water Conservation

or ERSC
3303

Approved Elective 2, 4 3

General Education Requirement 3

Units 13

Fourth Year

Term 1

NR/AGB 3326 Natural Resources Economics and Valuation 2-4

or NR 4418 or Applied Geographic Information System

or NR/SS 4431	or Spatial Data Analysis and Environmental Mapping	
or NR 4442	or Environmental Life-Cycle Analysis	
or NR 4445	or Systems Thinking in Environmental Management	
NR/CRP 4404	Environmental Law	4
or NR/CRP 4408	or Water Resource Law and Policy	
or NR 4416	or Environmental Impact Analysis and Management	
or AG 4452	or Leadership Seminar on Issues Affecting California Agriculture, Food Systems, and Natural Resources	
Approved Elective	2, 4	3
General Education Requirement		3
Free Elective	3, 4	3-4

Units

16

Term 2

NR 4460	Senior Project - Watershed Assessment and Protection	3-4
or NR 4462	or Senior Project - Applied Resource Analysis and Assessment	
or NR 4463	or Senior Project - Ecological Restoration	
or NR 4464	or Senior Project - Environmental Policy Analysis	
or NR 4465	or Senior Project - Ecosystem Management	
Approved Elective	2, 4	3
Approved Elective	2, 4	3

Free Elective ^{3, 4} 3-5

Units 14

Total Units 120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in a concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses/Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Sustainable Agriculture

Sustainable Agriculture Concentration - BS in Environmental Management and Protection

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
NR 1142	Environmental Management	4
BIO 1114	Plant Diversity and Ecology	4
or BOT 1121	or General Botany	
General Education Requirement (1A)		3

General Education Requirement (1C) 3

Free Elective^{3, 4} 3

Units 17

Term 2

MATH 1264 Calculus for Data Science I (2)¹ 3-4

or MATH 1267 or Business Calculus

SS 1120 Introductory Soil Science 4

General Education Requirement (1B) 3

Free Elective^{3, 4} 3

Free Elective	3, 4	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology	3
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or BIO 2215 or Biodiversity of California

or BIO 2217 or Wildlife Conservation Biology

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	¹ 4
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or CHEM
1122 or Fundamentals of Chemical Reactivity

STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
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BIO 1111	General Biology (5B & 5C)	4
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& BIO 1112	or Life: History and Diversity
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or BIO 1150	or Life: Molecules and Cells
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or BIO 1151

General Education Requirement	3
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Free Elective

3, 4

3-4

Units

14

Third Year

Term 1

NR 3308

Fire and Society (Upper-Division 4)

1

3-4

or NR 3323

or Human Dimensions in Natural Resources Management

or NR 3324

or Social Dimensions of Sustainable Food Systems

or NR 3328

or Environmental Leadership and Community Engagement

or NR 3335

or Conflict Management in Natural Resources

NR 3310	Global Climate Change (Upper-Division 2/5)	¹	3-4
or BOT 3311	or Plants, People and Civilization		
or BRAE 3348	or Energy for a Sustainable Society		
or ENVE 3324	or Introduction to Air Pollution		
NR 3363	Career Preparation and Practices in Natural Resources Fields		2
Approved Elective	^{2, 4}		3
General Education Requirement			3
Units			14

Term 2

NR 3304	Agroecology	4
or NR 3305	or Forest and Fire Ecology	
or NR 3306	or Natural Resource Ecology and Habitat Management	
NR 3319	Watershed Processes and Management	3-4
& NR 3320	or Forest Health and Disturbance Ecology	
or NR 4402	or Wetlands	
or NR/SS 4421	or Soil Morphology	
or SS 3321	or Soil Erosion and Water Conservation	
or ERSC 3303		
Approved Elective	2, 4	3
General Education Requirement		3

Fourth Year**Term 1**

NR/AGB 3326 Natural Resources Economics and Valuation 2-4

or NR 4418 or Applied Geographic Information System

or NR/SS
4431 or Spatial Data Analysis and Environmental Mapping

or NR 4442 or Environmental Life-Cycle Analysis

or NR 4445 or Systems Thinking in Environmental Management

NR/CRP 4404 Environmental Law 4

or NR/CRP
4408 or Water Resource Law and Policy

or NR 4416 or Environmental Impact Analysis and Management

or AG 4452 or Leadership Seminar on Issues Affecting California
Agriculture, Food Systems, and Natural Resources

Approved Elective 2, 4 3

General Education Requirement 3

Free Elective 3, 4 3-4

Units 16

Term 2

NR 4460 Senior Project - Watershed Assessment and Protection 3-4

or NR 4462 or Senior Project - Applied Resource Analysis and
Assessment

or NR 4463 or Senior Project - Ecological Restoration

or NR 4464

or NR 4465 or Senior Project - Environmental Policy Analysis

or Senior Project - Ecosystem Management

Approved Elective	2, 4	3
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Approved Elective	2, 4	3
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Free Elective	3, 4	3-5
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in a concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses/Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Sustainable Urban Development and Planning

Sustainable Urban Development and Planning Concentration - BS in Environmental Management and Protection

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NR 1142	Environmental Management	4
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BIO 1114	Plant Diversity and Ecology	4
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or BOT 1121	or General Botany	
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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Free Elective	3, 4	3
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Units	17
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Term 2

MATH 1264	Calculus for Data Science I (2)	1	3-4
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or MATH 1267 or Business Calculus

SS 1120	Introductory Soil Science	4
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General Education Requirement (1B)	3
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Free Elective	3, 4	3
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Free Elective	3, 4	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology		3
or BIO 2215	or Biodiversity of California		
or BIO 2217	or Wildlife Conservation Biology		
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	1	4
or CHEM 1122	or Fundamentals of Chemical Reactivity		
STAT 1110	Applied Statistical Concepts and Methods		3
General Education Requirement			3
General Education Requirement			3
Units			16

Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
BIO 1111	General Biology (5B & 5C) ¹	4
& BIO 1112	or Life: History and Diversity	
or BIO 1150	or Life: Molecules and Cells	
or BIO 1151		
General Education Requirement		3
Free Elective ^{3, 4}		3-4
Units		14

Third Year

Term 1

NR 3308	Fire and Society (Upper-Division 4)	¹	3-4
or NR 3323	or Human Dimensions in Natural Resources Management		
or NR 3324	or Social Dimensions of Sustainable Food Systems		
or NR 3328	or Environmental Leadership and Community Engagement		
or NR 3335	or Conflict Management in Natural Resources		
NR 3310	Global Climate Change (Upper-Division 2/5)	¹	3-4
or BOT 3311	or Plants, People and Civilization		
or BRAE 3348	or Energy for a Sustainable Society		
or ENVE 3324	or Introduction to Air Pollution		
NR 3363	Career Preparation and Practices in Natural Resources Fields		2

Approved Elective	2, 4	3
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General Education Requirement		3
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Units		14
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Term 2

NR 3304	Agroecology	4
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or NR 3305 or Forest and Fire Ecology

or NR 3306 or Natural Resource Ecology and Habitat Management

NR 3319	Watershed Processes and Management	3-4
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& NR 3320 or Forest Health and Disturbance Ecology

or NR 4402 or Wetlands

or NR/SS
4421

or Soil Morphology

or SS 3321

or Soil Erosion and Water Conservation

or ERSC
3303

Approved Elective 2, 4 3

General Education Requirement 3

Units 13

Fourth Year

Term 1

NR/AGB 3326 Natural Resources Economics and Valuation 2-4

or NR 4418 or Applied Geographic Information System

or NR/SS 4431	or Spatial Data Analysis and Environmental Mapping	
or NR 4442	or Environmental Life-Cycle Analysis	
or NR 4445	or Systems Thinking in Environmental Management	
NR/CRP 4404	Environmental Law	4
or NR/CRP 4408	or Water Resource Law and Policy	
or NR 4416	or Environmental Impact Analysis and Management	
or AG 4452	or Leadership Seminar on Issues Affecting California Agriculture, Food Systems, and Natural Resources	
Approved Elective	2, 4	3
General Education Requirement		3
Free Elective	3, 4	3-4

Units

16

Term 2

NR 4460	Senior Project - Watershed Assessment and Protection	3-4
or NR 4462	or Senior Project - Applied Resource Analysis and Assessment	
or NR 4463	or Senior Project - Ecological Restoration	
or NR 4464	or Senior Project - Environmental Policy Analysis	
or NR 4465	or Senior Project - Ecosystem Management	
Approved Elective	2, 4	3
Approved Elective	2, 4	3

Free Elective	3, 4	3-5
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in a concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses/Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Water Science and Management

Water Science and Management Concentration - BS in Environmental Management and Protection

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
NR 1142	Environmental Management	4
BIO 1114	Plant Diversity and Ecology	4
or BOT 1121	or General Botany	
General Education Requirement (1A)		3

General Education Requirement (1C) 3

Free Elective^{3, 4} 3

Units 17

Term 2

MATH 1264 Calculus for Data Science I (2)¹ 3-4

or MATH 1267 or Business Calculus

SS 1120 Introductory Soil Science 4

General Education Requirement (1B) 3

Free Elective^{3, 4} 3

Free Elective	3, 4	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology	3
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or BIO 2215 or Biodiversity of California

or BIO 2217 or Wildlife Conservation Biology

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	¹ 4
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or CHEM
1122 or Fundamentals of Chemical Reactivity

STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
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BIO 1111	General Biology (5B & 5C) ¹	4
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& BIO 1112	or Life: History and Diversity
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or BIO 1150	or Life: Molecules and Cells
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or BIO 1151

General Education Requirement	3
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Free Elective

3, 4

3-4

Units

14

Third Year

Term 1

NR 3308

Fire and Society (Upper-Division 4)

1

3-4

or NR 3323

or Human Dimensions in Natural Resources Management

or NR 3324

or Social Dimensions of Sustainable Food Systems

or NR 3328

or Environmental Leadership and Community Engagement

or NR 3335

or Conflict Management in Natural Resources

NR 3310	Global Climate Change (Upper-Division 2/5)	¹	3-4
or BOT 3311	or Plants, People and Civilization		
or BRAE 3348	or Energy for a Sustainable Society		
or ENVE 3324	or Introduction to Air Pollution		
NR 3363	Career Preparation and Practices in Natural Resources Fields		2
Approved Elective	^{2, 4}		3
General Education Requirement			3
Units			14

Term 2

NR 3304	Agroecology	4
or NR 3305	or Forest and Fire Ecology	
or NR 3306	or Natural Resource Ecology and Habitat Management	
NR 3319	Watershed Processes and Management	3-4
& NR 3320	or Forest Health and Disturbance Ecology	
or NR 4402	or Wetlands	
or NR/SS 4421	or Soil Morphology	
or SS 3321	or Soil Erosion and Water Conservation	
or ERSC 3303		
Approved Elective	2, 4	3
General Education Requirement		3

Fourth Year**Term 1**

NR/AGB 3326 Natural Resources Economics and Valuation 2-4

or NR 4418 or Applied Geographic Information System

or NR/SS
4431 or Spatial Data Analysis and Environmental Mapping

or NR 4442 or Environmental Life-Cycle Analysis

or NR 4445 or Systems Thinking in Environmental Management

NR/CRP 4404 Environmental Law 4

or NR/CRP
4408 or Water Resource Law and Policy

or NR 4416 or Environmental Impact Analysis and Management

or AG 4452 or Leadership Seminar on Issues Affecting California
Agriculture, Food Systems, and Natural Resources

Approved Elective 2, 4 3

General Education Requirement 3

Free Elective 3, 4 3-4

Units 16

Term 2

NR 4460 Senior Project - Watershed Assessment and Protection 3-4

or NR 4462 or Senior Project - Applied Resource Analysis and
Assessment

or NR 4463 or Senior Project - Ecological Restoration

or NR 4464

or NR 4465 or Senior Project - Environmental Policy Analysis

or Senior Project - Ecosystem Management

Approved Elective	2, 4	3
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Approved Elective	2, 4	3
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Free Elective	3, 4	3-5
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in a concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses/Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Experience and Event Management

Concentration Not Yet Declared - BS in Experience and Event Management
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.
Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

EIM 1101

Introduction to the Experience Industry

3

MATH 1004	Stretch College Algebra (2)	¹	3-5
or MATH 1006	or College Algebra		
or MATH 1007	or Precalculus		
or MATH 1267	or Business Calculus		
General Education Requirement (1A)			3
General Education Requirement (1C)			3
General Education Requirement			3
Units			15

Term 2

BUS 2212	Financial Accounting for Nonbusiness Majors	2	3
or AGB 2214	or Agribusiness Financial Accounting		
STAT 1110	Applied Statistical Concepts and Methods		3
or STAT 1210	or Business Statistics I		
General Education Requirement (1B)			3
General Education Requirement			3
Free Elective		3	1-3
Units			15

Second Year

Term 1

EIM 2210	Experience Design: Theories and Applications	4
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BUS 2215	Managerial Accounting ²	3
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or AGB 3323

or Decision Making with Agribusiness Accounting
Information

General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

EIM 2255	Leadership and Diverse Groups	3
	Concentration Course	3
	Concentration Course	3
	General Education Requirement	3
	General Education Requirement	3
	General Education Requirement	1
	Units	16

Third Year

Term 1

EIM 3313	Sustainability in the Experience Industry	3
EIM 3360	Assessment and Evaluation of Experience Industry Management	3
BUS 3346	Principles of Marketing	3
ENGL 3310	Corporate Communication (Upper-Division 4) ¹	3
Concentration Course		3
Units		15

Term 2

EIM 3370	Experiential Marketing Strategies for Experience Industry Management	3
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Concentration Course	3
Concentration Course	3
Concentration Course	3
General Education Requirement	3
Units	15

Fourth Year

Term 1

EIM 4405	Principles of Experience Industry Management	3
EIM 4416	Interpreting Experience Industry Data Analytics	3

EIM 4424	Financial Management in the Experience Industry	3
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EIM 4460	Senior Project: Experience Industry Management	3
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or EIM 4461	or Senior Project: Student Research	
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EIM 4463	Pre-Internship Seminar	1
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General Education Requirement		3
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Units		16
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Term 2

EIM 4465	Internship	12
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Units		12
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Choose either BUS or AGB route. [BUS 2212](#) is a prerequisite for [BUS 2215](#) and [AGB 2214](#) is a prerequisite for [AGB 3323](#).

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Event Planning and Management

Event Planning and Management Concentration - BS in Experience and Event Management

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

EIM 1101 Introduction to the Experience Industry 3

MATH 1004 Stretch College Algebra (2)¹ 3-5

or MATH
1006 or College Algebra

or MATH
1007 or Precalculus

or MATH
1267 or Business Calculus

General Education Requirement (1A) 3

General Education Requirement (1C) 3

General Education Requirement	3
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Units	15
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Term 2

BUS 2212	Financial Accounting for Nonbusiness Majors	2	3
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or AGB 2214

or Agribusiness Financial Accounting

STAT 1110	Applied Statistical Concepts and Methods	3
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or STAT 1210 or Business Statistics I

General Education Requirement (1B)	3
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General Education Requirement	3
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Free Elective	3	1-3
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Units	15
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Second Year

Term 1

EIM 2210	Experience Design: Theories and Applications	4
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BUS 2215	Managerial Accounting	2	3
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or AGB 3323

or Decision Making with Agribusiness Accounting
Information

General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

EIM 1114	Introduction to Hospitality and Travel	3
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EIM 2255	Leadership and Diverse Groups	3
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Approved Elective	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	1
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Units

16

Third Year

Term 1

EIM 3313	Sustainability in the Experience Industry	3
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EIM 3317	Hospitality, Convention and Meeting Management	3
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EIM 3320	Strategic Event Planning	3
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EIM 3360	Assessment and Evaluation of Experience Industry Management	3
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BUS 3346	Principles of Marketing	3
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Units

15

Term 2

EIM 3370	Experiential Marketing Strategies for Experience Industry Management	3
EIM 4420	Strategic Event Implementation	3
ENGL 3310	Corporate Communication (Upper-Division 4) ¹	3
Approved Elective		3
General Education Requirement		3
Units		15

Fourth Year

Term 1

EIM 4405	Principles of Experience Industry Management	3
EIM 4416	Interpreting Experience Industry Data Analytics	3
EIM 4424	Financial Management in the Experience Industry	3
EIM 4460	Senior Project: Experience Industry Management	3
or EIM 4461	or Senior Project: Student Research	
EIM 4463	Pre-Internship Seminar	1
General Education Requirement		3
Units		16

Term 2

EIM 4465	Internship	12
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Units	12
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Choose either BUS or AGB route. [BUS 2212](#) is a prerequisite for [BUS 2215](#) and [AGB 2214](#) is a prerequisite for [AGB 3323](#).

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Tourism, Hospitality, and Destination Management

***Tourism, Hospitality, and Destination Management Concentration - BS in
Experience and Event Management***

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
EIM 1101	Introduction to the Experience Industry	3
MATH 1004	Stretch College Algebra (2) ¹	3-5
or MATH 1006	or College Algebra	
or MATH 1007	or Precalculus	
or MATH 1267	or Business Calculus	
General Education Requirement (1A)		3

General Education Requirement (1C)	3
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General Education Requirement	3
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Units	15
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Term 2

BUS 2212	Financial Accounting for Nonbusiness Majors	2	3
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or AGB 2214

or Agribusiness Financial Accounting

STAT 1110	Applied Statistical Concepts and Methods	3
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or STAT 1210 or Business Statistics I

General Education Requirement (1B)	3
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General Education Requirement	3
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Free Elective	3	1-3
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Units	15
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Second Year

Term 1

EIM 2210	Experience Design: Theories and Applications	4
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BUS 2215	Managerial Accounting	2	3
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or AGB 3323

or Decision Making with Agribusiness Accounting
Information

General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

EIM 1114	Introduction to Hospitality and Travel	3
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EIM 2216	Resort and Lodging Operations	3
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EIM 2255	Leadership and Diverse Groups	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	1
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Units

16

Third Year

Term 1

EIM 3313	Sustainability in the Experience Industry	3
EIM 3317	Hospitality, Convention and Meeting Management	3
EIM 3360	Assessment and Evaluation of Experience Industry Management	3
BUS 3346	Principles of Marketing	3
ENGL 3310	Corporate Communication (Upper-Division 4) ¹	3

Units

15

Term 2

EIM 3318	Destination and Hospitality Marketing & Management	3
EIM 3370	Experiential Marketing Strategies for Experience Industry Management	3
Approved Elective		3
Approved Elective		3
General Education Requirement		3
Units		15

Fourth Year

Term 1

EIM 4405	Principles of Experience Industry Management	3
EIM 4416	Interpreting Experience Industry Data Analytics	3
EIM 4424	Financial Management in the Experience Industry	3
EIM 4460	Senior Project: Experience Industry Management	3
or EIM 4461	or Senior Project: Student Research	
EIM 4463	Pre-Internship Seminar	1
General Education Requirement		3
Units		16

Term 2

EIM 4465	Internship	12
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	Units	12
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	Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Choose either BUS or AGB route. [BUS 2212](#) is a prerequisite for [BUS 2215](#) and [AGB 2214](#) is a prerequisite for [AGB 3323](#).

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Sport and Recreation Management

Sport and Recreation Management Concentration - BS in Experience and Event Management

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
EIM 1101	Introduction to the Experience Industry	3
MATH 1004	Stretch College Algebra (2) ¹	3-5
or MATH 1006	or College Algebra	
or MATH 1007	or Precalculus	
or MATH 1267	or Business Calculus	
General Education Requirement (1A)		3

General Education Requirement (1C)	3
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General Education Requirement	3
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Units	15
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Term 2

BUS 2212	Financial Accounting for Nonbusiness Majors	2	3
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or AGB 2214	or Agribusiness Financial Accounting
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STAT 1110	Applied Statistical Concepts and Methods	3
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or STAT 1210	or Business Statistics I
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General Education Requirement (1B)	3
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General Education Requirement	3
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Free Elective

3

1-3

Units

15

Second Year

Term 1

EIM 1112	Introduction to Parks and Outdoor Recreation	3
or EIM 1160	or Introduction to Sport Management	
EIM 2210	Experience Design: Theories and Applications	4
BUS 2215	Managerial Accounting	2
or AGB 3323	or Decision Making with Agribusiness Accounting Information	3

General Education Requirement 3

General Education Requirement 3

Units 16

Term 2

EIM 2255 Leadership and Diverse Groups 3

EIM 2260 Community Relations and Sports-Based Youth Development 3

or EIM 2275
or Facilitation and Teambuilding Experiences

General Education Requirement 3

General Education Requirement 3

General Education Requirement	3
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General Education Requirement	1
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Units	16
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Third Year

Term 1

EIM 3313	Sustainability in the Experience Industry	3
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EIM 3323	Sport Marketing and the Fan Experience	3
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or EIM 3325	or Leadership in Outdoor Experiences	
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EIM 3360	Assessment and Evaluation of Experience Industry Management	3
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BUS 3346	Principles of Marketing	3
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ENGL 3310	Corporate Communication (Upper-Division 4) ¹	3
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Units	15
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Term 2

EIM 3370	Experiential Marketing Strategies for Experience Industry Management	3
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Approved Elective	3
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Approved Elective	3
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Approved Elective	3
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General Education Requirement	3
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Units

15

Fourth Year

Term 1

EIM 4405	Principles of Experience Industry Management	3
EIM 4416	Interpreting Experience Industry Data Analytics	3
EIM 4424	Financial Management in the Experience Industry	3
EIM 4460	Senior Project: Experience Industry Management	3
or EIM 4461	or Senior Project: Student Research	
EIM 4463	Pre-Internship Seminar	1

General Education Requirement	3
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Units	16
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Term 2

EIM 4465	Internship	12
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Units	12
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Choose either BUS or AGB route. [BUS 2212](#) is a prerequisite for [BUS 2215](#) and [AGB 2214](#) is a prerequisite for [AGB 3323](#).

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Facilities Engineering Technology - Division 1 & 2

Division 1 & 2 - BS in Facilities Engineering Technology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
EPO 1110	Plant Operations I	1
EPO 1125	Intro To Marine Engineering	4
& 1125L	and Intro To Marine Engineering Lab	
EPO 2213	Welding Lab	1
ET 1110	Introduction To Engineering Technology	1

CHE 1110	General Chemistry	4
& 1110L	and General Chemistry Laboratory (5A & 5C) ¹	
EGL 1110	Speech Communication (1C) ¹	3
ENG 1100	Engineering Graphics	2
PE 1101	Swim Competency Exam ²	0
General Education Requirement (3B)		3
Units		19

Term 2

CHE 2205	Chemistry of Power Plant Processes	3
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DL 1105	Marine Survival	2
& 1105L	and Marine Survival Laboratory	
DL 1105X	United States Coast Guard Lifeboatman's Exam	0
EGL 1100	English Composition (1A) ¹	3
FF 1100	Basic Marine Firefighting	0
GOV 2200	American Government (4A) ¹	3
LIB 1100	Information Fluency In The Digital World	2
MTH 2210	Calculus I (2) ¹	4
NAU 1104	Shipboard Security And Responsibility	1

Units

18

Term 3

CRU 1150

Sea Training I - Engine

8

Units

8

Second Year

Term 1

COM 2220L

Programming Applications for Engineering Technology Lab

1

EPO 2210

Plant Operations Ii

1

EPO 2214

Boilers

3

EPO 2215	Manufacturing Processes I	1
EPO 2230	Steam Plant System Operations	1
EGL 2220	Critical Thinking (1B) ¹	3
MTH 2211	Calculus Ii	4
PHY 2200	Engineering Physics I	4
& 2200L	and Engineering Physics I Lab	
Units		18

Term 2

EPO 2220	Diesel Engineering I	2
& 2220L	and Diesel Engineering I Lab	

EPO 2235	Steam Plant Watch Team Management	1
EPO 3312	Turbines	3
ET 2230	Properties Of Materials	2
ET 2232	Statics	3
PHY 2205	Engineering Physics Ii	4
& 2205L	and Engineering Physics Ii Lab	
General Education Requirement		3
Units		18

Term 3

CEP 2270	Facilities Engineering Technology Industry Cooperative I	3
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Units	3
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Third Year

Term 1

EPO 3319	Facilities Engineering Diagnostics Lab	1
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ET 2230L	Properties Of Materials Lab	1
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ET 2250	Electrical Circuits	4
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& 2250L	and Electrical Circuits Lab	
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ET 3330	Dynamics	3
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ET 3332	Strength Of Materials	3
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ET 3344	Thermodynamics	3
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General Education Requirement		3
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Units		18
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Term 2

EPO 3310	Plant Operations III	1
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EPO 3315	Manufacturing Processes Ii	1
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EPO 3321	Introduction To Power Generation Plants	1
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ET 3340	Fluid Mechanics	4
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& 3340L	and Fluid Mechanics Lab	
ET 3342	Refrigeration And A/C	3
& 3342L	and Refrigeration And A/C Lab	
ET 3370	Electronics	4
& 3370L	and Electronics Lab	
General Education Requirement		3
Units		17

Term 3

CEP 3370	Facilities Engineering Technology Industry Cooperative II	3
Units		3

Fourth Year

Term 1

ET 3350	Electrical Machinery	4
& 3350L	and Electrical Machinery Lab	
ET 4400	Inst & Measure Sys	4
& 4400L	and Inst & Measure Sys Lab	
ET 4442	Heating Ventilating A/C	3
& 4442L	and Heating Ventilating A/C Lab	
ET 4470	Engineering Management	3
General Education Requirement		3

General Education Requirement 3

Units 20

Term 2

ET 4460 Automation 4

& 4460L and Automation Lab

ET 4490 Power Engineering Tech 4

& 4490L and Power Engineering Tech Lab

ENG 3310 Engineering Ethics (Upper-Division 4)¹ 3

ENG 4472 Facilities Management 3

General Education Requirement 3

Units	17
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Total Units	159
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Swim assessments are required of all cadets during Orientation week. If [PE 1101](#) is not passed with a grade CR, then enrollment in [PE 1102](#) will be required.

Facilities Engineering Technology - Division 3 & 4

Division 3 & 4 - BS in Facilities Engineering Technology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
ET 1110	Introduction To Engineering Technology	1
CHE 1110	General Chemistry	4
& 1110L	and General Chemistry Laboratory (5A & 5C) ¹	
EGL 1110	Speech Communication (1C) ¹	3
ENG 1100	Engineering Graphics	2
FF 1100	Basic Marine Firefighting	0
GOV 2200	American Government (4A) ¹	3

PE 1101	Swim Competency Exam	2	0
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General Education Requirement			3
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Units			16
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Term 2

EPO 1110	Plant Operations I		1
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EPO 1125	Intro To Marine Engineering		4
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& 1125L	and Intro To Marine Engineering Lab		
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EPO 2213	Welding Lab		1
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CHE 2205	Chemistry of Power Plant Processes		3
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DL 1105	Marine Survival	2
& 1105L	and Marine Survival Laboratory	
DL 1105X	United States Coast Guard Lifeboatman's Exam	0
EGL 1100	English Composition (1A) ¹	3
MTH 2210	Calculus I (2) ¹	4
NAU 1104	Shipboard Security And Responsibility	1
Units		19

Term 3

CRU 1150	Sea Training I - Engine	8
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Units

8

Second Year

Term 1

COM 2220L	Programming Applications for Engineering Technology Lab	1
EPO 2210	Plant Operations Ii	1
EPO 2214	Boilers	3
EPO 2215	Manufacturing Processes I	1
EPO 2230	Steam Plant System Operations	1
EGL 2220	Critical Thinking (1B) ¹	3

MTH 2211	Calculus Ii	4
PHY 2200	Engineering Physics I	4
& 2200L	and Engineering Physics I Lab	
Units		18

Term 2

EPO 2220	Diesel Engineering I	2
& 2220L	and Diesel Engineering I Lab	
EPO 2235	Steam Plant Watch Team Management	1
EPO 3312	Turbines	3
ET 2230	Properties Of Materials	2

ET 2232	Statics	3
LIB 1100	Information Fluency In The Digital World	2
PHY 2205	Engineering Physics Ii	4
& 2205L	and Engineering Physics Ii Lab	
General Education Requirement		3
Units		20

Term 3

CEP 2270	Facilities Engineering Technology Industry Cooperative I	3
Units		3

Third Year

Term 1

EPO 3319	Facilities Engineering Diagnostics Lab	1
ET 2230L	Properties Of Materials Lab	1
ET 2250	Electrical Circuits	4
& 2250L	and Electrical Circuits Lab	
ET 3330	Dynamics	3
ET 3332	Strength Of Materials	3
ET 3344	Thermodynamics	3

General Education Requirement	3
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Units	18
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Term 2

EPO 3310	Plant Operations III	1
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EPO 3315	Manufacturing Processes Ii	1
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EPO 3321	Introduction To Power Generation Plants	1
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ET 3340	Fluid Mechanics	4
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& 3340L	and Fluid Mechanics Lab
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ET 3342	Refrigeration And A/C	3
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& 3342L	and Refrigeration And A/C Lab
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ET 3370	Electronics	4
& 3370L	and Electronics Lab	
General Education Requirement		3
Units		17

Term 3

CEP 3370	Facilities Engineering Technology Industry Cooperative II	3
Units		3

Fourth Year

Term 1

ET 3350	Electrical Machinery	4
& 3350L	and Electrical Machinery Lab	
ET 4400	Inst & Measure Sys	4
& 4400L	and Inst & Measure Sys Lab	
ET 4442	Heating Ventilating A/C	3
& 4442L	and Heating Ventilating A/C Lab	
ET 4470	Engineering Management	3
General Education Requirement		3
General Education Requirement		3
Units		20

ET 4460	Automation	4
& 4460L	and Automation Lab	
ET 4490	Power Engineering Tech	4
& 4490L	and Power Engineering Tech Lab	
ENG 3310	Engineering Ethics (Upper-Division 4) ¹	3
ENG 4472	Facilities Management	3
General Education Requirement		3
Units		17
Total Units		159

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Swim assessments are required of all cadets during Orientation week. If [PE 1101](#) is not passed with a grade CR, then enrollment in [PE 1102](#) will be required.

Food Science

Concentration Not Yet Declared - BS in Food Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

FDSC 1101	Orientation to the Food Science Major	1
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FDSC 1110	Introduction to Food Science and Sustainability	3
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or FSN 1111	or Elements of Food Processing	
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FSN 2250	Food and Nutrition: Culture and Customs (USCP)	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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General Education Requirement (1A)		3
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Units		14
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Term 2

FSN 2202	Introduction to Human Nutrition	3
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MATH 1261	Calculus I (2) ¹	4
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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General Education Requirement (1C)	3
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Units	14
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Second Year

Term 1

BIO 1111	General Biology (5B) ¹	3
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STAT 1110	Applied Statistical Concepts and Methods	3
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CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
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General Education Requirement (1B)	3
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General Education Requirement	3
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Units

16

Term 2

MCRO 2221

Introduction to Microbiology

4

PHYS 1121

College Physics I

4

General Education Requirement

3

General Education Requirement

3

Units

14

Third Year

Term 1

FDSC 3330	Food Processing and Engineering I	4
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CHEM 3350	Biochemistry: Fundamentals and Applications (Upper-Division 2/5) ¹	4
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Concentration Course		3
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Concentration Course		3
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General Education Requirement		3
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Units		17
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Term 2

FDSC 3310	Food Laws and Regulations	3
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FDSC 3340	Food Quality Assurance and Prerequisite Programs	4
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FDSC 3350	Food Chemistry	4
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STAT 3320	Statistical Methods for Food Science	3
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Units	14
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Fourth Year

Term 1

FDSC 3345	Food Safety and Sanitation	4
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FDSC 3355	Food Analysis	3
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FDSC 4420	Sensory Evaluation of Food	3
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MCRO 4421	Food Microbiology	3
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General Education Requirement	3
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Units	16
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Term 2

FDSC 4425	Food Product Development	3
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& FDSC 4426	and Food Science Exit Exam	
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Concentration Course	3
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Concentration Course	3
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Culinology®

Culinology® Concentration - BS in Food Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1	Units
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FDSC 1101	Orientation to the Food Science Major	1
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FDSC 1110	Introduction to Food Science and Sustainability	3
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or FSN 1111	or Elements of Food Processing	
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FSN 2250	Food and Nutrition: Culture and Customs (USCP)	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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General Education Requirement (1A)		3
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Units		14
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Term 2

FSN 2202	Introduction to Human Nutrition	3
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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MATH 1261	Calculus I (2) ¹	4
General Education Requirement (1C)		3
Units		14

Second Year

Term 1

FSN 1121	Fundamentals of Food	3
BIO 1111	General Biology (5B) ¹	3
PHYS 1121	College Physics I	4
STAT 1110	Applied Statistical Concepts and Methods	3

General Education Requirement (1B)	3
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Units	16
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Term 2

FSN 3323	Culinary Internship	1
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CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
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MCRO 2221	Introduction to Microbiology	4
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Third Year

Term 1

FDSC 3330	Food Processing and Engineering I	4
CHEM 3350	Biochemistry: Fundamentals and Applications (Upper-Division 2/5) ¹	4
MCRO 4421	Food Microbiology	3
General Education Requirement		3
Units		14

Term 2

FDSC 3310	Food Laws and Regulations	3
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FDSC 3340	Food Quality Assurance and Prerequisite Programs	4
FDSC 3350	Food Chemistry	4
FSN 3304	Culinology	3
STAT 3320	Statistical Methods for Food Science	3
Units		17

Fourth Year

Term 1

FDSC 3345	Food Safety and Sanitation	4
FDSC 3355	Food Analysis	3

FDSC 4420	Sensory Evaluation of Food	3
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General Education Requirement		3
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General Education Requirement		3
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Units		16
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Term 2

FDSC 4425	Food Product Development	3
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& FDSC 4426	and Food Science Exit Exam	
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FDSC 4462	Senior Project - Internship	2
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FSN 3346	Institutional Food Service	3
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General Education Requirement	3
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General Education Requirement	3
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Food Safety

Food Safety Concentration - BS in Food Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
FDSC 1101	Orientation to the Food Science Major	1
FDSC 1110	Introduction to Food Science and Sustainability	3
or FSN 1111	or Elements of Food Processing	
FSN 2250	Food and Nutrition: Culture and Customs	3
CHEM 1120	Fundamentals of Chemical Structure and Properties	4
General Education Requirement (1A)		3
Units		14
Term 2		

FSN 2202	Introduction to Human Nutrition	3
CHEM 1122	Fundamentals of Chemical Reactivity	4
MATH 1261	Calculus I	4
General Education Requirement (1C)		3
Units		14

Second Year

Term 1

BIO 1111	General Biology (5B) ¹	3
MCRO 2221	Introduction to Microbiology	4

PHYS 1121	College Physics I	4
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General Education Requirement (1B)	3
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Units	14
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Term 2

CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units

16

Third Year

Term 1

FDSC 3330	Food Processing and Engineering I	4
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CHEM 3350	Biochemistry: Fundamentals and Applications (Upper-Division 2/5) ¹	4
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MCRO 4421	Food Microbiology	3
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General Education Requirement		3
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Units

14

Term 2

FDSC 3310	Food Laws and Regulations	3
FDSC 3340	Food Quality Assurance and Prerequisite Programs	4
FDSC 3350	Food Chemistry	4
STAT 3320	Statistical Methods for Food Science	3
Approved Elective		3
Units		17

Fourth Year

Term 1

FDSC 3345	Food Safety and Sanitation	4
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FDSC 3355	Food Analysis	3
FDSC 4420	Sensory Evaluation of Food	3
General Education Requirement		3
General Education Requirement		3
Units		16

Term 2

FDSC 4425	Food Product Development	3
& FDSC 4426	and Food Science Exit Exam	
FDSC 4430	Food Processing and Engineering II	4

FDSC 4445	Food Safety Modernization Act: Human Food Safety and Produce Safety	3
FDSC 4460	Senior Project - Scientific and Technical Writing Course	2
or FDSC 4461	or Senior Project - Research	
or FDSC 4462	or Senior Project - Internship	
General Education Requirement		3
Units		15
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Sustainable Food Technology

Sustainable Food Technology Concentration - BS in Food Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
FDSC 1101	Orientation to the Food Science Major	1
FDSC 1110	Introduction to Food Science and Sustainability	3
or FSN 1111	or Elements of Food Processing	
FSN 2250	Food and Nutrition: Culture and Customs (USCP)	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4

General Education Requirement (1A)	3
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Units	14
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Term 2

FSN 2202	Introduction to Human Nutrition	3
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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MATH 1261	Calculus I (2) ¹	4
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General Education Requirement (1C)	3
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Units	14
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Second Year

Term 1

BIO 1111	General Biology (5B) ¹	3
STAT 1110	Applied Statistical Concepts and Methods	3
CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
General Education Requirement (1B)		3
General Education Requirement		3
Units		16

Term 2

MCRO 2221	Introduction to Microbiology	4
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PHYS 1121	College Physics I	4
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General Education Requirement	3
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General Education Requirement	3
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Units	14
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Third Year

Term 1

FDSC 3330	Food Processing and Engineering I	4
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CHEM 3350	Biochemistry: Fundamentals and Applications (Upper-Division 2/5) ¹	4
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Approved Elective	3
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General Education Requirement	3
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Units	14
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Term 2

FDSC 3310	Food Laws and Regulations	3
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FDSC 3340	Food Quality Assurance and Prerequisite Programs	4
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FDSC 3350	Food Chemistry	4
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STAT 3320	Statistical Methods for Food Science	3
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Approved Elective	3
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Units	17
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Fourth Year

Term 1

FDSC 3345	Food Safety and Sanitation	4
FDSC 3355	Food Analysis	3
FDSC 4420	Sensory Evaluation of Food	3
MCRO 4421	Food Microbiology	3
General Education Requirement		3
Units		16

Term 2

FDSC 4425	Food Product Development	3
& FDSC 4426	and Food Science Exit Exam	
FDSC 4430	Food Processing and Engineering II	4
FDSC 4460	Senior Project - Scientific and Technical Writing Course	2
or FDSC 4461	or Senior Project - Research	
or FDSC 4462	or Senior Project - Internship	
General Education Requirement		3
General Education Requirement		3
Units		15
Total Units		120

Required in Major or Support; also satisfies General Education (GE) requirement.

Forest and Fire Sciences

Concentration Not Yet Declared - BS in Forest and Fire Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NR 1141	Introduction to Forest Ecosystem Management	4
BIO 1114	Plant Diversity and Ecology (5B & 5C) ¹	4
or BOT 1121	or General Botany	
ECON 2001	Survey of Economics (4B) ¹	3

General Education Requirement (1A)	3
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Units	14
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Term 2

SS 1120	Introductory Soil Science	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1B)	3
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General Education Requirement (1C)	3
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General Education Requirement	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology	3
BRAE 2237	Introduction to Engineering Surveying	2
MATH 1264	Calculus for Data Science I (2) ¹	4
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
General Education Requirement		3
Free Elective	3	3
Free Elective	3	3
Units		16

Third Year

Term 1

NR 3305	Forest and Fire Ecology	4
NR 3308	Fire and Society (Upper-Division 4) ¹	3
Concentration Course ²		3
Concentration Course ²		3
Concentration Course ²		3
Units		16

Term 2

NR 3315	Forest Mensuration	4
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NR 3319	Watershed Processes and Management	3
& NR 3320	and Watershed Processes and Management Laboratory	
Concentration Course	²	3
General Education Requirement		3
General Education Requirement		3
Units		16

Fourth Year

Term 1

NR 4365	Silviculture and Fuels Management	4
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NR 4414	Sustainable Forest Management and Forest Operations	2
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NR 4417	Natural Resource Protection and Policy	3
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Concentration Course ²		4
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Units		13
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Term 2

NR 4402	Forest Health and Disturbance Ecology	3
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NR 4460	Senior Project - Watershed Assessment and Protection	3
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or NR 4466	or Senior Project - Wildland-Urban Interface Fire Protection	
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or NR 4475	or Senior Project - Forest Stewardship Practices	
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Concentration Course	²	3
Concentration Course	²	3
Free Elective	³	2
Units		14
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Forest Resources

Forest Resources Concentration - BS in Forest and Fire Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
NR 1141	Introduction to Forest Ecosystem Management	4
BIO 1114	Plant Diversity and Ecology (5B & 5C) ¹	4
or BOT 1121	or General Botany	

ECON 2001	Survey of Economics (4B)	¹	3
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General Education Requirement (1A)			3
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Units			14
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Term 2

SS 1120	Introductory Soil Science		4
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STAT 1110	Applied Statistical Concepts and Methods		3
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General Education Requirement (1B)			3
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General Education Requirement (1C)			3
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General Education Requirement			3
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Units

16

Second Year

Term 1

NR 2208 Dendrology 3

BRAE 2237 Introduction to Engineering Surveying 2

MATH 1264 Calculus for Data Science I (2)¹ 4

General Education Requirement 3

General Education Requirement 3

Units

15

Term 2

NR 2204	Wildland Fire Control	3
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NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
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General Education Requirement		3
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Free Elective ³		3
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Units		16
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Third Year

Term 1

NR 2350	Urban Forestry	3
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NR 3305	Forest and Fire Ecology	4
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NR 3308	Fire and Society (Upper-Division 4) ¹	3
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Approved Elective	2, 4	3
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Free Elective	3	3
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Units	16
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Term 2

NR 3315	Forest Mensuration	4
NR 3319	Watershed Processes and Management	3
& NR 3320	and Watershed Processes and Management Laboratory	
Approved Elective	2, 4	3
General Education Requirement		3
General Education Requirement		3
	Units	16

Fourth Year

Term 1

NR 4365	Silviculture and Fuels Management	4
NR 4414	Sustainable Forest Management and Forest Operations	2
NR 4417	Natural Resource Protection and Policy	3
Approved Elective	2, 4	3
Units		12

Term 2

NR 4402	Forest Health and Disturbance Ecology	3
NR 4403	Forest Health and Disturbance Ecology Laboratory	1
NR 4460	Senior Project - Watershed Assessment and Protection	3

or NR 4466 or Senior Project - Wildland-Urban Interface Fire Protection

or NR 4475 or Senior Project - Forest Stewardship Practices

Approved Elective	2, 4	3
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Approved Elective	2, 4	3
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Free Elective	3	2
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

Water Sciences

Water Sciences Concentration - BS in Forest and Fire Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
NR 1141	Introduction to Forest Ecosystem Management	4
BIO 1114	Plant Diversity and Ecology (5B & 5C) ¹	4
or BOT 1121	or General Botany	
ECON 2001	Survey of Economics (4B) ¹	3
General Education Requirement (1A)		3
Units		14

Term 2		
SS 1120	Introductory Soil Science	4

STAT 1110	Applied Statistical Concepts and Methods	3
General Education Requirement (1B)		3
General Education Requirement (1C)		3
General Education Requirement		3
Units		16

Second Year

Term 1

NR 2208	Dendrology	3
BRAE 2237	Introduction to Engineering Surveying	2

MATH 1264	Calculus for Data Science I (2)	¹ 4
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Term 2

NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	4
	1	

General Education Requirement	3
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Free Elective	3	3
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Free Elective	3	3
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Units	16
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Third Year

Term 1

NR 3305	Forest and Fire Ecology	4
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NR 3308	Fire and Society (Upper-Division 4)	3
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NR 3319	Watershed Processes and Management	3
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& NR 3320	and Watershed Processes and Management Laboratory	
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Approved Elective	2, 4	3
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Approved Elective	2, 4	3
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Units	16
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Term 2

NR 3315	Forest Mensuration	4
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ERSC 3303	Soil Erosion and Water Conservation	3
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Approved Elective	2, 4	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Fourth Year

Term 1

NR 4365	Silviculture and Fuels Management	4
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NR 4414	Sustainable Forest Management and Forest Operations	2
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NR 4417	Natural Resource Protection and Policy	3
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SS 3321	Soil Morphology	4
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or SS 4431	or Spatial Data Analysis and Environmental Mapping	
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Units

13

Term 2

NR 4402	Forest Health and Disturbance Ecology	3
NR 4418	Applied Geographic Information System	2
NR 4460	Senior Project - Watershed Assessment and Protection	3
or NR 4466	or Senior Project - Wildland-Urban Interface Fire Protection	
or NR 4475	or Senior Project - Forest Stewardship Practices	
Approved Elective	2, 4	4
Free Elective	3	2

Units	14
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Total Units	120
--------------------	------------

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

Wildland Fire

Wildland Fire Concentration - BS in Forest and Fire Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NR 1141	Introduction to Forest Ecosystem Management	4
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BIO 1114	Plant Diversity and Ecology (5B & 5C) ¹	4
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or BOT 1121

or General Botany

ECON 2001	Survey of Economics (4B) ¹	3
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General Education Requirement (1A)	3
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Units	14
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Term 2

SS 1120	Introductory Soil Science	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1B)	3
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General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement	3
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Units	16
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Second Year

Term 1

NR 2208	Dendrology	3
BRAE 2237	Introduction to Engineering Surveying	2
MATH 1264	Calculus for Data Science I (2) ¹	4
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

NR 2204	Wildland Fire Control	3
NR/LA 2218	Introduction to Geographic Information Systems (GIS)	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
General Education Requirement		3
Free Elective	3	3
Units		16

Third Year

Term 1

NR 2350	Urban Forestry	3
NR 3305	Forest and Fire Ecology	4
NR 3308	Fire and Society (Upper-Division 4) ¹	3
NR 3340	Wildland Fire Management	2
Free Elective ³		3
Units		15

Term 2

NR 3315	Forest Mensuration	4
NR 3319	Watershed Processes and Management	3

& NR 3320 and Watershed Processes and Management Laboratory

Approved Elective 2, 4 4

General Education Requirement 3

General Education Requirement 3

Units 17

Fourth Year

Term 1

NR 4365 Silviculture and Fuels Management 4

NR 4414 Sustainable Forest Management and Forest Operations 2

NR 4417	Natural Resource Protection and Policy	3
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Approved Elective	2, 4	4
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Units	13
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Term 2

NR 4402	Forest Health and Disturbance Ecology	3
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NR 4460	Senior Project - Watershed Assessment and Protection	3
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or NR 4466	or Senior Project - Wildland-Urban Interface Fire Protection
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or NR 4475	or Senior Project - Forest Stewardship Practices
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Approved Elective	2, 4	3
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Approved Elective	^{2, 4}	3
Free Elective	³	2
Units		14
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses may need to be at the 3000-4000 level to ensure completion of the requirement minimum of 40 units of upper-division.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Courses taken to meet a major course requirement cannot be double-counted in the concentration.

Individualized Course of Study

Individualized Course of Study - BS in General Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year			
Term 1			Units
ENGR 1110	Introduction to Engineering		1
IME 1140	Technical Graphics Communication for Design and Manufacturing		3
& IME 1143	and Introduction to Design and Manufacturing		
MATH 1261	Calculus I (2)	1	4

PHYS 1141	General Physics I	4
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General Education Requirement (1A)	3
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Units	15
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Term 2

CSC 1032	Programming for Scientists and Engineers	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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MATH 1262	Calculus II	4
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PHYS 1143	General Physics II	4
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General Education Requirement (1B)	3
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Units

18

Second Year

Term 1

ENGR 2211 Introduction to Mechanics 4

IME 2315 Financial Decision Making for Engineers 2

MATE 1220 Principles of Materials Engineering for Non-Majors 3

& MATE 1215 and Materials Laboratory I

MATH 2263 Calculus III 3

General Education Requirement (1C) 3

Units

15

Term 2

EE 2115	Circuits & Electronics for Non-Majors	4
& 2115L	and Circuits & Electronics Laboratory for Non-Majors	
ENGR 2212	Introduction to Engineering Dynamics	2
General Education Requirement (3A)		3
General Education Requirement (4A)		3
General Education Requirement (5B)		3
Units		15

Third Year

Term 1

STAT 3210	Engineering Statistics (Upper-Division 2/5)	¹	3
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Approved Elective		²	2
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General Education Requirement (3B)			3
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General Education Requirement (4B)			3
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General Education Requirement (6)			3
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Units			14
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Term 2

ENGR 3310	Ethical Engineering for the Anthropocene		2
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ME 3302	Thermodynamics		3
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WGQS 3350	Gender, Race, Culture, Science, and Technology	1	4
or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies		
Approved Elective		2	3
General Education Requirement (Upper-Division 3)			3
Units			15

Fourth Year

Term 1

ENGR 4460	Interdisciplinary Senior Project I		2
Approved Elective		2	3

Approved Elective²

3

Approved Elective²

3

General Education Requirement (Upper-Division 4)

3

Units

14

Term 2

ENGR 4461

Interdisciplinary Senior Project II

2

Approved Elective²

3

Approved Elective²

3

Approved Elective	2	3
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Approved Elective	2	3
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

The Individualized Course of Study consists of 26 units of technical electives with a minimum of 18 units at the 3000-4000 level.

Graphic Communication

Concentration Not Yet Declared - BS in Graphic Communication

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

GRC 1000 Introduction to Graphic Communication 3

GRC 1020 Introduction to Digital Applications 3

GRC 1100 Visual Literacy and Communication (3A)¹ 3

PHYS 1121 College Physics I (5A & 5C)¹ 4

General Education Requirement (1A) 3

Units 16

Term 2

GRC 1030	Graphic Communication Management, Manufacturing, and Strategy	3
GRC 1040	Materials for Graphic Communication Applications	3
GRC 1090	Web Development and Design Principles	3
STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1C)		3
Units		15

Second Year		
Term 1		
GRC 2000	Applied Graphic Communication Practices	1

GRC 2040	Print and Production Technology	3
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General Education Requirement (1B)		3
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General Education Requirement		3
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General Education Requirement		3
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General Education Requirement		3
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Units		16
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Term 2		
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GRC 2020	Digital File Preparation and Workflow	3
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GRC 2050	Digital Typography	3
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GRC 2070	Consumer Packaging	3
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Third Year

Term 1

GRC 3000	Experience Design	3
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GRC 3020	Digital Photography and Motion Graphics	3
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GRC 3100	Introduction to Applied Social Media in Graphic Communication	1
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Concentration Course or Elective	3
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Concentration Course or Elective	3
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Units	13
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Term 2

GRC 3030	Marketing and Sales Management for Graphic Communication	3
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GRC 3035	Operations and Project Management for Graphic Communication	3
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GRC 3280	Specialty Graphics and Printing	3
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Concentration Course or Elective	3
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Concentration Course or Elective	3
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Units 15

Fourth Year

Term 1

GRC 3060 Data-Driven Graphics 3

GRC 3200 Color Management (Upper-Division 2/5) ¹ 3

Concentration Course or Elective 3

Concentration Course or Elective 3

General Education Requirement 3

Units 15

Term 2

GRC 4060 Human Resource Management for Graphic Communication 3

GRC 4461 Senior Project - Independent Study 3

or GRC 4462 or Senior Project

or GRC 4463 or Senior Project - Applied Practices

Concentration Course or Elective 3

General Education Requirement 3

Free Elective² 3

Units 15

Total Units

120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Design Reproduction Technology

Design Reproduction Technology Concentration - BS in Graphic Communication
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

GRC 1000	Introduction to Graphic Communication	3
GRC 1020	Introduction to Digital Applications	3
GRC 1100	Visual Literacy and Communication (3A) ¹	3
PHYS 1121	College Physics I (5A & 5C) ¹	4
General Education Requirement (1A)		3
Units		16

Term 2

GRC 1030	Graphic Communication Management, Manufacturing, and Strategy	3
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GRC 1040	Materials for Graphic Communication Applications	3
GRC 1090	Web Development and Design Principles	3
STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1C)		3
Units		15

Second Year

Term 1

GRC 2000	Applied Graphic Communication Practices	1
GRC 2040	Print and Production Technology	3

General Education Requirement (1B)	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

GRC 2020	Digital File Preparation and Workflow	3
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GRC 2050	Digital Typography	3
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GRC 2070	Consumer Packaging	3
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Third Year

Term 1

GRC 3000	Experience Design	3
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GRC 3020	Digital Photography and Motion Graphics	3
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GRC 3100	Introduction to Applied Social Media in Graphic Communication	1
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GRC 3270	Packaging Graphics Technology and Design	3
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ART 1101	Fundamentals of Drawing	3
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	Units	13
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Term 2

GRC 3030	Marketing and Sales Management for Graphic Communication	3
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GRC 3035	Operations and Project Management for Graphic Communication	3
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GRC 3280	Specialty Graphics and Printing	3
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GRC 4500	Visual Communication Systems	3
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Concentration Elective	3	3
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Units 15

Fourth Year

Term 1

GRC 3060 Data-Driven Graphics 3

GRC 3200 Color Management (Upper-Division 2/5) ¹ 3

ART 1102 2D Design 3

Concentration Elective ³ 3

General Education Requirement 3

Units 15

Term 2

GRC 4060 Human Resource Management for Graphic Communication 3

GRC 4461 Senior Project - Independent Study 3

or GRC 4462 or Senior Project

or GRC 4463 or Senior Project - Applied Practices

Concentration Elective³ 3

General Education Requirement 3

Free Elective² 3

Units 15

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

May be substituted with an advisor approved course.

Graphic Communication Management

Graphic Communication Management Concentration - BS in Graphic Communication

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

GRC 1000 Introduction to Graphic Communication 3

GRC 1020 Introduction to Digital Applications 3

GRC 1100 Visual Literacy and Communication (3A)¹ 3

PHYS 1121 College Physics I (5A & 5C)¹ 4

General Education Requirement (1A) 3

Units 16

Term 2

GRC 1030	Graphic Communication Management, Manufacturing, and Strategy	3
GRC 1040	Materials for Graphic Communication Applications	3
GRC 1090	Web Development and Design Principles	3
STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1C)		3
Units		15

Second Year		
Term 1		
GRC 2000	Applied Graphic Communication Practices	1

GRC 2040	Print and Production Technology	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

GRC 2020	Digital File Preparation and Workflow	3
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GRC 2050	Digital Typography	3
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GRC 2070	Consumer Packaging	3
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General Education Requirement		3
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General Education Requirement		3
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Units		15
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Third Year

Term 1

GRC 3000	Experience Design	3
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GRC 3020	Digital Photography and Motion Graphics	3
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GRC 3100	Introduction to Applied Social Media in Graphic Communication	1
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BUS 2207	Legal Responsibilities of Business	3
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COMS 2213	Organizational Communication	3
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	Units	13
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Term 2		
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GRC 3030	Marketing and Sales Management for Graphic Communication	3
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GRC 3035	Operations and Project Management for Graphic Communication	3
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GRC 3280	Specialty Graphics and Printing	3
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BUS 2212	Financial Accounting for Nonbusiness Majors	3
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Concentration Elective

3

3

Units

15

Fourth Year

Term 1

GRC 3060

Data-Driven Graphics

3

GRC 3200

Color Management (Upper-Division 2/5)

1

3

Concentration Elective

3

3

Concentration Elective

3

3

General Education Requirement	3
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Units	15
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Term 2

GRC 4060	Human Resource Management for Graphic Communication	3
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GRC 4461	Senior Project - Independent Study	3
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or GRC 4462	or Senior Project
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or GRC 4463	or Senior Project - Applied Practices
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GRC 4600	Graphic Communication in Integrated Marketing Communications	3
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General Education Requirement	3
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Free Elective² 3

Units 15

Total Units 120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

May be substituted with an advisor approved course.

Graphics for Packaging

Graphics for Packaging Concentration - BS in Graphic Communication
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.
Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
GRC 1000	Introduction to Graphic Communication	3
GRC 1020	Introduction to Digital Applications	3
GRC 1100	Visual Literacy and Communication (3A) ¹	3
PHYS 1121	College Physics I (5A & 5C) ¹	4
General Education Requirement (1A)		3

Units

16

Term 2

GRC 1030	Graphic Communication Management, Manufacturing, and Strategy	3
GRC 1040	Materials for Graphic Communication Applications	3
GRC 1090	Web Development and Design Principles	3
STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1C)		3

Units

15

Second Year

Term 1

GRC 2000	Applied Graphic Communication Practices	1
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GRC 2040	Print and Production Technology	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

GRC 2020	Digital File Preparation and Workflow	3
GRC 2050	Digital Typography	3
GRC 2070	Consumer Packaging	3
General Education Requirement		3
General Education Requirement		3
Units		15

Third Year

Term 1

GRC 3000	Experience Design	3
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GRC 3020	Digital Photography and Motion Graphics	3
GRC 3080	Product Development for Cross-Device Interaction	3
GRC 3100	Introduction to Applied Social Media in Graphic Communication	1
ITP 2234	Packaging Design Fundamentals	3
Units		13

Term 2

GRC 3030	Marketing and Sales Management for Graphic Communication	3
GRC 3035	Operations and Project Management for Graphic Communication	3

GRC 3280	Specialty Graphics and Printing	3
GRC 4700	Advanced Consumer Packaging	3
ITP 3330	Packaging Fundamentals	3
Units		15

Fourth Year

Term 1

GRC 3060	Data-Driven Graphics	3
GRC 3200	Color Management (Upper-Division 2/5) ¹	3
Concentration Elective ³		3

Concentration Elective

3

3

General Education Requirement

3

Units

15

Term 2

GRC 4060

Human Resource Management for Graphic Communication

3

GRC 4461

Senior Project - Independent Study

3

or GRC 4462

or Senior Project

or GRC 4463

or Senior Project - Applied Practices

Concentration Elective

3

3

General Education Requirement	3
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Free Elective ²	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

May be substituted with an advisor approved course.

Immersive Experience Design

Immersive Experience Design Concentration - BS in Graphic Communication
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
GRC 1000	Introduction to Graphic Communication	3
GRC 1020	Introduction to Digital Applications	3
GRC 1100	Visual Literacy and Communication (3A) ¹	3
PHYS 1121	College Physics I (5A & 5C) ¹	4

General Education Requirement (1A)	3
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Units	16
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Term 2

GRC 1030	Graphic Communication Management, Manufacturing, and Strategy	3
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GRC 1040	Materials for Graphic Communication Applications	3
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GRC 1090	Web Development and Design Principles	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1C)	3
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Units	15
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Second Year

Term 1

GRC 2000	Applied Graphic Communication Practices	1
GRC 2040	Print and Production Technology	3
General Education Requirement (1B)		3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
Units		16

Term 2

GRC 2020	Digital File Preparation and Workflow	3
GRC 2050	Digital Typography	3
GRC 2070	Consumer Packaging	3
General Education Requirement		3
General Education Requirement		3
Units		15

Third Year

Term 1

GRC 3000	Experience Design	3
GRC 3020	Digital Photography and Motion Graphics	3
GRC 3100	Introduction to Applied Social Media in Graphic Communication	1
ART 4433	The Art of Mixed Reality	3
Concentration Elective	3	3
Units		13

Term 2

GRC 3030	Marketing and Sales Management for Graphic Communication	3
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GRC 3035	Operations and Project Management for Graphic Communication	3
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GRC 3080	Product Development for Cross-Device Interaction	3
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GRC 3280	Specialty Graphics and Printing	3
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Concentration Elective	3	3
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Units	15
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Fourth Year

Term 1

GRC 3060	Data-Driven Graphics	3
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GRC 3200	Color Management (Upper-Division 2/5) ¹	3
GRC 4290	User Experience Research Methods	3
GRC 4800	Immersive Computer Graphics Studio	3
General Education Requirement		3
Units		15

Term 2

GRC 4060	Human Resource Management for Graphic Communication	3
GRC 4461	Senior Project - Independent Study	3
or GRC 4462	or Senior Project	

or GRC 4463 or Senior Project - Applied Practices

Concentration Elective ³	3
General Education Requirement	3
Free Elective ²	3
Units	15
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

May be substituted with an advisor approved course.

User Experience/User Interface

User Experience/User Interface Concentration - BS in Graphic Communication
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
GRC 1000	Introduction to Graphic Communication	3
GRC 1020	Introduction to Digital Applications	3

GRC 1100	Visual Literacy and Communication (3A) ¹	3
PHYS 1121	College Physics I (5A & 5C) ¹	4
General Education Requirement (1A)		3
Units		16

Term 2

GRC 1030	Graphic Communication Management, Manufacturing, and Strategy	3
GRC 1040	Materials for Graphic Communication Applications	3
GRC 1090	Web Development and Design Principles	3

STAT 1110	Applied Statistical Concepts and Methods (2)	¹	3
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General Education Requirement (1C)			3
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Units			15
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Second Year

Term 1

GRC 2000	Applied Graphic Communication Practices		1
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GRC 2040	Print and Production Technology		3
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General Education Requirement (1B)			3
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General Education Requirement			3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

GRC 2020	Digital File Preparation and Workflow	3
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GRC 2050	Digital Typography	3
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GRC 2070	Consumer Packaging	3
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General Education Requirement	3
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General Education Requirement	3
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Units

15

Third Year

Term 1

GRC 3000	Experience Design	3
GRC 3020	Digital Photography and Motion Graphics	3
GRC 3100	Introduction to Applied Social Media in Graphic Communication	1
CSC/CPE 1024	Introduction to Computing	2
Concentration Elective	3	3-4

Units

13

Term 2

GRC 3030	Marketing and Sales Management for Graphic Communication	3
GRC 3035	Operations and Project Management for Graphic Communication	3
GRC 3280	Specialty Graphics and Printing	3
GRC 3990	Front-end Web Development	3
PHIL 3323	Ethics, Science, and Technology	3
Units		15

Fourth Year

Term 1

GRC 3060	Data-Driven Graphics	3
GRC 3200	Color Management (Upper-Division 2/5) ¹	3
GRC 4900	User Interface Innovations	3
Concentration Elective ³		3
General Education Requirement		3
Units		15

Term 2

GRC 4060	Human Resource Management for Graphic Communication	3
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GRC 4461	Senior Project - Independent Study	3
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or GRC 4462	or Senior Project	
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or GRC 4463	or Senior Project - Applied Practices	
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Concentration Elective	³	3
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General Education Requirement		3
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Free Elective	²	3
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Units		15
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Total Units		120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

May be substituted with an advisor approved course.

Individualized Course of Study

Individualized Course of Study - BS in Graphic Communication

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

GRC 1000

Introduction to Graphic Communication

3

GRC 1020	Introduction to Digital Applications	3
GRC 1100	Visual Literacy and Communication (3A) ¹	3
PHYS 1121	College Physics I (5A & 5C) ¹	4
General Education Requirement (1A)		3
Units		16

Term 2

GRC 1030	Graphic Communication Management, Manufacturing, and Strategy	3
GRC 1040	Materials for Graphic Communication Applications	3

GRC 1090	Web Development and Design Principles	3
STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1C)		3
Units		15

Second Year

Term 1

GRC 2000	Applied Graphic Communication Practices	1
GRC 2040	Print and Production Technology	3
General Education Requirement (1B)		3

General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

GRC 2020	Digital File Preparation and Workflow	3
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GRC 2050	Digital Typography	3
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GRC 2070	Consumer Packaging	3
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Third Year

Term 1

GRC 3000	Experience Design	3
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GRC 3020	Digital Photography and Motion Graphics	3
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GRC 3100	Introduction to Applied Social Media in Graphic Communication	1
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Individualized Course of Study Elective	3
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Individualized Course of Study Elective	3
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Units

13

Term 2

GRC 3030	Marketing and Sales Management for Graphic Communication	3
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GRC 3035	Operations and Project Management for Graphic Communication	3
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GRC 3280	Specialty Graphics and Printing	3
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Individualized Course of Study Elective		3
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Individualized Course of Study Elective		3
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Units

15

Fourth Year

Term 1

GRC 3060 Data-Driven Graphics 3

GRC 3200 Color Management (Upper-Division 2/5) ¹ 3

Individualized Course of Study Elective 3

Individualized Course of Study Elective 3

General Education Requirement 3

Units 15

Term 2

GRC 4060 Human Resource Management for Graphic
Communication 3

GRC 4461	Senior Project - Independent Study	3
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or GRC 4462	or Senior Project	
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or GRC 4463	or Senior Project - Applied Practices	
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Individualized Course of Study Elective	3
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General Education Requirement	3
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Free Elective ²	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

History

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year			
Term 1			Units
HIST 1100	Historical Inquiry		4
HIST 2201	United States History to 1877 (4A) ¹		3

General Education Requirement (1A) 3

Free Elective^{6, 7} 4

Units 14

Term 2

HIST 2222 World History to 1500 (4B)¹ 3

2000 level History Elective 3

General Education Requirement (1B) 3

General Education Requirement (1C) 3

Free Elective^{6, 7} 4

Units 16

Second Year

Term 1

HIST 2202 United States History Since 1877 3

2000 level History Elective 3

World Languages Requirement	5	4
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General Education Requirement (2)		3
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General Education Requirement		3
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Units		16
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Term 2		
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HIST 2223	World History since 1500	3
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General Education Requirement		3
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General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Free Elective^{6, 7} 3

Units 16

Third Year

Term 1

HIST 1101	Learn by Doing History	1
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HIST 3303	Research and Writing Seminar in History	4
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3000-4000 level History Elective	2, 3	4
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GE Upper-Division 4 course not in HIST	1	3
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General Education Requirement		3
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Units		15
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Term 2

HIST 1101	Learn by Doing History	1
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HIST 3304	Historiography	4
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Global Histories Elective	2, 4	4
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General Education Requirement		3
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Free Elective	6, 7	3
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Units		15
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Fourth Year

Term 1

HIST 4460 Senior Project I 2

Global Histories Elective^{2, 4} 4

General Education Requirement 3

Free Elective^{6, 7} 4

Units 13

Term 2

HIST 1101	Learn by Doing History	1
HIST 4461	Senior Project II	2
3000-4000 level History Elective	2, 3	4
Free Elective	6, 7	4
Free Elective	6, 7	4
Units		15

Total Units

120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A minimum of two 4000-level courses is required across History electives and Global Histories electives.

3

HIST 4400 and HIST 4467 may not count towards the History Electives.

4

Choose two courses (8 units) that focus on regions and themes outside of the U.S. and Europe. This includes courses on Africa, Asia, Latin America, and the Middle East, as

well as comparative or transregional courses that explore global themes such as urbanism, imperialism, or migration.

5

Additional prerequisites may be needed to complete this requirement.

6

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

7

Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Industrial Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1		Units
IME 1101	Introduction to Industrial and Manufacturing Engineering	1
IME 1140	Technical Graphics Communication for Design and Manufacturing	3
& IME 1143	and Introduction to Design and Manufacturing	
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4

MATH 1261	Calculus I (2) ¹	4
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General Education Requirement (1A)		3
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Units		15
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Term 2		
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IME 1141	Introduction to Metal Casting and Prototyping	1-2
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or IME 1142	or Materials Joining	
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or IME 1156	or Introduction to Modern Electronics Manufacturing	
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IME 1223	Process Improvement Fundamentals	4
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MATH 1262	Calculus II	4
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PHYS 1141

General Physics I

4

General Education Requirement (1C)

3

Units

16-17

Term 1

IME 2315

Financial Decision Making for Engineers

2

CSC 1032

Programming for Scientists and Engineers

3

MATH 1151

Linear Algebra²

3

or MATH 2341

or Linear Analysis

PHYS 1143	General Physics II	4
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General Education Requirement (1B)		3
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Units		15
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Term 2		
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IME 2212	Introduction to Enterprise Analytics and Database Systems	4
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ENGR 2211	Introduction to Mechanics ²	3-4
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or EE 2115 *and* EE

2115L

or Circuits & Electronics for Non-Majors *and*
Circuits & Electronics Laboratory for
Non-Majors

or EE 2201 *and* EE

2241

or Electric Circuits for Non-Majors *and*
Electric Circuit Analysis Laboratory I

or MATE 1220 *and*
MATE 1215

or Principles of Materials Engineering for
Non-Majors *and* Materials Laboratory I

MATH 2263	Calculus III	3
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STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
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General Education Requirement (3A)		3
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Units		16
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Term 1

IME 3302	Operations Research and Management	4
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IME 3326	Engineering Statistics and Six Sigma	4
ENGR 2211	Introduction to Mechanics ²	3-4
or EE 2115 <i>and</i> EE 2115L	or Circuits & Electronics for Non-Majors <i>and</i> Circuits & Electronics Laboratory for Non-Majors	
or EE 2201 <i>and</i> EE 2241	or Electric Circuits for Non-Majors <i>and</i> Electric Circuit Analysis Laboratory I	
or MATE 1220 <i>and</i> MATE 1215	or Principles of Materials Engineering for Non-Majors <i>and</i> Materials Laboratory I	
PSY 2201	Introductory Psychology (4B) ¹	3
General Education Requirement (4A)		3
	Units	17

Term 2

IME 3410	Production Planning and Control Systems	4
IME 3420	Process and System Simulation	4
IME 3443	Facilities Design and Warehousing	4
General Education Requirement (5B)		3
Units		15

Term 1

IME 4319	Human Factors and Ergonomics in Engineering and Systems Design	3
IME 4372	Applications of Enterprise Analytics	4
IME 4417	Supply Chain and Logistics Management	4
IME 4461	Senior Project - Design I	2
General Education Requirement (3B)		3
Units		16

Term 2

IME 4462	Senior Project - Design II	2
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Technical Elective	3
Technical Elective	0-3
General Education Requirement (6)	3
General Education Requirement (Upper-Division 3)	3
General Education Requirement (Upper-Division 4)	3
Units	17
Total Units	127-128

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Units in excess of total will be applied towards Technical Electives.

Industrial Technology and Packaging

Concentration Not Yet Declared - BS in Industrial Technology and Packaging

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1			Units
ITP 1100	Student Orientation, College Success, and Career Readiness		1
ITP 1125	Introduction to Industrial Technology and Packaging		1

MATH 1261	Calculus I (2) ¹	4
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or MATH 1267 <i>and</i> MATH 1277	or Business Calculus <i>and</i> Business Calculus Workshop	
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PHYS 1121	College Physics I	4
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General Education Requirement (1A)		3
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Units	13
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Term 2

ITP 1150	Power Systems and Renewable Energy	3
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ITP 2233	Product Modeling and Communication	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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or STAT 1210	or Business Statistics I	
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General Education Requirement (1C)		3
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Units		16
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Term 1

ITP 3330	Packaging Fundamentals (Upper-Division 2/5) 1	3
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ITP 3371	Supply Chain Management in Manufacturing and Services	3
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ECON 2001	Survey of Economics (4B) ¹	3
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Concentration Course		3
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General Education Requirement (1B)		3
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Units		15
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Term 2		
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ITP 3303	Lean Six Sigma Green Belt	3
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ITP 3326	Product Design and Development	3
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BUS 3391	Information Systems	3
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Concentration Course		3
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General Education Requirement (3A)		3
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Units		15
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Term 1		

ITP 4409	Packaging Machinery and Processes	3
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BUS 3346	Principles of Marketing	3
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Concentration Course 3

General Education Requirement (4A) 3

General Education Requirement (6) 3

Free Elective² 2-3

Units 17

Term 2

ITP 3341 Packaging Polymers and Processing 3

Concentration Course 3

Concentration Course

3

General Education Requirement (3B)

3

General Education Requirement (5B)

3

Units

15

Term 1

ITP 4411

Packaging Sustainability

3

Concentration Course

3

Concentration Course	3
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Concentration Course	3
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General Education Requirement (Upper-Division 3)	3
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Units	15
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Term 2	
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ITP 4464	Senior Project	3
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Concentration Course	3-4
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Concentration Course	3-4
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General Education Requirement (Upper-Division 4)	3
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Free Elective ²	0-2
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Industrial Technology

Industrial Technology Concentration - BS in Industrial Technology and Packaging
Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1			Units
ITP 1100	Student Orientation, College Success, and Career Readiness	1	
ITP 1125	Introduction to Industrial Technology and Packaging	1	
MATH 1261	Calculus I (2)	4	
or MATH 1267 <i>and</i> MATH 1277			

or Business Calculus *and* Business Calculus
Workshop

PHYS 1121	College Physics I	4
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General Education Requirement (1A)		3
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Units		13
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Term 2

ITP 1150	Power Systems and Renewable Energy	3
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ITP 2233	Product Modeling and Communication	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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STAT 1110 Applied Statistical Concepts and Methods 3

or STAT 1210

or Business Statistics I

General Education Requirement (1C)	3
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Units 16

Term 1

ITP 2260 Materials and Manufacturing Technology 3

ITP 3330	Packaging Fundamentals (Upper-Division 2/5)	3
	1	

ITP 3371	Supply Chain Management in Manufacturing and Services	3
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ECON 2001	Survey of Economics (4B) ¹	3
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General Education Requirement (1B)		3
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Units		15
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Term 2		
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ITP 3303	Lean Six Sigma Green Belt	3
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ITP 3326	Product Design and Development	3
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ITP 3390	Industrial Automation	3
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BUS 3391	Information Systems	3
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General Education Requirement (3A)		3
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Units		15
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Term 1		
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ITP 4403	Lean and Quality Systems Management	3
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ITP 4409	Packaging Machinery and Processes	3
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BUS 3346	Principles of Marketing	3
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General Education Requirement (4A)	3
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General Education Requirement (6)	3
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Free Elective ²	2-3
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Units	17
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Term 2	
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ITP 3341	Packaging Polymers and Processing	3
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ITP 4410	Operations Planning and Control	3
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ITP 4428	Commercialization of Innovation	3
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General Education Requirement (3B)

3

General Education Requirement (5B)

3

Units

15

Term 1

ITP 4411

Packaging Sustainability

3

ITP 4415

Supply Chain and Logistics

3

ITP 4496

Product Development and Manufacturing I

3

Approved Elective³

3-4

General Education Requirement (Upper-Division 3)

3

Units

16

Term 2

ITP 4464

Senior Project

3

ITP 4497

Product Development and Manufacturing II

3

Approved Elective³

3-4

General Education Requirement (Upper-Division 4)

3

Free Elective² 0-2

Units 13

Total Units 120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

Units in excess of total will be applied towards major's free electives.

Packaging

Packaging Concentration - BS in Industrial Technology and Packaging

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1		Units
ITP 1100	Student Orientation, College Success, and Career Readiness	1
ITP 1125	Introduction to Industrial Technology and Packaging	1
MATH 1261	Calculus I (2) ¹	4

or MATH 1267 *and*
MATH 1277

or Business Calculus *and* Business Calculus
Workshop

PHYS 1121	College Physics I	4
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General Education Requirement (1A)		3
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Units		13
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Term 2

ITP 1150	Power Systems and Renewable Energy	3
----------	------------------------------------	---

ITP 2233	Product Modeling and Communication	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
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STAT 1110	Applied Statistical Concepts and Methods	3
or STAT 1210	or Business Statistics I	

General Education Requirement (1C)	3
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Units	16
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Term 1

ITP 2234 Packaging Design Fundamentals 3

ITP 3330	Packaging Fundamentals (Upper-Division 2/5)	3
	1	

ITP 3371	Supply Chain Management in Manufacturing and Services	3
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ECON 2001	Survey of Economics (4B) ¹	3
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General Education Requirement (1B)		3
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Units		15
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Term 2		
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ITP 3303	Lean Six Sigma Green Belt	3
----------	---------------------------	---

ITP 3326	Product Design and Development	3
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ITP 3334	Structural Packaging Design	3
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BUS 3391	Information Systems	3
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General Education Requirement (3A)		3
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Units		15
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Term 1		
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ITP 2241	Industrial and Packaging Materials	3
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ITP 4409	Packaging Machinery and Processes	3
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BUS 3346	Principles of Marketing	3
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General Education Requirement (4A)	3
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General Education Requirement (6)	3
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Free Elective ²	2-3
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Units	17
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Term 2

ITP 3341	Packaging Polymers and Processing	3
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ITP 4408	Fiber-Based Packaging	3
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ITP 4475	Distribution Packaging Dynamics	3
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General Education Requirement (3B)

3

General Education Requirement (5B)

3

Units

15

Term 1

ITP 4411

Packaging Sustainability

3

ITP 4498

Packaging Development

3

Approved Elective

3

3

Approved Elective³

3-4

General Education Requirement (Upper-Division 3)

3

Units

16

Term 2

ITP 4430

Healthcare Packaging

3

ITP 4464

Senior Project

3

Approved Elective³

3-4

General Education Requirement (Upper-Division 4)

3

Free Elective² 0-2

Units 13

Total Units 120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

Units in excess of total will be applied towards major's free electives.

Interdisciplinary Studies

Concentration Not Yet Declared - BA in Interdisciplinary Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1		Units
ISLA 1101	Interdisciplinary Studies Major First Year Seminar	1
Introductory-level Interdisciplinary Course		3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
General Education Requirement (2)		3

Units

13

Term 2

ISLA 2201 Introduction to Interdisciplinary Studies 3

General Education Requirement (1B) 3

General Education Requirement 3

General Education Requirement 3

Free Elective² 3

Units

15

Term 1

ISLA 2255	Introduction to Interdisciplinary Theories	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		1
Free Elective ²		3

Units	16
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Term 2

Concentration Course ¹	3-4
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General Education Requirement	3
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General Education Requirement	3
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Free Elective ²	3
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Free Elective ²	3-4
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Units	16
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Term 1

Upper-Division ISLA Course¹ 3-4

Concentration Course¹ 3-4

Concentration Course¹ 3-4

General Education Requirement 3

Free Elective² 3

Units	15
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Term 2

ISLA 3355	Interdisciplinary Research Methods	4
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Upper-Division ISLA Course	¹	3-4
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Concentration Course	¹	3-4
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General Education Requirement		3
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Free Elective	²	3
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Units	16
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Term 1

ISLA 4440	Advanced Interdisciplinary Studies Seminar	3
ISLA 4480	Career Preparation for Interdisciplinary Studies Majors	1
Concentration Course	¹	3-4
General Education Requirement		3
Free Elective	²	3-4
	Units	14

Term 2

ISLA 4460 Senior Project Seminar 4

& ISLA 4461 and Senior Project

Concentration Course¹ 3-4

Free Elective² 3-4

Free Elective² 3-4

Units 15

Total Units 120

1

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Ethics, Law, and Social Justice

Ethics, Law, and Social Justice Concentration - BA in Interdisciplinary Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1	Units

ISLA 1101	Interdisciplinary Studies Major First Year Seminar	1
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Introductory-level Interdisciplinary Course	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement (2)	3
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Units	13
-------	----

Term 2

ISLA 2201	Introduction to Interdisciplinary Studies	3
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General Education Requirement (1B)

3

General Education Requirement

3

General Education Requirement

3

Free Elective²

3

Units

15

Term 1

ISLA 2255

Introduction to Interdisciplinary Theories

3

General Education Requirement 3

General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Free Elective² 3

Units 16

Term 2

Social Justice Course 3-4

General Education Requirement

3

General Education Requirement

3

Free Elective²

3

Free Elective²

3-4

Units

16

Term 1

Upper-Division ISLA Course¹

3-4

Ethics Course 3

Law Course 3-4

General Education Requirement 3

Free Elective² 3

Units 15

Term 2

ISLA 3355 Interdisciplinary Research Methods 4

Upper-Division ISLA Course¹ 3-4

Ethics Course

3

General Education Requirement

3

Free Elective²

3

Units

16

Term 1

ISLA 4440

Advanced Interdisciplinary Studies Seminar

3

ISLA 4480

Career Preparation for Interdisciplinary Studies Majors

1

Law Course 3-4

General Education Requirement 3

Free Elective² 3-4

Units 14

Term 2

ISLA 4460 Senior Project Seminar 4

& ISLA 4461 and Senior Project

Social Justice Course 3-4

Free Elective ²	3-4
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Free Elective ²	3-4
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Units	15
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Total Units	120
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1

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Global Citizenship and Social Sustainability

Global Citizenship and Social Sustainability Concentration - BA in Interdisciplinary Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1		Units
ISLA 1101	Interdisciplinary Studies Major First Year Seminar	1
Introductory-level Interdisciplinary Course		3
General Education Requirement (1A)		3
General Education Requirement (1C)		3

General Education Requirement (2) 3

Units 13

Term 2

ISLA 2201 Introduction to Interdisciplinary Studies 3

General Education Requirement (1B) 3

General Education Requirement 3

General Education Requirement 3

Free Elective² 3

Units

15

Term 1

ISLA 2255	Introduction to Interdisciplinary Theories	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		1

Free Elective²

3

Units

16

Term 2

Global Political Economy Course

3

General Education Requirement

3

General Education Requirement

3

Free Elective²

3

Free Elective²

3-4

Units

16

Term 1

Upper-Division ISLA Course¹ 3-4

Ethnicity and Culture Course 3-4

Global and Cross-Cultural Communication Course 3-4

General Education Requirement 3

Free Elective² 3

Units		15
Term 2		
ISLA 3355	Interdisciplinary Research Methods	4
Upper-Division ISLA Course	¹	3-4
Sustainability and Global Environment Course	¹	3-4
General Education Requirement		3
Free Elective	²	3
Units		16

Term 1

ISLA 4440	Advanced Interdisciplinary Studies Seminar	3
ISLA 4480	Career Preparation for Interdisciplinary Studies Majors	1
Global Citizenship and Social Sustainability Elective ¹		3-4
General Education Requirement		3
Free Elective ²		3-4
Units		14

Term 2

ISLA 4460 Senior Project Seminar 4

& ISLA 4461 and Senior Project

Global Citizenship and Social Sustainability Elective¹ 3-4

Free Elective² 3-4

Free Elective² 3-4

Units 15

Total Units 120

1

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Health and Society

Health and Society Concentration - BA in Interdisciplinary Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1	Units

ISLA 1101	Interdisciplinary Studies Major First Year Seminar	1
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Introductory-level Interdisciplinary Course	3
---	---

General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement (2)	3
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Units	13
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Term 2

ISLA 2201	Introduction to Interdisciplinary Studies	3
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General Education Requirement (1B) 3

General Education Requirement 3

General Education Requirement 3

Free Elective² 3

Units 15

Term 1

ISLA 2255 Introduction to Interdisciplinary Theories 3

General Education Requirement 3

General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Free Elective² 3

Units 16

Term 2

Gender, Race, Culture, and Health Course¹ 3-4

General Education Requirement 3

General Education Requirement 3

Free Elective² 3

Free Elective² 3-4

Units 16

Term 1

BIO/CHEM 3318 Genetic Engineering Technology 3

or HLTH 3305

or Drugs in Society

or POLS 3332

or World Food Systems

or POLS/WGQS 4457

or U.S. Reproductive Politics

Upper-Division ISLA Course	¹	3-4
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Environmental Health Course	¹	3-4
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General Education Requirement		3
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Free Elective	²	3
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Units		15
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Term 2

ISLA 3355 Interdisciplinary Research Methods 4

ANT 4401 Culture and Health 3-4

or SOC 3326 or Sociology of the Life Course

or SOC 4470 or Special Advanced Topics

Upper-Division ISLA Course¹ 3-4

General Education Requirement 3

Free Elective² 3

Units

16

Term 1

ISLA 4440	Advanced Interdisciplinary Studies Seminar	3
ISLA 4480	Career Preparation for Interdisciplinary Studies Majors	1
Health and Society Elective ¹		3-4
General Education Requirement		3
Free Elective ²		3-4

Units

14

Term 2

ISLA 4460

Senior Project Seminar

4

& ISLA 4461

and Senior Project

Health and Society Elective¹

3-4

Free Elective²

3-4

Free Elective²

3-4

Units

15

1

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Science, Technology, and Society

Science, Technology, and Society Concentration - BA in Interdisciplinary Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1	Units
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ISLA 1101	Interdisciplinary Studies Major First Year Seminar	1
	Introductory-level Interdisciplinary Course	3
	General Education Requirement (1A)	3
	General Education Requirement (1C)	3
	General Education Requirement (2)	3
	Units	13

ISLA 2201	Introduction to Interdisciplinary Studies	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Free Elective ²	3
----------------------------	---

Units	15
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Term 1

ISLA 2255	Introduction to Interdisciplinary Theories	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		1
Free Elective ²		3
Units		16

Term 2

Science Communication Course¹

3-4

General Education Requirement

3

General Education Requirement

3

Free Elective²

3

Free Elective²

3-4

Units

16

Term 1

Upper-Division ISLA Course ¹	3-4
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Gender, Race, Culture, Science, and Technology Course ¹	3-4
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Philosophy of Science and Technology Course	3
---	---

General Education Requirement	3
-------------------------------	---

Free Elective ²	3
----------------------------	---

Units	15
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Term 2	
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ISLA 3355	Interdisciplinary Research Methods	4
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Upper-Division ISLA Course¹

3-4

Science, Technology, and Society Elective¹

3-4

General Education Requirement

3

Free Elective²

3

Units

16

Term 1		

ISLA 4440

Advanced Interdisciplinary Studies Seminar

3

ISLA 4480	Career Preparation for Interdisciplinary Studies Majors	1
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Science, Technology, and Society Elective	¹	3-4
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General Education Requirement		3
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Free Elective	²	3-4
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Units		14
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Term 2		
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ISLA 4460	Senior Project Seminar	4
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& ISLA 4461	and Senior Project	
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Science, Technology, and Society Elective ¹	3-4
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Free Elective ²	3-4
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Free Elective ²	3-4
----------------------------	-----

Units	15
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Total Units	120
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1

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Visual, Media, and Cultural Studies

Visual, Media, and Cultural Studies Concentration - BA in Interdisciplinary Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1		Units

ISLA 1101	Interdisciplinary Studies Major First Year Seminar	1
Introductory-level Interdisciplinary Course		3

General Education Requirement (1A)	3
------------------------------------	---

General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement (2)	3
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Units	13
--------------	-----------

Term 2

ISLA 2201	Introduction to Interdisciplinary Studies	3
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General Education Requirement (1B)	3
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General Education Requirement	3
-------------------------------	---

General Education Requirement	3
-------------------------------	---

Free Elective ² 3

Units	15
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Term 1

ISLA 2255 Introduction to Interdisciplinary Theories 3

General Education Requirement	3
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General Education Requirement	3
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General Education Requirement 3

General Education Requirement 1

Free Elective² 3

Units 16

Term 2

Art, Dance, Music, and Theatre Course 3

General Education Requirement 3

General Education Requirement 3

Free Elective²

3

Free Elective²

3-4

Units

16

Term 1

Upper-Division ISLA Course¹

3-4

Literature Course

3

Popular Culture Course¹

3-4

General Education Requirement

3

Free Elective²

3

Units

15

Term 2

ISLA 3355

Interdisciplinary Research Methods

4

Upper-Division ISLA Course¹

3-4

Technology and Human Expression Course¹

3-4

General Education Requirement

3

Free Elective²

3

Units

16

Term 1		

ISLA 4440

Advanced Interdisciplinary Studies Seminar

3

ISLA 4480

Career Preparation for Interdisciplinary Studies Majors

1

Visual, Media, and Cultural Studies Elective¹

3-4

General Education Requirement

3

Free Elective²

3-4

Units

14

Term 2

ISLA 4460

Senior Project Seminar

4

& ISLA 4461

and Senior Project

Visual, Media, and Cultural Studies Elective¹

3-4

Free Elective ²	3-4
----------------------------	-----

Free Elective ²	3-4
----------------------------	-----

Units	15
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Total Units	120
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1

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

International Strategy and Security

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

Term 1		Units
GMA 1110	First Year Seminar	1
GMA 1115	Intro To Security Studies	3
GMA 2240	World Geography	3

EGL 1100	English Composition (1A) ¹	3
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MTH 1101	College Algebra	3
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Units	13
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Term 2

GMA 1100	International Relations (4B) ¹	3
----------	---	---

EGL 2220	Critical Thinking (1B) ¹	3
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MTH 1107	Elementary Statistics (2)	¹	3
General Education Requirement			3
General Education Requirement			3
General Education Requirement			1
Units			16



Term 1

GMA 2215 Comparative Politics 3

ECO 1100 Macroeconomics 3

EGL 1110 Speech Communication (1C)¹ 3

GOV 2200 American Government (4A)¹ 3

General Education Requirement 3

Units

15

Term 2

GMA 2230	U. S. Maritime Policy	3
GMA 2255	Strategic Policy Leadership	3
ECO 1101	Microeconomics	3
MPM 1190	TSGB/International Experience Preparation	1
GMA Course		3

General Education Requirement

3

Units

16

Term 3

CEP 3330

International Strategy and Security Cooperative

3

MPM 1195

Tsgb/International Experience Special Topics

3

Units

6

Term 1

GMA 3300	U.S. Foreign Policy (Upper-Division 4)	¹	3
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GMA 3370	International Political Economy		3
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EGL 3300	Advanced Writing (GWR)		3
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General Education Requirement			3
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GMA Course			3
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Units

15

Term 2

GMA 3330	Maritime Security	3
GMA 3375	Research Methods	3
HIS 3300	Maritime History Of The U. S.	3
GMA Course		3
General Education Requirement		3

Units

15

Term 1

GMA 4415	Strategy And War	3
GMA 4460	Senior Thesis	3
GMA Course		3
Free Elective		3

Units

12

Term 2

GMA 3320

Ocean Environmental Management

3

GMA 4420

Cybersecurity

3

GMA 4425

Civil Conflict

3

Free Elective

3

Units

12

Total Units

120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Journalism

Concentration Not Yet Declared - BS in Journalism

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units

JOUR 1102

Introduction to Journalism

1

JOUR 2203	News Reporting and Writing	3
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
------------------------------------	--	---

General Education Requirement		3
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Units		13
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Term 2		
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JOUR 2228	Media, Self and Society (4B) ¹	3
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JOUR 2285	Introduction to Multimedia Journalism	3
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STAT 1000	Statistical and Data Literacy (2) ¹	3
or STAT 1110	or Applied Statistical Concepts and Methods	
or STAT 1210	or Business Statistics I	
General Education Requirement (1B)		3
General Education Requirement		3
	Units	15

Second Year		
Term 1		

JOUR 2219	Mass Media in a Cross-Cultural Society	3
-----------	--	---

or JOUR 3319 or Global Communication

JOUR 3300	Public Relations	3
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Concentration Course or Elective	3
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Term 2

JOUR 3303	Video Storytelling	3
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Journalism Elective ²	3
Concentration Course or Elective	3
General Education Requirement	3
General Education Requirement	1
Free Elective ³	3
Units	16

Third Year

Term 1

JOUR 3302	Mass Media Law	3
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JOUR 3305	Journalism Ethics	3
-----------	-------------------	---

Concentration Course or Elective	3
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General Education Requirement	3
-------------------------------	---

Free Elective ³	3
----------------------------	---

Units	15
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Term 2

JOUR 3334	Editing for Online and Print Publication	3
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JOUR 3390

Visual Communication for the Mass Media

3

Concentration Course or Elective

3

General Education Requirement

3

Free Elective³

3

Units

15

Fourth Year

Term 1

Journalism Elective²

3

Concentration Course or Elective

3

General Education Requirement

3

Free Elective³

4

Free Elective³

3

Units

16

Term 2

JOUR 4444Media Internship

2

Journalism Elective²

3

General Education Requirement	3
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Free Elective ³	4
----------------------------	---

Free Elective ³	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses can only be used once for major degree credit.

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Media Innovation

Media Innovation Concentration - BS in Journalism

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
JOUR 1102	Introduction to Journalism	1
JOUR 2203	News Reporting and Writing	3

General Education Requirement (1A)	3
------------------------------------	---

General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	13
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Term 2	
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JOUR 2228	Media, Self and Society (4B) ¹	3
-----------	---	---

JOUR 2285	Introduction to Multimedia Journalism	3
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STAT 1000	Statistical and Data Literacy (2) ¹	3
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or STAT 1110 or Applied Statistical Concepts and Methods

or STAT 1210 or Business Statistics I

General Education Requirement (1B) 3

General Education Requirement 3

Units **15**

Second Year

Term 1

JOUR 2219 Mass Media in a Cross-Cultural Society 3

or JOUR 3319 or Global Communication

JOUR 3334	Editing for Online and Print Publication	3
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JOUR 3350	Data Journalism	3
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General Education Requirement		3
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General Education Requirement		3
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Units		15
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Term 2		
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JOUR 3303	Video Storytelling	3
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JOUR 3310	Advanced Digital Journalism	3
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or JOUR 3345	or Social Media for Strategic Communication	
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Journalism Elective ²	3
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General Education Requirement	3
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General Education Requirement	1
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Free Elective ³	3
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Units	16
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Third Year

Term 1

JOUR 3300	Public Relations	3
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JOUR 3302	Mass Media Law	3
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JOUR 3305	Journalism Ethics	3
-----------	-------------------	---

General Education Requirement	3
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Free Elective ³	3
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Units	15
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Term 2

JOUR 3352	Advanced News Reporting Practicum	3
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or JOUR 3353	or Advanced Broadcast Journalism Practicum
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JOUR 3385	Media Innovation and Entrepreneurship	3
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JOUR 3390	Visual Communication for the Mass Media	3
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General Education Requirement		3
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Free Elective	3	3
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Units		15
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Fourth Year

Term 1

JOUR 4462	Senior Project: Media Practicum	3
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Journalism Elective	2	3
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General Education Requirement		3
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Free Elective	3	4
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Free Elective	3	3
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Units		16
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Term 2		
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JOUR 4444	Media Internship	2
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Journalism Elective ²	3
General Education Requirement	3
Free Elective ³	4
Free Elective ³	3
Units	15
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses can only be used once for major degree credit.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

News

News Concentration - BS in Journalism

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

JOUR 1102	Introduction to Journalism	1
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JOUR 2203	News Reporting and Writing	3
-----------	----------------------------	---

General Education Requirement (1A)	3
------------------------------------	---

General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement	3
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Units	13
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Term 2

JOUR 2228	Media, Self and Society (4B) ¹	3
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JOUR 2285	Introduction to Multimedia Journalism	3
STAT 1000	Statistical and Data Literacy (2) ¹	3
or STAT 1110	or Applied Statistical Concepts and Methods	
or STAT 1210	or Business Statistics I	
General Education Requirement (1B)		3
General Education Requirement		3
Units		15

Second Year

Term 1

JOUR 2219	Mass Media in a Cross-Cultural Society	3
or JOUR 3319	or Global Communication	
JOUR 3304	Public Affairs Reporting	3
JOUR 3334	Editing for Online and Print Publication	3
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

JOUR 3303	Video Storytelling	3
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Journalism Elective ²	3
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Writing/Digital Path or Audio/Visual Path Elective	3
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General Education Requirement	3
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General Education Requirement	1
-------------------------------	---

Free Elective ³	3
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Units	16
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Third Year	
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Term 1	
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JOUR 3302	Mass Media Law	3
-----------	----------------	---

JOUR 3305	Journalism Ethics	3
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Writing/Digital Path or Audio/Visual Path Elective	3
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General Education Requirement	3
-------------------------------	---

Free Elective ³	3
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Units	15
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Term 2

JOUR 3300	Public Relations	3
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JOUR 3390	Visual Communication for the Mass Media	3
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Writing/Digital Path or Audio/Visual Path Elective	3
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General Education Requirement	3
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Free Elective ³	3
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Units	15
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Fourth Year

Term 1

JOUR 4462	Senior Project: Media Practicum	3
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Journalism Elective	2	3
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General Education Requirement		3
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Free Elective	3	4
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Free Elective	3	3
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Units		16
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Term 2		
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JOUR 4444	Media Internship	2
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Journalism Elective ²	3
General Education Requirement	3
Free Elective ³	4
Free Elective ³	3
Units	15
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses can only be used once for major degree credit.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Public Relations

Public Relations Concentration - BS in Journalism

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

JOUR 1102	Introduction to Journalism	1
JOUR 2203	News Reporting and Writing	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
General Education Requirement		3
Units		13

Term 2		
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JOUR 2228	Media, Self and Society (4B) ¹	3
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JOUR 2285	Introduction to Multimedia Journalism	3
STAT 1000	Statistical and Data Literacy (2) ¹	3
or STAT 1110	or Applied Statistical Concepts and Methods	
or STAT 1210	or Business Statistics I	
General Education Requirement (1B)		3
General Education Requirement		3
Units		15

Second Year

Term 1

JOUR 2219	Mass Media in a Cross-Cultural Society	3
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or JOUR 3319	or Global Communication	
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JOUR 3300	Public Relations	3
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General Education Requirement		3
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General Education Requirement		3
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Free Elective ³		3
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Units		15
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Term 2

JOUR 3303	Video Storytelling	3
JOUR 3342	Public Relations Writing and Editing	3
JOUR 3347	Public Relations Research	3
Journalism Elective ²		3
General Education Requirement		3
General Education Requirement		1
Units		16

Third Year

Term 1

JOUR 3302 Mass Media Law 3

JOUR 3305 Journalism Ethics 3

JOUR 3313 Public Relations Campaigns 3

General Education Requirement 3

Free Elective³ 3

Units 15

Term 2

JOUR 3314	Public Relations and Crisis Management	3
or JOUR 3315	or Public Relations and Advertising Production	
or JOUR 3345	or Social Media for Strategic Communication	
JOUR 3334	Editing for Online and Print Publication	3
JOUR 3390	Visual Communication for the Mass Media	3
General Education Requirement		3
Free Elective	3	3
Units		15

Fourth Year

Term 1

JOUR 4464 Senior Project: Public Relations Practicum 3

Journalism Elective² 3

General Education Requirement 3

Free Elective³ 4

Free Elective³ 3

Units 16

Term 2

JOUR 4444	Media Internship	2
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Journalism Elective ²	3
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General Education Requirement	3
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Free Elective ³	4
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Free Elective ³	3
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Units	15
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Total Units	120
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Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses can only be used once for major degree credit.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Individualized Course of Study

Individualized Course of Study - BS in Journalism

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
JOUR 1102	Introduction to Journalism	1
JOUR 2203	News Reporting and Writing	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
General Education Requirement		3
Units		13

Term 2	
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JOUR 2228	Media, Self and Society (4B) ¹	3
JOUR 2285	Introduction to Multimedia Journalism	3
STAT 1000	Statistical and Data Literacy (2) ¹	3
or STAT 1110	or Applied Statistical Concepts and Methods	
or STAT 1210	or Business Statistics I	
General Education Requirement (1B)		3
General Education Requirement		3
	Units	15

Second Year

Term 1

JOUR 2219 Mass Media in a Cross-Cultural Society 3

or JOUR 3319 or Global Communication

JOUR 3300 Public Relations 3

General Education Requirement 3

General Education Requirement 3

Free Elective³ 3

Units

15

Term 2

JOUR 3303 Video Storytelling 3

JOUR 3304 Public Affairs Reporting 3

or JOUR 3347 or Public Relations Research

or JOUR 3385 or Media Innovation and Entrepreneurship

JOUR 3307 Feature Writing 3

or JOUR 3333 or Broadcast News

or JOUR 3342 or Public Relations Writing and Editing

Journalism Elective	2	3
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General Education Requirement		3
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General Education Requirement		1
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Units		16
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Third Year		
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Term 1		
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JOUR 3302	Mass Media Law	3
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JOUR 3305	Journalism Ethics	3
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Media Technology Elective	3
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General Education Requirement	3
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Free Elective ³	3
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Units	15
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Term 2	
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JOUR 3313	Public Relations Campaigns	3
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or JOUR 3352	or Advanced News Reporting Practicum
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or JOUR 3353	or Advanced Broadcast Journalism Practicum
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JOUR 3334	Editing for Online and Print Publication	3
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JOUR 3390	Visual Communication for the Mass Media	3
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General Education Requirement		3
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Free Elective	3	3
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Units		15
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Fourth Year

Term 1

JOUR 4462	Senior Project: Media Practicum	3
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or JOUR 4464	or Senior Project: Public Relations Practicum	
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Journalism Elective	2	3
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General Education Requirement		3
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Free Elective	3	4
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Free Elective	3	3
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Units		16
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Term 2		
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JOUR 4444	Media Internship	2
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Journalism Elective ²	3
General Education Requirement	3
Free Elective ³	4
Free Elective ³	3
Units	15
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Courses can only be used once for major degree credit.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Kinesiology

Concentration Not Yet Declared - BS in Kinesiology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

HLTH 1155

Multicultural Perspectives and Health (4B) ¹

3

or HLTH 1160

or Women's Health and Society

KINE 1180	Introduction to Kinesiology	1
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BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
-----------	--	---

MATH 1007	Precalculus (2) ¹	3-4
-----------	------------------------------	-----

or MATH 1261

or Calculus I

Units	15
-------	----

Term 2

PHYS 1121	College Physics I	4
-----------	-------------------	---

STAT 1110

Applied Statistical Concepts and Methods

3

General Education Requirement (1A)

3

General Education Requirement (1C)

3

Free Elective³

2

Units

15

Second Year

Term 1

KINE 2266

Sport and Performance Psychology

3

KINE 3312	Motor Learning	3
BIO 2231	Human Anatomy and Physiology I	4
General Education Requirement (1B)		3
General Education Requirement (3A)		3
Units		16

Term 2

KINE 3319	Introduction to Research Methods in Kinesiology	4
BIO 2232	Human Anatomy and Physiology II	4

General Education Requirement (3B)

3

General Education Requirement (4A)

3

Free Elective³

3

Units

17

Third Year

Term 1

KINE 3303

Physiology of Exercise

4

KINE 3323

Sport and Gender²

3

or KINE 3324	or Sports, Media, and United States Popular Culture	
or KINE 3325	or Sport and Physical Activity Throughout Civilizations	
General Education Requirement (6)		3
Free Elective ³		2
Units		12
Term 2		
KINE 3307	Adapted Physical Activity	3
Concentration Course		3

Concentration Course	3
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General Education Requirement (Upper-Division 2/5)	3
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General Education Requirement (Upper-Division 3)	3
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Units	15
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Fourth Year

Term 1

KINE 4403	Biomechanics	4
-----------	--------------	---

KINE 4412	Physical Activity and Public Health	3
-----------	-------------------------------------	---

Concentration Course	3
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Concentration Course	2-3
----------------------	-----

Free Elective ³	2
----------------------------	---

Units	15
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Term 2

KINE 4451	Nutrition for Fitness and Sport	4
-----------	---------------------------------	---

KINE 4460	Senior Project - Experiential	2
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or KINE 4461	or Senior Project - Report
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or KINE 4462	or Senior Project - Research
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or KINE 4463 or Senior Project - Internship

Concentration Course	3
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Concentration Course	3
----------------------	---

General Education Requirement (Upper-Division 4)	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

KINE 3325 is required in the Sport Science concentration. A course taken to fulfill a major course requirement cannot be double-counted in the concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Exercise Science

Exercise Science Concentration - BS in Kinesiology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

HLTH 1155	Multicultural Perspectives and Health (4B) ¹	3
or HLTH 1160	or Women's Health and Society	
KINE 1180	Introduction to Kinesiology	1
BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
MATH 1007	Precalculus (2) ¹	3-4
or MATH 1261	or Calculus I	
Units		15

Term 2

PHYS 1121	College Physics I	4
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STAT 1110	Applied Statistical Concepts and Methods	3
-----------	--	---

General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

Free Elective ³	2
----------------------------	---

Units	15
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Second Year

Term 1

KINE 2266	Sport and Performance Psychology	3
-----------	----------------------------------	---

KINE 3312	Motor Learning	3
-----------	----------------	---

BIO 2231	Human Anatomy and Physiology I	4
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General Education Requirement (1B)		3
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General Education Requirement (3A)		3
------------------------------------	--	---

Units		16
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Term 2

KINE 3307	Adapted Physical Activity	3
-----------	---------------------------	---

BIO 2232	Human Anatomy and Physiology II	4
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General Education Requirement (3B)		3
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General Education Requirement (4A)		3
------------------------------------	--	---

Free Elective ³		3
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Units		16
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Third Year	
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Term 1	
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KINE 3303	Physiology of Exercise	4
-----------	------------------------	---

KINE 3319	Introduction to Research Methods in Kinesiology	4
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KINE 3323	Sport and Gender ²	3
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or KINE 3324

or Sports, Media, and United States Popular Culture

or KINE 3325

or Sport and Physical Activity Throughout
Civilizations

General Education Requirement (6)	3
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Units	14
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Term 2

KINE 3349	Exercise Testing and Prescription	3
-----------	-----------------------------------	---

KINE 4406	Motor Control	3
-----------	---------------	---

General Education Requirement (Upper-Division 2/5)	3
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General Education Requirement (Upper-Division 3)	3
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Free Elective ³	2
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Units	14
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Fourth Year

Term 1

KINE 4403	Biomechanics	4
-----------	--------------	---

KINE 4404	Clinical Exercise Physiology	4
-----------	------------------------------	---

KINE 4412	Physical Activity and Public Health	3
-----------	-------------------------------------	---

Concentration Elective		3
------------------------	--	---

Free Elective ³		2
----------------------------	--	---

Units		16
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Term 2

KINE 3330	Group Exercise and Personal Training	2
-----------	--------------------------------------	---

KINE 4451	Nutrition for Fitness and Sport	4
-----------	---------------------------------	---

KINE 4460	Senior Project - Experiential	2
-----------	-------------------------------	---

or KINE 4461	or Senior Project - Report	
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or KINE 4462	or Senior Project - Research	
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or KINE 4463	or Senior Project - Internship	
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Concentration Elective		3
------------------------	--	---

General Education Requirement (Upper-Division 4)		3
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Units		14
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Total Units		120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

KINE 3325 is required in the Sport Science concentration. A course taken to fulfill a major course requirement cannot be double-counted in the concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Health Promotion

Health Promotion Concentration - BS in Kinesiology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

HLTH 1155	Multicultural Perspectives and Health (4B) ¹	3
or HLTH 1160	or Women's Health and Society	
KINE 1180	Introduction to Kinesiology	1
BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
MATH 1007	Precalculus (2) ¹	3-4
or MATH 1261	or Calculus I	
Units		15

Term 2

PHYS 1121	College Physics I	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

Free Elective ³	2
----------------------------	---

Units	15
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Second Year

Term 1

KINE 2266	Sport and Performance Psychology	3
-----------	----------------------------------	---

KINE 3312	Motor Learning	3
-----------	----------------	---

BIO 2231	Human Anatomy and Physiology I	4
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General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement (3A)	3
------------------------------------	---

Units	16
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Term 2

BIO 2232	Human Anatomy and Physiology II	4
----------	---------------------------------	---

KINE 3307	Adapted Physical Activity	3
-----------	---------------------------	---

General Education Requirement (3B)	3
------------------------------------	---

General Education Requirement (4A)	3
------------------------------------	---

Free Elective ³	3
----------------------------	---

Units	16
-------	----

Third Year

Term 1

KINE 3303	Physiology of Exercise	4
-----------	------------------------	---

KINE 3319	Introduction to Research Methods in Kinesiology	4
-----------	---	---

KINE 3323	Sport and Gender ²	3
-----------	-------------------------------	---

or KINE 3324

or Sports, Media, and United States Popular Culture

or KINE 3325

or Sport and Physical Activity Throughout
Civilizations

General Education Requirement (6)	3
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Units	14
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Term 2

HLTH 2261	Social Determinants of Health	3
-----------	-------------------------------	---

Concentration Elective

4

General Education Requirement (Upper-Division 2/5)

3

General Education Requirement (Upper-Division 3)

3

Free Elective³

2

Units

15

Fourth Year

Term 1

KINE 4401

Leadership in Health and Physical Activity Programs

3

KINE 4403	Biomechanics	4
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KINE 4412	Physical Activity and Public Health	3
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HLTH 4434	Health Promotion Program Planning	4
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Free Elective ³		2
----------------------------	--	---

Units	16
-------	----

Term 2

KINE 4451	Nutrition for Fitness and Sport	4
-----------	---------------------------------	---

KINE 4460	Senior Project - Experiential	2
-----------	-------------------------------	---

or KINE 4461	or Senior Project - Report
or KINE 4462	or Senior Project - Research
or KINE 4463	or Senior Project - Internship

Concentration Elective	4
General Education Requirement (Upper-Division 4)	3
Units	13
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

KINE 3325 is required in the Sport Science concentration. A course taken to fulfill a major course requirement cannot be double-counted in the concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Sport Science

Sport Science Concentration - BS in Kinesiology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

HLTH 1155	Multicultural Perspectives and Health (4B) ¹	3
or HLTH 1160	or Women's Health and Society	
KINE 1180	Introduction to Kinesiology	1
BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
MATH 1007	Precalculus (2) ¹	3-4
or MATH 1261	or Calculus I	
Units		15

Term 2

PHYS 1121	College Physics I	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

Free Elective ³	2
----------------------------	---

Units	15
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Second Year

Term 1

KINE 2266	Sport and Performance Psychology	3
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KINE 3312	Motor Learning	3
-----------	----------------	---

BIO 2231	Human Anatomy and Physiology I	4
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General Education Requirement (1B)		3
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General Education Requirement (3A)		3
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Units		16
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Term 2

KINE 3307	Adapted Physical Activity	3
-----------	---------------------------	---

BIO 2232	Human Anatomy and Physiology II	4
----------	---------------------------------	---

General Education Requirement (3B)		3
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General Education Requirement (4A)		3
------------------------------------	--	---

Free Elective ³		3
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Units		16
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Third Year	
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Term 1	
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KINE 3303	Physiology of Exercise	4
-----------	------------------------	---

KINE 3319	Introduction to Research Methods in Kinesiology	4
-----------	---	---

Concentration Elective		3
------------------------	--	---

General Education Requirement (6)		3
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Free Elective ³		2
----------------------------	--	---

Units		16
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Term 2

KINE 3323	Sport and Gender ²	3
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or KINE 3324

or Sports, Media, and United States Popular Culture

or KINE 3325

or Sport and Physical Activity Throughout
Civilizations

KINE 3330	Group Exercise and Personal Training	2
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KINE 3366	Applied Sport and Performance Psychology	3
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General Education Requirement (Upper-Division 2/5)	3
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General Education Requirement (Upper-Division 3)	3
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Units	14
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Fourth Year

Term 1

KINE 3325	Sport and Physical Activity Throughout Civilizations	2	3
KINE 4401	Leadership in Health and Physical Activity Programs		3
KINE 4403	Biomechanics		4
KINE 4412	Physical Activity and Public Health		3
Units			13

Term 2

KINE 4451	Nutrition for Fitness and Sport		4
KINE 4460	Senior Project - Experiential		2

or KINE 4461	or Senior Project - Report
or KINE 4462	or Senior Project - Research
or KINE 4463	or Senior Project - Internship

Concentration Elective	3
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General Education Requirement (Upper-Division 4)	3
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Free Elective ²	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

KINE 3325 required in the Sport Science concentration. A course taken to fulfill a major course requirement cannot be double-counted in the concentration.

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Landscape Architecture

Suggested Five-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units

LA 1104	History of Landscape Architecture (3A)	1	3
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LA 1110	Landscape Representation I	3
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EDES 1123	Place, People, and the Built Environment (4B) ¹	3
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ENGL 1133	Multilingual Writing and Rhetoric (1A) ¹	3
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or ENGL 1134	or Writing and Rhetoric	
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MATH 1007	Precalculus (2) ¹	3
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Units	15
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Term 2	
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LA 1112	Landscape Representation II	4
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LA 1120	Studio I - Beginning Design	3
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BIO 1114	Plant Diversity and Ecology (5B & 5C) ¹	4
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or BOT 1121	or General Botany	
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COMS 1101	Public Speaking (1C) ¹	3
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or COMS 1102	or Public Speaking in Digital Spaces	
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Units	14
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Second Year

Term 1

LA 2220	Studio II - Site Based Design	3
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PLSC 2234	Introduction to Plant Materials	3
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Major Support 1B Course ¹	3
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Designated Elective	3-4
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General Education Requirement	3
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Units	15-16
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Term 2

LA 2210	Materials and Techniques of Landscape Construction	4
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LA 2212	Introduction to Planting Design	3
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LA 2222	Studio III - Design for Implementation	3
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General Education Requirement		3
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General Education Requirement		3
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Units		16
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Third Year		
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Term 1		
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LA 2203	Landscape Ecology: Concepts, Issues, and Interrelationships	3
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LA 2214	Site Engineering Techniques and Applications	4
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LA 3310	Landscape Representation III	3
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LA 3320	Studio IV - Landscape Performance and Design	4
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	Units	14
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Term 2		
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LA 3303	Theory for Landscape Architects	3
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LA 3312	Geographic Information Systems (GIS) Applications for Design Projects	3
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LA 3322	Studio V - Environmental Planning and Design	4
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Designated Elective		3
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General Education Requirement

3

Units

16

Fourth Year

Term 1

LA 3304

Contemporary Issues in Cultural Landscapes

3

LA 3306

Professional Practice of Landscape Architecture

3

LA 4420

Studio VI - Community and Social Design

4

Designated Elective

3-4

Units

13-14

Term 2

LA 4371 Landscape Architecture Internship 4

LA 4410 Sustainability, Resilience, and Climate Ecology in Design 3

LA 4422 Interdisciplinary Design Studio 4

or LA 4424 or Contemporary Issues in Landscape Architecture
Design Studio

CAED Designated Elective 3

General Education Requirement 3

Units 17

Fifth Year

Term 1

LA 4413 Social Equity and Design 3

LA 4461 Senior Project Design Studio I 4

CAED Designated Elective 3

General Education Requirement 3

Units 13

Term 2

LA 4414 Advanced Landscape Construction 3

or LA 4415 or Advanced Landscape Representation

or LA 4416	or Advanced Planting Design	
or LA 4417	or Social and Behavioral Factors for Landscape Architecture	
or LA 4418	or Contemporary Issues in Landscape Architecture	
LA 4462	Senior Project Design Studio II	4
CAED Designated Elective		3
General Education Requirement		3
	Units	13
	Total Units	146-148

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Liberal Arts and Engineering Studies

Concentration Not Yet Declared - BS in Liberal Arts and Engineering Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4
MATH 1261	Calculus I (2) ³	4
General Education Requirement (1A)		3

General Education Requirement (1C)	3
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Units	14
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Term 2	
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MATH 1262	Calculus II	4
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PHYS 1141	General Physics I	4
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General Education Requirement (1B)	3
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General Education Requirement	3
-------------------------------	---

Units	14
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Second Year

Term 1

MATH 2263	Calculus III	3
PHYS 1143	General Physics II	4
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
Units		16

Term 2

Engineering Concentration Course	1, 2	3
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Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Concentration Course	1, 2	3
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General Education Requirement		3
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General Education Requirement		3
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Units		15
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Third Year

Term 1

LAES 4301	Liberal Arts and Engineering Studies Capstone I	4
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STAT 3210	Engineering Statistics (Upper-Division 2/5)	3
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Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Concentration Course	1, 2	3
-----------------------------------	------	---

Units	16
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Term 2

LAES 4302

Liberal Arts and Engineering Studies Capstone II

4

Engineering Concentration Course^{1, 2}

2-3

Study Abroad or Upper-Division Global Perspectives^{2, 4}

5

General Education Requirement

3

Units

15

Fourth Year

Term 1

LAES 4461

Senior Project I in Liberal Arts and Engineering Studies

3

Engineering Concentration Course	1, 2	2-3
----------------------------------	------	-----

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Concentration Course	1, 2	3
-----------------------------------	------	---

Liberal Arts Concentration Course	1, 2	3
-----------------------------------	------	---

Units	14
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Term 2

LAES 4462	Senior Project II in Liberal Arts and Engineering Studies	3
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Engineering Concentration Course	1, 2	3
Engineering Concentration Course	1, 2	3
Liberal Arts Concentration Course	1, 2	4
General Education Requirement		3
Units		16
Total Units		120

1

A minimum of 10 units must be taken at the 3000-4000 level.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be required to complete the total units required for the degree.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

A list of Global Perspectives courses is provided on the [LAES website](#). All Global Perspective courses are 3000-4000 level (upper-division).

Computer Science

Computer Science Concentration - BS in Liberal Arts and Engineering Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C)	3	4
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MATH 1261	Calculus I (2)	3	4
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General Education Requirement (1A)			3
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General Education Requirement (1C)			3
------------------------------------	--	--	---

Units			14
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Term 2

MATH 1262	Calculus II	4
-----------	-------------	---

PHYS 1141	General Physics I	4
-----------	-------------------	---

General Education Requirement (1B)	3
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General Education Requirement	3
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Units	14
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Second Year	
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Term 1	
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MATH 2263	Calculus III	3
-----------	--------------	---

PHYS 1143	General Physics II	4
-----------	--------------------	---

General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory
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CSC/CPE/SE Elective	5	3
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Liberal Arts Concentration Course	1, 2	3
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General Education Requirement		3
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General Education Requirement		3
-------------------------------	--	---

Units		16
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Third Year		
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Term 1		
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CSC 2001	Data Structures	4
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& 2001L and Data Structures Laboratory

LAES 4301 Liberal Arts and Engineering Studies Capstone I 4

STAT 3210 Engineering Statistics (Upper-Division 2/5) ³ 3

CSC/CPE/SE Elective ⁵ 3

Liberal Arts Concentration Course ^{1, 2} 3

Units 17

Term 2

LAES 4302 Liberal Arts and Engineering Studies Capstone II 4

CSC/CPE/SE Elective	5	3
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Study Abroad or Upper-Division Global Perspectives	2, 4	5
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General Education Requirement		3
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Units		15
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Fourth Year		
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Term 1		
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LAES 4461	Senior Project I in Liberal Arts and Engineering Studies	3
-----------	--	---

CSC/CPE/SE Elective	5	3
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CSC/CPE/SE Elective	5	3
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Liberal Arts Concentration Course	1, 2	3
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Liberal Arts Concentration Course	1, 2	3
-----------------------------------	------	---

Units	15
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Term 2

LAES 4462	Senior Project II in Liberal Arts and Engineering Studies	3
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CSC/CPE/SE Elective	5	3
---------------------	---	---

Liberal Arts Concentration Course	1, 2	4
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General Education Requirement		3
-------------------------------	--	---

Units		13
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Total Units		120
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1

A minimum of 10 units must be taken at the 3000-4000 level.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be required to complete the total units required for the degree.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

A list of Global Perspectives courses is provided on the [LAES website](#). All Global Perspective courses are 3000-4000 level (upper-division).

5

Suggested electives to pursue studies in areas such as usability studies, game design, full stack web development, or computer graphics are provided on the [LAES website](#).

Electrical Engineering

Electrical Engineering Concentration - BS in Liberal Arts and Engineering Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4
MATH 1261	Calculus I (2) ³	4
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		14

Term 2		
MATH 1262	Calculus II	4

PHYS 1141	General Physics I	4
-----------	-------------------	---

General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	14
-------	----

Second Year

Term 1

MATH 2263	Calculus III	3
-----------	--------------	---

PHYS 1143	General Physics II	4
-----------	--------------------	---

General Education Requirement

3

General Education Requirement

3

General Education Requirement

3

Units

16

Term 2

EE 2211

Electric Circuit Analysis I

4

& EE 2241

and Electric Circuit Analysis Laboratory I

Liberal Arts Concentration Course

1, 2

3

General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Units	13
-------	----

Third Year

Term 1

LAES 4301	Liberal Arts and Engineering Studies Capstone I	4
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MATH 2341	Linear Analysis	4
-----------	-----------------	---

or MATH 2343	or Differential Equations
--------------	---------------------------

STAT 3210	Engineering Statistics (Upper-Division 2/5)	3	3
-----------	---	---	---

CPE/EE Elective		5	3
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Liberal Arts Concentration Course		1, 2	3
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Units			17
-------	--	--	----

Term 2			
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EE 2212	Electric Circuit Analysis II		3
---------	------------------------------	--	---

LAES 4302	Liberal Arts and Engineering Studies Capstone II		4
-----------	--	--	---

Study Abroad or Upper-Division Global Perspectives

2, 4

5

General Education Requirement

3

Units

15

Fourth Year

Term 1

LAES 4461

Senior Project I in Liberal Arts and Engineering Studies

3

CPE/EE Elective

5

3

CPE/EE Elective

5

3

Liberal Arts Concentration Course	1, 2	3
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Liberal Arts Concentration Course	1, 2	3
-----------------------------------	------	---

Units	15
-------	----

Term 2		
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LAES 4462	Senior Project II in Liberal Arts and Engineering Studies	3
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CPE/EE Elective	5	3
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CPE/EE Elective	5	3-4
-----------------	---	-----

Liberal Arts Concentration Course	1, 2	4
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General Education Requirement		3
-------------------------------	--	---

Units		16
-------	--	----

Total Units		120
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1

A minimum of 10 units must be taken at the 3000-4000 level.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be required to complete the total units required for the degree.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

A list of Global Perspectives courses is provided on the [LAES website](#). All Global Perspective courses are 3000-4000 level (upper-division).

5

Suggested electives to pursue studies in areas such as animatronics, audio engineering, or power systems are provided on the [LAES website](#).

Industrial Engineering

Industrial Engineering Concentration - BS in Liberal Arts and Engineering Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4
MATH 1261	Calculus I (2) ³	4
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		14

Term 2		
MATH 1262	Calculus II	4

PHYS 1141	General Physics I	4
-----------	-------------------	---

General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	14
-------	----

Second Year	
-------------	--

Term 1	
--------	--

MATH 2263	Calculus III	3
-----------	--------------	---

PHYS 1143	General Physics II	4
-----------	--------------------	---

General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	16
-------	----

Term 2	
--------	--

CSC 1032	Programming for Scientists and Engineers	3
----------	--	---

IME 1223	Process Improvement Fundamentals	4
----------	----------------------------------	---

IME 2315	Financial Decision Making for Engineers	2
----------	---	---

Liberal Arts Concentration Course

1, 2

3

General Education Requirement

3

Units

15

Third Year

Term 1

IME 1140

Technical Graphics Communication for Design and Manufacturing

3

& IME 1143

and Introduction to Design and Manufacturing

LAES 4301

Liberal Arts and Engineering Studies Capstone I

4

STAT 3210	Engineering Statistics (Upper-Division 2/5)	3	3
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IE Elective		5	3
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Liberal Arts Concentration Course		1, 2	3
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Units			16
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Term 2			
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IME 2212	Introduction to Enterprise Analytics and Database Systems		4
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LAES 4302	Liberal Arts and Engineering Studies Capstone II		4
-----------	--	--	---

Study Abroad or Upper-Division Global Perspectives	2, 4	5
--	------	---

General Education Requirement		3
-------------------------------	--	---

Units		16
-------	--	----

Fourth Year		
-------------	--	--

Term 1		
--------	--	--

LAES 4461	Senior Project I in Liberal Arts and Engineering Studies	3
-----------	--	---

IE Elective	5	4
-------------	---	---

Liberal Arts Concentration Course	1, 2	3
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Liberal Arts Concentration Course	1, 2	3
-----------------------------------	------	---

Units	13
-------	----

Term 2		
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LAES 4462	Senior Project II in Liberal Arts and Engineering Studies	3
-----------	---	---

IE Elective	5	3
-------------	---	---

Liberal Arts Concentration Course	1, 2	4
-----------------------------------	------	---

General Education Requirement		3
-------------------------------	--	---

General Education Requirement		3
-------------------------------	--	---

Units	16
-------	----

Total Units	120
-------------	-----

1

A minimum of 10 units must be taken at the 3000-4000 level.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be required to complete the total units required for the degree.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

A list of Global Perspectives courses is provided on the [LAES website](#). All Global Perspective courses are 3000-4000 level (upper-division).

5

Suggested electives to pursue studies in areas such as project management, process improvement, or production engineering are provided on the [LAES website](#).

Engineering Individualized Course of Study

Engineering Individualized Course of Study - BS in Liberal Arts and Engineering Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C)	3	4
-----------	---	---	---

MATH 1261	Calculus I (2)	3	4
-----------	----------------	---	---

General Education Requirement (1A)			3
------------------------------------	--	--	---

General Education Requirement (1C)			3
------------------------------------	--	--	---

Units			14
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Term 2			
--------	--	--	--

MATH 1262	Calculus II		4
-----------	-------------	--	---

PHYS 1141	General Physics I		4
-----------	-------------------	--	---

General Education Requirement (1B)

3

General Education Requirement

3

Units

14

Second Year

Term 1

MATH 2263

Calculus III

3

PHYS 1143

General Physics II

4

General Education Requirement

3

General Education Requirement

3

General Education Requirement

3

Units

16

Term 2

Engineering Elective 5, 6, 7

3

Engineering Elective 5, 6, 7

3

Liberal Arts Concentration Course 1, 2

3

General Education Requirement

3

General Education Requirement

3

Units

15

Third Year

Term 1

LAES 4301

Liberal Arts and Engineering Studies Capstone I

4

STAT 3210

Engineering Statistics (Upper-Division 2/5)

3

3

Engineering Elective

5, 6, 7

3

Engineering Elective

5, 6, 7

3

Liberal Arts Concentration Course

1, 2

3

Units

16

Term 2

LAES 4302

Liberal Arts and Engineering Studies Capstone II

4

Engineering Elective

5, 6, 7

2-3

Study Abroad or Upper-Division Global Perspectives

2, 4

5

General Education Requirement

3

Units

15

Fourth Year

Term 1

LAES 4461 Senior Project I in Liberal Arts and Engineering Studies 3

Engineering Elective 5, 6, 7 2-3

Engineering Elective 5, 6, 7 3

Liberal Arts Concentration Course 1, 2 3

Liberal Arts Concentration Course 1, 2 3

Units

14

Term 2

LAES 4462 Senior Project II in Liberal Arts and Engineering Studies 3

Engineering Elective 5, 6, 7 3

Engineering Elective 5, 6, 7 3

Liberal Arts Concentration Course 1, 2 4

General Education Requirement 3

Units

16

1

A minimum of 10 units must be taken at the 3000-4000 level.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be required to complete the total units required for the degree.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

A list of Global Perspectives courses is provided on the [LAES website](#). All Global Perspective courses are 3000-4000 level (upper-division).

5

LAES director approved electives from courses offered by the College of Engineering.

6

A minimum of 10 units must be at the 3000-4000 level.

7

The student must provide written justification for how this collection of courses constitutes a cohesive and effective hybrid integration with the student's other concentration courses selected from the College of Liberal Arts.

Liberal Arts Focused Course of Study

Liberal Arts Focused Course of Study - BS in Liberal Arts and Engineering Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C)	3	4
-----------	---	---	---

MATH 1261	Calculus I (2)	3	4
-----------	----------------	---	---

General Education Requirement (1A)			3
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General Education Requirement (1C)			3
------------------------------------	--	--	---

Units			14
-------	--	--	----

Term 2

MATH 1262

Calculus II

4

PHYS 1141

General Physics I

4

General Education Requirement (1B)

3

General Education Requirement

3

Units

14

Second Year

Term 1

MATH 2263

Calculus III

3

PHYS 1143	General Physics II	4
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	16
-------	----

Term 2

Engineering Concentration Course	1, 2	3
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Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Minor Course	5, 6, 7, 8	3
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General Education Requirement		3
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General Education Requirement		3
-------------------------------	--	---

Units		15
-------	--	----

Third Year		
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Term 1		
--------	--	--

LAES 4301	Liberal Arts and Engineering Studies Capstone I	4
-----------	---	---

STAT 3210	Engineering Statistics (Upper-Division 2/5)	3
-----------	---	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Minor Course	5, 6, 7, 8	3
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Units	16
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Term 2		
--------	--	--

LAES 4302	Liberal Arts and Engineering Studies Capstone II	4
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Engineering Concentration Course	1, 2	2-3
----------------------------------	------	-----

Study Abroad or Upper-Division Global Perspectives

2, 4

5

General Education Requirement

3

Units

15

Fourth Year

Term 1

LAES 4461

Senior Project I in Liberal Arts and Engineering Studies

3

Engineering Concentration Course

1, 2

2-3

Engineering Concentration Course

1, 2

3

Liberal Arts Minor Course	5, 6, 7, 8	3
---------------------------	------------	---

Liberal Arts Minor Course	5, 6, 7, 8	3
---------------------------	------------	---

Units	14
-------	----

Term 2

LAES 4462	Senior Project II in Liberal Arts and Engineering Studies	3
-----------	---	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Minor Course	5, 6, 7, 8	4
---------------------------	------------	---

General Education Requirement		3
-------------------------------	--	---

Units		16
-------	--	----

Total Units		120
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1

A minimum of 10 units must be taken at the 3000-4000 level.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be required to complete the total units required for the degree.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

A list of Global Perspectives courses is provided on the [LAES website](#). All Global Perspective courses are 3000-4000 level (upper-division).

5

The student must provide written justification for how this minor constitutes a cohesive and effective hybrid integration with the student's other concentration courses selected from the College of Engineering.

6

Students must follow all requirements of an approved minor in one of the Liberal Arts academic program. However, if the Liberal Arts minor requires, for completion, more units beyond the minimum 16 units required for the LAES Liberal Arts Minor concentration, LAES students are not required to take those additional units to complete the concentration as part of the LAES degree.

7

A minimum of 10 units at the 3000-4000 level must be taken toward the requirements for the selected minor.

8

The Credit/No Credit policy for minors will not apply to minors taken to complete this concentration.

Technical and Professional Communication

Technical and Professional Communication Concentration - BS in Liberal Arts and Engineering Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C)	3	4
-----------	---	---	---

MATH 1261	Calculus I (2)	3	4
-----------	----------------	---	---

General Education Requirement (1A)			3
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General Education Requirement (1C)			3
------------------------------------	--	--	---

Units			14
-------	--	--	----

Term 2			
--------	--	--	--

MATH 1262	Calculus II		4
-----------	-------------	--	---

PHYS 1141	General Physics I		4
-----------	-------------------	--	---

General Education Requirement (1B)

3

General Education Requirement

3

Units

14

Second Year

Term 1

MATH 2263

Calculus III

3

PHYS 1143

General Physics II

4

General Education Requirement

3

General Education Requirement	3
-------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	16
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Term 2	
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Engineering Concentration Course ^{1, 2}	3
--	---

Engineering Concentration Course ^{1, 2}	3
--	---

General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

General Education Requirement

3

Units

15

Third Year

Term 1

ENGL 2222

Introduction to Technical and Professional Communication

4

LAES 4301

Liberal Arts and Engineering Studies Capstone I

4

STAT 3210

Engineering Statistics (Upper-Division 2/5)

3

3

Engineering Concentration Course

1, 2

2-3

Liberal Arts Concentration Elective

4

Units

17

Term 2

LAES 4302

Liberal Arts and Engineering Studies Capstone II

4

Engineering Concentration Course

1,2

3

Engineering Concentration Course

1, 2

2-3

Study Abroad or Upper-Division Global Perspectives

2, 4

5

Units

15

Fourth Year

Term 1

LAES 4461 Senior Project I in Liberal Arts and Engineering Studies 3

Engineering Concentration Course 1, 2 3

Engineering Concentration Course 1, 2 3

Liberal Arts Concentration Elective 4

Units 13

Term 2

LAES 4462	Senior Project II in Liberal Arts and Engineering Studies	3
ENGL 4491	Practicum in Technical and Professional Communication	4
Engineering Concentration Course ^{1, 2}		3
Engineering Concentration Course ^{1, 2}		3
General Education Requirement		3
Units		16
Total Units		120

A minimum of 10 units must be taken at the 3000-4000 level.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be required to complete the total units required for the degree.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

A list of Global Perspectives courses is provided on the [LAES website](#). All Global Perspective courses are 3000-4000 level (upper-division).

Liberal Arts Individualized Course of Study

Liberal Arts Individualized Course of Study - BS in Liberal Arts and Engineering Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4
MATH 1261	Calculus I (2) ³	4
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		14

Term 2

MATH 1262	Calculus II	4
-----------	-------------	---

PHYS 1141	General Physics I	4
-----------	-------------------	---

General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	14
-------	----

Second Year

Term 1

MATH 2263	Calculus III	3
-----------	--------------	---

PHYS 1143	General Physics II	4
-----------	--------------------	---

General Education Requirement	3
-------------------------------	---

General Education Requirement	3
-------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	16
-------	----

Term 2

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Individualized Course of Study Elective	5, 6, 7	3
--	---------	---

General Education Requirement		3
-------------------------------	--	---

General Education Requirement		3
-------------------------------	--	---

Units		15
-------	--	----

Third Year		
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Term 1		
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LAES 4301	Liberal Arts and Engineering Studies Capstone I	4
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STAT 3210	Engineering Statistics (Upper-Division 2/5)	3
-----------	---	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Individualized Course of Study Elective	5, 6, 7	3
--	---------	---

Units	16
-------	----

Term 2		
--------	--	--

LAES 4302	Liberal Arts and Engineering Studies Capstone II	4
-----------	--	---

Engineering Concentration Course	1, 2	2-3
----------------------------------	------	-----

Study Abroad or Upper-Division Global Perspectives

2, 4

5

General Education Requirement

3

Units

15

Fourth Year

Term 1

LAES 4461

Senior Project I in Liberal Arts and Engineering Studies

3

Engineering Concentration Course

1, 2

2-3

Engineering Concentration Course

1, 2

3

Liberal Arts Individualized Course of Study Elective	5, 6, 7	3
--	---------	---

Liberal Arts Individualized Course of Study Elective	5, 6, 7	3
--	---------	---

Units		14
-------	--	----

Term 2		
--------	--	--

LAES 4462	Senior Project II in Liberal Arts and Engineering Studies	3
-----------	---	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Engineering Concentration Course	1, 2	3
----------------------------------	------	---

Liberal Arts Individualized Course of Study Elective	5, 6, 7	4
--	---------	---

General Education Requirement		3
-------------------------------	--	---

Units		16
-------	--	----

Total Units		120
-------------	--	-----

1

A minimum of 10 units must be taken at the 3000-4000 level.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be required to complete the total units required for the degree.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

A list of Global Perspectives courses is provided on the [LAES website](#). All Global Perspective courses are 3000-4000 level (upper-division).

5

LAES director approved electives from courses offered by the College of Liberal Arts.

6

A minimum of 10 units must be at the 3000-4000 level.

7

The student must provide written justification for how this collection of courses constitutes a cohesive and effective hybrid integration with the student's other concentration courses selected from the College of Engineering.

Liberal Studies

Concentration Not Yet Declared - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

LS 1201	Orientation to Liberal Studies	1
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LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3
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HIST 2201	United States History to 1877 (4A) ¹	3
-----------	---	---

PSC 1101	Physical Science for Elementary Educators I (5A & 5C) ¹	4
----------	--	---

General Education Requirement (1A)		3
------------------------------------	--	---

Units		14
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Term 2

LS 2230 Field Experience in the Elementary Classroom I 2

HIST 2210 World History to 1800 (4B)¹ 3

PSC 1102 Physical Science for Elementary Educators II 4

STAT 1000 Statistical and Data Literacy (2)¹ 3

or STAT
1110 or Applied Statistical Concepts and Methods

General Education Requirement (1C) 3

Units **15**

Second Year

Term 1

LS 3211	Arts in the Elementary Classroom	3
LS 3214	Constitutional Issues and Social/Historical Foundations of U.S. and California Education	4
PSY 2201	Introductory Psychology	3
General Education Requirement (1B)		3
General Education Requirement (5B)		3
Units		16

Term 2

LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
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LS 3250	Field Experience in the Elementary Classroom II	2
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EDUC 2207	Children's Learning and Development in Educational Settings	3
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MATH 3481	Mathematics for Elementary Teaching I	4
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General Education Requirement		3
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Units		15
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Third Year

Term 1

LS 3301	Liberal Studies Career Preparation	1
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KINE 3310	Concepts and Applications in Elementary Physical Education	3
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MATH 3482	Mathematics for Elementary Teaching II	4
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PHIL 2230	Philosophical Classics: Knowledge and Reality	3
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or PHIL 2231	or Philosophical Classics: Ethics and Political Philosophy	
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Concentration Course or Elective		3
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Units		14
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Term 2

LS 3310	Storytelling: Modern Applications of Traditional Narrative	3
BIO 3210	Biology of Plants and Animals for Future Teachers	4
ENGL 3391	Applied Linguistics	3
MATH 3483	Mathematics for Elementary Teaching III	3
Concentration Course or Elective		3
Units		16

Fourth Year

Term 1

LS 4411	Advanced History/Social Sciences Integration for the K-8 Classroom	3
or LS 4412	or Advanced Arts Integration for the K-8 Classroom	
or LS 4413	or Advanced Science/Engineering Integration for the K-8 Classroom	
EDUC 4427	Culturally Responsive Methods and Inquiry for Teaching Emergent Bilinguals	3
Concentration Course or Elective		3
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

LS 4461 Senior Project Seminar 3

or LS 4462 or Senior Project Research

EDUC 4440 Educating Individuals with Exceptional Needs 3

Concentration Course or Elective 3

General Education Requirement 3

Free Elective² 1-3

Units 15

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Environmental Education

Environmental Education Concentration - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
LS 1201	Orientation to Liberal Studies	1
LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3
HIST 2201	United States History to 1877 (4A) ¹	3
PSC 1101	Physical Science for Elementary Educators I (5A & 5C) ¹	4
General Education Requirement (1A)		3
Units		14
Term 2		

LS 2230	Field Experience in the Elementary Classroom I	2
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HIST 2210	World History to 1800 (4B) ¹	3
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PSC 1102	Physical Science for Elementary Educators II	4
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STAT 1000	Statistical and Data Literacy (2) ¹	3
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or STAT

1110	or Applied Statistical Concepts and Methods	
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General Education Requirement (1C)		3
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Units		15
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Second Year

Term 1

LS 3211	Arts in the Elementary Classroom	3
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LS 3214	Constitutional Issues and Social/Historical Foundations of U.S. and California Education	4
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PSY 2201	Introductory Psychology	3
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General Education Requirement (1B)		3
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General Education Requirement (5B)		3
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Units		16
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Term 2

LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
LS 3250	Field Experience in the Elementary Classroom II	2
EDUC 2207	Children's Learning and Development in Educational Settings	3
MATH 3481	Mathematics for Elementary Teaching I	4
General Education Requirement		3
Units		15

Third Year

Term 1

LS 3301

Liberal Studies Career Preparation

1

KINE 3310

Concepts and Applications in Elementary Physical Education

3

MATH 3482

Mathematics for Elementary Teaching II

4

PHIL 2230

Philosophical Classics: Knowledge and Reality

3

or PHIL 2231

or Philosophical Classics: Ethics and Political Philosophy

Cultural and Social Elective³

3-4

Units

14

Term 2

LS 3310	Storytelling: Modern Applications of Traditional Narrative	3
BIO 3210	Biology of Plants and Animals for Future Teachers	4
ENGL 3391	Applied Linguistics	3
MATH 3483	Mathematics for Elementary Teaching III	3
Ecological and Environmental Elective ³		3-4
Units		16

Fourth Year

Term 1

LS 4411	Advanced History/Social Sciences Integration for the K-8 Classroom	3
or LS 4412	or Advanced Arts Integration for the K-8 Classroom	
or LS 4413	or Advanced Science/Engineering Integration for the K-8 Classroom	
EDUC 4427	Culturally Responsive Methods and Inquiry for Teaching Emergent Bilinguals	3
Education and Communication Elective	4	3
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

LS 4461 Senior Project Seminar 3

or LS 4462 or Senior Project Research

EDUC 4440 Educating Individuals with Exceptional Needs 3

Capstone Elective 3

General Education Requirement 3

Free Elective² 1-3

Units 15

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

Units in excess of total will be applied towards major's free electives.

4

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Approved Elective.

Human Development

Human Development Concentration - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
LS 1201	Orientation to Liberal Studies	1
LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3
HIST 2201	United States History to 1877 (4A) ¹	3

PSC 1101	Physical Science for Elementary Educators I (5A & 5C)	¹	4
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General Education Requirement (1A)			3
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Units			14
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Term 2			
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LS 2230	Field Experience in the Elementary Classroom I		2
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HIST 2210	World History to 1800 (4B)	¹	3
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PSC 1102	Physical Science for Elementary Educators II		4
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STAT 1000	Statistical and Data Literacy (2)	¹	3
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or STAT or Applied Statistical Concepts and Methods
1110

General Education Requirement (1C) 3

Units 15

Second Year

Term 1

LS 3211 Arts in the Elementary Classroom 3

LS 3214 Constitutional Issues and Social/Historical Foundations of
U.S. and California Education 4

PSY 2201 Introductory Psychology 3

General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
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LS 3250	Field Experience in the Elementary Classroom II	2
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EDUC 2207	Children's Learning and Development in Educational Settings	3
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MATH 3481	Mathematics for Elementary Teaching I	4
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General Education Requirement

3

Units

15

Third Year

Term 1

LS 3301

Liberal Studies Career Preparation

1

KINE 3310

Concepts and Applications in Elementary Physical Education

3

MATH 3482

Mathematics for Elementary Teaching II

4

PHIL 2230

Philosophical Classics: Knowledge and Reality

3

or PHIL
2231

or Philosophical Classics: Ethics and Political Philosophy

Child Development and Growth Elective

4

Units

15

Term 2

LS 3310

Storytelling: Modern Applications of Traditional Narrative

3

BIO 3210

Biology of Plants and Animals for Future Teachers

4

ENGL 3391

Applied Linguistics

3

MATH 3483

Mathematics for Elementary Teaching III

3

Applications to Education and Advocacy Elective

3

3-4

Fourth Year

Term 1

LS 4411 Advanced History/Social Sciences Integration for the K-8 Classroom 3

or LS 4412

or Advanced Arts Integration for the K-8 Classroom

or LS 4413

or Advanced Science/Engineering Integration for the K-8 Classroom

EDUC 4427 Culturally Responsive Methods and Inquiry for Teaching Emergent Bilinguals 3

Social Context and Relationship Elective ³

3-4

General Education Requirement 3

General Education Requirement 3

Units 15

Term 2

LS 4461 Senior Project Seminar 3

or LS 4462 or Senior Project Research

EDUC 4440 Educating Individuals with Exceptional Needs 3

Child Development Course 2

General Education Requirement 3

Free Elective	2	1-3
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

Units in excess of total will be applied towards major's free electives.

Mathematics

Mathematics Concentration - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

LS 1201	Orientation to Liberal Studies	1
LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3
HIST 2201	United States History to 1877 (4A) ¹	3

PSC 1101	Physical Science for Elementary Educators I (5A & 5C)	¹	4
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General Education Requirement (1A)			3
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Units			14
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Term 2			
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LS 2230	Field Experience in the Elementary Classroom I		2
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HIST 2210	World History to 1800 (4B)	¹	3
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PSC 1102	Physical Science for Elementary Educators II		4
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STAT 1000	Statistical and Data Literacy (2)	¹	3
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or STAT or Applied Statistical Concepts and Methods
1110

General Education Requirement (1C) 3

Units 15

Second Year

Term 1

LS 3211 Arts in the Elementary Classroom 3

LS 3214 Constitutional Issues and Social/Historical Foundations of
U.S. and California Education 4

PSY 2201 Introductory Psychology 3

General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
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LS 3250	Field Experience in the Elementary Classroom II	2
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EDUC 2207	Children's Learning and Development in Educational Settings	3
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MATH 3481	Mathematics for Elementary Teaching I	4
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General Education Requirement

3

Units

15

Third Year

Term 1

LS 3301

Liberal Studies Career Preparation

1

KINE 3310

Concepts and Applications in Elementary Physical Education

3

MATH 1261

Calculus I

4

MATH 3482

Mathematics for Elementary Teaching II

4

PHIL 2230	Philosophical Classics: Knowledge and Reality	3
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or PHIL 2231	or Philosophical Classics: Ethics and Political Philosophy	
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Units	15
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Term 2

LS 3310	Storytelling: Modern Applications of Traditional Narrative	3
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BIO 3210	Biology of Plants and Animals for Future Teachers	4
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ENGL 3391	Applied Linguistics	3
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MATH 1262	Calculus II	4
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MATH 3483	Mathematics for Elementary Teaching III	3
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Fourth Year

Term 1

LS 4411	Advanced History/Social Sciences Integration for the K-8 Classroom	3
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or LS 4412

or Advanced Arts Integration for the K-8 Classroom

or LS 4413

or Advanced Science/Engineering Integration for the K-8 Classroom

EDUC 4427	Culturally Responsive Methods and Inquiry for Teaching Emergent Bilinguals	3
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MATH 2031	Transition to Advanced Mathematics	3
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General Education Requirement 3

General Education Requirement 3

Units 15

Term 2

LS 4461 Senior Project Seminar 3

or LS 4462 or Senior Project Research

EDUC 4440 Educating Individuals with Exceptional Needs 3

MATH 3971 Technology in Mathematics Education 3

or MATH 3051 or Combinatorics I

or MATH 3055	or Graph Theory	
or MATH 3111	or Number Theory	
	or Euclidean Geometry	
or MATH 3511	or Teaching Statistics: Pedagogy, Content, Technology, and Assessment	
or STAT 3710		
General Education Requirement		3
Free Elective ²		1-3
Units		13
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

English

English Concentration - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

LS 1201	Orientation to Liberal Studies	1
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LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3
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HIST 2201	United States History to 1877 (4A) ¹	3
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PSC 1101	Physical Science for Elementary Educators I (5A & 5C) ¹	4
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General Education Requirement (1A)		3
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Units		14
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Term 2		
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LS 2230	Field Experience in the Elementary Classroom I	2
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HIST 2210	World History to 1800 (4B) ¹	3
PSC 1102	Physical Science for Elementary Educators II	4
STAT 1000	Statistical and Data Literacy (2) ¹	3
or STAT 1110	or Applied Statistical Concepts and Methods	
General Education Requirement (1C)		3
Units		15

Second Year

Term 1

LS 3211	Arts in the Elementary Classroom	3
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LS 3214	Constitutional Issues and Social/Historical Foundations of U.S. and California Education	4
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PSY 2201	Introductory Psychology	3
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General Education Requirement (1B)		3
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General Education Requirement		3
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Units		16
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Term 2		
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LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
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LS 3250	Field Experience in the Elementary Classroom II	2
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EDUC 2207	Children's Learning and Development in Educational Settings	3
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MATH 3481	Mathematics for Elementary Teaching I	4
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General Education Requirement		3
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Units		15
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Third Year		
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Term 1		
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LS 3301	Liberal Studies Career Preparation	1
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ENGL 3311	Writing With Style	3
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KINE 3310	Concepts and Applications in Elementary Physical Education	3
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MATH 3482	Mathematics for Elementary Teaching II	4
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PHIL 2230	Philosophical Classics: Knowledge and Reality	3
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or PHIL 2231	or Philosophical Classics: Ethics and Political Philosophy	
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Units	14
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Term 2

LS 3310	Storytelling: Modern Applications of Traditional Narrative	3
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BIO 3210	Biology of Plants and Animals for Future Teachers	4
ENGL 3391	Applied Linguistics	3
ENGL 3392	Reading Instruction for the Teaching of Young Adult Literature	4
MATH 3483	Mathematics for Elementary Teaching III	3
Units		17

Fourth Year

Term 1

LS 4411	Advanced History/Social Sciences Integration for the K-8 Classroom	3
or LS 4412		

or LS 4413 or Advanced Arts Integration for the K-8 Classroom

or Advanced Science/Engineering Integration for the K-8
Classroom

EDUC 4427 Culturally Responsive Methods and Inquiry for Teaching 3
Emergent Bilinguals

ENGL 3393 Theory and Practice of Tutoring Writing 4

or ENGL or Teaching English Language Arts in Secondary Schools
4424

General Education Requirement 3

General Education Requirement 3

Units 16

Term 2

LS 4461 Senior Project Seminar 3

or LS 4462 or Senior Project Research

EDUC 4440 Educating Individuals with Exceptional Needs 3

English Concentration Approved Elective 3

General Education Requirement 3

Free Elective² 1-3

Units 13

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Science

Science Concentration - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
LS 1201	Orientation to Liberal Studies	1
LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3
HIST 2201	United States History to 1877 (4A) ¹	3
PSC 1101	Physical Science for Elementary Educators I (5A & 5C) ¹	4
General Education Requirement (1A)		3
Units		14
Term 2		

LS 2230	Field Experience in the Elementary Classroom I	2
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HIST 2210	World History to 1800 (4B) ¹	3
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PSC 1102	Physical Science for Elementary Educators II	4
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STAT 1000	Statistical and Data Literacy (2) ¹	3
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or STAT

1110	or Applied Statistical Concepts and Methods	
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General Education Requirement (1C)		3
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Units		15
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Second Year

Term 1

LS 3211 Arts in the Elementary Classroom 3

LS 3214 Constitutional Issues and Social/Historical Foundations of
U.S. and California Education 4

PSY 2201 Introductory Psychology 3

General Education Requirement (1B) 3

General Education Requirement 3

Units 16

Term 2

LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
LS 3250	Field Experience in the Elementary Classroom II	2
EDUC 2207	Children's Learning and Development in Educational Settings	3
MATH 3481	Mathematics for Elementary Teaching I	4
General Education Requirement		3
Units		15

Third Year

Term 1

LS 3301	Liberal Studies Career Preparation	1
KINE 3310	Concepts and Applications in Elementary Physical Education	3
MATH 3482	Mathematics for Elementary Teaching II	4
PHIL 2230	Philosophical Classics: Knowledge and Reality	3
or PHIL 2231	or Philosophical Classics: Ethics and Political Philosophy	
Chemistry or Physics Course ^{3,4}		3-4
Units		14

Term 2

LS 3310	Storytelling: Modern Applications of Traditional Narrative	3
BIO 3210	Biology of Plants and Animals for Future Teachers	4
ENGL 3391	Applied Linguistics	3
MATH 3483	Mathematics for Elementary Teaching III	3
Chemistry or Physics Course ^{3,4}		3-4
Units		17

Fourth Year

Term 1

LS 4411	Advanced History/Social Sciences Integration for the K-8 Classroom	3
or LS 4412	or Advanced Arts Integration for the K-8 Classroom	
or LS 4413	or Advanced Science/Engineering Integration for the K-8 Classroom	
EDUC 4427	Culturally Responsive Methods and Inquiry for Teaching Emergent Bilinguals	3
Science Concentration Approved Elective	4,5	1-3
General Education Requirement		3
General Education Requirement		3
Units		14

Term 2

LS 4461 Senior Project Seminar 3

or LS 4462 or Senior Project Research

EDUC 4440 Educating Individuals with Exceptional Needs 3

Science Concentration Approved Elective^{4,5} 3

General Education Requirement 3

Free Elective² 1-3

Units 15

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

A minimum of 3 units in CHEM is required. A two-course sequence in Chemistry fulfills the authorization requirement; a Physics course is not needed if this sequence is completed.

4

Courses can only be used once for major degree credit.

Units in excess of total will be applied towards major's free electives.

Social Science

Social Science Concentration - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
LS 1201	Orientation to Liberal Studies	1
LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3

HIST 2201	United States History to 1877 (4A) ¹	3
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PSC 1101	Physical Science for Elementary Educators I (5A & 5C) ¹	4
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General Education Requirement (1A)		3
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Units		14
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Term 2

LS 2230	Field Experience in the Elementary Classroom I	2
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HIST 2210	World History to 1800 (4B) ¹	3
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PSC 1102	Physical Science for Elementary Educators II	4
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STAT 1000

Statistical and Data Literacy (2)

1

3

or STAT
1110

or Applied Statistical Concepts and Methods

General Education Requirement (1C)

3

Units

15

Second Year

Term 1

LS 3211

Arts in the Elementary Classroom

3

LS 3214

Constitutional Issues and Social/Historical Foundations of
U.S. and California Education

4

PSY 2201	Introductory Psychology	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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Units	16
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Term 2	
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LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
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LS 3250	Field Experience in the Elementary Classroom II	2
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EDUC 2207	Children's Learning and Development in Educational Settings	3
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MATH 3481

Mathematics for Elementary Teaching I

4

General Education Requirement

3

Units

15

Third Year

Term 1

LS 3301

Liberal Studies Career Preparation

1

KINE 3310

Concepts and Applications in Elementary Physical Education

3

MATH 3482

Mathematics for Elementary Teaching II

4

PHIL 2230	Philosophical Classics: Knowledge and Reality	3
or PHIL 2231	or Philosophical Classics: Ethics and Political Philosophy	

U.S. Government Elective	3
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Units	14
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Term 2

LS 3310	Storytelling: Modern Applications of Traditional Narrative	3
BIO 3210	Biology of Plants and Animals for Future Teachers	4
ENGL 3391	Applied Linguistics	3
HIST 2208	Survey of California History	3

MATH 3483

Mathematics for Elementary Teaching III

3

Units

16

Fourth Year

Term 1

LS 4411

Advanced History/Social Sciences Integration for the K-8 Classroom

3

or LS 4412

or Advanced Arts Integration for the K-8 Classroom

or LS 4413

or Advanced Science/Engineering Integration for the K-8 Classroom

EDUC 4427

Culturally Responsive Methods and Inquiry for Teaching Emergent Bilinguals

3

GEOG 1150	Human Geography	3
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or GEOG 3308	or Global Geography
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Term 2

LS 4461	Senior Project Seminar	3
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or LS 4462	or Senior Project Research
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EDUC 4440	Educating Individuals with Exceptional Needs	3
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Ethnic Studies, History, Sociology, Women, Gender & Queer Studies	3
Elective ³	

General Education Requirement	3
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Free Elective ²	1-3
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Units	15
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Total Units	120
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1

Required in Major or Support: also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

Units in excess of total will be applied towards major's free electives.

Teaching English to Speakers of Other Languages

Teaching English to Speakers of Other Languages Concentration - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

LS 1201

Orientation to Liberal Studies

1

LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3
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HIST 2201	United States History to 1877 (4A) ¹	3
-----------	---	---

PSC 1101	Physical Science for Elementary Educators I (5A & 5C) ¹	4
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General Education Requirement (1A)		3
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Units	14
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Term 2

LS 2230	Field Experience in the Elementary Classroom I	2
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HIST 2210	World History to 1800 (4B) ¹	3
PSC 1102	Physical Science for Elementary Educators II	4
STAT 1000	Statistical and Data Literacy (2) ¹	3
or STAT 1110	or Applied Statistical Concepts and Methods	
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

LS 3211	Arts in the Elementary Classroom	3
LS 3214	Constitutional Issues and Social/Historical Foundations of U.S. and California Education	4
PSY 2201	Introductory Psychology	3
General Education Requirement (1C)		3
General Education Requirement		3
Units		16

Term 2

LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
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LS 3250	Field Experience in the Elementary Classroom II	2
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EDUC 2207	Children's Learning and Development in Educational Settings	3
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MATH 3481	Mathematics for Elementary Teaching I	4
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General Education Requirement		3
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Units		15
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Third Year		
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Term 1		
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LS 3301	Liberal Studies Career Preparation	1
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ENGL 2290	Introduction to Linguistics	3
or ENGL 3390	or The Linguistic Structure of Modern English	
KINE 3310	Concepts and Applications in Elementary Physical Education	3
MATH 3482	Mathematics for Elementary Teaching II	4
PHIL 2230	Philosophical Classics: Knowledge and Reality	3
or PHIL 2231	or Philosophical Classics: Ethics and Political Philosophy	
Units		14

Term 2

LS 3310	Storytelling: Modern Applications of Traditional Narrative	3
BIO 3210	Biology of Plants and Animals for Future Teachers	4
COMS 3316	Intercultural Communication	3
ENGL 3391	Applied Linguistics	3
MATH 3483	Mathematics for Elementary Teaching III	3
Units		16

Fourth Year

Term 1

LS 4411	Advanced History/Social Sciences Integration for the K-8 Classroom	3
or LS 4412	or Advanced Arts Integration for the K-8 Classroom	
or LS 4413	or Advanced Science/Engineering Integration for the K-8 Classroom	
EDUC 4427	Culturally Responsive Methods and Inquiry for Teaching Emergent Bilinguals	3
ENGL 4497	Theories of Language Learning and Teaching	3
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

LS 4461 Senior Project Seminar 3

or LS 4462 or Senior Project Research

EDUC 4440 Educating Individuals with Exceptional Needs 3

ENGL 4498 Approaches to Teaching English to Speakers of Other Languages 3

ENGL 4499 Practicum in Teaching English to Speakers of Other Languages 1

Free Elective² 1-3

General Education Requirement 3

Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Individualized Course of Study

Individualized Course of Study - BS in Liberal Studies

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

LS 1201	Orientation to Liberal Studies	1
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LS 2370	Performing Arts for Children: Theory and Practice (3A) ¹	3
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HIST 2201	United States History to 1877 (4A) ¹	3
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PSC 1101	Physical Science for Elementary Educators I (5A & 5C) ¹	4
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General Education Requirement (1A)		3
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Units		14
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Term 2

LS 2230 Field Experience in the Elementary Classroom I 2

HIST 2210 World History to 1800 (4B)¹ 3

PSC 1102 Physical Science for Elementary Educators II 4

STAT 1000 Statistical and Data Literacy (2)¹ 3

or STAT
1110 or Applied Statistical Concepts and Methods

General Education Requirement (1C) 3

Units **15**

Second Year

Term 1

LS 3211	Arts in the Elementary Classroom	3
LS 3214	Constitutional Issues and Social/Historical Foundations of U.S. and California Education	4
PSY 2201	Introductory Psychology	3
General Education Requirement (1B)		3
General Education Requirement		3
Units		16

Term 2

LS 2255	Children's Literature in a Diverse Society (3B) ¹	3
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LS 3250	Field Experience in the Elementary Classroom II	2
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EDUC 2207	Children's Learning and Development in Educational Settings	3
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MATH 3481	Mathematics for Elementary Teaching I	4
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General Education Requirement		3
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Units		15
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Third Year

Term 1

LS 3301 Liberal Studies Career Preparation 1

KINE 3310 Concepts and Applications in Elementary Physical Education 3

MATH 3482 Mathematics for Elementary Teaching II 4

PHIL 2230 Philosophical Classics: Knowledge and Reality 3

or PHIL 2231 or Philosophical Classics: Ethics and Political Philosophy

Individualized Course of Study Elective 3

Units 14

Term 2

LS 3310	Storytelling: Modern Applications of Traditional Narrative	3
BIO 3210	Biology of Plants and Animals for Future Teachers	4
ENGL 3391	Applied Linguistics	3
MATH 3483	Mathematics for Elementary Teaching III	3
Individualized Course of Study Elective		3
Units		16

Fourth Year

Term 1

LS 4411	Advanced History/Social Sciences Integration for the K-8 Classroom	3
or LS 4412	or Advanced Arts Integration for the K-8 Classroom	
or LS 4413	or Advanced Science/Engineering Integration for the K-8 Classroom	
EDUC 4427	Culturally Responsive Methods and Inquiry for Teaching Emergent Bilinguals	3
Individualized Course of Study Elective		3
General Education Requirement		3
General Education Requirement		3
Units		15

Term 2

LS 4461 Senior Project Seminar 3

or LS 4462 or Senior Project Research

EDUC 4440 Educating Individuals with Exceptional Needs 3

Individualized Course of Study Elective 3

General Education Requirement 3

Free Elective² 1-3

Units 15

Total Units

120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Manufacturing Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

IME 1101	Introduction to Industrial and Manufacturing Engineering	1
IME 1140	Technical Graphics Communication for Design and Manufacturing	3
& IME 1143	and Introduction to Design and Manufacturing	
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
MATH 1261	Calculus I (2) ¹	4
General Education Requirement (1A)		3
Units		15

Term 2

IME 1156	Introduction to Modern Electronics Manufacturing	2
IME 1223	Process Improvement Fundamentals	4
MATH 1262	Calculus II	4
PHYS 1141	General Physics I	4
General Education Requirement (1B)		3
Units		17

Second Year

Term 1

IME 1141	Introduction to Metal Casting and Prototyping	1
IME 1142	Materials Joining	1
ENGR 2211	Introduction to Mechanics	4
MATH 1151	Linear Algebra	3-4
or MATH 2341	or Linear Analysis	
PHYS 1143	General Physics II	4
General Education Requirement (1C)		3
Units		16-17

Term 2

IME 2243	Intermediate Design and Manufacturing	2
IME 2315	Financial Decision Making for Engineers	2
ENGR 2212	Introduction to Engineering Dynamics	2
MATE 1210	Principles of Materials Engineering for Majors	4
& MATE 1215	and Materials Laboratory I	
MATH 2263	Calculus III	3
General Education Requirement (3A)		3
Units		16

Third Year

Term 1

IME 3330	Fundamentals of Manufacturing Engineering	4
CSC 1032	Programming for Scientists and Engineers	3
EE 2115	Circuits & Electronics for Non-Majors	4
& 2115L	and Circuits & Electronics Laboratory for Non-Majors	
STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
General Education Requirement (4A)		3
Units		17

Term 2

IME 3327	Test Design and Analysis in Manufacturing Engineering	4
IME 3356	Manufacturing and Process Automation	4
IME 4450	Computer-Aided Manufacturing and Process Analysis	4
General Education Requirement (5B)		3
Units		15

Fourth Year

Term 1

IME 4418	Product and Process Development	4
IME 4461	Senior Project - Design I	2

Technical Elective	4
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General Education Requirement (3B)	3
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General Education Requirement (4B)	3
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Units	16
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Term 2

IME 4462	Senior Project - Design II	2
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Technical Elective	4
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Technical Elective	1
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General Education Requirement (Upper-Division 3)	3
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General Education Requirement (Upper-Division 4)	3
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General Education Requirement (6)	3
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Units	16
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Total Units	128-129
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Marine Engineering Technology - Division 1 & 2

Division 1 & 2 - BS in Marine Engineering Technology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

EPO 1110	Plant Operations I	1
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EPO 1125	Intro To Marine Engineering	4
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& 1125L	and Intro To Marine Engineering Lab	
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EPO 2213	Welding Lab	1
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ET 1110	Introduction To Engineering Technology	1
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ENG 1100	Engineering Graphics	2
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CHE 1110	General Chemistry	4
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& 1110L	and General Chemistry Laboratory (5A & 5C)	1
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PE 1101	Swim Competency Exam	2	0
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General Education Requirement	3
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Units	16
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Term 2

DL 1105 Marine Survival 2

& 1105L and Marine Survival Laboratory

DL 1105X United States Coast Guard Lifeboatman's Exam 0

EGL 1100 English Composition (1A)¹ 3

FF 1100 Basic Marine Firefighting 0

GOV 2200	American Government (4A) ¹	3
LIB 1100	Information Fluency In The Digital World	2
MTH 2210	Calculus I (2) ¹	4
NAU 1104	Shipboard Security And Responsibility	1
General Education Requirement		3
Units		18

Term 3

CRU 1150	Sea Training I - Engine	8
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Units	8
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Second Year

Term 1

COM 2220L	Programming Applications for Engineering Technology Lab	1
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EPO 2210	Plant Operations Ii	1
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EPO 2214	Boilers	3
EPO 2215	Manufacturing Processes I	1
EPO 2230	Steam Plant System Operations	1
EGL 2220	Critical Thinking (1B) ¹	3
MTH 2211	Calculus II	4
PHY 2200	Engineering Physics I	4
& 2200L	and Engineering Physics I Lab	

Units

18

Term 2

EPO 2220	Diesel Engineering I	2
& 2220L	and Diesel Engineering I Lab	
EPO 2235	Steam Plant Watch Team Management	1
EPO 3312	Turbines	3
ET 2230	Properties Of Materials	2

ET 2232	Statics	3
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EGL 1110	Speech Communication (1C) ¹	3
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PHY 2205	Engineering Physics Ii	4
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& 2205L	and Engineering Physics Ii Lab	
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Units	18
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Term 3

CRU 2250	Sea Training II - Engine	8
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Units

8

Third Year

Term 1

ET 2230L	Properties Of Materials Lab	1
ET 2250	Electrical Circuits	4
& 2250L	and Electrical Circuits Lab	
ET 3330	Dynamics	3

ET 3332	Strength Of Materials	3
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ET 3344	Thermodynamics	3
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General Education Requirement	3
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Units	17
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Term 2

EPO 3310	Plant Operations III	1
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EPO 3315	Manufacturing Processes Ii	1
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EPO 3322	Diesel Engineering Ii/Simulator	2
& 3322L	and Diesel Engineering Ii/Simulator Lab	
ET 3340	Fluid Mechanics	4
& 3340L	and Fluid Mechanics Lab	
ET 3342	Refrigeration And A/C	3
& 3342L	and Refrigeration And A/C Lab	
ET 3370	Electronics	4
& 3370L	and Electronics Lab	

General Education Requirement

3

Units

18

Term 3

CRU 3350

Sea Training III - Engine

8

Units

8

Fourth Year

Term 1

ET 3350	Electrical Machinery	4
& 3350L	and Electrical Machinery Lab	
ET 4400	Inst & Measure Sys	4
& 4400L	and Inst & Measure Sys Lab	
ET 4470	Engineering Management	3
ENG 4430	Naval Architecture	3
General Education Requirement		3

Units

17

Term 2

ENG 3310	Engineering Ethics	3
EPO 2217	Shipboard Medical	1
ET 4460	Automation	4
& 4460L	and Automation Lab	
ET 4490	Power Engineering Tech	4

& 4490L and Power Engineering Tech Lab

FF 3300 Advanced Marine Firefighting 0

General Education Requirement 3

General Education Requirement 3

Units 18

Total Units 164

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Swim assessments are required of all cadets during Orientation week. If [PE 1101](#) is not passed with a grade CR, then enrollment in [PE 1102](#) will be required.

Marine Engineering Technology - Division 3 & 4

Division 3 & 4 - BS in Marine Engineering Technology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
ET 1110	Introduction To Engineering Technology	1
CHE 1110	General Chemistry	4
& 1110L	and General Chemistry Laboratory (5A & 5C)	1
ENG 1100	Engineering Graphics	2
FF 1100	Basic Marine Firefighting	0

GOV 2200	American Government (4A) ¹	3
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PE 1101	Swim Competency Exam ²	0
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General Education Requirement		3
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General Education Requirement		3
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Units		16
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Term 2

EPO 1110	Plant Operations I	1
EPO 1125	Intro To Marine Engineering	4
& 1125L	and Intro To Marine Engineering Lab	
EPO 2213	Welding Lab	1
DL 1105	Marine Survival	2
& 1105L	and Marine Survival Laboratory	
DL 1105X	United States Coast Guard Lifeboatman's Exam	0

EGL 1100	English Composition (1A) ¹	3
LIB 1100	Information Fluency In The Digital World	2
MTH 2210	Calculus I (2) ¹	4
NAU 1104	Shipboard Security And Responsibility	1
Units		18

Term 3

CRU 1150	Sea Training I - Engine	8
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Units	8
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Second Year

Term 1

COM 2220L	Programming Applications for Engineering Technology Lab	1
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EPO 2210	Plant Operations Ii	1
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EPO 2214	Boilers	3
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EPO 2215	Manufacturing Processes I	1
EPO 2230	Steam Plant System Operations	1
EGL 2220	Critical Thinking (1B) ¹	3
MTH 2211	Calculus Ii	4
PHY 2200	Engineering Physics I	4
& 2200L	and Engineering Physics I Lab	
	Units	18

Term 2

EPO 2220	Diesel Engineering I	2
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& 2220L	and Diesel Engineering I Lab	
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EPO 2235	Steam Plant Watch Team Management	1
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EPO 3312	Turbines	3
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ET 2230	Properties Of Materials	2
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ET 2232	Statics	3
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EGL 1110	Speech Communication (1C) ¹	3
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PHY 2205	Engineering Physics Ii	4
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& 2205L	and Engineering Physics Ii Lab	
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Units	18
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Term 3

CRU 2250	Sea Training II - Engine	8
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Units	8
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Third Year

Term 1

ET 2230L	Properties Of Materials Lab	1
ET 2250	Electrical Circuits	4
& 2250L	and Electrical Circuits Lab	
ET 3330	Dynamics	3
ET 3332	Strength Of Materials	3

ET 3344	Thermodynamics	3
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General Education Requirement	3
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Units	17
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Term 2	
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EPO 3310	Plant Operations III	1
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EPO 3315	Manufacturing Processes Ii	1
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EPO 3322	Diesel Engineering Ii/Simulator	2
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& 3322L and Diesel Engineering Ii/Simulator Lab

ET 3340 Fluid Mechanics 4

& 3340L and Fluid Mechanics Lab

ET 3342 Refrigeration And A/C 3

& 3342L and Refrigeration And A/C Lab

ET 3370 Electronics 4

& 3370L and Electronics Lab

General Education Requirement

3

Units

18

Term 3

CRU 3350

Sea Training III - Engine

8

Units

8

Fourth Year

Term 1

ET 3350	Electrical Machinery	4
& 3350L	and Electrical Machinery Lab	
ET 4400	Inst & Measure Sys	4
& 4400L	and Inst & Measure Sys Lab	
ET 4470	Engineering Management	3
ENG 4430	Naval Architecture	3
General Education Requirement		3

Term 2

EPO 2217	Shipboard Medical	1
ET 4460	Automation	4
& 4460L	and Automation Lab	
ET 4490	Power Engineering Tech	4
& 4490L	and Power Engineering Tech Lab	

ENG 3310	Engineering Ethics	3
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FF 3300	Advanced Marine Firefighting	0
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General Education Requirement	3
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General Education Requirement	3
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Units	18
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Total Units	164
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Swim assessments are required of all cadets during Orientation week. If [PE 1101](#) is not passed with a grade CR, then enrollment in [PE 1102](#) will be required.

Marine Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
MSCI 1100	Orientation to Marine Sciences	1
BIO 1150	Life: History and Diversity (5C) ¹	4

CHEM 1120	Fundamentals of Chemical Structure and Properties	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1A)		3
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Units		15
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Term 2		
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MSCI 1111	The Oceans (5B) ¹	3
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MSCI 1112	The Oceans Laboratory	1
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BIO 1151	Life: Molecules and Cells	4
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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General Education Requirement (1C)		3
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Units		15
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Second Year

Term 1

PSC 2201	Physical Oceanography (5A) ¹	3
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PHYS 1121	College Physics I ⁶	4
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or PHYS 1141

or General Physics I

MATH 1261	Calculus I (2)	1,5	4
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or MATH 1264

or Calculus for Data Science I

General Education Requirement (1B)	3
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General Education Requirement	3
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Units	17
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Term 2

MSCI 3300	Marine Ecology	4
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PHYS 1123	College Physics II	6	4
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or PHYS 1143

or General Physics II

General Education Requirement	3
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General Education Requirement	3
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Units	14
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Third Year	
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Term 1	
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MSCI 3301	Biological Oceanography	3
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CHEM 2240	Organic Chemistry: Fundamentals and Applications	4-5
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or CHEM 2242	or Organic Chemistry I
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Marine Science Elective	3,4	4
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General Education Requirement	3
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Units	14
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Term 2	
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CHEM 3370	Marine Chemistry	3
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Marine Science Elective	3,4	3
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Approved Elective	2,7,8	4
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General Education Requirement	3
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Free Elective	9	2
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Units

15

Fourth Year

Term 1

Marine Science Elective^{3,4}

4

Approved Elective^{2,7,8}

4

General Education Requirement

3

General Education Requirement

3

Units

14

Term 2

BIO 4461 Senior Project - Research Proposal 2 2

or BIO 4462

or Senior Project - Research Experience

or BIO 4463

or Senior Project - Meta-analysis in Biology

Marine Science Elective 3,4 4

Approved Elective 2,7,8 4

General Education Requirement 3

Free Elective 9 2-3

Units	16
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If [BIO 4461](#), [BIO 4462](#), [BIO 4463](#), [CHEM 4461](#), [PHYS 4461](#), or [PHYS 4462](#) is used to meet the senior project requirement, it cannot be double-counted as an elective. If a student takes more than one of these courses, the additional units can be applied to electives.

3

Courses taken to meet a Major or Support requirement cannot be double-counted in Approved Electives.

4

Excess units will be applied to Approved Electives.

5

Students emphasizing in Chemistry, Engineering, or Physics should take [MATH 1261](#).

6

Students emphasizing in Physics should take [PHYS 1141](#) and [PHYS 1143](#) instead of [PHYS 1121](#) and [PHYS 1123](#). Note: [MATH 1261](#) is required for [PHYS 1141](#).

7

Maximum of 6 units may be applied toward Approved Electives: [BIO 2200](#), [BIO 3300](#), [BIO 4400](#), [BIO 4450](#), [BIO 4466](#), [BIO 4485](#), [BIO 4495](#), [ENGR 4400](#), [ENVE 4400](#), [GEOL 2200](#), [GEOL 4400](#), [PHYS 4400](#).

8

A maximum of 3 units from [COMS 3390](#), [COMS 3395](#), SCM/[ENGR 3302](#), [MSCI 4401](#).

9

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Marine Transportation - Division 1 & 2

Division 1 & 2 - BS in Marine Transportation

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

DL 1105 Marine Survival 2

& 1105L and Marine Survival Laboratory

DL 1105X United States Coast Guard Lifeboatman's Exam 0

DL 1115 Marlinspike 1

NAU 1103 Marine Transportation 3

NAU 1104 Shipboard Security And Responsibility 1

NAU 1105 Ship Structure 2

NAU 1109	Industrial Equipment And Safety	2
EGL 1100	English Composition (1A) ¹	3
FF 1100	Basic Marine Firefighting	0
MTH 1100	College Algebra And Trigonometry	4
PE 1101	Swim Competency Exam ²	0
Units		18

Term 2

DL 1100	Small Craft Operations	1
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DL 1110	Ship Operations I	1
DL 1120	Cargo Operations	1
NAU 1102	Navigation I	4
& 1102L	and Navigation I Lab	
NAU 1110	Seamanship	3
ECO 1100	Macroeconomics (4B) ¹	3
EGL 1110	Speech Communication (1C) ¹	3
Units		16

Term 3

CRU 1100	Sea Training I - Deck	8
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Units	8
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Second Year

Term 1

DL 2225	Radar / Arpa	4
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& 2225L	and RADAR/ARPA Laboratory	
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NAU 2230	Rules Of The Road	2
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EGL 2220	Critical Thinking (1B) ¹	3
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GOV 2200	American Government (4A) ¹	3
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PHY 1100	General Physics I	4
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& 1100L	and General Physics I Laboratory (5A & 5C) ¹	
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Units	16
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Term 2

NAU 1120	Marine Engineering	3
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NAU 2205	Ship Stability (2) ¹	3
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NAU 3335	Ecdis	3
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& 3335L and Ecdis Lab

ENG 1120L Marine Engineering Lab 1

General Education Requirement (3A) 3

General Education Requirement (6) 3

Units **16**

Term 3

CRU 2200 Sea Training II - Deck 8

& 2200L and Sea Training II Lab (Deck)

Units **8**

Third Year

Term 1

DL 2240	Global Maritime Distress Safety System	3
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& 2240L	and Global Maritime Distress Safety System Lab	
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DL 3301	Navigation Piloting Lab	1
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DL 3310	Marine Supervisory Lab	1
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NAU 3302	Advanced Navigation	3
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& 3302L	and Advanced Navigation Lab	
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NAU 3320	Tank Vessel Operations	3
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NAU 3330	Meteorology (Upper-Division 2/5)	1	3
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FF 3300	Advanced Marine Firefighting		0
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	Units		14
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Term 2

DL 3311	Marine Management Lab		1
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DL 3320	Intro To Bridge Simulator		2
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NAU 3300	Celestial Navigation		4
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& 3300L	and Celestial Navigation Lab		
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NAU 3325	Port & Cargo Operations	3
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General Education Requirement (5B)		3
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General Education Requirement (Upper-Division 3, GWR)		3
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Units		16
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Term 3		
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CRU 3300	Sea Training III - Deck	8
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Units		8
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Fourth Year		
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Term 1		
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DL 4405	Shipboard Medical	2
& 4405L	and Shipboard Medical Lab	
DL 4410	Ship Handling	1
DL 4420	Watchstanding Simulation	2
NAU 4410	License Seminar	2
& 4410L	and License Seminar Lab	
Approved Elective		3
General Education Requirement (3B)		3
Units		13

Term 2

DL 3305	Tug And Barge	1
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NAU 4415	Transportation Security	2
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NAU 4435	Marine Transportation: People, Planet And The Profession	3
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LAW 3315	Admiralty Law (Upper-Division 4)	3
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Approved Elective		3
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Units		12
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Total Units		145
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Required in Major or Support; also satisfies General Education (GE) requirement.

2

Swim assessments are required of all cadets during Orientation week. If [PE 1101](#) is not passed with a grade CR, then enrollment in [PE 1102](#) will be required.

Marine Transportation - Division 3 & 4

Division 3 & 4 - BS in Marine Transportation

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

DL 1105

Marine Survival

2

& 1105L	and Marine Survival Laboratory	
DL 1105X	United States Coast Guard Lifeboatman's Exam	0
DL 1115	Marlinspike	1
NAU 1103	Marine Transportation	3
NAU 1104	Shipboard Security And Responsibility	1
NAU 1105	Ship Structure	2
NAU 1109	Industrial Equipment And Safety	2
EGL 1100	English Composition (1A) ¹	3

MTH 1100	College Algebra And Trigonometry	4
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PE 1101	Swim Competency Exam ²	0
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Units	18
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Term 2

DL 1100	Small Craft Operations	1
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DL 1110	Ship Operations I	1
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DL 1120	Cargo Operations	1
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NAU 1102	Navigation I	4
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& 1102L	and Navigation I Lab	
NAU 1110	Seamanship	3
ECO 1100	Macroeconomics (4B) ¹	3
EGL 1110	Speech Communication (1C) ¹	3
FF 1100	Basic Marine Firefighting	0
Units		16

Term 3

CRU 1100	Sea Training I - Deck	8
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Second Year

Term 1

DL 2225	Radar/Arpa	4
& 2225L	and RADAR/ARPA Laboratory	
NAU 2230	Rules Of The Road	2
EGL 2220	Critical Thinking (1B) ¹	3
GOV 2200	American Government (4A) ¹	3

PHY 1100	General Physics I	4
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& 1100L	and General Physics I Laboratory (5A & 5C)	1
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Units	16
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Term 2

NAU 1120	Marine Engineering	3
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NAU 2205	Ship Stability (2)	3
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NAU 3335	Ecdis	3
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& 3335L	and Ecdis Lab	
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ENG 1120L	Marine Engineering Lab	1
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General Education Requirement (3B)	3
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General Education Requirement (6)	3
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Units	16
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Term 3

CRU 2200	Sea Training II - Deck	8
----------	------------------------	---

& 2200L	and Sea Training Ii Lab (Deck)
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Units	8
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Third Year

Term 1

DL 2240	Global Maritime Distress Safety System	3
& 2240L	and Global Maritime Distress Safety System Lab	
DL 3310	Marine Supervisory Lab	1
NAU 3302	Advanced Navigation	3
& 3302L	and Advanced Navigation Lab	
NAU 3320	Tank Vessel Operations	3
NAU 3330	Meteorology (Upper-Division 2/5) ¹	3
FF 3300	Advanced Marine Firefighting	0

General Education Requirement (Upper-Division 3, GWR)	3
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Units	16
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Term 2	
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DL 3301	Navigation Piloting Lab	1
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DL 3311	Marine Management Lab	1
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DL 3320	Intro To Bridge Simulator	2
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NAU 3300	Celestial Navigation	4
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& 3300L	and Celestial Navigation Lab	
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NAU 3325	Port & Cargo Operations	3
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General Education Requirement (5B)

3

Units

14

Term 3

CRU 3300

Sea Training III - Deck

8

Units

8

Fourth Year

Term 1

DL 3305

Tug And Barge

1

DL 4420

Watchstanding Simulation

2

NAU 4410	License Seminar	2
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& 4410L	and License Seminar Lab	
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NAU 4415	Transportation Security	2
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Approved Elective		3
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General Education Requirement (3A)		3
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Units		13
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Term 2		
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DL 4405	Shipboard Medical	2
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& 4405L	and Shipboard Medical Lab	
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DL 4410	Ship Handling	1
NAU 4435	Marine Transportation: People, Planet And The Profession	3
LAW 3315	Admiralty Law (Upper-Division 4) ¹	3
Approved Elective		3
Units		12
Total Units		145

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Swim assessments are required of all cadets during Orientation week. If [PE 1101](#) is not passed with a grade CR, then enrollment in [PE 1102](#) will be required.

Materials Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
MATE 1110	Introduction to Materials Engineering	1
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ¹	4
IME 1140	Technical Graphics Communication for Design and Manufacturing	1

MATH 1261	Calculus I (2) ¹	4
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		16

Term 2

MATE 1210	Principles of Materials Engineering for Majors	3
MATE 1215	Materials Laboratory I	1
ENGL 1148	Critical Thinking for Technical Writers (1B) ¹	3

IME 1142	Materials Joining	1
MATH 1262	Calculus II	4
PHYS 1141	General Physics I	4
Units		16

Second Year

Term 1

MATE 2225	Materials Microscopy Laboratory	1
MATE 2245	Engineering Analysis and Technical Communication	1
MATE 2280	Materials Thermodynamics and Kinetics	4

MATH 2341	Linear Analysis	4
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PHYS 1143	General Physics II	4
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General Education Requirement (4A)		3
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Units		17
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Term 2		
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MATE 2235	Materials Spectroscopy Laboratory	1
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MATE 3360	Metallurgical Materials Systems	4
-----------	---------------------------------	---

EE 2201	Electric Circuits for Non-Majors	3
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& EE 2241	and Electric Circuit Analysis Laboratory I	
-----------	--	--

ENGR 2211	Introduction to Mechanics	4
-----------	---------------------------	---

STAT 3210	Engineering Statistics (Upper-Division 2/5) ¹	3
-----------	--	---

Units	15
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Third Year

Term 1

MATE 3340	Electronic Materials Systems	4
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MATE 3480	Composite Material Systems	4
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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or CHEM 2240

or Organic Chemistry: Fundamentals and Applications

IME 3326 Engineering Statistics and Six Sigma 3-4

or ME 3234 or Design Thinking and Creativity

General Education Requirement (3A) 3

Units 18-19

Term 2

MATE 3232 Materials Ethics, Diversity, and Society (Upper-Division 3
4) 1

MATE 3310 Polymeric Material Systems 4

MATE 3401	Advanced Materials Characterization	1
MATE 3403	Computational Materials Engineering	2
General Education Requirement (4B)		3
General Education Requirement (5B)		3
Units		16

Fourth Year		
Term 1		

MATE 4300	Materials Selection for the Life Cycle	3
MATE 4422	Ceramic and Glass Materials Systems	3

MATE 4461	Senior Project I	1
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Technical Elective	2, 3	3
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General Education Requirement (3B)		3
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Units		13
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Term 2

MATE 4462	Senior Project II	2
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Technical Elective	2, 3	3
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Professional Development Elective	2, 3	3-4
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General Education Requirement (6)	3
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General Education Requirement (Upper-Division 3)	3
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Units	14-15
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Total Units	125-127
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet this requirement, it cannot be double-counted to satisfy another Major or Support requirement.

3

Maximum of 8 units of [MATE 4400](#) may count towards Technical Electives or Professional Development Electives.

Mathematics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
MATH 1261	Calculus I (2) ¹	4
CSC 1001	Fundamentals of Computer Science	4
& 1001L	and Fundamentals of Computer Science Laboratory	
PHYS 1141	General Physics I (5A & 5C) ¹	4
General Education Requirement (1A)		3

Units

15

Term 2

MATH 1151	Linear Algebra	3
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MATH 1262	Calculus II	4
-----------	-------------	---

CSC 2001	Data Structures	3-4
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& 2001L	or Computing with Data	
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or CSC 2600	or Mathematical Programming	
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or MATH 3681	or Computational Physics	
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or PHYS 4202	or Introduction to Probability and Simulation	
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or STAT 2610		
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General Education Requirement (1C)

3

Units

13

Second Year

Term 1

MATH 2001

Mathematics Orientation

1

MATH 2031

Transition to Advanced Mathematics

3

MATH 2263

Calculus III

3

STAT 1510

Statistics I

3

General Education Requirement (1B)

3

General Education Requirement	3
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Units	16
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Term 2	
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MATH 2343	Differential Equations	3
-----------	------------------------	---

MATH 3152	Advanced Linear Algebra	4
-----------	-------------------------	---

General Education Requirement	3
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General Education Requirement	3
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Free Elective	6	3
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Third Year

Term 1

MATH 3051	Combinatorics I (Upper-Division 2/5) ¹	3
or MATH 3111	or Number Theory	
or MATH 3301	or Complex Analysis	
MATH 4201	Abstract Algebra I	4
Track Elective ^{2, 3, 4, 5}		3
General Education Requirement		3

Free Elective	6	3
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Units	16
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Term 2

MATH 4202	Abstract Algebra II	3
-----------	---------------------	---

Track Elective	2, 3, 4, 5	3
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Track Elective	2, 3, 4, 5	3
----------------	------------	---

General Education Requirement	3
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General Education Requirement	3
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Units

15

Fourth Year

Term 1

MATH 4264

Real Analysis I

4

MATH 4463

Senior Project Seminar

3

or MATH 4464

or Senior Project Applied Seminar

Track Elective 2, 3, 4, 5

3

General Education Requirement

3

Units

13

Term 2

Track Elective 2, 3, 4, 5

3

Track Elective 2, 3, 4, 5

3

Track Elective 2, 3, 4, 5

3-4

General Education Requirement

3

Free Elective 6

0-6

Units

16

1

Required in Major or Support; also satisfied General Education (GE) requirement.

2

A maximum of 14 units may be at the 1000-2000-3000 level.

3

A maximum of 4 units may be at the 1000-2000 level.

4

A maximum of 8 units may be from non-MATH prefix courses.

5

Courses can only be used once for major degree credit.

6

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

General Curriculum in Mechanical Engineering

General Curriculum in Mechanical Engineering - BS in Mechanical Engineering (San Luis Obispo Campus)

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ME 1125	Introduction to Mechanical Engineering ¹	1
ME 1148	Engineering Design Communication ²	2
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4

IME 1143	Introduction to Design and Manufacturing	2	2
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MATH 1261	Calculus I (2)	3	4
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General Education Requirement (1A)			3
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Units			16
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Term 2			
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IME 1141	Introduction to Metal Casting and Prototyping		1
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or IME 1142	or Materials Joining		
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or IME 1149	or Introduction to Manufacturing Processes: Metal Casting and Joining		
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MATH 1262	Calculus II	4
-----------	-------------	---

PHYS 1141	General Physics I	4
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General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement (1C)	3
------------------------------------	---

Units	15
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Second Year

Term 1

ME 2240	Applied Programming for Mechanical Engineering	1
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ME 2248	Design Using Solid Modeling	1
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
PHYS 1143	General Physics II	4
General Education Requirement (3A)		3
Units		16

Term 2		
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ME 2212	Engineering Dynamics	3
---------	----------------------	---

EE 2115	Circuits & Electronics for Non-Majors	4
& 2115L	and Circuits & Electronics Laboratory for Non-Majors	
MATE 1220	Principles of Materials Engineering for Non-Majors	3
& MATE 1215	and Materials Laboratory I	
MATH 2341	Linear Analysis	4
General Education Requirement (4A)		3
Units		17

Third Year

Term 1

ME 3234	Design Thinking and Creativity (Upper-Division 4) ³	3
ME 3236	Engineering Measurement and Data Analysis (Upper-Division 2/5) ³	3
ME 3302	Thermodynamics	3
ME 3328	Design for Strength and Stiffness	4
General Education Requirement (3B)		3
Units		16

Term 2

ME 3317	Vibrations and System Modeling	4
---------	--------------------------------	---

ME 3329	Mechanical Systems Design	3
ME 3341	Fluid Mechanics	4
& ME 3342	and Fluid Mechanics Laboratory	4
BIO 1111	General Biology (5B)	3
or BIO 2213	or Life Science for Engineers	
or BIO 2215	or Biodiversity of California	
or BIO 2217	or Wildlife Conservation Biology	
General Education Requirement (4B)		3
Units		17

Fourth Year

Term 1

ME 3343	Heat Transfer	4
ME 4417	Mechanical Controls and Implementations	3
ME 4460	Senior Design Project I	2
Technical Elective	6, 7, 8	3
Technical Elective	6, 7, 8	3
General Education Requirement (6)		3

Units

18

Term 2

ME 4440

Thermal System Design

3

ME 4461

Senior Design Project II⁵

2

Technical Elective^{6, 7, 8}

3

Technical Elective^{6, 7, 8}

2-3

General Education Requirement (Upper-Division 3)

3

Units

13-14

Total Units

128-12

9

1

Transfer students and change of major students take [ME 2225](#) in lieu of [ME 1125](#).

2

[ME 1148](#) and [IME 1143](#) intended to be taken concurrently in the same term.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

[ME 3341](#) and [ME 3342](#) intended to be taken concurrently in the same term.

5

ENGR 4460 and ENGR 4461 (4), or ENGR 4463 and ENGR 4464 (4) may substitute for ME 4460 and ME 4461 (4).

6

Consultation with advisor is recommended prior to selecting Technical Electives. 3000 level Technical Electives cannot be used for graduate credit in the blended BS+MS Mechanical Engineering program.

7

ME 4470, ME 4471, ME 5570, and ME 5571 are variable topic courses and may or may not count as ME Technical Electives. Contact the instructor for additional information. A course substitution form may be required.

8

ME 4400 and ME 5500 are independent study courses and may be acceptable for Technical Elective credit. A course substitution form is required.

Energy Resources

Energy Resources Concentration - BS in Mechanical Engineering (San Luis Obispo Campus)

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
ME 1125	Introduction to Mechanical Engineering ¹	1
ME 1148	Engineering Design Communication ²	2
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4

IME 1143	Introduction to Design and Manufacturing	2	2
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MATH 1261	Calculus I (2)	3	4
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General Education Requirement (1A)			3
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Units			16
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Term 2			
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IME 1141	Introduction to Metal Casting and Prototyping		1
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or IME 1142	or Materials Joining		
-------------	----------------------	--	--

or IME 1149	or Introduction to Manufacturing Processes: Metal Casting and Joining		
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MATH 1262	Calculus II	4
-----------	-------------	---

PHYS 1141	General Physics I	4
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General Education Requirement (1B)	3
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General Education Requirement (1C)	3
------------------------------------	---

Units	15
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Second Year

Term 1

ME 2240	Applied Programming for Mechanical Engineering	1
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ME 2248	Design Using Solid Modeling	1
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
PHYS 1143	General Physics II	4
General Education Requirement (3A)		3
Units		16

Term 2		
--------	--	--

ME 2212	Engineering Dynamics	3
---------	----------------------	---

EE 2115	Circuits & Electronics for Non-Majors	4
& 2115L	and Circuits & Electronics Laboratory for Non-Majors	
MATE 1220	Principles of Materials Engineering for Non-Majors	3
& MATE 1215	and Materials Laboratory I	
MATH 2341	Linear Analysis	4
General Education Requirement (4A)		3
Units		17

Third Year

Term 1

ME 3234	Design Thinking and Creativity (Upper-Division 4) ³	3
ME 3236	Engineering Measurement and Data Analysis (Upper-Division 2/5) ³	3
ME 3302	Thermodynamics	3
ME 3328	Design for Strength and Stiffness	4
General Education Requirement (3B)		3
Units		16

Term 2

ME 3315	Energy Conversion	3
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ME 3329	Mechanical Systems Design	3
ME 3341	Fluid Mechanics	4
& ME 3342	and Fluid Mechanics Laboratory	4
BIO 1111	General Biology (5B)	3
or BIO 2213	or Life Science for Engineers	
or BIO 2215	or Biodiversity of California	
or BIO 2217	or Wildlife Conservation Biology	
General Education Requirement (4B)		3
Units		16

Fourth Year

Term 1

ME 3317 Vibrations and System Modeling 4

ME 3343 Heat Transfer 4

ME 4460 Senior Design Project I 2

Technical Elective 4

General Education Requirement (6) 3

Units 17

Term 2

ME 4417	Mechanical Controls and Implementations	3
ME 4440	Thermal System Design	3
ME 4461	Senior Design Project II	2
Technical Elective		4-5
General Education Requirement (Upper-Division 3)		3
Units		15-16
Total Units		128-129

1

Transfer students and change of major students take [ME 2225](#) in lieu of [ME 1125](#).

2

ME 1148 and IME 1143 intended to be taken concurrently in the same term.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

ME 3341 and ME 3342 intended to be taken concurrently in the same term.

Sustainable Technology for the Built Environment (HVAC & R)

***Sustainable Technology for the Built Environment (HVAC & R) Concentration - BS
in Mechanical Engineering (San Luis Obispo Campus)***

*Suggested Four-Year Flowcharts are planning guides and do not represent a required
course sequence.*

*Courses may be completed in a different order, provided requisite requirements are
satisfied.*

First Year

Term 1

Units

ME 1125	Introduction to Mechanical Engineering ¹	1
ME 1148	Engineering Design Communication ²	2
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4
IME 1143	Introduction to Design and Manufacturing ²	2
MATH 1261	Calculus I (2) ³	4

General Education Requirement (1A)	3
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Units	16
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Term 2	
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IME 1141	Introduction to Metal Casting and Prototyping	1
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or IME 1142	or Materials Joining
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or IME 1149	or Introduction to Manufacturing Processes: Metal Casting and Joining
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MATH 1262	Calculus II	4
-----------	-------------	---

PHYS 1141	General Physics I	4
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General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement (1C)

3

Units

15

Second Year

Term 1

ME 2240

Applied Programming for Mechanical Engineering

1

ME 2248

Design Using Solid Modeling

1

ENGR 2211

Introduction to Mechanics

4

MATH 2263

Calculus III

3

PHYS 1143	General Physics II	4
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General Education Requirement (3A)	3
------------------------------------	---

Units	16
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Term 2

ME 2212	Engineering Dynamics	3
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EE 2115	Circuits & Electronics for Non-Majors	4
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& 2115L	and Circuits & Electronics Laboratory for Non-Majors	
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MATE 1220	Principles of Materials Engineering for Non-Majors	3
-----------	--	---

& MATE 1215	and Materials Laboratory I	
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MATH 2341	Linear Analysis	4
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General Education Requirement (4A)	3
------------------------------------	---

Units	17
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Third Year

Term 1

ME 3234	Design Thinking and Creativity (Upper-Division 4) ³	3
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ME 3236	Engineering Measurement and Data Analysis (Upper-Division 2/5) ³	3
---------	--	---

ME 3302	Thermodynamics	3
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ME 3328	Design for Strength and Stiffness	4
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General Education Requirement (3B)		3
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Units		16
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Term 2		
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ME 3329	Mechanical Systems Design	3
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ME 3341	Fluid Mechanics	4
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& ME 3342	and Fluid Mechanics Laboratory	4
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ME 3355	Introduction to Sustainable Energy Usage in Buildings	3
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BIO 1111	General Biology (5B)	3	3
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or BIO 2213

or Life Science for Engineers

or BIO 2215

or Biodiversity of California

or BIO 2217

or Wildlife Conservation Biology

General Education Requirement (4B)			3
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Units			16
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Fourth Year

Term 1

ME 3317	Vibrations and System Modeling		4
---------	--------------------------------	--	---

ME 3343	Heat Transfer	4
ME 4456	HVAC&R Air and Water Distribution for Sustainable Building Environments	3
ME 4457	Environmentally Efficient and Sustainable Refrigeration Systems	3
ME 4465	HVAC&R Senior Design Project I	2
Units		16

Term 2

ME 4417	Mechanical Controls and Implementations	3
ME 4440	Thermal System Design	3

ME 4455	Building Energy Performance and Modeling	3
ME 4466	HVAC&R Senior Design Project II	2
General Education Requirement (Upper-Division 3)		3
General Education Requirement (6)		3
Units		17
Total Units		129

1

Transfer students and change of major students take [ME 2225](#) in lieu of [ME 1125](#).

2

ME 1148 and IME 1143 intended to be taken concurrently in the same term.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

ME 3341 and ME 3342 intended to be taken concurrently in the same term.

Mechatronics

Mechatronics Concentration - BS in Mechanical Engineering (San Luis Obispo Campus)

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
ME 1125	Introduction to Mechanical Engineering ¹	1
ME 1148	Engineering Design Communication ²	2
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4
IME 1143	Introduction to Design and Manufacturing ²	2
MATH 1261	Calculus I (2) ³	4
General Education Requirement (1A)		3

Units

16

Term 2

IME 1141	Introduction to Metal Casting and Prototyping	1
or IME 1142	or Materials Joining	
or IME 1149	or Introduction to Manufacturing Processes: Metal Casting and Joining	
MATH 1262	Calculus II	4
PHYS 1141	General Physics I	4
General Education Requirement (1B)		3
General Education Requirement (1C)		3

Units

15

Second Year

Term 1

ME 2240	Applied Programming for Mechanical Engineering	1
ME 2248	Design Using Solid Modeling	1
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
PHYS 1143	General Physics II	4

General Education Requirement (3A)	3
------------------------------------	---

Units	16
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Term 2	
--------	--

ME 2212	Engineering Dynamics	3
---------	----------------------	---

EE 2115	Circuits & Electronics for Non-Majors	4
---------	---------------------------------------	---

& 2115L	and Circuits & Electronics Laboratory for Non-Majors	
---------	--	--

MATE 1220	Principles of Materials Engineering for Non-Majors	3
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& MATE 1215	and Materials Laboratory I	
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MATH 2341	Linear Analysis	4
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General Education Requirement (4A)

3

Units

17

Third Year

Term 1

ME 3234

Design Thinking and Creativity (Upper-Division 4)

3

3

ME 3236

Engineering Measurement and Data Analysis

(Upper-Division 2/5)

3

3

ME 3302

Thermodynamics

3

ME 3328

Design for Strength and Stiffness

4

General Education Requirement (3B)	3
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Units	16
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Term 2	
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ME 3305	Mechatronics I	4
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ME 3329	Mechanical Systems Design	3
---------	---------------------------	---

ME 3341	Fluid Mechanics	4
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& ME 3342	and Fluid Mechanics Laboratory	4
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BIO 1111	General Biology (5B)	3
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or BIO 2213		
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or BIO 2215 or Life Science for Engineers

or BIO 2217 or Biodiversity of California

or Wildlife Conservation Biology

General Education Requirement (4B) 3

Units 17

Fourth Year

Term 1

ME 3319 Introduction to System Dynamics 4

ME 3343 Heat Transfer 4

ME 4305	Mechatronics II	3
ME 4460	Senior Design Project I	2
General Education Requirement (6)		3
Units		16

Term 2

ME 4419	Advanced Control Systems	3
ME 4440	Thermal System Design	3
ME 4461	Senior Design Project II	2

Technical Elective	3-4
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Technical Elective	0-2
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General Education Requirement (Upper-Division 3)	3
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Units	15-16
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Total Units	128-129
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1

Transfer students and change of major students take [ME 2225](#) in lieu of [ME 1125](#).

2

[ME 1148](#) and [IME 1143](#) intended to be taken concurrently in the same term.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

ME 3341 and ME 3342 intended to be taken concurrently in the same term.

Manufacturing

Manufacturing Concentration - BS in Mechanical Engineering (San Luis Obispo Campus)

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

ME 1125	Introduction to Mechanical Engineering ¹	1
ME 1148	Engineering Design Communication ²	2
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A & 5C) ³	4
IME 1143	Introduction to Design and Manufacturing ²	2
MATH 1261	Calculus I (2) ³	4
General Education Requirement (1A)		3
Units		16

Term 2

IME 1141	Introduction to Metal Casting and Prototyping	1
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or IME 1142	or Materials Joining	
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or IME 1149	or Introduction to Manufacturing Processes: Metal Casting and Joining	
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MATH 1262	Calculus II	4
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PHYS 1141	General Physics I	4
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General Education Requirement (1B)		3
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General Education Requirement (1C)		3
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Units		15
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Second Year

Term 1

ME 2240	Applied Programming for Mechanical Engineering	1
ME 2248	Design Using Solid Modeling	1
ENGR 2211	Introduction to Mechanics	4
MATH 2263	Calculus III	3
PHYS 1143	General Physics II	4
General Education Requirement (3A)		3

Units**16****Term 2**

ME 2212	Engineering Dynamics	3
EE 2115	Circuits & Electronics for Non-Majors	4
& 2115L	and Circuits & Electronics Laboratory for Non-Majors	
MATE 1220	Principles of Materials Engineering for Non-Majors	3
& MATE 1215	and Materials Laboratory I	
MATH 2341	Linear Analysis	4
General Education Requirement (4A)		3

Third Year

Term 1

ME 3234	Design Thinking and Creativity (Upper-Division 4) ³	3
ME 3236	Engineering Measurement and Data Analysis (Upper-Division 2/5) ³	3
ME 3302	Thermodynamics	3
ME 3328	Design for Strength and Stiffness	4
General Education Requirement (3B)		3

Units

16

Term 2

ME 3317	Vibrations and System Modeling	4
ME 3329	Mechanical Systems Design	3
ME 3341	Fluid Mechanics	4
& ME 3342	and Fluid Mechanics Laboratory	4
BIO 1111	General Biology (5B)	3
or BIO 2213	or Life Science for Engineers	
or BIO 2215	or Biodiversity of California	

or BIO 2217 or Wildlife Conservation Biology

General Education Requirement (4B) 3

Units 17

Fourth Year

Term 1

ME 3343 Heat Transfer 4

ME 4417 Mechanical Controls and Implementations 3

ME 4460 Senior Design Project I 2

IME 3327 Test Design and Analysis in Manufacturing Engineering 4

General Education Requirement (6)	3
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Units	16
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Term 2	
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ME 4440	Thermal System Design	3
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ME 4461	Senior Design Project II	2
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IME 3330	Fundamentals of Manufacturing Engineering	4
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Manufacturing Elective	3-4
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General Education Requirement (Upper-Division 3)	3
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Units 15-16

Total Units 128-12

9

1

Transfer students and change of major students take [ME 2225](#) in lieu of [ME 1125](#).

2

[ME 1148](#) and [IME 1143](#) intended to be taken concurrently in the same term.

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

[ME 3341](#) and [ME 3342](#) intended to be taken concurrently in the same term.

Microbiology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

BIO 1150

Life: History and Diversity (5B & 5C) ¹

4

MCRO 1100

Introduction to Microbiology Research

2

CHEM 1120

Fundamentals of Chemical Structure and Properties (5A)
¹

4

STAT 1110

Applied Statistical Concepts and Methods

3

General Education Requirement (1A)

3

Units

16

Term 2

BIO 1151	Life: Molecules and Cells	4
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CHEM 1122	Fundamentals of Chemical Reactivity	4
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MATH/DATA 1264	Calculus for Data Science I (2) ¹	4
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General Education Requirement (1C)	3
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Units	15
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Second Year

Term 1

MCRO 2224	General Microbiology I	4
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CHEM 2240	Organic Chemistry: Fundamentals and Applications	4-5
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or CHEM 2242	or Organic Chemistry I	
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General Education Requirement (1B)		3
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General Education Requirement		3
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Units		14
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Term 2		
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MCRO 2227	General Microbiology II	4
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PHYS 1121	College Physics I	4
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Approved Elective	3, 4, 5, 6, 7	4
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General Education Requirement		3
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Units		15
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Third Year

Term 1

MCRO 3351	Microbial Genetics	3
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MCRO 3352	Microbial Genetics Laboratory	2
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CHEM 3350	Biochemistry: Fundamentals and Applications	4
	(Upper-Division 2/5)	
or CHEM 3352		

1, 8

or Biochemistry

Approved Elective	3, 4, 5, 6, 7	4
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General Education Requirement		3
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Units		16
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Term 2

Approved Elective	3, 4, 5, 6, 7	4
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Restricted Elective	2	4
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Restricted Elective	2	4
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General Education Requirement

3

Units

15

Fourth Year

Term 1

Approved Elective^{3, 4, 5, 6, 7}

3

Restricted Elective²

4

General Education Requirement

3

Free Elective⁹

3-4

Units

14

Term 2

BIO 4461

Senior Project - Research Proposal

2

or BIO 4462

or Senior Project - Research Experience

or BIO 4463

or Senior Project - Meta-analysis in Biology

Restricted Elective²

3

General Education Requirement

3

General Education Requirement

3

Free Elective⁹

4

Units	15
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Total Units	120
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1

Required in Major or Support; also satisfied General Education (GE) requirement.

2

Excess units will be applied to Approved Electives

3

May be substituted with an advisor approved course.

4

Consultation with advisor is recommended prior to selecting electives; bear in mind your selections may impact pursuit of post-baccalaureate studies and/or goals.

5

Students planning to attend graduate or professional schools are strongly advised to meet with their advisors to ensure they meet the prerequisites for entry into these programs. Additional classes in math and chemistry may be necessary.

6

Maximum of 6 units may be applied toward Approved Electives: [BIO 2200](#), [BIO 3300](#), [BIO 4400](#), [BIO 4450](#), [BIO 4466](#), [BIO 4485](#), [BIO 4495](#), or [MSCI 4401](#).

7

If [BIO 4462](#) or [BIO 4463](#) is used to meet the senior project requirement, it cannot also be counted as an Approved Elective.

8

[CHEM 3352](#) is suggested for students who plan to pursue graduate school or a health professions career.

9

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Music

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

MU 1100 Introduction to Music Studies 1

MU 1101 Music Fundamentals 3-4

or MU 1103 or Materials and Structures of Music I

MU 1104 Musicianship I 2

MU 1150 Applied Music I 1

Major Ensemble¹

1

General Education Requirement (1A)

3

General Education Requirement (1C)

3

Free Elective (Recommended MU 1161)^{4, 5}

1

Units

15

Term 2

MU 1103	Materials and Structures of Music I	4
or MU 2203	or Materials and Structures of Music II	
MU 1122	Ethnomusicology and World Music I	4
MU 1150	Applied Music I	1
MU 1106	Musicianship II	2
Major Ensemble ¹		1
General Education Requirement (1B)		3

Free Elective (Recommended MU 1162)^{4, 5}

1

Units

16

Second Year

Term 1

MU 2203

Materials and Structures of Music II

4

MU 2222

Ethnomusicology and World Music II

4

MU 2250	Applied Music II	1
Major Ensemble	¹	1
General Education Requirement (2)		3
General Education Requirement		3
Free Elective	^{4, 5}	1
Units		17

Term 2

MU 2250	Applied Music II	1
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MU 2221	Jazz Styles ²	3-4
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or MU 2227

or Popular Music of the United States

Major Ensemble ¹	1
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General Education Requirement	3
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General Education Requirement	1
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General Education Requirement	3
-------------------------------	---

General Education Requirement	3
-------------------------------	---

Free Elective ^{4, 5}	0-1
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Units	16
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Third Year	
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Term 1

MU 3311	Introduction to Music Technology and Composition	4
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MU 3350	Applied Music III	1
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Major Ensemble	¹	1
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General Education Requirement		3
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General Education Requirement		3
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Units

12

Term 2

MU 3331

Historical Musicology I

4

MU 3350

Applied Music III

1

Major Ensemble¹

1

MU Elective 3000-4000 level³

4

General Education Requirement	3
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Free Elective	4, 5	3
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Units	16
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Fourth Year

Term 1

MU 4431	Historical Musicology II	4
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MU Elective 3000-4000 level	3	4
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General Education Requirement		3
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Free Elective	4, 5	4
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Units		15
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Term 2		
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MU 4461	Senior Project	2
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MU Elective 3000-4000 level	3	4
General Education Requirement		3
Free Elective	4, 5	4
Units		13
Total Units		120

1

At least two units must be upper-division to complete the Major Ensemble requirement.

2

Required in Major or Support; also satisfies General Education (GE) requirement.

3

No more than 2 units of music electives may be satisfied by the following courses: [MU 3368](#), [MU 3370](#), [MU 3371](#), [MU 3372](#), [MU 3373](#), [MU 3374](#), [MU 3375](#), [MU 3377](#), [MU 3378](#), [MU 3380](#), [MU 3381](#), [MU 3382](#), [MU 3383](#), [MU 3384](#), [MU 3387](#), [MU 3388](#), or [MU 3389](#).

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Concentration Not Yet Declared - BS in Nutrition

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
NUTR 1101	Orientation to the Nutrition Major	1
FSN 2202	Introduction to Human Nutrition	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
General Education Requirement (1A)		3

General Education Requirement (1C)	3
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Units	14
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Term 2	
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BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
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FSN 1121	Fundamentals of Food	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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Units

16

Second Year

Term 1

NUTR 3315	Nutrition in Aging	2
CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
MCRO 2224	General Microbiology I	4
or MCRO 2221	or Introduction to Microbiology	
General Education Requirement		3
General Education Requirement		3

Units

16

Term 2

NUTR 3310 Maternal and Child Nutrition 3

FSN 2250 Food and Nutrition: Culture and Customs (USCP) 3

BIO 2231 Human Anatomy and Physiology I 4

CHEM 3350 Biochemistry: Fundamentals and Applications 4
(Upper-Division 2/5) ¹

Units

14

Third Year

Term 1

BIO 2232	Human Anatomy and Physiology II	4
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NUTR 3331	Macronutrient Metabolism	3
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Concentration Course		3
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Concentration Course		3
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General Education Requirement		3
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Units		16
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Term 2

NUTR 3332	Micronutrient Metabolism	3
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NUTR 3333	Nutrient Metabolism Lab	1
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Concentration Course		3
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General Education Requirement		3
-------------------------------	--	---

Free Elective ²		3
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Units		13
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Fourth Year

Term 1

NUTR 4415	Nutrition Counseling & Education	3
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NUTR 4431	Medical Nutrition Therapy I	4
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Concentration Course		3
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General Education Requirement		3
-------------------------------	--	---

Free Elective ²		3
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Units		16
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Term 2		
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NUTR 4416	Community Nutrition	3
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NUTR 4432	Medical Nutrition Therapy II	4
-----------	------------------------------	---

NUTR 4460	Senior Project - Scientific and Technical Writing	2
or NUTR 4461	or Senior Project - Research	
or NUTR 4462	or Senior Project - Internship	
Concentration Course		3
General Education Requirement		3
	Units	15
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Dietetics

Dietetics Concentration - BS in Nutrition

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NUTR 1101	Orientation to the Nutrition Major	1
FSN 2202	Introduction to Human Nutrition	3

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A)	4
	1	

General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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Units	14
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Term 2

FSN 1121	Fundamentals of Food	3
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FSN 2250	Food and Nutrition: Culture and Customs (USCP)	3
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BIO 1151	Life: Molecules and Cells (5B & 5C)	4
	1	

STAT 1110	Applied Statistical Concepts and Methods (2)	1	3
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General Education Requirement (1B)			3
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Units			16
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Second Year			
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Term 1			
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NUTR 3321	Experimental Foods		3
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CHEM 2240	Organic Chemistry: Fundamentals and Applications		4
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MCRO 2224	General Microbiology I		4
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or MCRO 2221 or Introduction to Microbiology

General Education Requirement 3

Units 14

Term 2

NUTR 3315 Nutrition in Aging 2

CHEM 3350 Biochemistry: Fundamentals and Applications 4
(Upper-Division 2/5)¹

PSY 2201 Introductory Psychology 3

General Education Requirement 3

General Education Requirement

3

Units

15

Third Year

Term 1

NUTR 3310

Maternal and Child Nutrition

3

BIO 2231

Human Anatomy and Physiology I

4

General Education Requirement

3

General Education Requirement

3

General Education Requirement	3
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Units	16
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Term 2	
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FSN 3346	Institutional Food Service	3
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NUTR 3331	Macronutrient Metabolism	3
-----------	--------------------------	---

BIO 2232	Human Anatomy and Physiology II	4
----------	---------------------------------	---

Approved Elective	3
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General Education Requirement	3
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Fourth Year

Term 1

NUTR 3332	Micronutrient Metabolism	3
NUTR 3333	Nutrient Metabolism Lab	1
NUTR 4415	Nutrition Counseling & Education	3
NUTR 4426	Nutrition and Foodservice Systems Management	3
NUTR 4431	Medical Nutrition Therapy I	4

Units**14****Term 2**

NUTR 4416 Community Nutrition 3

NUTR 4432 Medical Nutrition Therapy II 4

NUTR 4460 Senior Project - Scientific and Technical Writing 2

or NUTR 4461 or Senior Project - Research

or NUTR 4462 or Senior Project - Internship

Free Elective 3

Free Elective 3

Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Nutrition and Pre-Health Sciences

Nutrition and Pre-Health Sciences Concentration - BS in Nutrition

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

NUTR 1101	Orientation to the Nutrition Major	1
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
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FSN 2202	Introduction to Human Nutrition	3
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		14
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Term 2

FSN 1121	Fundamentals of Food	3
BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
General Education Requirement (1B)		3
General Education Requirement		3
Units		16

Second Year

Term 1

NUTR 3315	Nutrition in Aging	2
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CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
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MCRO 2224	General Microbiology I	4
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or MCRO 2221	or Introduction to Microbiology	
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General Education Requirement		3
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General Education Requirement		3
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Units		16
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Term 2

NUTR 3310	Maternal and Child Nutrition	3
FSN 2250	Food and Nutrition: Culture and Customs (USCP)	3
BIO 2231	Human Anatomy and Physiology I	4
CHEM 3350	Biochemistry: Fundamentals and Applications (Upper-Division 2/5) ¹	4
Units		14

Third Year		
Term 1		

FSN 3305	Nutrition and Exercise for Health and Disease Prevention	3
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NUTR 3331	Macronutrient Metabolism	3
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BIO 2232	Human Anatomy and Physiology II	4
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Concentration Course		3
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General Education Requirement		3
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Units		16
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Term 2		
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NUTR 3332	Micronutrient Metabolism	3
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NUTR 3333	Nutrient Metabolism Lab	1
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Concentration Course	3
----------------------	---

General Education Requirement	3
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Free Elective ²	3
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Units	13
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Fourth Year

Term 1

NUTR 4415	Nutrition Counseling & Education	3
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NUTR 4431	Medical Nutrition Therapy I	4
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Concentration Course	3
----------------------	---

General Education Requirement	3
-------------------------------	---

Free Elective ²	3
----------------------------	---

Units	16
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Term 2

NUTR 4416	Community Nutrition	3
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NUTR 4432	Medical Nutrition Therapy II	4
-----------	------------------------------	---

NUTR 4460	Senior Project - Scientific and Technical Writing	2
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or NUTR 4461 or Senior Project - Research

or NUTR 4462 or Senior Project - Internship

Concentration Course	3
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General Education Requirement	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Oceanography

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

OCN 1100

Marine Biology

4

& 1100L

and Marine Biology Lab (5B & 5C) ¹

CHE 1110	General Chemistry	4
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& 1110L	and General Chemistry Laboratory (5A)	1
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EGL 1100	English Composition (1A)	3
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PE 1101	Swim Competency Exam	0
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GMA 1110	First Year Seminar	1
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Units		12
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Term 2

OCN 1110 Marine Ecology 4

& 1110L and Marine Ecology Lab

CHE 2210 General Chemistry II 4

& 2210L and General Chemistry II Laboratory

EGL 2220 Critical Thinking (1B)¹ 3

MTH 2210 Calculus I 4

Second Year

Term 1

OCN 2200	Oceanography I: Geology And Chemistry	4
& 2200L	and Introduction To Oceanography Lab	
EGL 1110	Speech Communication (1C) ¹	3

GOV 2200	American Government (4A) ¹	3
MTH 1107	Elementary Statistics (2) ¹	3
PHY 1100	General Physics I	4
& 1100L	and General Physics I Laboratory	
Units		17

Term 2

OCN 2210	Oceanography Ii: Physics And Biology	4
& 2210L	and Introduction To Oceanography Laboratory Ii	
COM 1107	Programming in R	1
PHY 1105	General Physics Ii	4
& 1105L	and General Physics Ii Laboratory	
General Education Requirement		3
General Education Requirement		3

Third Year

Term 1

OCN 3305	Introduction To Ocean Science Research	1
OCN 3320	Oceans And Climate (Upper-Division 2/5) ¹	3
OCN 3330	Marine Microbial Ecology	3

COM 2210	Engineering Computer Programming	3
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& 2210L	and Oceanography Computer Programming Laboratory	
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Free Elective		3
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Units		13
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Term 2		
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OCN 3310	Oceanographic Instruments And Analysis	3
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& 3310L	and Oceanographic Instruments And Analysis Lab	
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OCN 3360	Marine Geology	3
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GMA 3320	Ocean Environmental Management (Upper-Division 4) ¹	3
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General Education Requirement	3
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Free Elective	3
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Units	15
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Term 3

OCN 3380	Directed Research/Co-Op Experience	3
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& 3380L	and Directed Research Lab/Co-Op	
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OCN 3300	Global Education	3
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Units	6
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Fourth Year

Term 1

EGL 3301	Project Based Writing For Science (GWR)	3
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OCN 3340	Chemical Oceanography	3
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Upper-Division OCN Elective	3
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Free Elective	3
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Units	12
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Term 2

OCN 3350	Physical Oceanography	3
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Upper-Division OCN Elective	3
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Upper-Division OCN Elective	3
-----------------------------	---

General Education Requirement	3
-------------------------------	---

Free Elective	3
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Swim assessments are required of all cadets during Orientation week. If [PE 1101](#) is not passed with a grade CR, then enrollment in [PE 1102](#) will be required.

Philosophy

Concentration Not Yet Declared - BA in Philosophy

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
PHIL 1101	Introduction to Philosophy	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3

General Education Requirement (2)	3
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General Education Requirement	3
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Units	15
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Term 2

PHIL 1126	Logic and Argumentative Writing (1B) 1	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

General Education Requirement

3

General Education Requirement

1

Units

16

Second Year

Term 1

PHIL 2230

Philosophical Classics: Knowledge and Reality (3B)¹

3

PHIL 2231

Philosophical Classics: Ethics and Political Philosophy

3

PHIL 2241

Symbolic Logic

3

General Education Requirement

3

Free Elective⁴

3

Units

15

Term 2

PHIL 3309

Early Greek Philosophy through Plato

3

or PHIL 3310

or Aristotle and Hellenistic Philosophy

or PHIL 3312

or Medieval Philosophy

PHIL 3331

Ethics

3

Free Elective	4	3
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Free Elective	4	3
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Free Elective	4	3
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Units	15
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Third Year

Term 1

PHIL 3300	Philosophical Writing and Research	3
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PHIL 3313	Early Modern Rationalism	3
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or PHIL 3314 or Early Modern Empiricism

or PHIL 3315 or Kant and 19th Century European Philosophy

PHIL 3321 Philosophy of Science 3

Concentration Course or Elective² 3

General Education Requirement 3

Units 15

Term 2

PHIL 3317 History of Analytic Philosophy 3

or PHIL 3318 or Phenomenology

or PHIL 3319 or Existentialism

PHIL 4429 OR History of Philosophy Elective 3

Arts and Humanities Breadth Course 3

Concentration Course or Elective 2 3

General Education Requirement 3

Units 15

Fourth Year

Term 1

PHIL 4411	Metaphysics	3
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Concentration Course or Elective	2	3
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Concentration Course or Elective	2	3
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Free Elective	4	3
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Free Elective	4	3
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Units	15
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Term 2

PHIL 4412	Epistemology	3
PHIL 4459	Senior Project Preparation	1
PHIL 4460	Senior Project	2
Concentration Course or Elective ²		3
Free Elective ⁴		3
Free Elective ⁴		2
Units		14
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration, and at least 9 units must be 4000-level if the PHIL Electives option is chosen rather than declaring a concentration.

3

If a Religious Studies (RELS) course is taken to meet the Arts and Humanities Breadth requirement, it cannot be double-counted in the Philosophy and Religion concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Ethics and Society

Ethics and Society Concentration - BA in Philosophy

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
PHIL 1101	Introduction to Philosophy	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
General Education Requirement (2)		3

General Education Requirement	3
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Units	15
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Term 2	
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PHIL 1126	Logic and Argumentative Writing (1B) 1	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement

1

Units

16

Second Year

Term 1

PHIL 2230

Philosophical Classics: Knowledge and Reality (3B)¹

3

PHIL 2231

Philosophical Classics: Ethics and Political Philosophy

3

PHIL 2241

Symbolic Logic

3

General Education Requirement

3

Free Elective	4	3
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Units	15
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Term 2		
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PHIL 3309	Early Greek Philosophy through Plato	3
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or PHIL 3310	or Aristotle and Hellenistic Philosophy
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or PHIL 3312	or Medieval Philosophy
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PHIL 3331	Ethics	3
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Free Elective	4	3
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Free Elective	4	3
---------------	---	---

Free Elective	4	3
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Units	15
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Third Year

Term 1

PHIL 3300	Philosophical Writing and Research	3
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PHIL 3313	Early Modern Rationalism	3
-----------	--------------------------	---

or PHIL 3314	or Early Modern Empiricism
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or PHIL 3315 or Kant and 19th Century European Philosophy

PHIL 3321 Philosophy of Science 3

Ethics Elective 3

General Education Requirement 3

Units 15

Term 2

PHIL 3317 History of Analytic Philosophy 3

or PHIL 3318 or Phenomenology

or PHIL 3319 or Existentialism

PHIL 4429 OR History of Philosophy Elective

3

Ethics Elective

3

Arts and Humanities Breadth Course³

3

General Education Requirement

3

Units

15

Fourth Year

Term 1

PHIL 4411Metaphysics

3

PHIL 4439	Topics in Ethics and Political Philosophy	3
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Ethics Elective		3
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Free Elective ⁴		3
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Free Elective ⁴		3
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Units		15
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Term 2		
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PHIL 4412	Epistemology	3
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PHIL 4459	Senior Project Preparation	1
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PHIL 4460	Senior Project	2
Technology Ethics Elective		3
Free Elective ⁴		3
Free Elective ⁴		2
Units		14
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

3

If a Religious Studies (RELS) course is taken to meet the Arts and Humanities Breadth requirement, it cannot be double-counted in the Philosophy and Religion concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Ethics of Science and Technology

Ethics of Science and Technology Concentration - BA in Philosophy

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

PHIL 1101	Introduction to Philosophy	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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General Education Requirement (2)	3
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General Education Requirement	3
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Units	15
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Term 2

PHIL 1126	Logic and Argumentative Writing (1B) ¹	3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		3
General Education Requirement		1
Units		16

Second Year

Term 1

PHIL 2230	Philosophical Classics: Knowledge and Reality (3B) ¹	3
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PHIL 2231	Philosophical Classics: Ethics and Political Philosophy	3
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PHIL 2241	Symbolic Logic	3
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General Education Requirement		3
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Free Elective ⁴		3
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Units		15
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Term 2

PHIL 3309	Early Greek Philosophy through Plato	3
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or PHIL 3310	or Aristotle and Hellenistic Philosophy	
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or PHIL 3312	or Medieval Philosophy	
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PHIL 3331	Ethics	3
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Free Elective ⁴		3
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Free Elective ⁴		3
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Free Elective ⁴		3
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Units		15
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Third Year

Term 1

PHIL 3300	Philosophical Writing and Research	3
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PHIL 3313	Early Modern Rationalism	3
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or PHIL 3314	or Early Modern Empiricism	
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or PHIL 3315	or Kant and 19th Century European Philosophy	
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PHIL 3321	Philosophy of Science	3
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PHIL 3323	Ethics, Science, and Technology	3
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General Education Requirement		3
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Units

15

Term 2

PHIL 3317 History of Analytic Philosophy 3

or PHIL 3318 or Phenomenology

or PHIL 3319 or Existentialism

PHIL 4429 OR History of Philosophy Elective 3

Ethics and Philosophy of Science and Technology Elective 3

Arts and Humanities Breadth Course ³ 3

General Education Requirement 3

Units

15

Fourth Year

Term 1

PHIL 4411	Metaphysics	3
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Ethics and Philosophy of Science and Technology Elective	3
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Ethics Elective	3
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Free Elective ⁴	3
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Free Elective ⁴	3
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Units

15

Term 2

PHIL 4412	Epistemology	3
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PHIL 4421	Philosophy of Space, Time, and Matter	3
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or PHIL 4422	or Philosophy of Mind
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or PHIL 4439	or Topics in Ethics and Political Philosophy
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PHIL 4459	Senior Project Preparation	1
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PHIL 4460	Senior Project	2
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Free Elective ⁴		3
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Free Elective	4	2
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Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

3

If a Religious Studies (RELS) course is taken to meet the Arts and Humanities Breadth requirement, it cannot be double-counted in the Philosophy and Religion concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Philosophy and Religion

Philosophy and Religion Concentration - BA in Philosophy

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

PHIL 1101	Introduction to Philosophy	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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General Education Requirement (2)	3
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General Education Requirement	3
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Units	15
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Term 2

PHIL 1126	Logic and Argumentative Writing (1B) ¹	3
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General Education Requirement	3
General Education Requirement	3
General Education Requirement	3
General Education Requirement	3
General Education Requirement	1
Units	16

Second Year

Term 1

PHIL 2230	Philosophical Classics: Knowledge and Reality (3B) ¹	3
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PHIL 2231	Philosophical Classics: Ethics and Political Philosophy	3
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PHIL 2241	Symbolic Logic	3
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General Education Requirement		3
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Free Elective ⁴		3
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Units		15
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Term 2

PHIL 3309	Early Greek Philosophy through Plato	3
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or PHIL 3310	or Aristotle and Hellenistic Philosophy	
or PHIL 3312	or Medieval Philosophy	
PHIL 3331	Ethics	3
Free Elective ⁴		3
Free Elective ⁴		3
Free Elective ⁴		3
Units		15

Third Year

Term 1

PHIL 3300 Philosophical Writing and Research 3

PHIL 3313 Early Modern Rationalism 3

or PHIL 3314 or Early Modern Empiricism

or PHIL 3315 or Kant and 19th Century European Philosophy

PHIL 3342 Philosophy of Religion 3

General Education Requirement 3

Units 12

Term 2

PHIL 3317	History of Analytic Philosophy	3
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or PHIL 3318	or Phenomenology	
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or PHIL 3319	or Existentialism	
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PHIL 3321	Philosophy of Science	3
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PHIL 4429 OR History of Philosophy Elective		3
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Asian Philosophy and Religion Elective		3
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General Education Requirement		3
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Units		15
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Fourth Year	
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Term 1

PHIL 4411	Metaphysics	3
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Religion Elective	3
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Senior Level Philosophy Elective	3
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Free Elective ⁴	3
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Free Elective ⁴	3
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Units	15
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Term 2

PHIL 4412	Epistemology	3
PHIL 4459	Senior Project Preparation	1
PHIL 4460	Senior Project	2
Religion Elective		3
Free Elective ⁴		3
Free Elective ⁴		2
	Units	14
	Total Units	117

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a course is taken to meet a Major or Support requirement, it cannot be double-counted as an Elective for the concentration.

3

If a Religious Studies (RELS) course is taken to meet the Arts and Humanities Breadth requirement, it cannot be double-counted in the Philosophy and Religion concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Physics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

PHYS 1100	Introduction to the Physics Major	1
CHEM 1120	Fundamentals of Chemical Structure and Properties	4
MATH 1261	Calculus I (2) ¹	4
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		15

Term 2

PHYS 1141	General Physics I (5A & 5C) ¹	4
CSC 1001	Fundamentals of Computer Science	4
& 1001L	and Fundamentals of Computer Science Laboratory	
MATH 1262	Calculus II	4
General Education Requirement (1B)		3
Units		15

Second Year

Term 1

PHYS 1143	General Physics II	4
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MATH 2263	Calculus III	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

PHYS 2211	General Physics III: Modern Physics	4
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MATH 2341	Linear Analysis	4
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General Education Requirement		3
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General Education Requirement		3
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Units		14
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Third Year		
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Term 1		
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PHYS 3305	Classical Mechanics I (Upper-Division 2/5) ¹	3
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PHYS 3316	Instrumentation and Techniques of Experimental Physics ²	4
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PHYS 3320	Methods of Theoretical Physics	4
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General Education Requirement		3
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Units		14
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Term 2		
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PHYS 3301	Statistical Mechanics	3
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PHYS 3339	Communicating Physics	1
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PHYS 3340	Quantum Physics Laboratory I ²	1
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PHYS 4408	Electromagnetic Fields and Waves I	3
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General Education Requirement

3

Free Elective

2, 3, 4

4

Units

15

Fourth Year

Term 1

PHYS 3341

Quantum Physics Laboratory II

2

1

PHYS 4405

Quantum Mechanics I

3

PHYS 4461

Senior Project I

2

or PHYS 4461 *and* PHYS 4462 or Senior Project I *and* Senior Project II

General Education Requirement 3

Free Elective 2, 3, 4 3

Free Elective 2, 3, 4 3

Units 15

Term 2

Free Elective 2, 3, 4 4

Free Elective	2, 3, 4	4
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Free Elective	2, 3, 4	4
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Free Elective	2, 3, 4	4
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Units	16
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Major courses with lab component may not be taken as CR/NC grading: [PHYS 1143](#), [PHYS 3316](#), [PHYS 3323](#), [PHYS 3340](#), [PHYS 3341](#), [PHYS 4428](#), [ASTR 4444](#), or [EE 4422](#).

3

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

4

Free Electives may need to be at the 3000-4000 level to ensure completion of the required minimum of 40 units of upper-division courses.

Physics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units

PHYS 1100

Introduction to the Physics Major

1

CHEM 1120	Fundamentals of Chemical Structure and Properties	4
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MATH 1261	Calculus I (2) ¹	4
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		15
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Term 2		
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PHYS 1141	General Physics I (5A & 5C) ¹	4
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CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory	
MATH 1262	Calculus II	4
General Education Requirement (1B)		3
Units		15

Second Year		
Term 1		

PHYS 1143	General Physics II ²	4
MATH 2263	Calculus III	3

General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2	
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PHYS 2211	General Physics III: Modern Physics	4
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MATH 2341	Linear Analysis	4
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement

3

Units

17

Third Year

Term 1

PHYS 3305	Classical Mechanics I (Upper-Division 2/5) ¹	3
PHYS 3316	Instrumentation and Techniques of Experimental Physics ²	4
PHYS 3320	Methods of Theoretical Physics	4

Technical Elective

2, 3

4

Units

15

Term 2

PHYS 3301

Statistical Mechanics

3

PHYS 3339

Communicating Physics

1

PHYS 3340

Quantum Physics Laboratory I

2

1

PHYS 4408

Electromagnetic Fields and Waves I

3

General Education Requirement

3

Free Elective	4	3
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Units	14
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Fourth Year

Term 1

PHYS 3341	Quantum Physics Laboratory II	2	1
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PHYS 4405	Quantum Mechanics I	3
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PHYS 4461	Senior Project I	2
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or PHYS 4461 <i>and</i> PHYS 4462	or Senior Project I <i>and</i> Senior Project II
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Technical Elective	2, 3	3
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Technical Elective	2, 3	3
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General Education Requirement		3
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Units		15
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Term 2

PHYS 3323	Optics	2	4
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or PHYS 4425

or Solid State Physics

or PHYS 4428

or Nonlinear Dynamics and Chaos

or ASTR 4444

or Observational Astronomy

Technical Elective	2, 3	3
Technical Elective	2, 3	3
Technical Elective	2, 3	3
	Units	13
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

Major courses with lab component may not be taken as CR/NC grading: [PHYS 1143](#), [PHYS 3316](#), [PHYS 3323](#), [PHYS 3340](#), [PHYS 3341](#), [PHYS 4428](#), [ASTR 4444](#), or [EE 4422](#).

3

Total combined elective credit in PHYS 4400, PHYS 4404, ASTR 4400, ASTR 4404, GEOL 4400, and GEOL 4404 limited to 4 units.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Plant Sciences

Concentration Not Yet Declared - BS in Plant Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
PLSC 1101	Orientation to Plant Sciences	1

PLSC 1120	Principles of Plant Sciences	3
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& 1120L	and Principles of Plant Sciences Lab	
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BOT 1121	General Botany (5B & 5C) ¹	4
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MATH 1006	College Algebra (2) ¹	3
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General Education Requirement (1A)		3
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Units		14
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Term 2		
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PLSC 1124	Plant Propagation	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
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General Education Requirement (1B)		3
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General Education Requirement (1C)		3
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Units		13
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Second Year

Term 1

SS 1120	Introductory Soil Science	4
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Concentration Course		3
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Concentration Course	3
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Concentration Course	3
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General Education Requirement	3
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Units	16
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Term 2	
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PLSC 3321	Weed Biology and Management	4
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SS 2221	Soil Health and Plant Nutrition	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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Concentration Course	3
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General Education Requirement	3
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Units	17
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Third Year

Term 1

PLSC 3313	Agricultural Entomology	3
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PLSC 3323	Plant Pathology	3
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PLSC 3351	Experimental Techniques and Analysis	3
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BRAE 3340	Irrigation Water Management (Upper-Division 2/5)	1	3
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Units	12
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Term 2

PLSC 3304	Introduction to Plant Breeding	3
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Concentration Course	3
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Concentration Course	3
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Concentration Course	3
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General Education Requirement	3
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Fourth Year

Term 1

PLSC 4461	Senior Project I	1
Concentration Course		3
Concentration Course		3
General Education Requirement		3
General Education Requirement		3

General Education Requirement	3
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Units	16
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Term 2	
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PLSC 4410	Crop Physiology	3
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PLSC 4462	Senior Project II	1
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Concentration Course	4
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Concentration Course	3
----------------------	---

Concentration Course	3
----------------------	---

General Education Requirement	3
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Units	17
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Fruit and Crop Science

Fruit and Crop Science Concentration - BS in Plant Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
PLSC 1101	Orientation to Plant Sciences	1
PLSC 1120	Principles of Plant Sciences	3
& 1120L	and Principles of Plant Sciences Lab	
BOT 1121	General Botany (5B & 5C) ¹	4
MATH 1006	College Algebra (2) ¹	3
General Education Requirement (1A)		3
Units		14

Term 2

PLSC 1124	Plant Propagation	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
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General Education Requirement (1B)	3
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General Education Requirement (1C)	3
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Units	13
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Second Year

Term 1

PLSC 1150	California Row Crop Production	4
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SS 1120	Introductory Soil Science	4
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Approved Elective ²		2-3
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Approved Elective ²		3
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General Education Requirement		3
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Units		16
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Term 2		
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PLSC 1132	Introduction to Fruit Crop Production	4
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PLSC 3321	Weed Biology and Management	4
SS 2221	Soil Health and Plant Nutrition	4
STAT 1110	Applied Statistical Concepts and Methods	3
General Education Requirement		3
Units		18

Third Year

Term 1

PLSC 3313	Agricultural Entomology	3
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PLSC 3323	Plant Pathology	3
PLSC 3351	Experimental Techniques and Analysis	3
BRAE 3340	Irrigation Water Management (Upper-Division 2/5) ¹	3
Units		12

Term 2

PLSC 2244	Precision Farming	3
PLSC 3304	Introduction to Plant Breeding	3
PLSC 3360	Advanced Fruit Crop Production	4

Approved Elective	2	3
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General Education Requirement		3
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Units		16
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Fourth Year		
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Term 1		
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PLSC 2205	Orchard and Vegetable Enterprise Project	2
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or PLSC 3333	or Greenhouse Vegetable Production	
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PLSC 4420	Organic Crop Production Systems	3
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PLSC 4461	Senior Project I	1
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Units	15
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Term 2

PLSC 4410	Crop Physiology	3
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PLSC 4462	Senior Project II	1
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Approved Elective ²	3
Approved Elective ²	3
Approved Elective ²	3
General Education Requirement	3
Units	16
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

A minimum of 9 units must be taken at the 3000-4000 level.

Environmental Horticultural Science

Environmental Horticultural Science Concentration - BS in Plant Sciences

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
PLSC 1101	Orientation to Plant Sciences	1
PLSC 1120	Principles of Plant Sciences	3

& 1120L and Principles of Plant Sciences Lab

BOT 1121 General Botany (5B & 5C) ¹ 4

MATH 1006 College Algebra (2) ¹ 3

General Education Requirement (1A) 3

Units 14

Term 2

PLSC 1124 Plant Propagation 3

CHEM 1120 Fundamentals of Chemical Structure and Properties (5A) ¹ 4

General Education Requirement (1B)	3
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General Education Requirement (1C)	3
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Units	13
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Second Year	
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Term 1	
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PLSC 1123	Introduction to Sustainable Site Horticulture	3
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PLSC 2234	Introduction to Plant Materials	3
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SS 1120	Introductory Soil Science	4
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Approved Elective	3
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General Education Requirement	3
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Units	16
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Term 2

PLSC 3301	Horticultural Production Techniques	3
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PLSC 3321	Weed Biology and Management	4
-----------	-----------------------------	---

SS 2221	Soil Health and Plant Nutrition	4
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement

3

Units

17

Third Year

Term 1

PLSC 3313

Agricultural Entomology

3

PLSC 3323

Plant Pathology

3

PLSC 3351

Experimental Techniques and Analysis

3

BRAE 3340

Irrigation Water Management (Upper-Division 2/5)¹

3

Units

12

Term 2

PLSC 3304 Introduction to Plant Breeding 3

PLSC 3332 Sustainable Site Design and Systems 3

PLSC 3334 Advanced Plant Materials 3

PLSC 3340 Principles of Greenhouse Environments 3

General Education Requirement 3

Units

15

Fourth Year

Term 1

PLSC 4461	Senior Project I	1
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Approved Elective	3
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Approved Elective	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units

16

Term 2

PLSC 4410	Crop Physiology	3
-----------	-----------------	---

PLSC 4427	Disease and Pest Control Systems for Ornamental Plants	4
-----------	--	---

PLSC 4462	Senior Project II	1
-----------	-------------------	---

Approved Elective		3
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Approved Elective		3
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General Education Requirement		3
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Units	17
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

Plant Protection Science

Plant Protection Science Concentration - BS in Plant Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

PLSC 1101	Orientation to Plant Sciences	1
PLSC 1120	Principles of Plant Sciences	3
& 1120L	and Principles of Plant Sciences Lab	
BOT 1121	General Botany (5B & 5C) ¹	4
MATH 1006	College Algebra (2) ¹	3
General Education Requirement (1A)		3
Units		14

Term 2

PLSC 1124	Plant Propagation	3
-----------	-------------------	---

CHEM 1120	Fundamentals of Chemical Structure and Properties	4
	(5A) ¹	

General Education Requirement (1B)	3
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General Education Requirement (1C)	3
------------------------------------	---

Units	13
-------	----

Second Year

Term 1

SS 1120	Introductory Soil Science	4
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Approved Elective	3
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Approved Elective	3
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Approved Elective	3
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General Education Requirement	3
-------------------------------	---

Units	16
-------	----

Term 2

PLSC 3321	Weed Biology and Management	4
-----------	-----------------------------	---

SS 2221	Soil Health and Plant Nutrition	4
---------	---------------------------------	---

STAT 1110	Applied Statistical Concepts and Methods	3
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Approved Elective		3
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General Education Requirement		3
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Units		17
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Third Year	
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Term 1	
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PLSC 3313	Agricultural Entomology	3
-----------	-------------------------	---

PLSC 3323	Plant Pathology	3
-----------	-----------------	---

PLSC 3351	Experimental Techniques and Analysis	3
-----------	--------------------------------------	---

BRAE 3340	Irrigation Water Management (Upper-Division 2/5) ¹	3
-----------	---	---

Units	12
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Term 2

PLSC 3304	Introduction to Plant Breeding	3
-----------	--------------------------------	---

PLSC 4406	Advanced Weed Management	4
-----------	--------------------------	---

PLSC 4427	Disease and Pest Control Systems for Ornamental Plants	4
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Approved Elective	3
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General Education Requirement

3

Units

17

Fourth Year

Term 1

PLSC 2205

Orchard and Vegetable Enterprise Project

2

or PLSC 2212

or Environmental Horticulture Enterprise Project

or PLSC 3333

or Greenhouse Vegetable Production

PLSC 4428

Advanced Plant Pathology

3

PLSC 4441

Biological Control for Pest Management

3

PLSC 4461	Senior Project I	1
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Term 2

PLSC 4410	Crop Physiology	3
-----------	-----------------	---

PLSC 4431	Integrated Pest Management for Insects	3
-----------	--	---

PLSC 4462	Senior Project II	1
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Approved Elective	3
General Education Requirement	3
General Education Requirement	3
Units	16
Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

Political Science

Concentration Not Yet Declared - BA in Political Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

POLS 1112

U.S. and California Government (4A)¹

3

POLS 1180

Political Inquiry

3

POLS 2229

Introduction to Comparative Politics

3

General Education Requirement (1A)

3

General Education Requirement (1C)

3

Units

15

Term 2

POLS 2225	Introduction to International Relations	3
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POLS 2230	Basic Concepts of Political Thought	3
-----------	-------------------------------------	---

ANT 2201	Cultural Anthropology (4B) ¹	3
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or GEOG 1150

or Human Geography

or HIST 2222

or World History to 1500

or HIST 2223

or World History since 1500

or SOC 1110

or Comparative Societies

General Education Requirement (1B)	3
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General Education Requirement

3

Units

15

Second Year

Term 1

STAT 1110

Applied Statistical Concepts and Methods (2)

1

3

3000-4000 level POLS Elective

2, 3

3

General Education Requirement

3

General Education Requirement

3

General Education Requirement	3
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Units	15
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Term 2	
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3000-4000 level POLS Elective ^{2, 3}	3
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3000-4000 level POLS Elective ^{2, 3}	3
---	---

General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

General Education Requirement	1
-------------------------------	---

Free Elective⁴

3

Units

16

Third Year

Term 1

POLS 3359

Research Design

3

POLS Diversity, Equity, and Inclusion Elective³

3

Concentration Course or Elective^{2, 3}

3

Concentration Course or Elective

2, 3

3

General Education Requirement

3

Units

15

Term 2

POLS 3361

Data Analysis in Political Science

3

3000-4000 level POLS Elective

2, 3

3

Concentration Course or Elective

2, 3

3

Concentration Course or Elective	2, 3	3
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General Education Requirement		3
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Units		15
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Fourth Year		
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Term 1		
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POLS 4461	Senior Project I	2
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Concentration Course or Elective	2, 3	3
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Concentration Course or Elective	2, 3	3
----------------------------------	------	---

Free Elective	4	3
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Free Elective	4	3
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Units	14
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Term 2	
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POLS 4462	Senior Project II	2
-----------	-------------------	---

Concentration Course or Elective	2, 3	3
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Free Elective	4	3
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Free Elective	4	3
---------------	---	---

Free Elective	4	4
---------------	---	---

Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Concentration. A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Upper-Division Electives.

3

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Global Politics

Global Politics Concentration - BA in Political Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

POLS 1112	U.S. and California Government (4A) ¹	3
-----------	--	---

POLS 1180	Political Inquiry	3
-----------	-------------------	---

POLS 2229	Introduction to Comparative Politics	3
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		15
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Term 2		
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POLS 2225	Introduction to International Relations	3
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POLS 2230	Basic Concepts of Political Thought	3
-----------	-------------------------------------	---

ANT 2201	Cultural Anthropology (4B) ¹	3
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or GEOG 1150

or Human Geography

or HIST 2222

or World History to 1500

or HIST 2223

or World History since 1500

or SOC 1110

or Comparative Societies

General Education Requirement (1B)	3
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General Education Requirement	3
-------------------------------	---

Units	15
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Second Year

Term 1

STAT 1110	Applied Statistical Concepts and Methods (2)	¹	3
3000-4000 level POLS Elective		^{2, 3}	3
General Education Requirement			3
General Education Requirement			3
General Education Requirement			3
Units			15

Term 2

3000-4000 level POLS Elective^{2, 3} 3

3000-4000 level POLS Elective^{2, 3} 3

General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Free Elective⁴ 3

Units 16

Third Year

Term 1

POLS 3308	Political Violence and Conflict Resolution	3	3
or POLS 3381	or Peace and War		
POLS 3321	Comparative Political Culture		3
or POLS 4426	or International Organizations and Law		
or POLS 4427	or Politics of the Global Economy		
POLS 3359	Research Design		3
POLS Diversity, Equity, and Inclusion Elective		3	3

General Education Requirement	3
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Units	15
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Term 2

POLS 3328	Politics of Developing Areas	3
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or POLS 3383	or Politics of the European Union
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POLS 3361	Data Analysis in Political Science	3
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POLS 3382	Comparative Foreign Policy	3
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or POLS 4420	or Contemporary U.S. Foreign Policy
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3000-4000 level POLS Elective	2, 3	3
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General Education Requirement 3

Units 15

Fourth Year

Term 1

POLS 4428 Issues and Topics in Comparative Politics 3

or POLS 4429 or Issues and Topics in International Relations

POLS 4461 Senior Project I 2

Concentration Approved Elective 2, 3, 5 3

Free Elective	4	3
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Free Elective	4	3
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Units	14
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Term 2

POLS 4462	Senior Project II	2
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Concentration Approved Elective	2, 3, 5	3
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Free Elective	4	3
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Free Elective	4	3
---------------	---	---

Free Elective	4	4
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Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A maximum of 3 units of either [POLS 3386](#) or [POLS 338](#) may count towards the Concentration. A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Upper-Division Electives.

3

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

Consultation with advisor is recommended prior to selecting Approved Electives; bear in mind that your selection may impact pursuit of post-baccalaureate studies and/or goals.

Pre-Law

Pre-Law Concentration - BA in Political Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

POLS 1112	U.S. and California Government (4A) ¹	3
-----------	--	---

POLS 1180	Political Inquiry	3
-----------	-------------------	---

POLS 2229	Introduction to Comparative Politics	3
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		15
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Term 2

POLS 2225 Introduction to International Relations 3

POLS 2230 Basic Concepts of Political Thought 3

ANT 2201 Cultural Anthropology (4B) ¹ 3

or GEOG 1150

or Human Geography

or HIST 2222

or World History to 1500

or HIST 2223

or World History since 1500

or SOC 1110

or Comparative Societies

General Education Requirement (1B) 3

General Education Requirement

3

Units

15

Second Year

Term 1

STAT 1110

Applied Statistical Concepts and Methods (2)

1

3

3000-4000 level POLS Elective

2, 3

3

General Education Requirement

3

General Education Requirement

3

General Education Requirement	3
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Units	15
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Term 2	
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3000-4000 level POLS Elective ^{2, 3}	3
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3000-4000 level POLS Elective ^{2, 3}	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	1
-------------------------------	---

Free Elective	4	3
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Units	16
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Third Year

Term 1

POLS 2245	Judicial Process	3
-----------	------------------	---

POLS 3341	U.S. Constitutional Law	3
-----------	-------------------------	---

POLS 3359	Research Design	3
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POLS Diversity, Equity, and Inclusion Elective	3	3
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General Education Requirement 3

Units 15

Term 2

POLS 3343 Civil Rights in the U.S. 3

POLS 3344 Civil Liberties 3

POLS 3361 Data Analysis in Political Science 3

3000-4000 level POLS Elective^{2, 3} 3

General Education Requirement 3

Units

15

Fourth Year

Term 1

POLS 3334	Jurisprudence	3
POLS 4461	Senior Project I	2
Concentration Approved Elective ^{2, 3, 5}		3
Free Elective ⁴		3
Free Elective ⁴		3

Units

14

Term 2

POLS 4462

Senior Project II

2

Concentration Approved Elective 2, 3, 5

3

Free Elective 4

3

Free Elective 4

3

Free Elective 4

4

Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Concentration. A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Upper-Division Electives.

3

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

Consultation with advisor is recommended prior to selecting Approved Electives; bear in mind that your selection may impact pursuit of post-baccalaureate studies and/or goals.

U.S. Politics

U.S. Politics Concentration - BA in Political Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

POLS 1112	U.S. and California Government (4A) ¹	3
POLS 1180	Political Inquiry	3
POLS 2229	Introduction to Comparative Politics	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		15

Term 2

POLS 2225	Introduction to International Relations	3
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POLS 2230	Basic Concepts of Political Thought	3
ANT 2201	Cultural Anthropology (4B) ¹	3
or GEOG 1150	or Human Geography	
or HIST 2222	or World History to 1500	
or HIST 2223	or World History since 1500	
or SOC 1110	or Comparative Societies	
General Education Requirement (1B)		3
General Education Requirement		3
Units		15

Second Year

Term 1

STAT 1110	Applied Statistical Concepts and Methods (2)	1	3
3000-4000 level POLS Elective		2, 3	3
General Education Requirement			3
General Education Requirement			3
General Education Requirement			3
Units			15

Term 2

3000-4000 level POLS Elective^{2, 3} 3

3000-4000 level POLS Elective^{2, 3} 3

General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Free Elective⁴ 3

Units 16

Third Year

Term 1

POLS 3359	Research Design	3
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POLS Diversity, Equity, and Inclusion Elective	3	3
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Concentration Required Course	2, 3	3
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Concentration Required Course	2, 3	3
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General Education Requirement		3
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Units		15
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Term 2

POLS 3361 Data Analysis in Political Science 3

3000-4000 level POLS Elective^{2, 3} 3

Concentration Required Course^{2, 3} 3

Concentration Required Course^{2, 3} 3

General Education Requirement 3

Units 15

Fourth Year

Term 1

POLS 4461

Senior Project I

2

Concentration Required Course^{2, 3}

3

Concentration Required Course^{2, 3}

3

Free Elective⁴

3

Free Elective⁴

3

Units

14

Term 2

POLS 4462

Senior Project II

2

Concentration Approved Elective^{2, 3, 5}

3

Free Elective⁴

3

Free Elective⁴

3

Free Elective⁴

4

Units

15

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Concentration. A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Upper-Division Electives.

3

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

Consultation with advisor is recommended prior to selecting Approved Electives; bear in mind that your selection may impact pursuit of post-baccalaureate studies and/or goals.

Individualized Course of Study

Individualized Course of Study - BA in Political Science

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

POLS 1112	U.S. and California Government (4A) ¹	3
POLS 1180	Political Inquiry	3
POLS 2229	Introduction to Comparative Politics	3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
Units		15

Term 2

POLS 2225	Introduction to International Relations	3
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POLS 2230	Basic Concepts of Political Thought	3
ANT 2201	Cultural Anthropology (4B) ¹	3
or GEOG 1150	or Human Geography	
or HIST 2222	or World History to 1500	
or HIST 2223	or World History since 1500	
or SOC 1110	or Comparative Societies	
General Education Requirement (1B)		3
General Education Requirement		3
Units		15

Second Year

Term 1

STAT 1110	Applied Statistical Concepts and Methods (2)	1	3
3000-4000 level POLS Elective		2, 3	3
General Education Requirement			3
General Education Requirement			3
General Education Requirement			3
Units			15

Term 2

3000-4000 level POLS Elective^{2, 3} 3

3000-4000 level POLS Elective^{2, 3} 3

General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Free Elective⁴ 3

Units 16

Third Year

Term 1

POLS 3359	Research Design	3
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POLS Diversity, Equity, and Inclusion Elective	3	3
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Individualized Course of Study Elective	3
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Individualized Course of Study Elective	3
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General Education Requirement	3
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Units	15
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Term 2

POLS 3361	Data Analysis in Political Science	3
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3000-4000 level POLS Elective	2, 3	3
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Individualized Course of Study Elective		3
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Individualized Course of Study Elective		3
---	--	---

General Education Requirement		3
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Units		15
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Fourth Year

Term 1

POLS 4461	Senior Project I	2
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Individualized Course of Study Elective	3
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Individualized Course of Study Elective	3
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Free Elective ⁴	3
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Free Elective ⁴	3
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Units	14
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Term 2

POLS 4462	Senior Project II	2
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Individualized Course of Study Elective	3
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Free Elective ⁴	3
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Free Elective ⁴	3
----------------------------	---

Free Elective ⁴	4
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Units	15
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Total Units	120
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Required in Major or Support; also satisfies General Education (GE) requirement.

2

A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Concentration. A maximum of 3 units of either [POLS 3386](#) or [POLS 3387](#) may count towards the Upper-Division Electives.

3

If a course is taken to meet a Major or Support requirement, it cannot be double-counted in the concentration.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Psychology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

PSY 1102	Orientation to the Psychology Major	2
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PSY 2201	Introductory Psychology (4B) ¹	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		14
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Term 2

PSY 2205	Personality	3
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or PSY 2252	or Social Psychology	
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or PSY 2256	or Developmental Psychology	
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PSY 2240	Biopsychology	3
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PSY/CD 2229	Research Methods in Psychology	3
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General Education Requirement (1B)		3
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General Education Requirement		3
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Units		15
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Second Year

Term 1

PSY 3302	Organizational Behavior	4
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or PSY 3324	or Psychology of Gender
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or PSY 3352	or Psychology of Conflict and Justice
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or PSY 3360	or Applied Social Psychology
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or PSY 4419	or Applied Personality
-------------	------------------------

or PSY 4465	or Cross-Cultural International Psychology
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or PSY 4475	or The Social Psychology of Prejudice
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PSY 3372	Multicultural Psychology	4
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General Education Requirement	3
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General Education Requirement	1
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General Education Requirement	3
-------------------------------	---

Units	15
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Term 2	
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PSY 3320	Health Psychology	4
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or PSY 3356	or Child and Adolescent Mental Health
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or PSY 3370	or Introduction to Clinical and Counseling Psychology
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or PSY 4405	or Adult Mental Health and Psychopathology
-------------	--

or PSY 4460	or Child Abuse and Neglect
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PSY 3333	Advanced Research Methods	4
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General Education Requirement	3
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General Education Requirement	3
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Units	14
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Third Year

Term 1

PSY 3357	Cognition	4
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or PSY 4430	or Sensation and Perception
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or PSY 4440	or Learning and Memory
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or PSY 4480	or Cognitive Neuroscience
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Diversity, Equity and Inclusion Course (Upper-Division 4)	¹	3-4
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Professional Skills Support Course		3
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Free Elective	²	3
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Free Elective	²	3
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Units		17
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Term 2

PSY Elective 3000-4000 level		4
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Upper-Division Science Course (Upper-Division 2/5)	¹	3-4
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General Education Requirement		3
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Free Elective	²	3
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Free Elective	²	3
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Units		16
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Fourth Year		
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Term 1		
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PSY 4448	Research Internship I	3
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or PSY 4453	or Supervised Fieldwork Internship I	
PSY 4461	Senior Project Seminar	2
Free Elective ²		3
Free Elective ²		3
Free Elective ²		3
Units		14

Term 2

PSY 4449	Research Internship II	3
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or PSY 4454 or Supervised Fieldwork Internship II

PSY 4462 Senior Project 2

Free Elective² 3

Free Elective² 3

PSY Elecitive 3000-4000 level 4

Units 15

Total Units 120

1

Required in Major or Support, also satisfies General Education (GE) requirement.

If a General Education (GE) course is used to satisfy a Major or Support Requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Public Health

Concentration Not Yet Declared - BS in Public Health

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
HLTH 1101	Introduction to Public Health	3
HLTH 1155	Multicultural Perspectives and Health (4B) ¹	3
or HLTH 1160		

or Women's Health and Society

ANT 2201	Cultural Anthropology	3
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or SOC 1110	or Comparative Societies
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BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
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General Education Requirement (1A)	3
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Units	16
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Term 2

HLTH 2261	Social Determinants of Health	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
-----------	--	---

PSY 2201	Introductory Psychology	3
----------	-------------------------	---

STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
-----------	---	---

General Education Requirement (1C)		3
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Units		16
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Second Year

Term 1

BIO 2231	Human Anatomy and Physiology I	4
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STAT 3520	Statistics II	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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Units	13
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Term 2

HLTH 3316	Environmental Health	3
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BIO 2232	Human Anatomy and Physiology II	4
----------	---------------------------------	---

FSN 2202	Introduction to Human Nutrition	3
----------	---------------------------------	---

MCRO 2221	Introduction to Microbiology	4
-----------	------------------------------	---

Units	14
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Third Year

Term 1

HLTH 3318	Applied Epidemiology	4
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HLTH 3322	Public Health Policy and Advocacy (Upper-Division 4) ¹	3
-----------	---	---

Concentration Course	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

HLTH 3344	Health Care Systems	3
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Concentration Course	3
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Free Elective ²	4
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Units

16

Fourth Year

Term 1

HLTH 4402	Research Methods in Public Health Settings	4
HLTH 4434	Health Promotion Program Planning	4
Concentration Course		3
Concentration Course		3

Units

14

Term 2

HLTH 4410	Global Health	3
HLTH 4435	Health Promotion Program Implementation and Evaluation	3
HLTH 4460	Senior Project - Experiential	2
or HLTH 4461	or Senior Project - Report	
or HLTH 4462	or Senior Project - Research	
or HLTH 4463	or Senior Project - Internship	
General Education Requirement		3
Free Elective ²		4
Units		15

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Community Health Promotion

Community Health Promotion Concentration - BS in Public Health

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
HLTH 1101	Introduction to Public Health	3
HLTH 1155	Multicultural Perspectives and Health (4B) ¹	3
or HLTH 1160	or Women's Health and Society	
ANT 2201	Cultural Anthropology	3
or SOC 1110	or Comparative Societies	
BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
General Education Requirement (1A)		3

Units

16

Term 2

HLTH 2261	Social Determinants of Health	3
-----------	-------------------------------	---

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
-----------	--	---

PSY 2201	Introductory Psychology	3
----------	-------------------------	---

STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
-----------	---	---

General Education Requirement (1C)		3
------------------------------------	--	---

Units

16

Second Year

Term 1

BIO 2231	Human Anatomy and Physiology I	4
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STAT 3520	Statistics II	3
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General Education Requirement (1B)	3
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General Education Requirement	3
-------------------------------	---

Free Elective ²	3
----------------------------	---

Units	16
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Term 2

HLTH 3316	Environmental Health	3
BIO 2232	Human Anatomy and Physiology II	4
FSN 2202	Introduction to Human Nutrition	3
MCRO 2221	Introduction to Microbiology	4
Units		14

Third Year

Term 1

HLTH 3305	Drugs in Society	3
-----------	------------------	---

HLTH 3318	Applied Epidemiology	4
-----------	----------------------	---

HLTH 3322	Public Health Policy and Advocacy (Upper-Division 4) ¹	3
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Units	16
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Term 2

HLTH 3310	Injury Prevention	3
-----------	-------------------	---

HLTH 3344	Health Care Systems	3
-----------	---------------------	---

General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Units	12
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Fourth Year

Term 1

HLTH 3348	Public Health and Mental Health	3
-----------	---------------------------------	---

HLTH 4402	Research Methods in Public Health Settings	4
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HLTH 4434	Health Promotion Program Planning	4
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Concentration Elective	3
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Free Elective ²	2
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Units	16
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Term 2

HLTH 4410	Global Health	3
-----------	---------------	---

HLTH 4435	Health Promotion Program Implementation and Evaluation	3
-----------	--	---

HLTH 4460	Senior Project - Experiential	2
-----------	-------------------------------	---

or HLTH 4461	or Senior Project - Report
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or HLTH 4462 or Senior Project - Research

or HLTH 4463 or Senior Project - Internship

General Education Requirement 3

Free Elective² 3

Units 14

Total Units 120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Health Equity and Global Health

Health Equity and Global Health Concentration - BS in Public Health

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
HLTH 1101	Introduction to Public Health	3
HLTH 1155	Multicultural Perspectives and Health (4B) ¹	3
or HLTH 1160	or Women's Health and Society	

ANT 2201	Cultural Anthropology	3
----------	-----------------------	---

or SOC 1110	or Comparative Societies	
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BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
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General Education Requirement (1A)		3
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Units		16
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Term 2		
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HLTH 2261	Social Determinants of Health	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
-----------	--	---

PSY 2201	Introductory Psychology	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1C)		3
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Units		16
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Second Year

Term 1

BIO 2231	Human Anatomy and Physiology I	4
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STAT 3520	Statistics II	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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Free Elective ²	2
----------------------------	---

Units	15
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Term 2

HLTH 3316	Environmental Health	3
-----------	----------------------	---

BIO 2232	Human Anatomy and Physiology II	4
----------	---------------------------------	---

FSN 2202	Introduction to Human Nutrition	3
----------	---------------------------------	---

MCRO 2221	Introduction to Microbiology	4
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Units	14
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Third Year

Term 1

HLTH 3318	Applied Epidemiology	4
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HLTH 3322	Public Health Policy and Advocacy (Upper-Division 4) ¹	3
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HLTH 3348	Public Health and Mental Health	3
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General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

HLTH 3344	Health Care Systems	3
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Concentration Elective	3	3-4
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Free Elective	2	3
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Fourth Year

Term 1

HLTH 4402	Research Methods in Public Health Settings	4
HLTH 4413	Health Promotion for Special Populations	3
HLTH 4434	Health Promotion Program Planning	4
HLTH 4444	Comparative Health Care Systems	3

Term 2

HLTH 4410	Global Health	3
HLTH 4435	Health Promotion Program Implementation and Evaluation	3
HLTH 4460	Senior Project - Experiential	2
or HLTH 4461	or Senior Project - Report	
or HLTH 4462	or Senior Project - Research	
or HLTH 4463	or Senior Project - Internship	
General Education Requirement		3
Free Elective ²		3
Units		14

Total Units

120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

Units in excess of total will be applied towards major's free electives.

Health Management and Administration

Health Management and Administration Concentration - BS in Public Health

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

HLTH 1101	Introduction to Public Health	3
HLTH 1155	Multicultural Perspectives and Health (4B) ¹	3
or HLTH 1160	or Women's Health and Society	
ANT 2201	Cultural Anthropology	3
or SOC 1110	or Comparative Societies	

BIO 1151	Life: Molecules and Cells (5B & 5C) ¹	4
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General Education Requirement (1A)		3
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Units		16
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Term 2

HLTH 2261	Social Determinants of Health	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
-----------	--	---

PSY 2201	Introductory Psychology	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1C)		3
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Units		16
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Second Year		
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Term 1		
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BIO 2231	Human Anatomy and Physiology I	4
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STAT 3520	Statistics II	3
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General Education Requirement (1B)		3
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General Education Requirement	3
-------------------------------	---

Free Elective ²	3
----------------------------	---

Units	16
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Term 2	
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HLTH 3316	Environmental Health	3
-----------	----------------------	---

BIO 2232	Human Anatomy and Physiology II	4
----------	---------------------------------	---

FSN 2202	Introduction to Human Nutrition	3
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MCRO 2221	Introduction to Microbiology	4
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Units

14

Third Year

Term 1

BUS 2212	Financial Accounting for Nonbusiness Majors	3
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or ECON 2001	or Survey of Economics
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HLTH 3318	Applied Epidemiology	4
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HLTH 3322	Public Health Policy and Advocacy (Upper-Division 4)	3
	1	

General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Units	16
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Term 2

HLTH 3344	Health Care Systems	3
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Concentration Elective	3	3-4
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Free Elective	2	2
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Units

14

Fourth Year

Term 1

HLTH 4402	Research Methods in Public Health Settings	4
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HLTH 4434	Health Promotion Program Planning	4
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KINE 4401	Leadership in Health and Physical Activity Programs	3
-----------	---	---

Free Elective	2	3
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Units

14

Term 2

HLTH 4410	Global Health	3
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HLTH 4435	Health Promotion Program Implementation and Evaluation	3
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HLTH 4444	Comparative Health Care Systems	3
-----------	---------------------------------	---

HLTH 4460	Senior Project - Experiential	2
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or HLTH 4461	or Senior Project - Report
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or HLTH 4462	or Senior Project - Research
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or HLTH 4463	or Senior Project - Internship
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General Education Requirement	3
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Units 14

Total Units 120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

3

Units in excess of total will be applied towards major's free electives.

Sociology

Concentration Not Yet Declared - BA in Sociology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

SOC 1101	Orientation to the Sociology Major	1
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SOC 1110	Comparative Societies (4B)	3
----------	----------------------------	---

Lower-Division Social Sciences Elective	3	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
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General Education Requirement	3
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Units16

Term 2

SOC 1111Social Problems3

SOC 2216U.S. Race and Ethnic Relations3

STAT 1110Applied Statistical Concepts and Methods (2)13

General Education Requirement (1B)3

General Education Requirement3

Units15

Second Year

Term 1

SOC 2218	International Political Economy	3
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SOC/WGQS 3311	Sociology of Gender and Sexuality	4
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SOC Elective ²		3-4
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General Education Requirement		3
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General Education Requirement		3
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Units		16
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Term 2

SOC 2222 Classical Social Theory 3

Concentration Course 3-4

General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Units 14

Third Year

Term 1

SOC 3315	Global Race and Ethnic Relations	1	3
or SOC 3321	or Migration		
or SOC 3326	or Sociology of the Life Course		
or SOC 3343	or Sociology of the Global South		
SOC 3353	Research Methods for Sociology		4
Concentration Course			3-4
General Education Requirement			3
Free Elective		4	3
Units			17

Term 2

SOC 4464

Professional Development for Sociologists

1

SOC Elective²

3-4

Concentration Course

3-4

General Education Requirement

3

Free Elective⁴

3

Units

14

Fourth Year

Term 1

SOC 4461 Senior Project I 2

SOC Elective² 3-4

Concentration Course 3-4

Free Elective⁴ 3

Free Elective⁴ 3

Units 15

Term 2

SOC 4462

Senior Project II

1

SOC Elective²

3-4

SOC Elective²

0-3

Concentration Course

0-4

Free Elective⁴

3

Free Elective⁴

3

Units

13

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

12 units must be at the 3000-4000 level.

3

Courses from the following disciplines will be acceptable towards the lower-division Social Sciences course: Anthropology, Ethnic Studies, Geography, Sociology, Political Science, Psychology, or Women's, Gender, and Queer Studies.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Criminal Justice

Criminal Justice Concentration - BA in Sociology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
SOC 1101	Orientation to the Sociology Major	1
SOC 1110	Comparative Societies (4B) ¹	3
Lower-Division Social Sciences Elective ³		3
General Education Requirement (1A)		3

General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	16
-------	----

Term 2

SOC 1111	Social Problems	3
----------	-----------------	---

SOC 2216	U.S. Race and Ethnic Relations	3
----------	--------------------------------	---

STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
-----------	---	---

General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement

3

Units

15

Second Year

Term 1

SOC 2218

International Political Economy

3

SOC 3302

Criminology

4

SOC/WGQS 3311

Sociology of Gender and Sexuality

4

General Education Requirement

3

General Education Requirement	3
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Units	17
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Term 2	
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SOC 2222	Classical Social Theory	3
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SOC Elective	3-4
--------------	-----

General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

General Education Requirement	1
-------------------------------	---

Third Year

Term 1

SOC 3315	Global Race and Ethnic Relations	1	3
or SOC 3321	or Migration		
or SOC 3326	or Sociology of the Life Course		
or SOC 3343	or Sociology of the Global South		
SOC 3353	Research Methods for Sociology		4
SOC 4402	Gender, Crime, and Violence		4

or SOC 4412 or Contemporary Issues in Criminal Justice

General Education Requirement 3

Free Elective⁴ 3

Units 17

Term 2

SOC 4464 Professional Development for Sociologists 1

SOC Elective² 3-4

Concentration Course 3-4

General Education Requirement

3

Free Elective⁴

3

Units

14

Fourth Year

Term 1

SOC 4461

Senior Project I

2

SOC Elective²

3-4

Concentration Course

3-4

Free Elective⁴

3

Free Elective⁴

3

Units

15

Term 2

SOC 4462

Senior Project II

1

SOC Elective²

3-4

SOC Elective²

0-3

Concentration Course	0-2
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Free Elective ⁴	3
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Free Elective ⁴	3
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Units	13
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

12 units must be at the 3000-4000 level.

3

Courses from the following disciplines will be acceptable towards the lower-division Social Sciences course: Anthropology, Ethnic Studies, Geography, Sociology, Political Science, Psychology, or Women's, Gender, and Queer Studies.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Organizations

Organizations Concentration - BA in Sociology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
SOC 1101	Orientation to the Sociology Major	1
SOC 1110	Comparative Societies (4B) ¹	3
Lower-Division Social Sciences Elective ³		3
General Education Requirement (1A)		3
General Education Requirement (1C)		3
General Education Requirement		3
Units		16

Term 2

SOC 1111	Social Problems	3
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SOC 2216	U.S. Race and Ethnic Relations	3
----------	--------------------------------	---

STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
-----------	---	---

General Education Requirement (1B)	3
------------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	15
-------	----

Second Year

Term 1

SOC 2218	International Political Economy	3
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SOC/WGQS 3311	Sociology of Gender and Sexuality	4
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SOC Elective ²		3-4
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General Education Requirement		3
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General Education Requirement		3
-------------------------------	--	---

Units		16
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Term 2

SOC 2222	Classical Social Theory	3
SOC Elective ²		3-4
General Education Requirement		3
General Education Requirement		3
General Education Requirement		1
Units		13

Third Year
Term 1

SOC 3310	Self, Organizations, and Society	3-4
or COMS 3320	or Intergroup Communication	
or PSY 2252	or Social Psychology	
SOC 3315	Global Race and Ethnic Relations ¹	3
or SOC 3321	or Migration	
or SOC 3326	or Sociology of the Life Course	
or SOC 3343	or Sociology of the Global South	
SOC 3353	Research Methods for Sociology	4
General Education Requirement		3

Free Elective	4	3
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Units	17
-------	----

Term 2		
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SOC 3395	Sociology of Complex Organizations	4
----------	------------------------------------	---

or SOC 4423	or Gender and Work
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SOC 4464	Professional Development for Sociologists	1
----------	---	---

Concentration Course	3-4
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General Education Requirement	3
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Free Elective	4	3
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Units	15
-------	----

Fourth Year

Term 1

SOC 4461	Senior Project I	2
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SOC Elective	2	3-4
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Concentration Course	3-4
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Free Elective	4	3
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Free Elective⁴

3

Units

15

Term 2

SOC 4462

Senior Project II

1

SOC Elective²

3-4

SOC Elective²

0-3

Concentration Course

0-3

Free Elective ⁴	3
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Free Elective ⁴	3
----------------------------	---

Units	13
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

12 units must be at the 3000-4000 level.

3

Courses from the following disciplines will be acceptable towards the lower-division Social Sciences course: Anthropology, Ethnic Studies, Geography, Sociology, Political Science, Psychology, or Women's, Gender, and Queer Studies.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

At least one concentration course must be in Sociology. A minimum of 16 units is required for the concentration, at least 10 of which must be upper-division. Units in excess of total will be applied towards major's free electives.

Social Justice

Social Justice Concentration - BA in Sociology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

SOC 1101	Orientation to the Sociology Major	1
----------	------------------------------------	---

SOC 1110	Comparative Societies (4B)	3
----------	----------------------------	---

Lower-Division Social Sciences Elective	3	3
---	---	---

General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement	3
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Units16

Term 2

SOC 1111Social Problems3

SOC 2216U.S. Race and Ethnic Relations3

STAT 1110Applied Statistical Concepts and Methods (2)13

General Education Requirement (1B)3

General Education Requirement3

Units15

Second Year

Term 1

SOC 2218	International Political Economy	3
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SOC/WGQS 3311	Sociology of Gender and Sexuality	4
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SOC Elective ²		3-4
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General Education Requirement		3
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General Education Requirement		3
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Units		16
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Term 2

SOC 2222	Classical Social Theory	3
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Concentration Course	5	3-4
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

General Education Requirement	1
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Units	14
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Third Year

Term 1

SOC 3315

Global Race and Ethnic Relations¹

3

or SOC 3321

or Migration

or SOC 3326

or Sociology of the Life Course

or SOC 3343

or Sociology of the Global South

SOC 3353

Research Methods for Sociology

4

Concentration Course⁵

3-4

General Education Requirement

3

Free Elective	4	3
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Units	17
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Term 2		
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SOC 4464	Professional Development for Sociologists	1
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SOC Elective	2	3-4
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Concentration Course	5	3-4
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General Education Requirement		3
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Free Elective⁴

3

Units

14

Fourth Year

Term 1

SOC 4461

Senior Project I

2

SOC Elective²

3-4

Concentration Course⁵

3-4

Free Elective⁴

3

Free Elective⁴

3

Units

15

Term 2

SOC 4462

Senior Project II

1

SOC Elective²

3-4

SOC Elective²

0-3

Concentration Course	5	0-4
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Free Elective	4	3
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Free Elective	4	3
---------------	---	---

Units	13
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

12 units must be at the 3000-4000 level.

3

Courses from the following disciplines will be acceptable towards the lower-division Social Sciences course: Anthropology, Ethnic Studies, Geography, Sociology, Political Science, Psychology, or Women's, Gender, and Queer Studies.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

5

A minimum of 16 units is required for the concentration. Any additional units may be applied towards major's free electives.

Social Services

Social Services Concentration - BA in Sociology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

SOC 1101	Orientation to the Sociology Major	1
SOC 1110	Comparative Societies (4B) ¹	3
Lower-Division Social Sciences Elective ³		3
General Education Requirement (1A)		3
General Education Requirement (1C)		3

General Education Requirement	3
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Units	16
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Term 2

SOC 1111	Social Problems	3
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SOC 2216	U.S. Race and Ethnic Relations	3
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STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1B)	3
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General Education Requirement	3
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Units

15

Second Year

Term 1

SOC 2218	International Political Economy	3
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SOC/WGQS 3311	Sociology of Gender and Sexuality	4
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SOC Elective ²		3-4
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General Education Requirement		3
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General Education Requirement		3
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Units

16

Term 2

SOC 2222

Classical Social Theory

3

SOC Elective²

3-4

General Education Requirement

3

General Education Requirement

3

General Education Requirement

1

Units

13

Third Year

Term 1

SOC 3303	Social Work, Social Advocacy, and Social Service Agencies	4
SOC 3315	Global Race and Ethnic Relations ¹	3
or SOC 3321	or Migration	
or SOC 3326	or Sociology of the Life Course	
or SOC 3343	or Sociology of the Global South	
SOC 3353	Research Methods for Sociology	4
General Education Requirement		3

Free Elective	4	3
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Units	17
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Term 2

SOC 4414	Theories of Social Work in Counseling Agencies	4
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SOC 4464	Professional Development for Sociologists	1
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Concentration Course	3-4
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Concentration Course	3-4
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Free Elective	4	3
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Units

16

Fourth Year

Term 1

SOC 4461

Senior Project I

2

SOC Elective²

3-4

General Education Requirement

3

Free Elective⁴

3

Free Elective⁴

3

Units

14

Term 2

SOC 4462

Senior Project II

1

SOC Elective²

3-4

SOC Elective²

0-3

Concentration Course

0-2

Free Elective⁴

3

Free Elective	4	3
Units		13
Total Units		120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

12 units must be at the 3000-4000 level.

3

Courses from the following disciplines will be acceptable towards the lower-division Social Sciences course: Anthropology, Ethnic Studies, Geography, Sociology, Political Science, Psychology, or Women's, Gender, and Queer Studies.

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Individualized Course of Study

Individualized Course of Study - BA in Sociology

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
SOC 1101	Orientation to the Sociology Major	1

SOC 1110	Comparative Societies (4B)	¹	3
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Lower-Division Social Sciences Elective	³	3
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement	3
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Units	16
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Term 2

SOC 1111	Social Problems	3
----------	-----------------	---

SOC 2216	U.S. Race and Ethnic Relations	3
----------	--------------------------------	---

STAT 1110	Applied Statistical Concepts and Methods (2) ¹	3
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General Education Requirement (1B)		3
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General Education Requirement		3
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Units		15
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Second Year

Term 1

SOC 2218	International Political Economy	3
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SOC/WGQS 3311

Sociology of Gender and Sexuality

4

SOC Elective²

3-4

General Education Requirement

3

General Education Requirement

3

Units

16

Term 2

SOC 2222

Classical Social Theory

3

Individual Course of Study Elective

3-4

General Education Requirement 3

General Education Requirement 3

General Education Requirement 1

Units 14

Third Year

Term 1

SOC 3315 Global Race and Ethnic Relations 1 3

or SOC 3321

or Migration

or SOC 3326

or Sociology of the Life Course

or SOC 3343

or Sociology of the Global South

SOC 3353

Research Methods for Sociology

4

Individual Course of Study Elective

3-4

General Education Requirement

3

Free Elective⁴

3

Units

17

Term 2

SOC 4464

Professional Development for Sociologists

1

SOC Elective²

3-4

Individual Course of Study Elective

3-4

General Education Requirement

3

Free Elective⁴

3

Units

14

Fourth Year

Term 1

SOC 4461

Senior Project I

2

SOC Elective	2	3-4
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Individual Course of Study Elective		3-4
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Free Elective	4	3
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Free Elective	4	3
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Units		15
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Term 2		
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SOC 4462	Senior Project II	1
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SOC Elective	2	3-4
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SOC Elective	2	0-3
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Individual Course of Study Elective		0-4
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Free Elective	4	3
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Free Elective	4	3
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Units		13
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Total Units		120
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Required in Major or Support; also satisfies General Education (GE) requirement.

2

12 units must be at the 3000-4000 level.

3

Courses from the following disciplines will be acceptable towards the lower-division Social Sciences course: Anthropology, Ethnic Studies, Geography, Sociology, Political Science, Psychology, or Women's, Gender, and Queer Studies.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Software Engineering

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1		Units
CSC/CPE 1000	Computing Majors Orientation	1
CSC/CPE 1024	Introduction to Computing	2
MATH 1261	Calculus I (2) ³	4
or MATH 1264	or Calculus for Data Science I	
Life Science Support Elective (5B) ³		4
General Education Requirement (1A)		3
Units		14

Term 2

CSC 1001 Fundamentals of Computer Science 4

& 1001L and Fundamentals of Computer Science Laboratory

MATH 1262 Calculus II 4

or MATH 1265 or Calculus for Data Science II

PHYS 1141 General Physics I (5A & 5C)³ 4

or CHEM 1120 or Fundamentals of Chemical Structure and Properties

General Education Requirement (1C) 3

Units 15

Second Year

Term 1

CSC 2001	Data Structures	4
& 2001L	and Data Structures Laboratory	
MATH 1151	Linear Algebra	3
PSY 2201	Introductory Psychology (4B)	3
General Education Requirement (1B)		3
Units		13

Term 2

CSC/CPE 2050	System Software Mechanics	3
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COMS 2217	Small Group Collaboration and Creativity ⁴	3-4
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or PSY 3350	or Teamwork	
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MATH 2031	Transition to Advanced Mathematics	3
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General Education Requirement (3A)		3
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General Education Requirement (4A)		3
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Units		15
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Third Year

Term 1

CSC 3001	Modern Application Development	4
CSC 3100	Software Engineering	4
CSC/CPE 3201	Introduction to Computer Security	3
General Education Requirement (3B)		3
General Education Requirement (6)		3
Units		17

Term 2

CSC 3449	Algorithms and Complexity	4
----------	---------------------------	---

CSC 3660	Introduction to Databases ¹	2-4
or CSC 3665	or Introduction to Database Management Systems	
STAT 3210	Engineering Statistics (Upper-Division 2/5) ³	3
Software Engineering Elective ²		3-4
General Education Requirement (Upper-Division 4)		3
Units		15

Fourth Year

Term 1

CSC 3300	Programming Languages	3
CSC 4160	Software Engineering Capstone I	4
PHIL 3323	Ethics, Science, and Technology (Upper-Division 3)	3
WGQS 3350	Gender, Race, Culture, Science, and Technology	4
or WGQS 3351	or Gender, Race, Class, Nation: Critical Computing and Engineering Studies	
Mathematics/Statistics/Science Elective		3
Units		17

Term 2

CSC 4161	Senior Project - Software Engineering Capstone II	4
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Mathematics/Statistics/Science Elective	3
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Technical Elective ²	3
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Technical Elective ²	0-4
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Units	14
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Total Units	120
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1

CSC 3665 will satisfy this requirement and 2 units of Technical Electives.

2

A combined maximum of 6 units may be taken from [CPE 4491](#), [CPE 4492](#), [CPE 4493](#), [CSC 4091](#), [CSC 4092](#), [CSC 4093](#), [CSC 4191](#), [CSC 4192](#), [CSC 4193](#), [CSC 4291](#), [CSC 4292](#), [CSC 4293](#), [CSC 4400](#), [CSC 4495](#), [CSC 4691](#), [CSC 4692](#), [CSC 4693](#), [CSC 4791](#), [CSC 4792](#), [CSC 4793](#), [CSC 4891](#), [CSC 4892](#), [CSC 4893](#), [CSC 4991](#), [CSC 4992](#), [CSC 4993](#), [DATA 4720](#), [ENGR 2995](#), and [ENGR 4995](#); of which up to a combined 4 units may be taken from [CPE 4491](#), [CSC 4091](#), [CSC 4191](#), [CSC 4291](#), [CSC 4400](#), [CSC 4495](#), [CSC 4691](#), [CSC 4791](#), [CSC 4891](#), [CSC 4991](#), and [DATA 4720](#).

3

Required in Major or Support; also satisfies General Education (GE) requirement.

4

[PSY 3350](#) will satisfy this requirement and 1 unit of Technical Electives.

Spanish

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

SPAN 2201	Intermediate Spanish I (3B) ¹	4
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General Education Requirement (1A)	3
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General Education Requirement (1C)	3
------------------------------------	---

General Education Requirement	3
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Units	13
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Term 2

SPAN 2202	Intermediate Spanish II	3-4
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or SPAN 2206	or Spanish for Heritage Speakers
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General Education Requirement (1B)	3
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General Education Requirement (2)

3

Free Elective⁴

3-4

Free Elective⁴

3

Units

16

Second Year

Term 1

SPAN 2207

Introduction to Spanish Linguistics

3

SPAN 2233

Introduction to Hispanic Literature

3

General Education Requirement

3

General Education Requirement

3

Free Elective⁴

3

Units

15

Term 2

SPAN 3301

Advanced Writing in Spanish

3

SPAN 3300 Elective²

3

General Education Requirement

3

General Education Requirement

1

General Education Requirement

3

Free Elective⁴

3

Units

16

Third Year

Term 1

SPAN 3302

Spanish Conversation, Composition, and Advocacy

3

SPAN 3300 Elective²

3

SPAN 3300 Elective	2	3
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General Education Requirement		3
-------------------------------	--	---

Free Elective	4	3
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Units		15
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Term 2

WLC 3360	Research Methods in World Languages and Cultures	3
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SPAN 3300 Elective	2	3
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Approved Language/Culture Elective 3300-4400^{2,3}

3

General Education Requirement

3

Free Elective⁴

3-4

Units

15

Fourth Year

Term 1

SPAN 4400 Elective²

3

Approved Language/Culture Elective 3300-4400	2, 3	3
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General Education Requirement		3
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Free Elective	4	3
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Free Elective	4	3
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Units		15
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Term 2		
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WLC 4460	Senior Project	3
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SPAN 4400 Elective ²	3
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Approved Language/Culture Elective 3300-4400 ^{2,3}	3
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Free Elective ⁴	3
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Free Elective ⁴	3
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Units	15
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Total Units	120
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¹ Required in Major or Support; also satisfies General Education (GE) requirement. ²
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Credit may be earned more than once when course is taken with different subtitles or topics.

3

Courses from this list may not be double-counted in the major. Repeatable courses may be applied when taken with different subtopics.

4

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Statistics

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

DATA/STAT 1000

Statistical and Data Literacy (2) ¹

3

STAT 1510	Statistics I	3
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MATH/DATA 1264	Calculus for Data Science I	4
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		16
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Term 2		
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DATA/STAT 1810	Introduction to Statistical Computing with R	3
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STAT 3520	Statistics II	3
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MATH/DATA 1265	Calculus for Data Science II	4
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General Education Requirement (1B)	3
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General Education Requirement	3
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General Education Requirement	1
-------------------------------	---

Units	17
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Second Year	
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Term 1	
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STAT 3540	Statistical Methods for Study Design and Analysis	4
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DATA/STAT 3820	Intermediate Statistical Computing with R	3
----------------	---	---

MATH 1151	Linear Algebra	3
-----------	----------------	---

General Education Requirement	3
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General Education Requirement	3
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Units	16
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Term 2

STAT 2610	Introduction to Probability and Simulation	3
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STAT 3530	Applied Linear Models	4
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CSC 1001	Fundamentals of Computer Science	4
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& 1001L	and Fundamentals of Computer Science Laboratory	
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General Education Requirement	3
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General Education Requirement

3

Units

17

Third Year

Term 1

DATA/STAT 3800

Introduction to Statistical Computing with SAS and SQL

3-4

or DATA 3301

or Introduction to Data Science

STAT 4610

Probability Theory

3

STAT Electives List A

3

General Education Requirement

3

General Education Requirement	3
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Units	15
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Term 2	
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STAT 4366	Statistical Communication, Collaboration, and Consulting	5
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STAT 4620	Statistical Theory	3
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STAT Elective List B	3
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General Education Requirement	3
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Units	14
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Fourth Year

Term 1

STAT 4460	Senior Project: Statistics Capstone	2
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STAT Elective List A		3
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STAT Elective List A		3
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General Education Requirement		3
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Free Elective ²		1-2
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Units		13
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Term 2

STAT Elective List B	3
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STAT Elective List B	3
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General Education Requirement	3
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Free Elective ²	3
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Units	12
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Total Units	120
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1

Required in Major or Support: also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Theatre Arts

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
TH 1145	Mainstage Rehearsal and Performance I	3
or TH 1155	or Mainstage Production I	
TH 2201	First Year Theatre Experience	3
TH 2227	Theatre History I (3A) ¹	3

or TH 2228 or Theatre History II

TH 2290 Dramaturgy and Playwriting 3

General Education Requirement (1A) 3

General Education Requirement (1C) 3

Units **18**

Term 2

TH 2220 Acting I 3

TH 2225 Costume Construction I 3

TH 2295 Foundations in Theatrical Design 3

General Education Requirement (1B)	3
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General Education Requirement (2)	3
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Units	15
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Second Year	
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Term 1	
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TH 2230	Stagecraft I	3
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Lower-Division Major Elective	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	1
-------------------------------	---

General Education Requirement	3
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Units	16
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Term 2	
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Lower-Division Major Elective	3
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Lower-Division Major Elective	3
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General Education Requirement	3
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General Education Requirement	3
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General Education Requirement	3
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Units

15

Third Year

Term 1

TH 3305	Diversity in U.S. Theatre (Upper-Division 3, USCP, GWR)	3
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TH 3325	Costume Construction II	3
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or TH 3330	or Stagecraft II
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TH 4432	Costume Design	3
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or TH 4430	or Scenic Design
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Upper-Division Major Elective		3
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General Education Requirement 3

Units 15

Term 2

ENGL 3339 Introduction to Shakespeare 3

Upper-Division Major Elective 3

Upper-Division Major Elective 3

General Education Requirement 3

Free Elective² 2-3

Units

14

Fourth Year

Term 1

TH 4450

Directing

3

Upper-Division Major Elective

3

Upper-Division Major Elective

3

Free Elective²

3

Free Elective²

3

Free Elective	2	3
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Units	18
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Term 2		
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TH 4461	Senior Project - Seminar	3
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Upper-Division Major Elective	3
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Free Elective	2	3
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Units	9
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Wine and Viticulture

Concentration Not Yet Declared - BS in Wine and Viticulture

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

WVIT 1102

Global Wine and Viticulture

3

CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
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SS 1120	Introductory Soil Science	4
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		17
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Term 2

WVIT 2202	Fundamentals of Enology	3
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BOT 1121	General Botany (5B & 5C) ¹	4
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MATH 1261	Calculus I (2) ¹	3-4
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or MATH 1267	or Business Calculus	
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STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1B)		3
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Units		16
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Second Year

Term 1

WVIT 2233	Basic Viticulture	3
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ECON 2040	Macroeconomics (4B)	3
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Concentration Course		3
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General Education Requirement		3
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General Education Requirement		3
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Units		15
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Term 2

WVIT 3343	Branded Wine Marketing	3
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AGB 2214	Agribusiness Financial Accounting	3
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or BUS 2214	or Financial Accounting	
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Concentration Course	3
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General Education Requirement	3
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General Education Requirement	3
-------------------------------	---

Units	15
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Third Year

Term 1

WVIT 3331	Advanced Viticulture - Fall	4
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BRAE 3340	Irrigation Water Management (Upper-Division 2/5) ¹	3
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Concentration Course	3
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Concentration Course	3
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General Education Requirement	3
-------------------------------	---

Units	16
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Term 2

WVIT 3339	Internship in Wine and Viticulture	3
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Concentration Course	3
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Concentration Course	3
----------------------	---

Concentration Course	3
----------------------	---

General Education Requirement	3
-------------------------------	---

Units	15
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Fourth Year

Term 1

WVIT 4442	Sensory Evaluation of Wine	3
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Concentration Course	3
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Concentration Course	3
----------------------	---

Free Elective

2

0-3

Units

12

Term 2

WVIT 4423

Wine Law and Compliance

3

WVIT 4463

Issues, Trends, and Careers in the Grape and Wine Industry

2

Concentration Course

3

Concentration Course

3

Concentration Course or Free Elective

2

2-3

Units	14
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Enology

Enology Concentration - BS in Wine and Viticulture

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

WVIT 1102	Global Wine and Viticulture	3
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CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4
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SS 1120	Introductory Soil Science	4
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
------------------------------------	--	---

Units		17
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Term 2

WVIT 2202	Fundamentals of Enology	3
-----------	-------------------------	---

BOT 1121	General Botany (5B & 5C) ¹	4
----------	---------------------------------------	---

MATH 1261	Calculus I (2) ¹	3-4
-----------	-----------------------------	-----

or MATH 1267

or Business Calculus

STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1B)		3
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Units		16
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Second Year

Term 1

WVIT 2233	Basic Viticulture	3
-----------	-------------------	---

ECON 2040	Macroeconomics (4B)	3
-----------	---------------------	---

CHEM 1122	Fundamentals of Chemical Reactivity	4
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General Education Requirement		3
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Units		13
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Term 2

WVIT 3343	Branded Wine Marketing	3
-----------	------------------------	---

AGB 2214	Agribusiness Financial Accounting	3
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or BUS 2214	or Financial Accounting	
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CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
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MCRO 2221	Introduction to Microbiology	4
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Units		14
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Third Year

Term 1

WVIT 3331	Advanced Viticulture - Fall	4
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WVIT 3339	Internship in Wine and Viticulture	3
-----------	------------------------------------	---

CHEM 3350	Biochemistry: Fundamentals and Applications	4
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General Education Requirement		3
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Units		14
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Term 2		
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WVIT/MCRO 3301	Wine Microbiology	3
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WVIT 3365	Wine Analysis and Amelioration	3
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WVIT 4423	Wine Law and Compliance	3
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General Education Requirement		3
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General Education Requirement

3

Units

15

Fourth Year

Term 1

WVIT 4404

Winemaking I

4

WVIT 4442

Sensory Evaluation of Wine

3

BRAE 3340

Irrigation Water Management (Upper-Division 2/5)¹

3

General Education Requirement

3

Free Elective²

0-3

Units

16

Term 2

WVIT 4405	Winemaking II	4
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WVIT 4463	Issues, Trends, and Careers in the Grape and Wine Industry	2
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WVIT 4464	Senior Project - Enology and Viticulture	3
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or WVIT 4465	or Senior Project - Research Experience in Enology or Viticulture	
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WVIT 4477	Advanced Wine Sensory Analysis	3
-----------	--------------------------------	---

General Education Requirement	3
-------------------------------	---

Units	15
-------	----

Total Units	120
-------------	-----

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Viticulture

Viticulture Concentration - BS in Wine and Viticulture

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year

Term 1

Units

WVIT 1102	Global Wine and Viticulture	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) 1	4
SS 1120	Introductory Soil Science	4
General Education Requirement (1A)		3
General Education Requirement (1C)		3

Units

17

Term 2

WVIT 2202	Fundamentals of Enology	3
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BOT 1121	General Botany (5B & 5C) ¹	4
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MATH 1261	Calculus I (2) ¹	3-4
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or MATH 1267

or Business Calculus

STAT 1110	Applied Statistical Concepts and Methods	3
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General Education Requirement (1B)		3
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Second Year

Term 1

WVIT 2233	Basic Viticulture	3
CHEM 2240	Organic Chemistry: Fundamentals and Applications	4
ECON 2040	Macroeconomics (4B)	3
General Education Requirement		3
General Education Requirement		3

Units

16

Term 2

WVIT 3343 Branded Wine Marketing 3

AGB 2214 Agribusiness Financial Accounting 3

or BUS 2214 or Financial Accounting

SS 2221 Soil Health and Plant Nutrition 4

General Education Requirement 3

General Education Requirement 3

Units

16

Third Year

Term 1

WVIT 3331	Advanced Viticulture - Fall	4
BRAE 3340	Irrigation Water Management (Upper-Division 2/5) ¹	3
PLSC 3323	Plant Pathology	3
General Education Requirement		3
Units		13

Term 2

WVIT 3302	Wine Fermentation Laboratory	2
WVIT 3332	Advanced Viticulture - Spring	4
WVIT 3339	Internship in Wine and Viticulture	3
PLSC 3313	Agricultural Entomology	3
PLSC 3321	Weed Biology and Management	4
Units		16

Fourth Year

Term 1

WVIT 4442	Sensory Evaluation of Wine	3
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WVIT 4414	Grape Pest Management	3
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General Education Requirement	3
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Free Elective ²	2-3
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Free Elective ²	0-3
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Units	15
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Term 2

WVIT 4423	Wine Law and Compliance	3
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WVIT 4428	Winegrape Vineyard Management	3
WVIT 4463	Issues, Trends, and Careers in the Grape and Wine Industry	2
WVIT 4464	Senior Project - Enology and Viticulture	3
or WVIT 4465	or Senior Project - Research Experience in Enology or Viticulture	
	Units	11
	Total Units	120

1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.

Wine Business

Wine Business Concentration - BS in Wine and Viticulture

Suggested Four-Year Flowcharts are planning guides and do not represent a required course sequence.

Courses may be completed in a different order, provided requisite requirements are satisfied.

First Year		
Term 1		Units
WWIT 1102	Global Wine and Viticulture	3
CHEM 1120	Fundamentals of Chemical Structure and Properties (5A) ¹	4

SS 1120	Introductory Soil Science	4
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General Education Requirement (1A)		3
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General Education Requirement (1C)		3
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Units		17
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Term 2		
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WVIT 2202	Fundamentals of Enology	3
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BOT 1121	General Botany (5B & 5C) ¹	4
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MATH 1261	Calculus I (2) ¹	3-4
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or MATH 1267 or Business Calculus

STAT 1110 Applied Statistical Concepts and Methods 3

General Education Requirement (1B) 3

Units 16

Second Year

Term 1

WVIT 2233 Basic Viticulture 3

AGB 2212 Agricultural Economics 3

or ECON 2030 or Microeconomics

ECON 2040	Macroeconomics (4B)	3
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General Education Requirement		3
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General Education Requirement		3
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Units		15
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Term 2		
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WVIT 3343	Branded Wine Marketing	3
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AGB 2214	Agribusiness Financial Accounting	3
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or BUS 2214	or Financial Accounting	
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AGB 2260	Agribusiness Data Literacy	3
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General Education Requirement	3
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General Education Requirement	3
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Units	15
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Third Year

Term 1

WVIT 3331	Advanced Viticulture - Fall	4
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AGB 3308	Introduction to Agribusiness Finance	3
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AGB 3323	Decision Making with Agribusiness Accounting Information	3
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BRAE 3340	Irrigation Water Management (Upper-Division 2/5)	1	3
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General Education Requirement			3
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Units			16
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Term 2

WVIT 3302	Wine Fermentation Laboratory		2
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WVIT 3339	Internship in Wine and Viticulture		3
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AGB 3369	Agricultural Personnel Management		3
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or BUS 3384	or Human Resources Management		
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EIM 3320	Strategic Event Planning	3
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General Education Requirement		3
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Units		14
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Fourth Year		
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Term 1		
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WVIT 4423	Wine Law and Compliance	3
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WVIT 4442	Sensory Evaluation of Wine	3
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WVIT 4460	Senior Project - Wine Business	3
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Free Elective

2

0-3

Units

12

Term 2

WVIT 3345	Wine Marketing Research and Market Analysis	3
WVIT 4433	Wine Industry Sales	4
WVIT 4447	Logistics for the Global Wine Industry	3
WVIT 4450	Wine Business Plan Development	3
WVIT 4463	Issues, Trends, and Careers in the Grape and Wine Industry	2

Units	15
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Total Units	120
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1

Required in Major or Support; also satisfies General Education (GE) requirement.

2

If a General Education (GE) course is used to satisfy a Major or Support requirement, additional units of Free Electives may be needed to complete the total units required for the degree.